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(12) **United States Patent**
Morioka et al.

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(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ELECTROPHOTOGRAPHIC IMAGE FORMING APPARATUS, CARTRIDGE AND DRUM UNIT**

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(21) Appl. No.: **17/407,213**

(22) Filed: **Aug. 20, 2021**

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. PCT/JP2020/012811, filed on Mar. 17, 2020.

(30) **Foreign Application Priority Data**

Mar. 18, 2019 (JP) JP2019-050355

(51) **Int. Cl.**
G03G 21/16 (2006.01)
G03G 21/18 (2006.01)

(52) **U.S. Cl.**
CPC **G03G 21/1814** (2013.01); **G03G 21/1857** (2013.01); **G03G 21/186** (2013.01)

(58) **Field of Classification Search**

CPC G03G 21/1814; G03G 21/1857; G03G 21/186; G03G 15/757; G03G 21/1864; G03G 2221/1657

See application file for complete search history.

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Primary Examiner — Hoang X Ngo

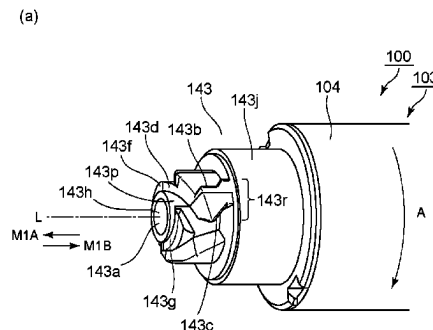
(74) *Attorney, Agent, or Firm* — Venable LLP

(57) **ABSTRACT**

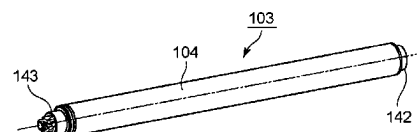
[Problem to be solved] To further develop conventional technique.

[Solution] a cartridge includes a casing, a photosensitive drum and a coupling. the coupling includes a driving force receiving portion for receiving a driving force by engagement with a driving force application member, a braking force receiving portion for receiving in braking force by engagement with a braking force application member, and a guide for moving the braking force application member relative to the driving force application member.

89 Claims, 110 Drawing Sheets



(b)



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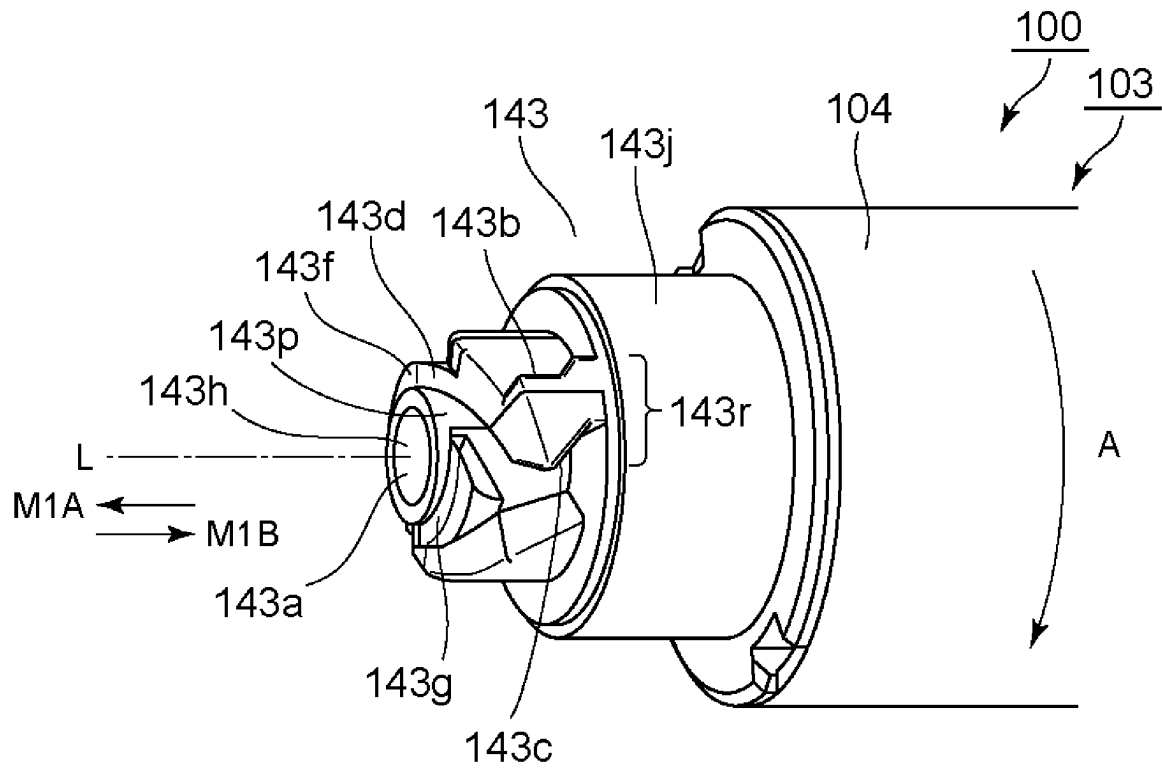
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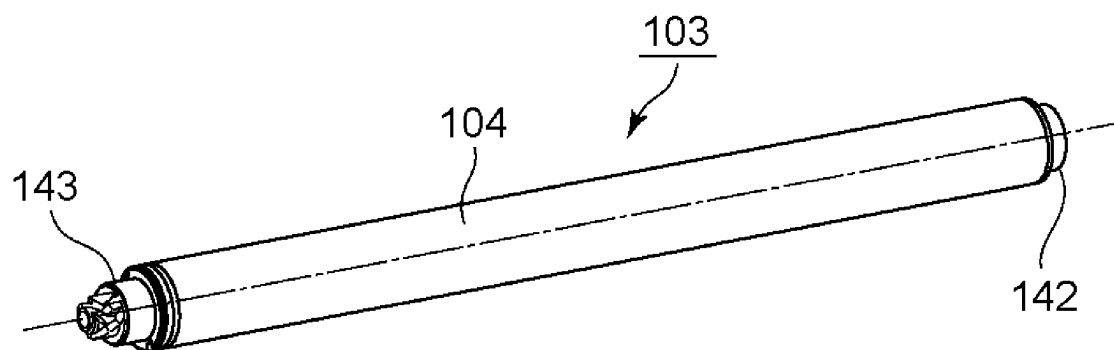


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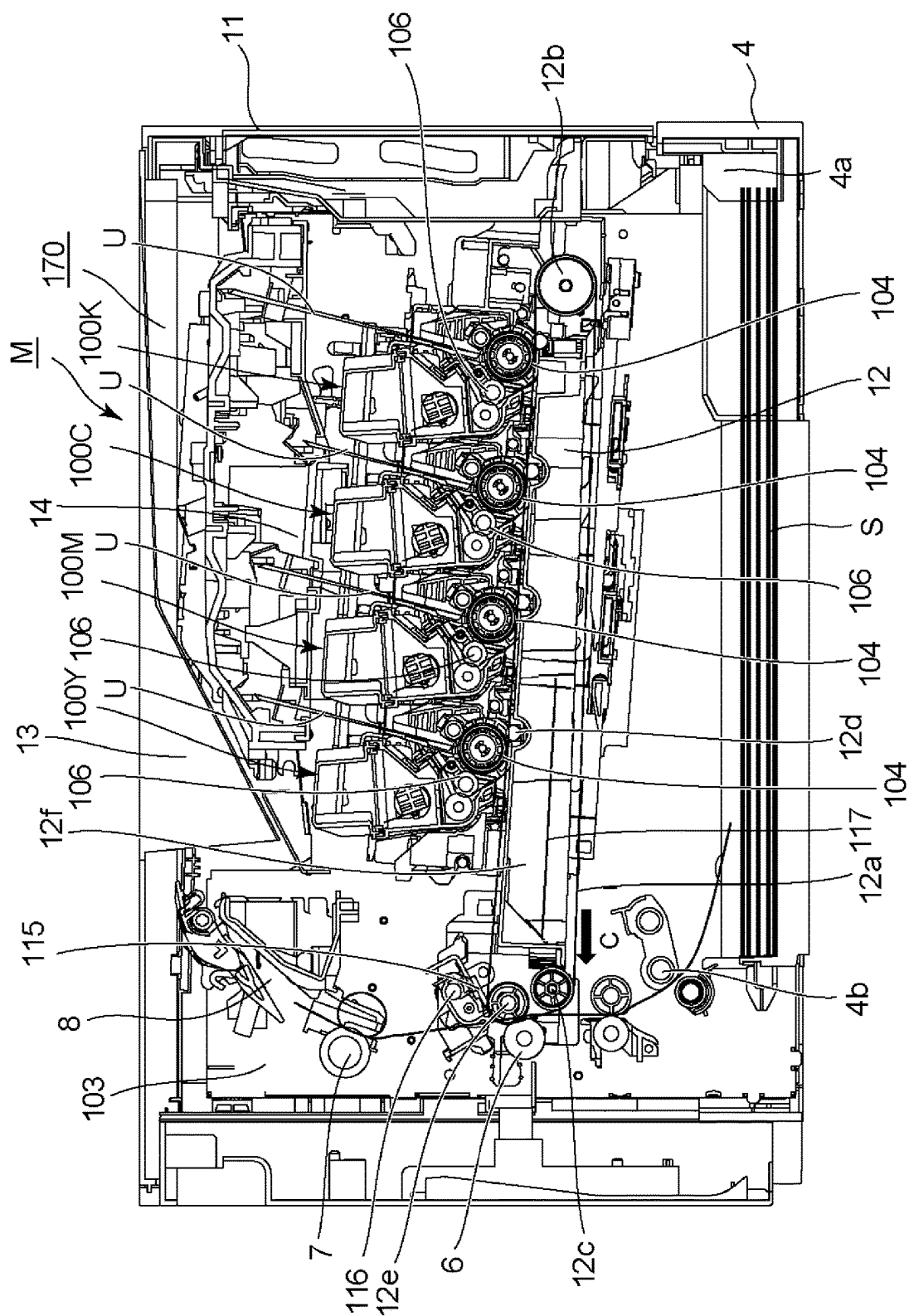


Fig. 2

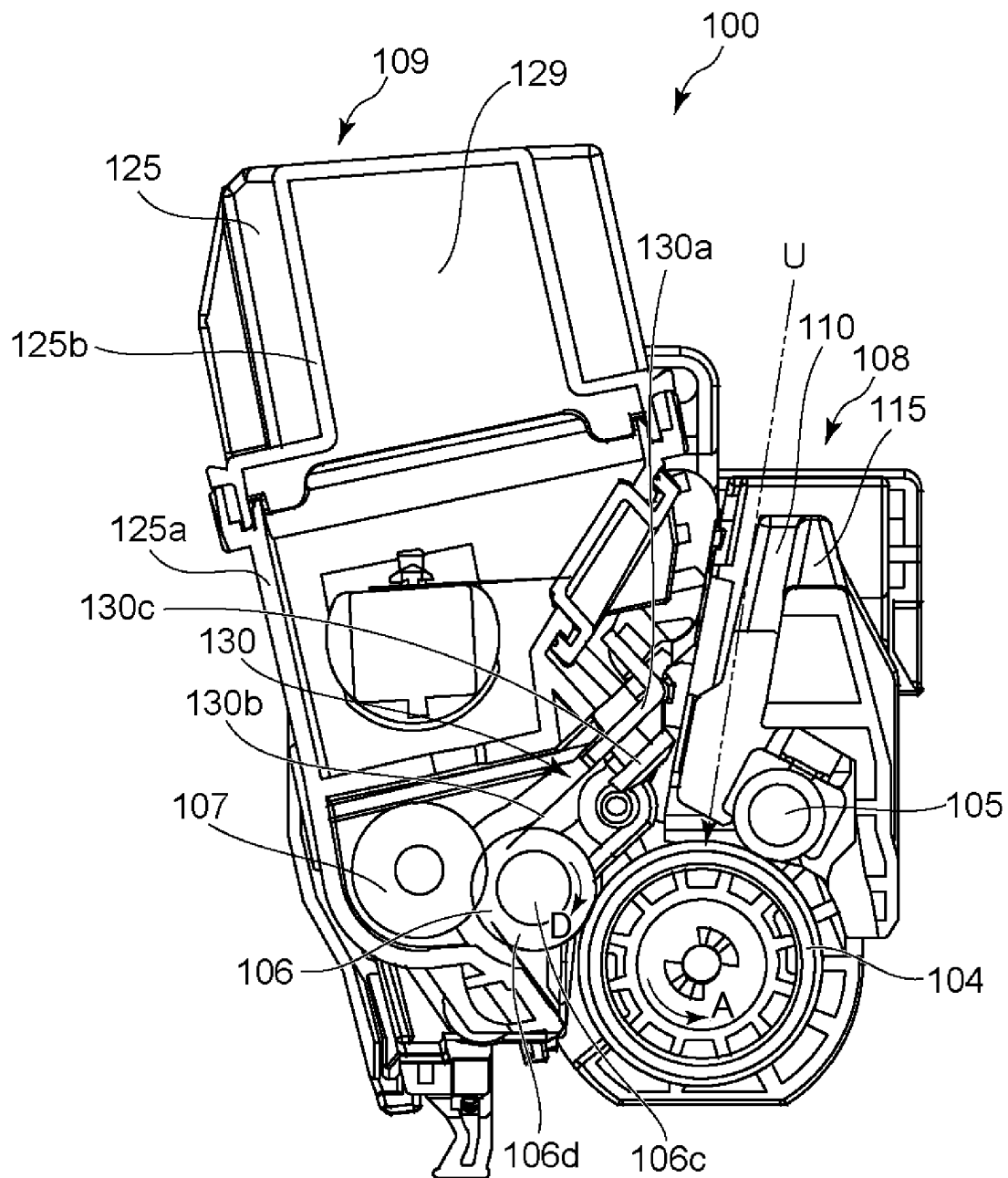


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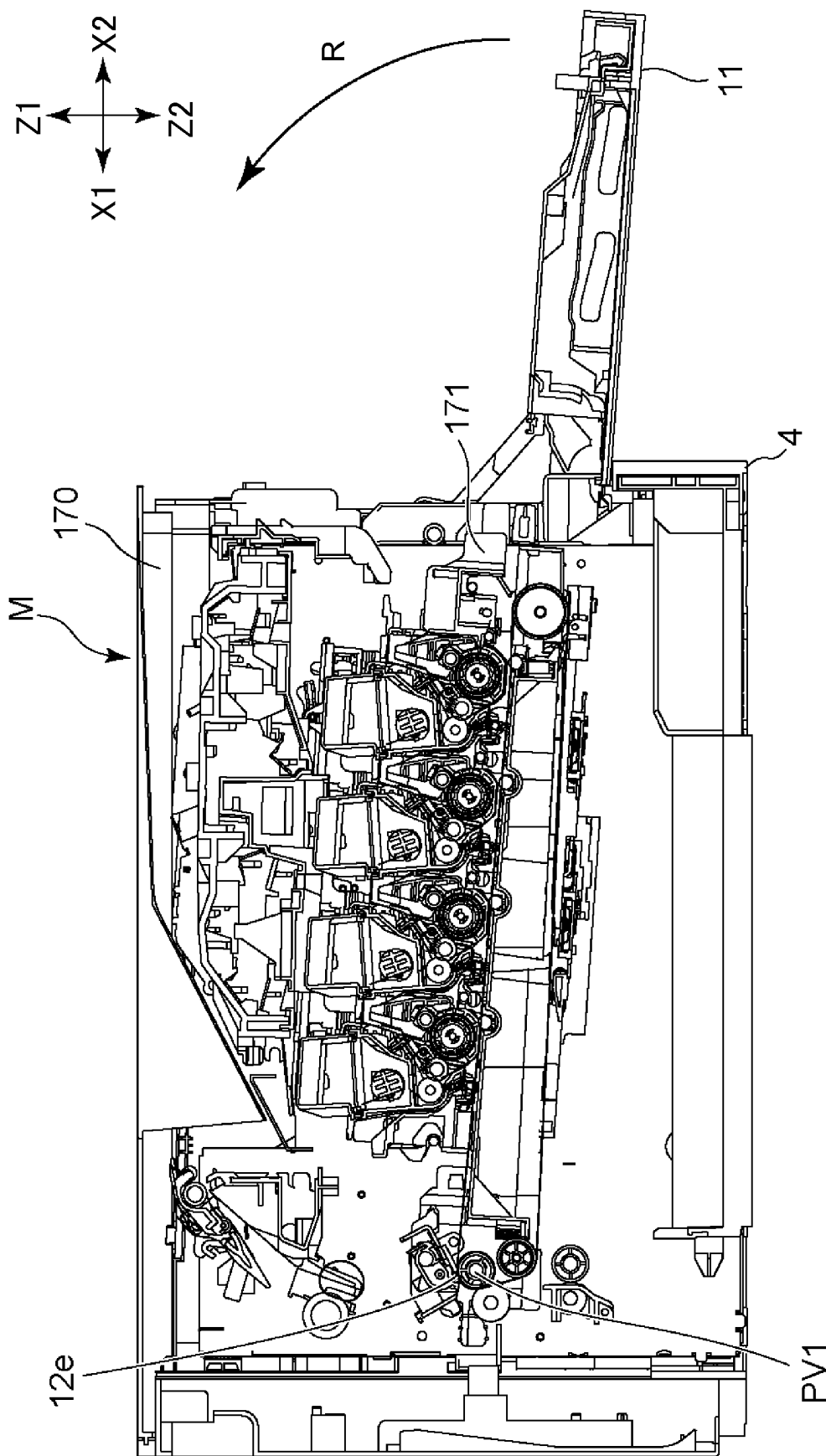


Fig. 4

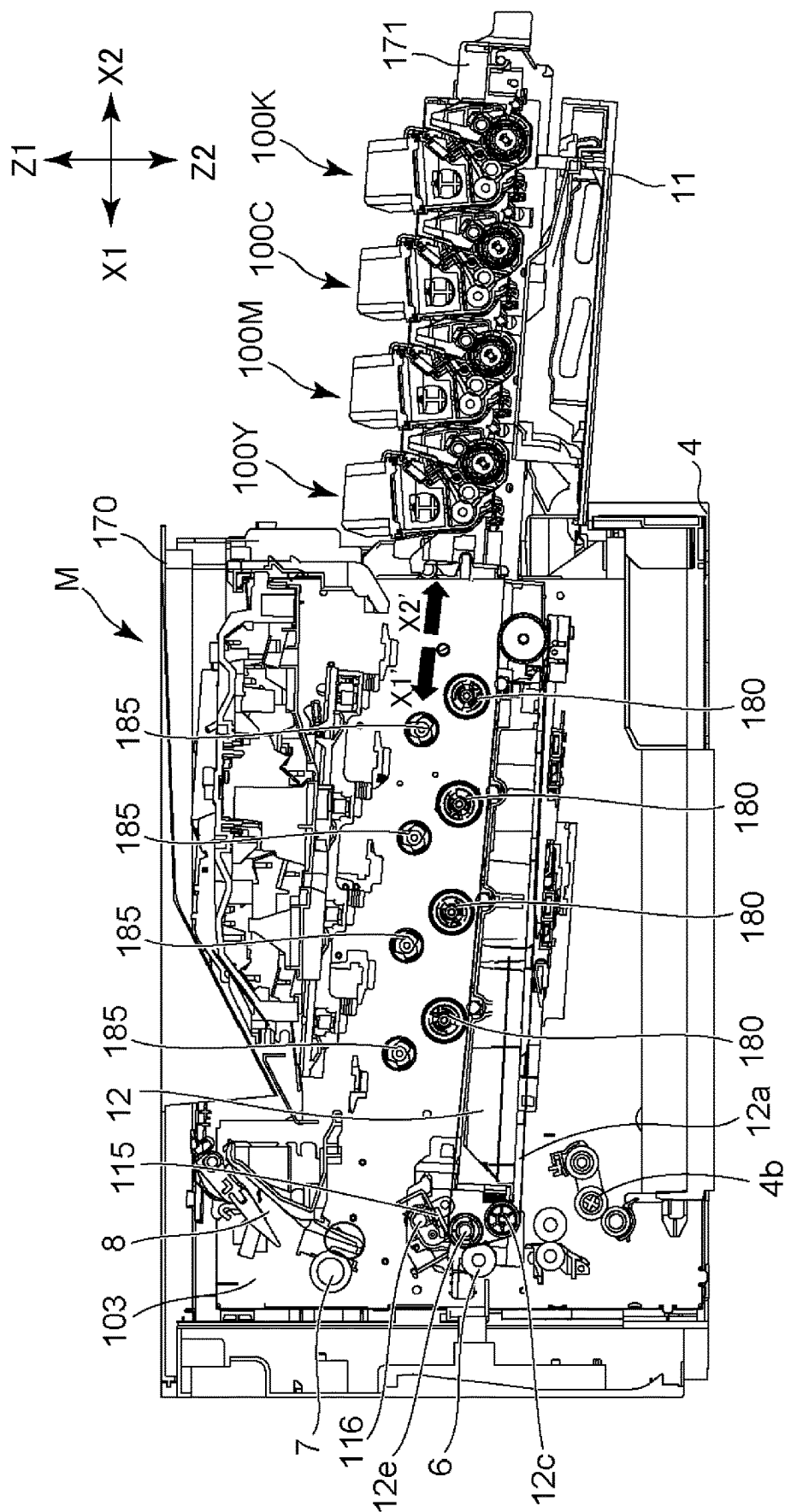


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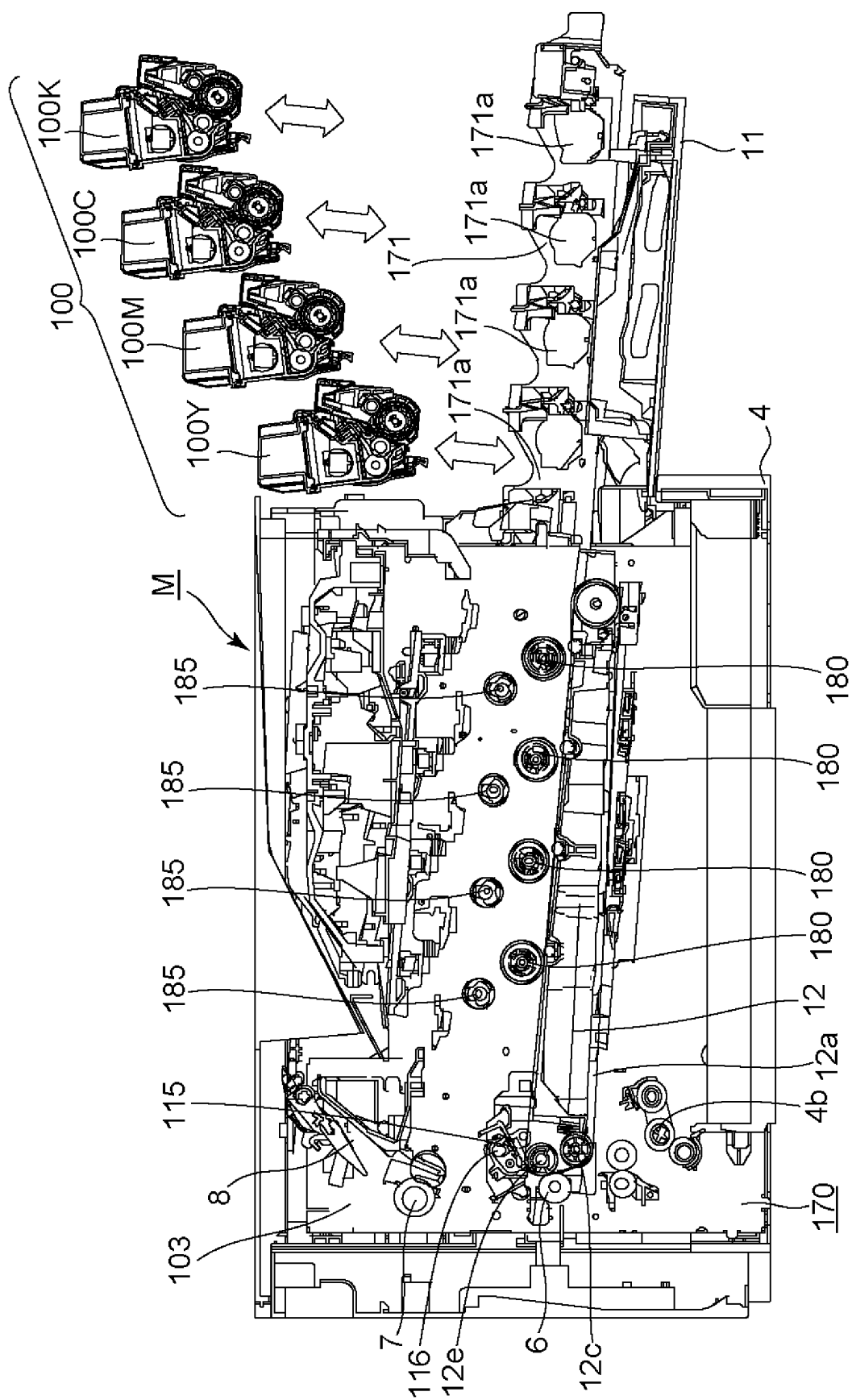


Fig. 6

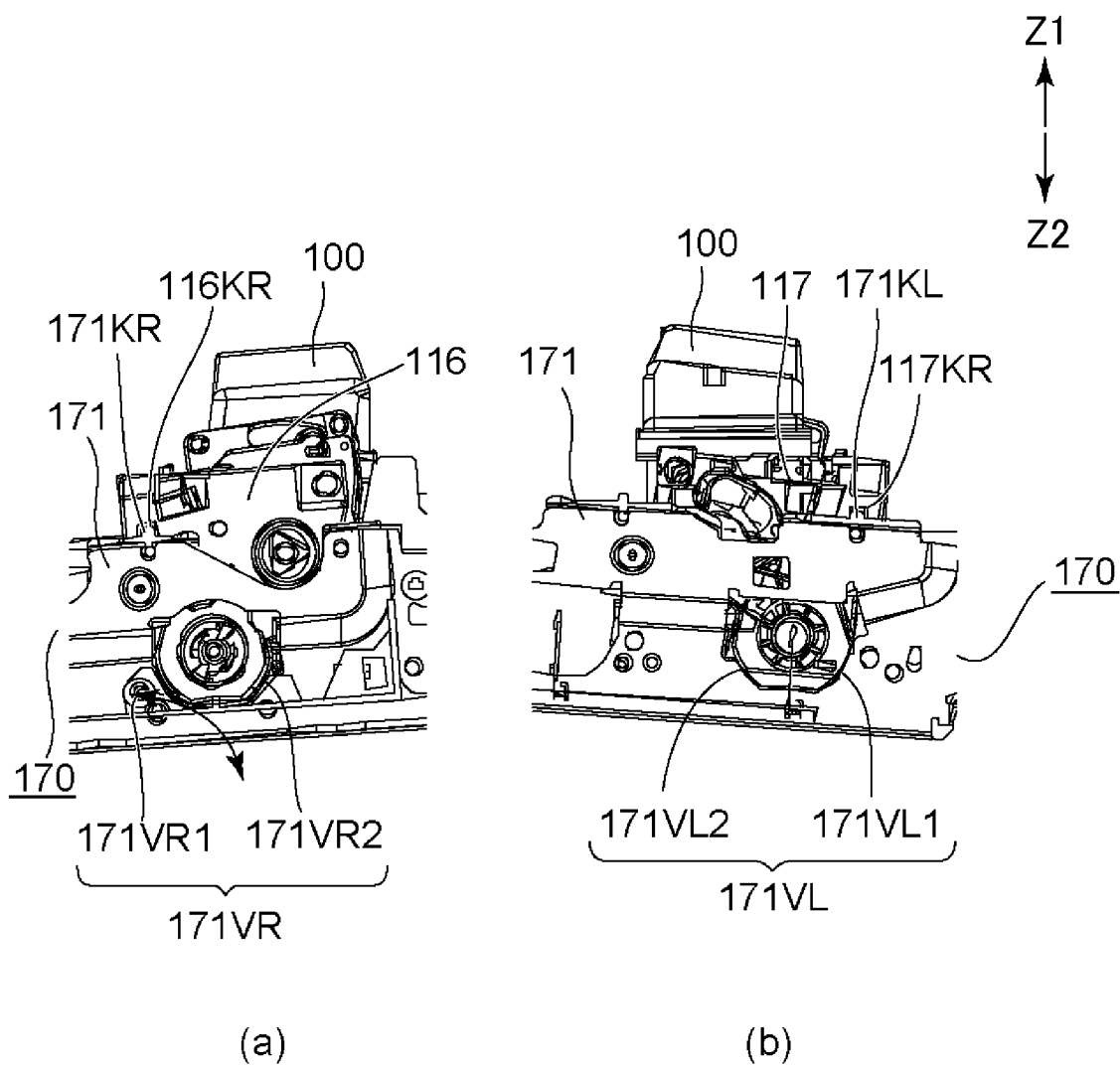
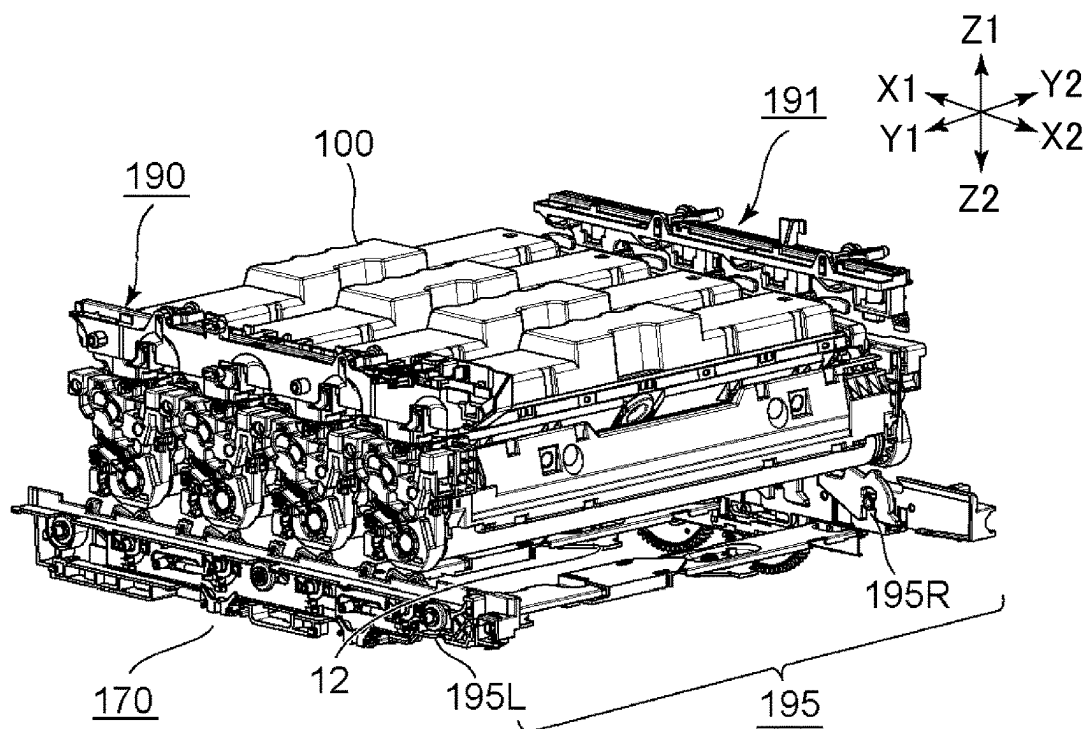
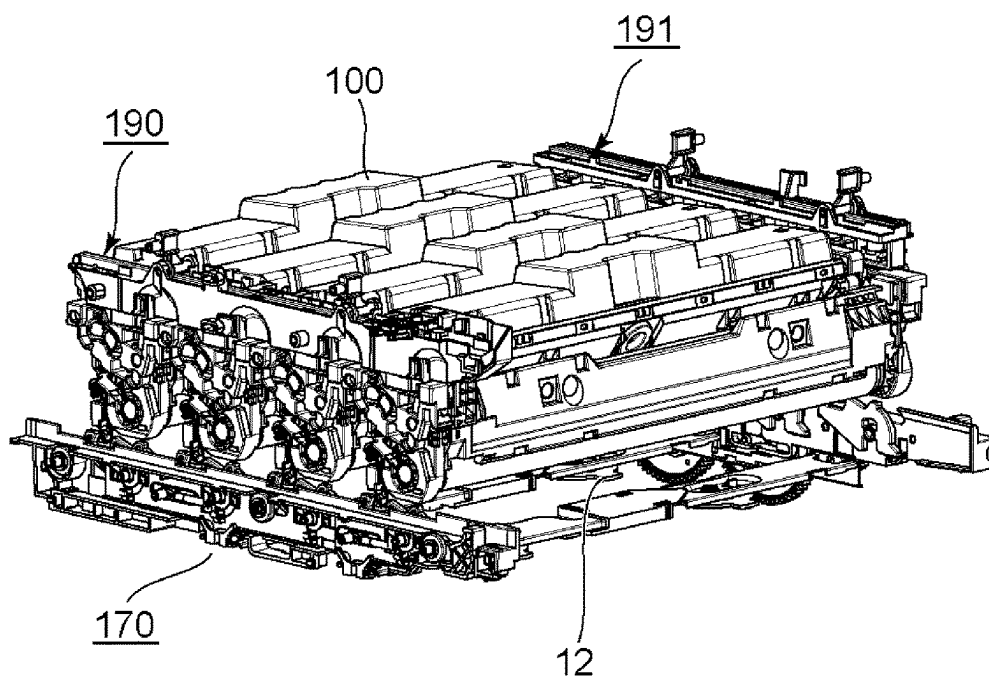


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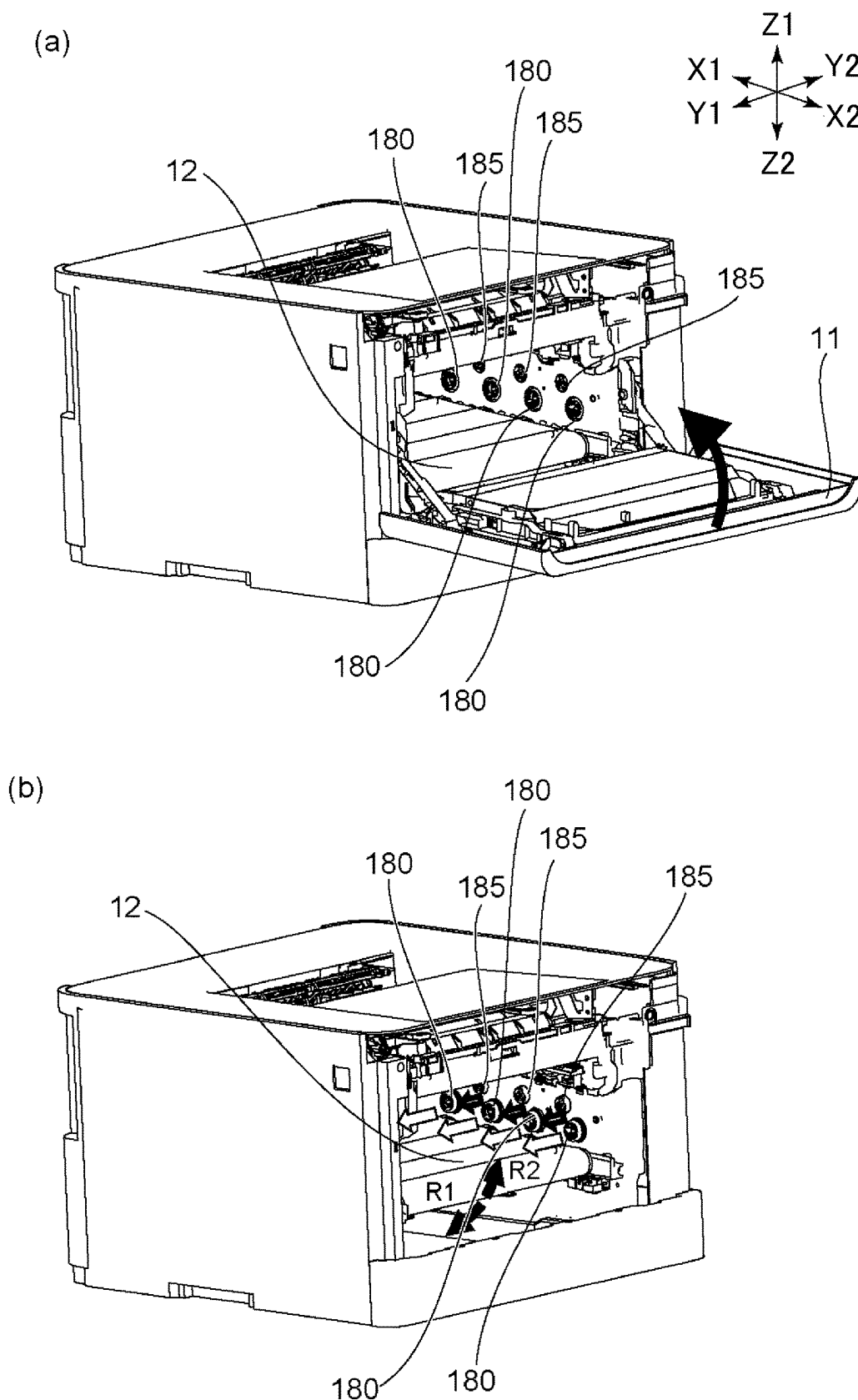


(a)



(b)

Fig. 8



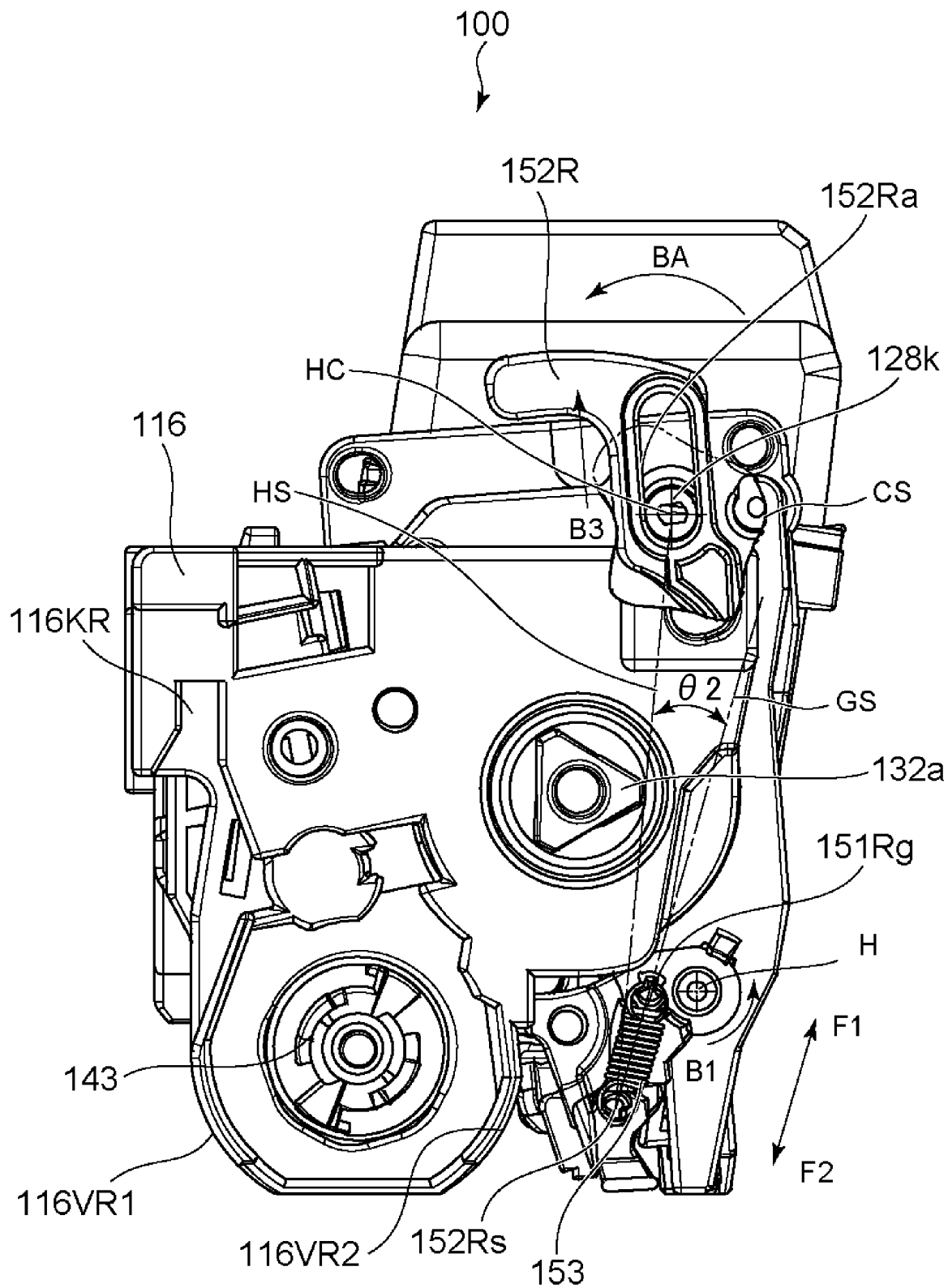


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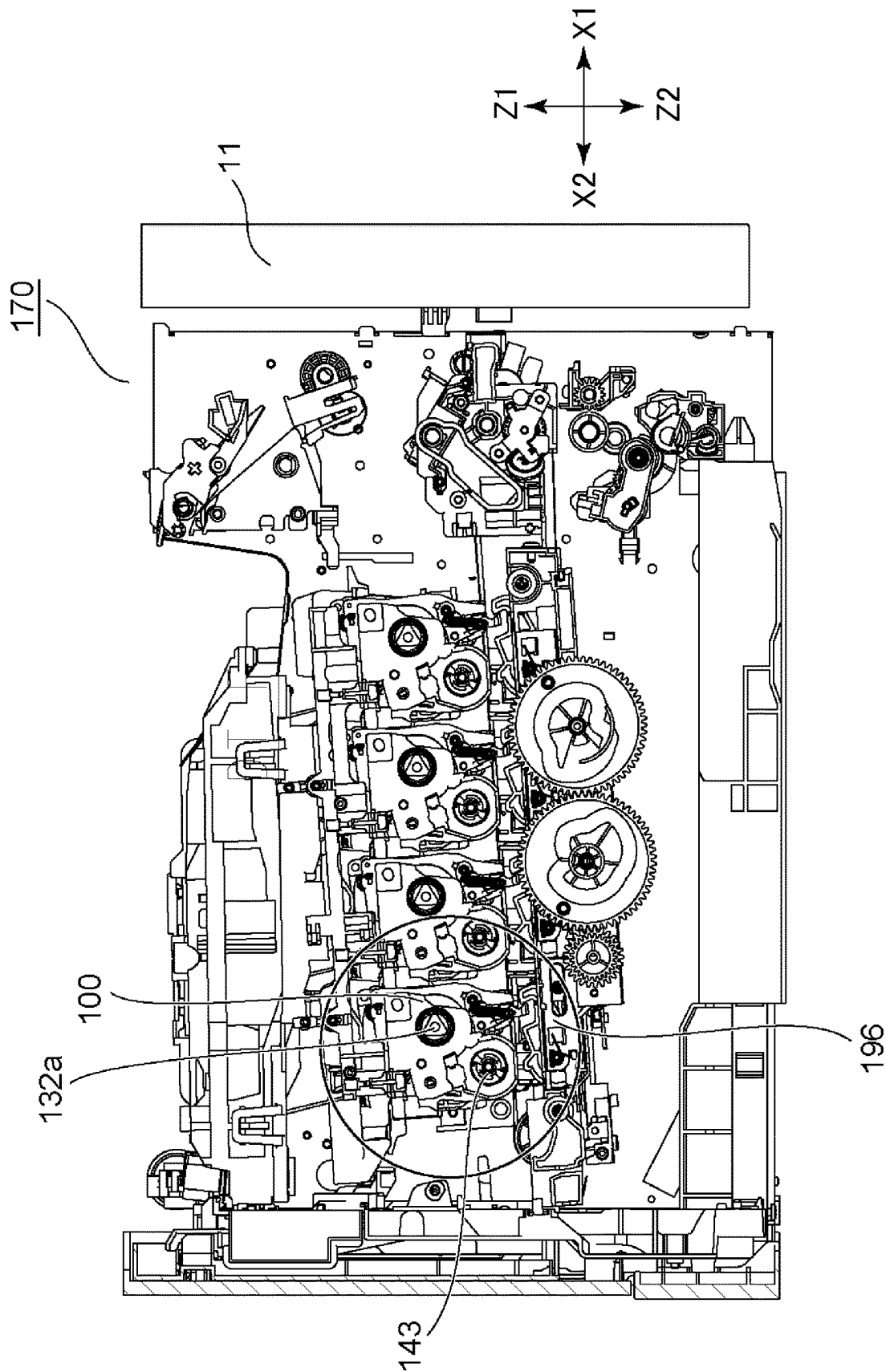


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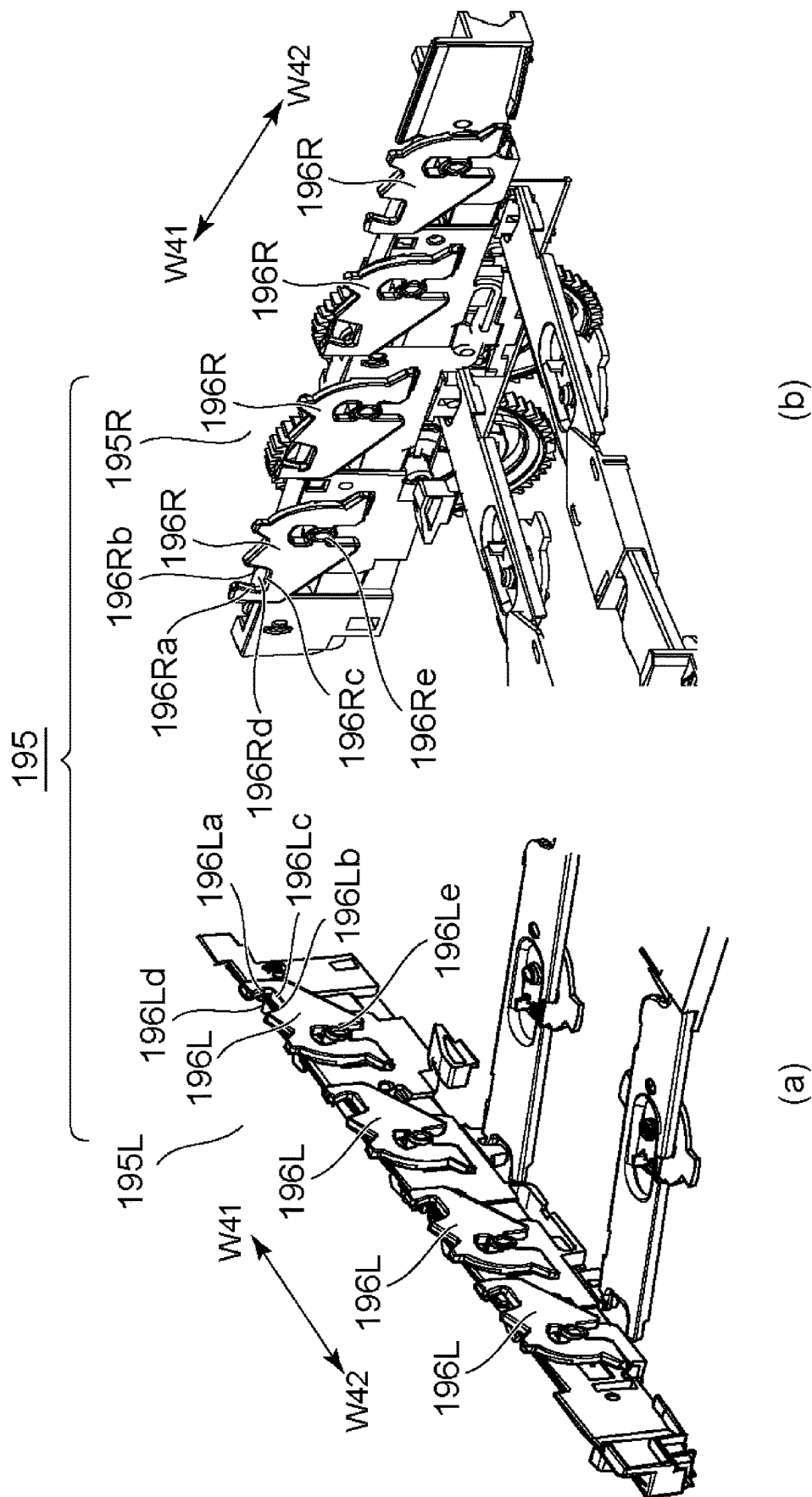


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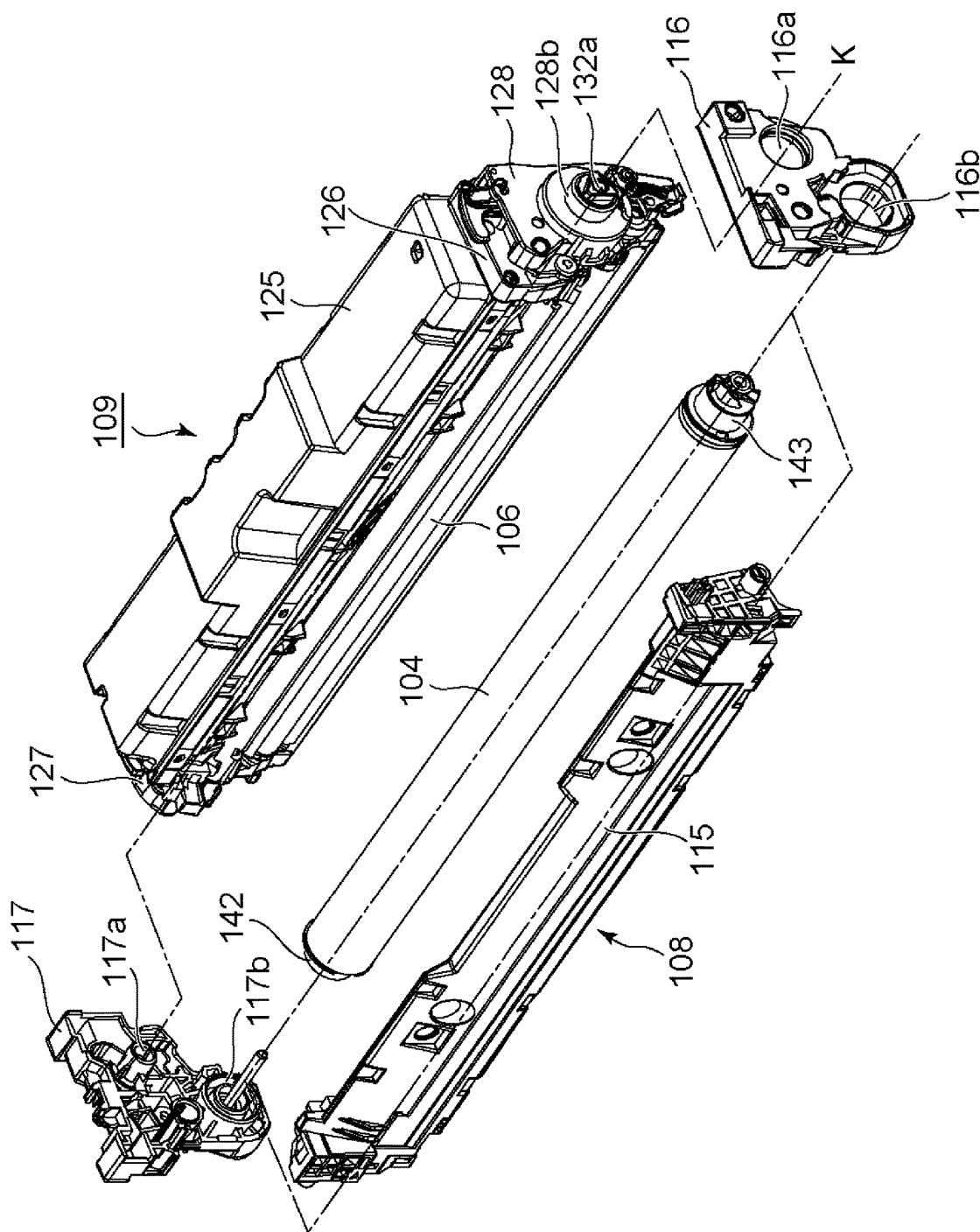


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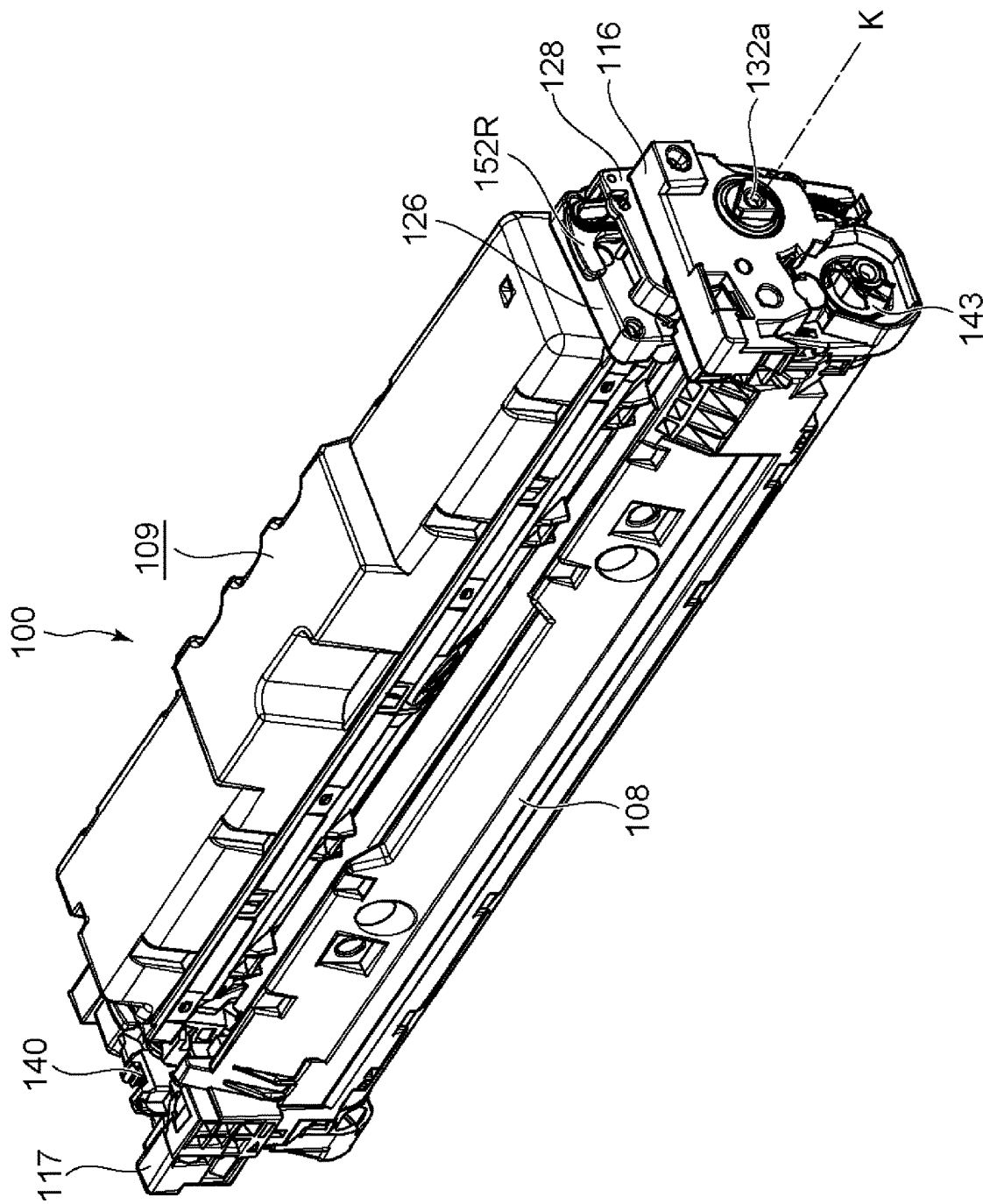


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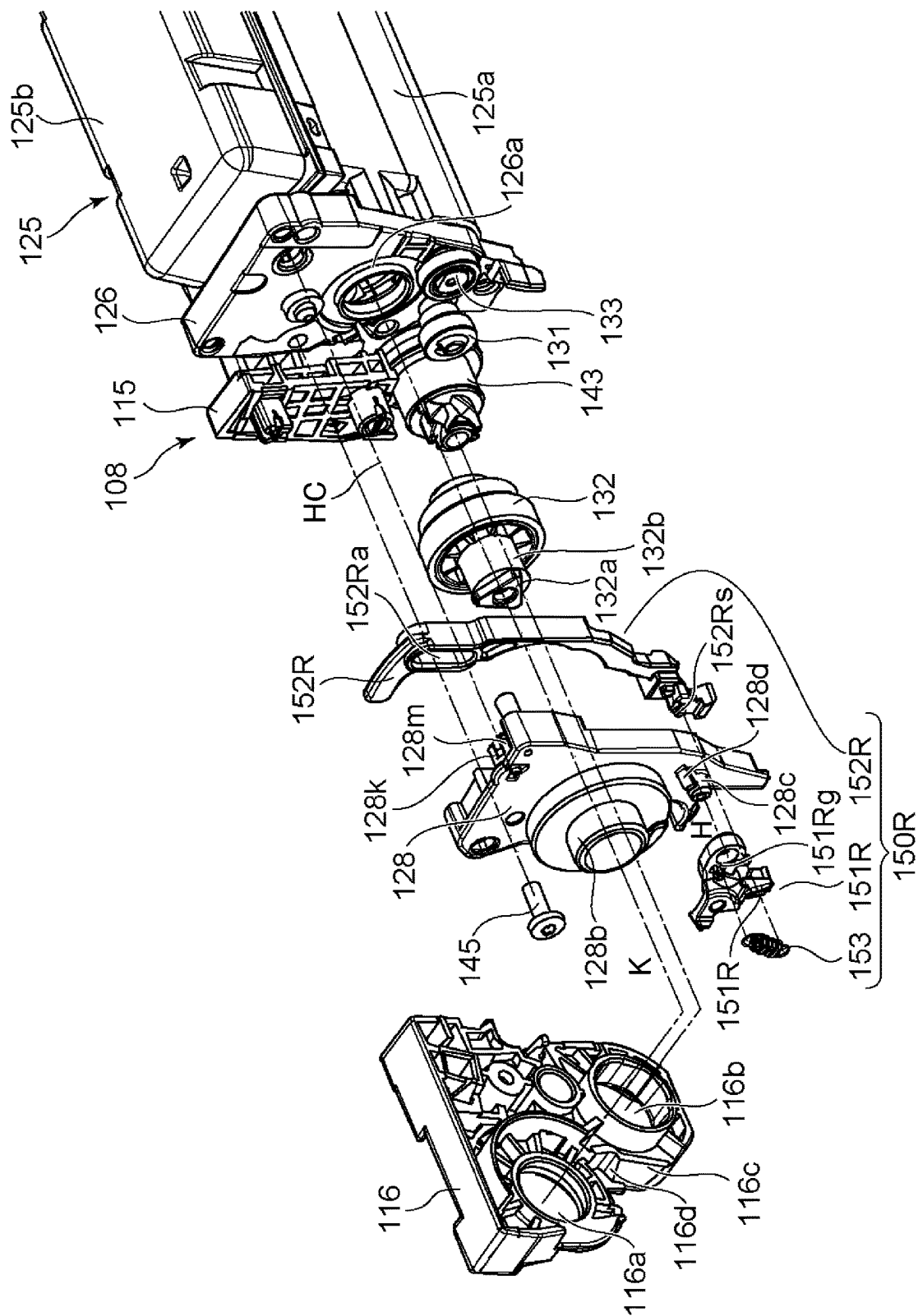


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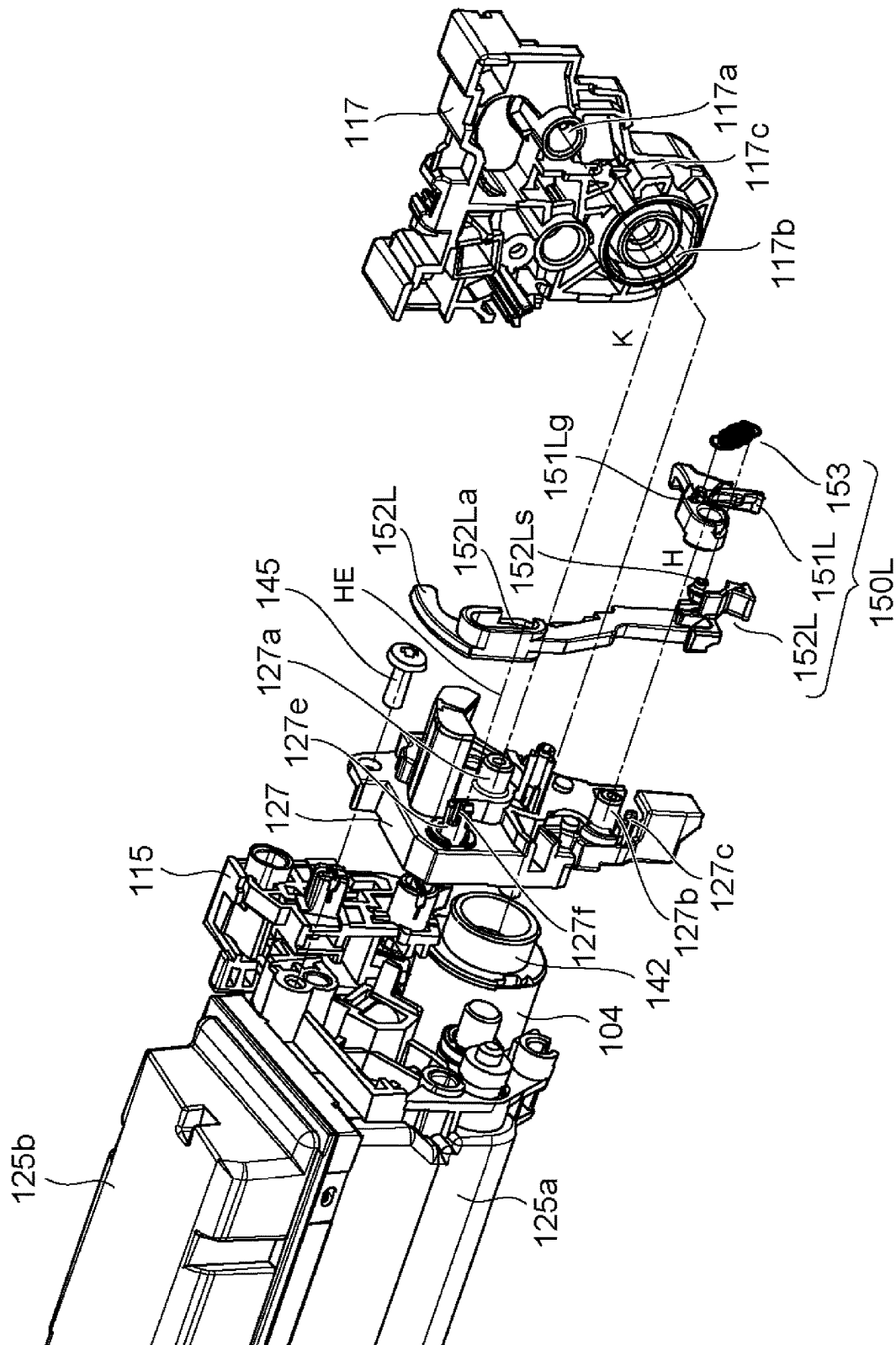


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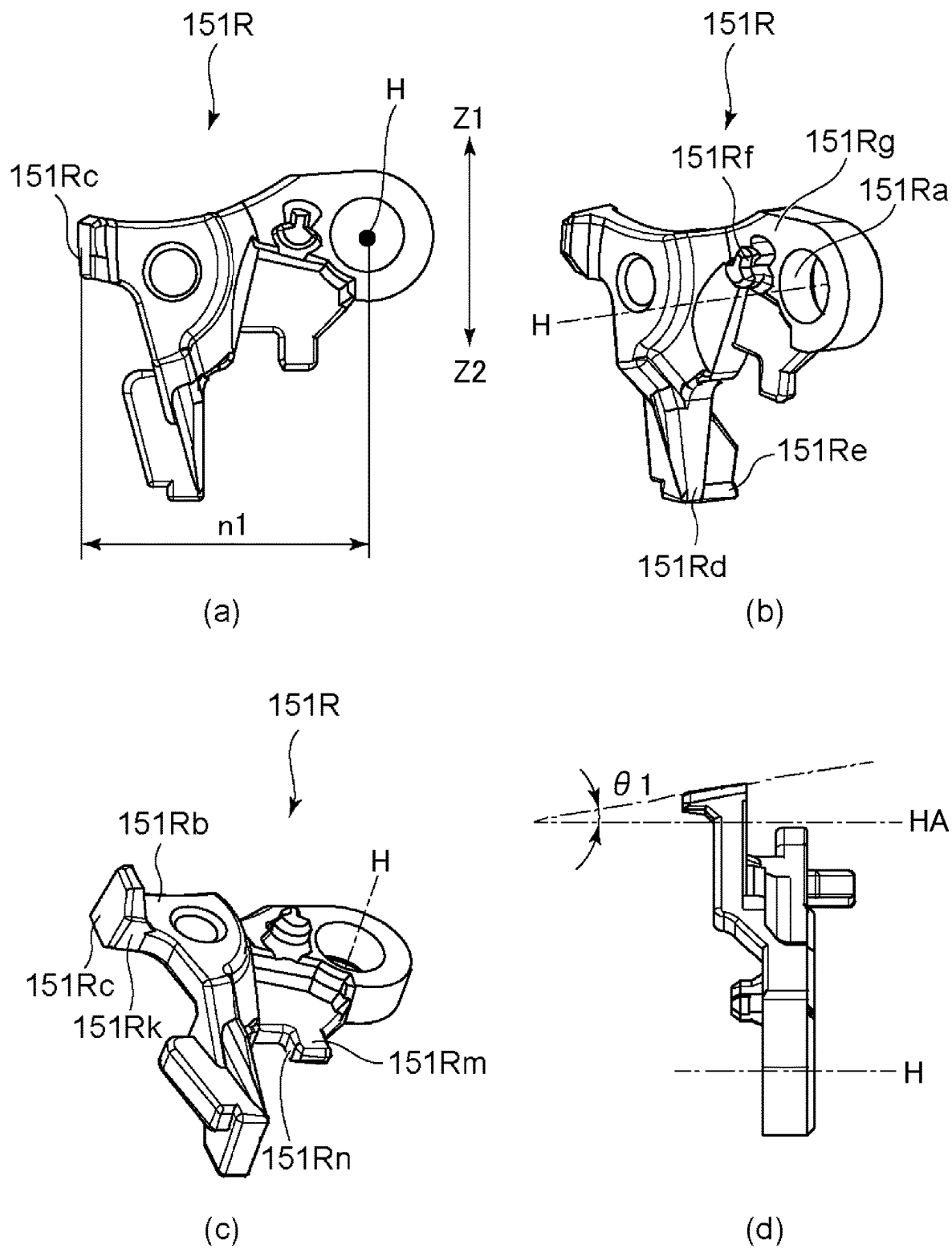


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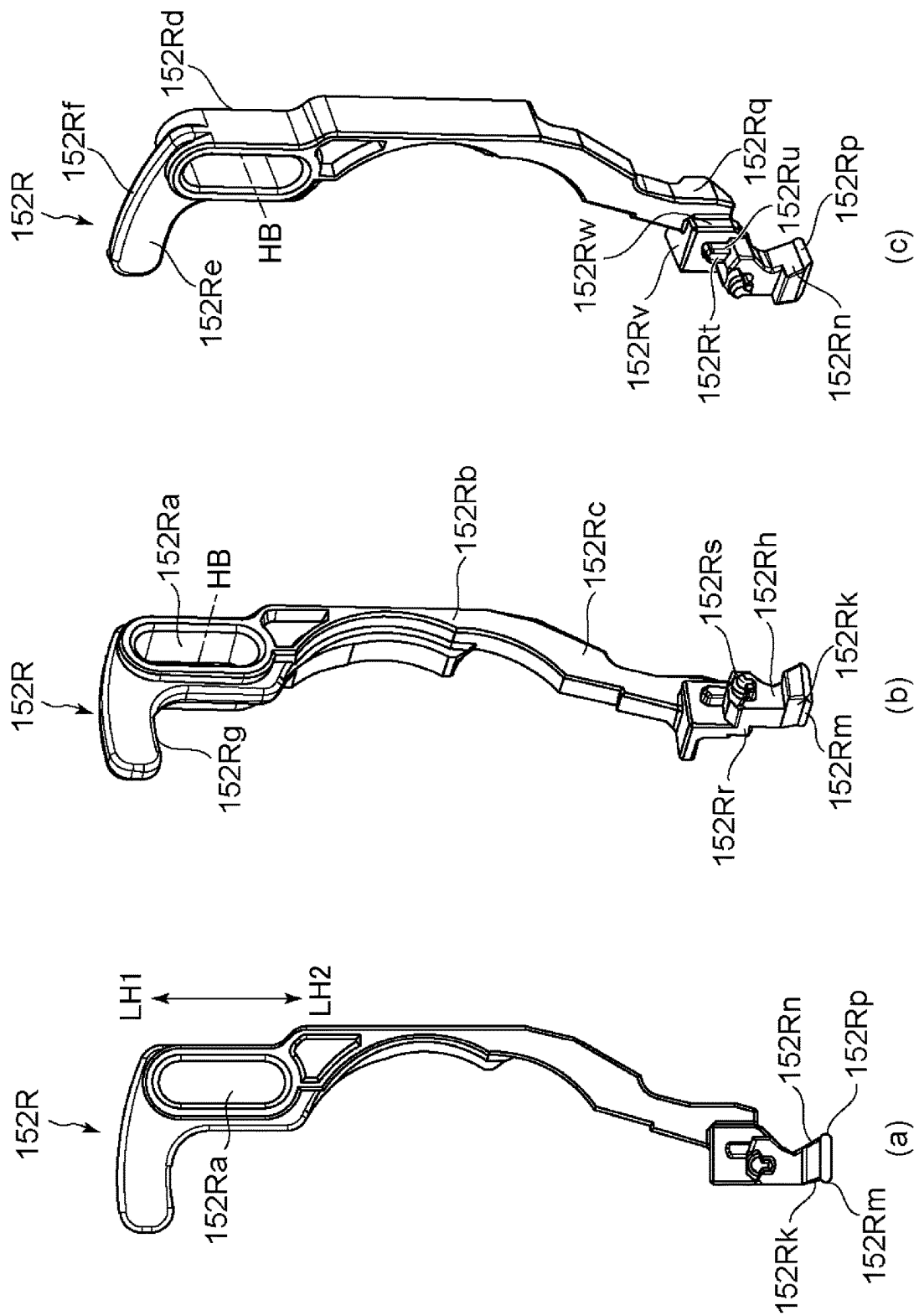


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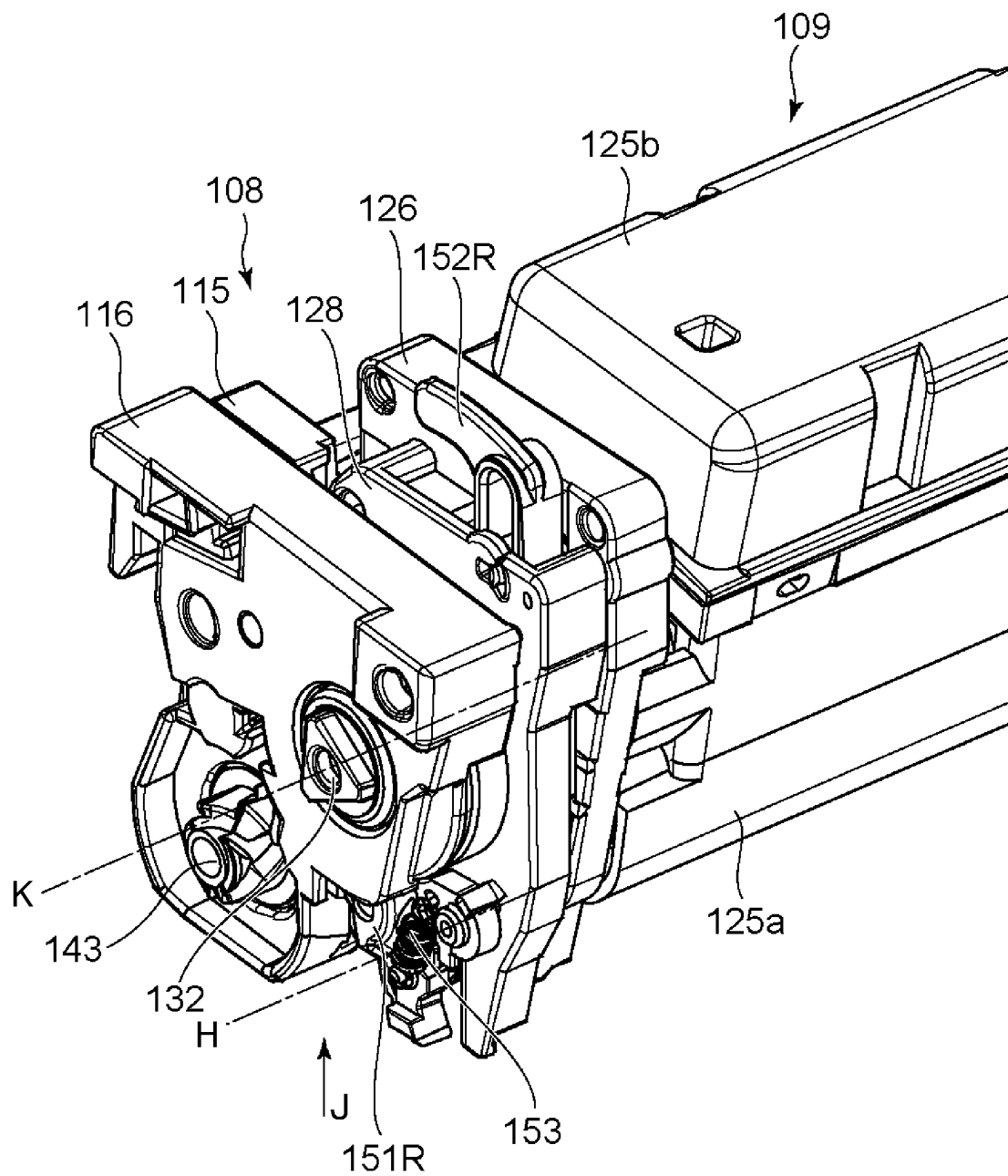


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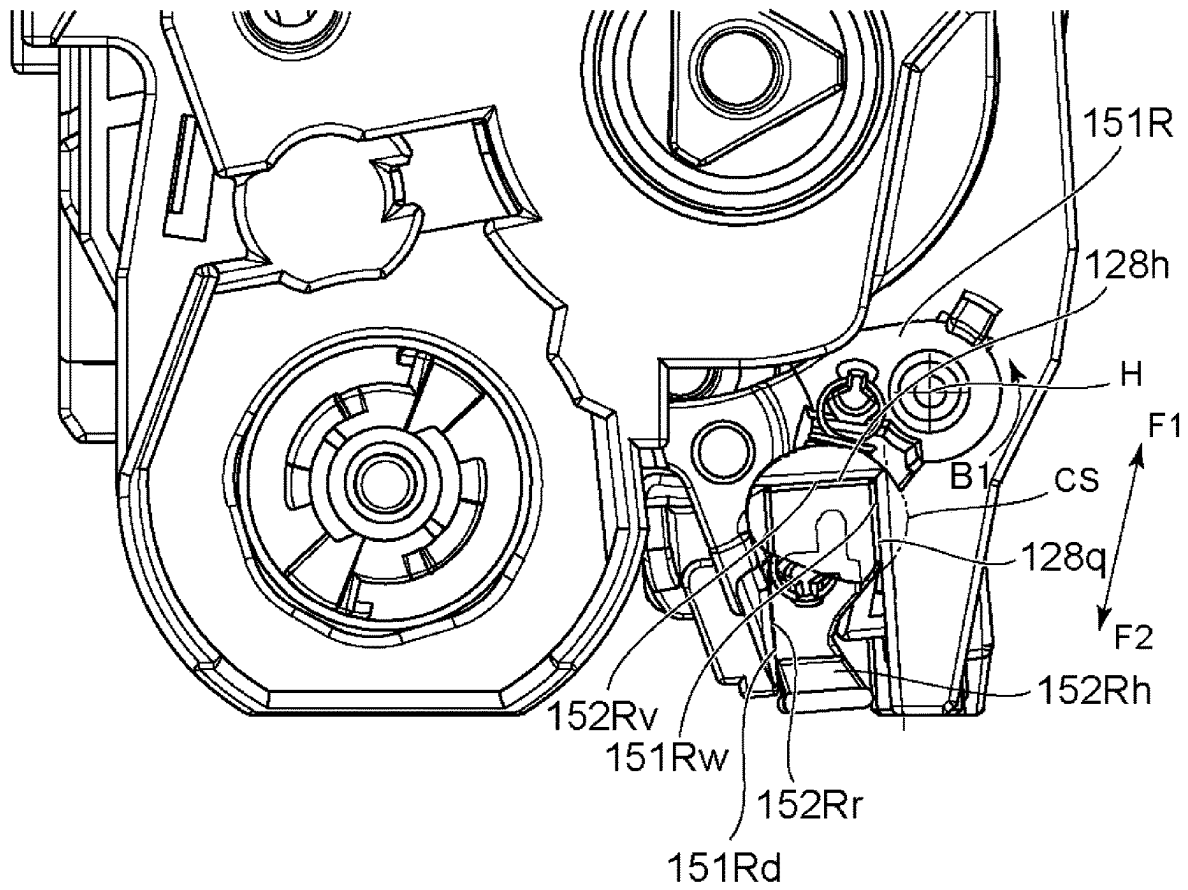


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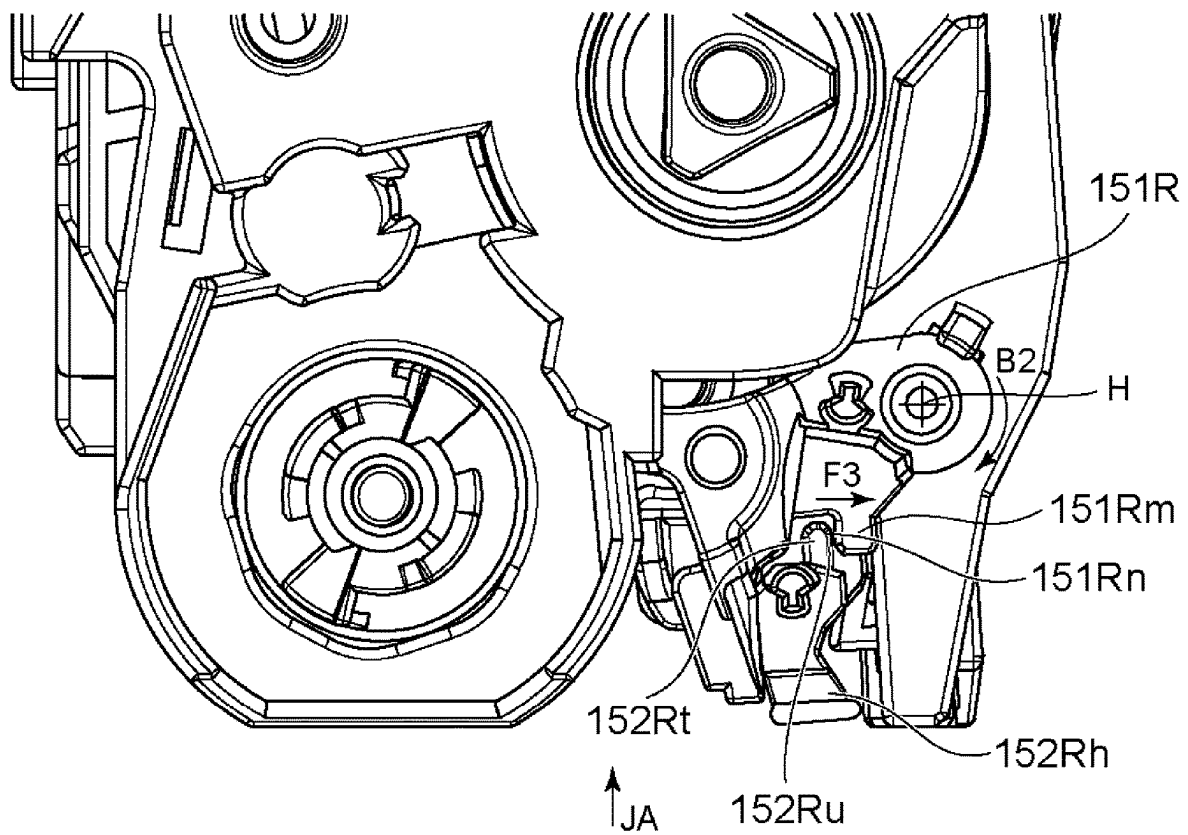


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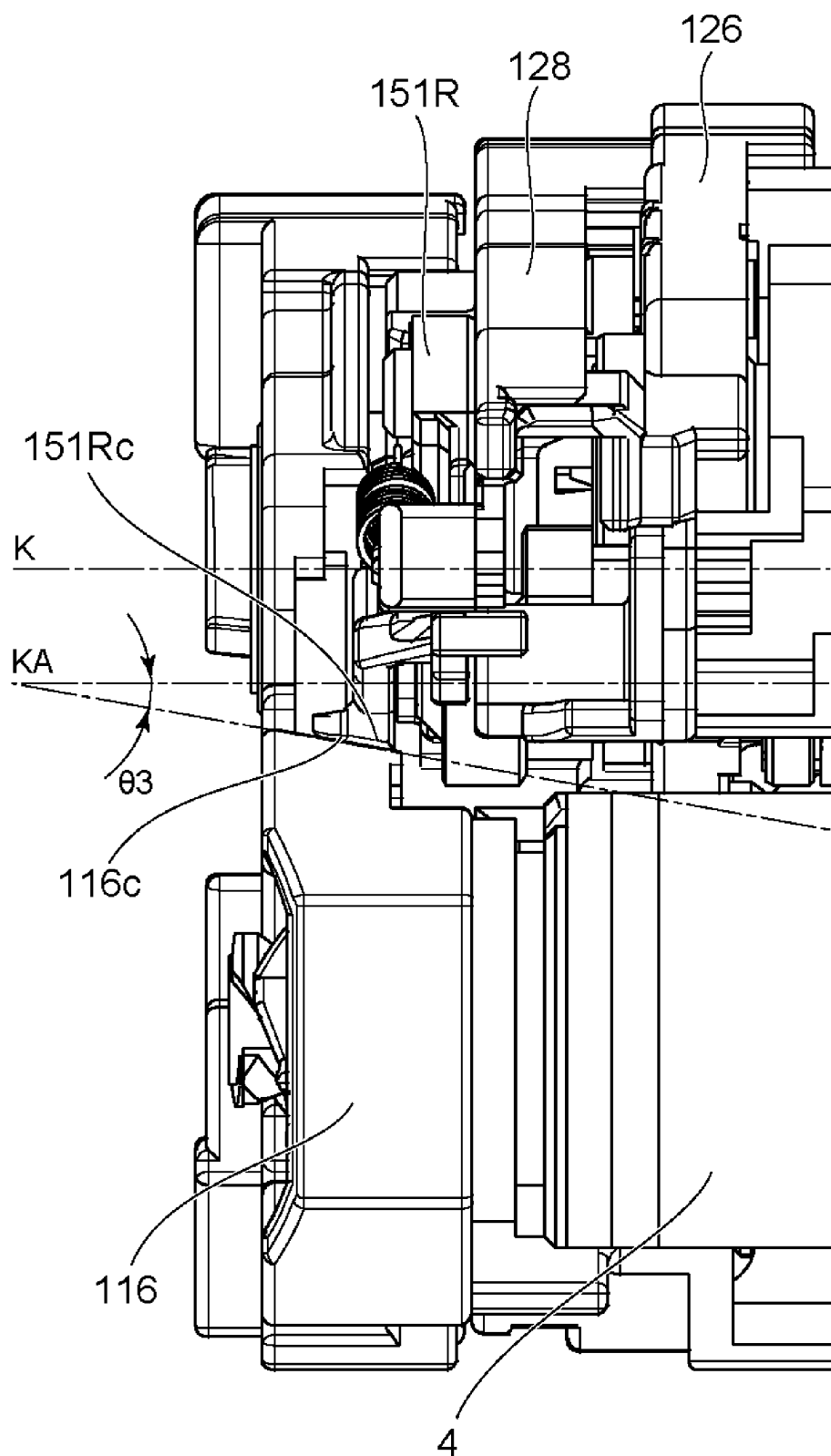


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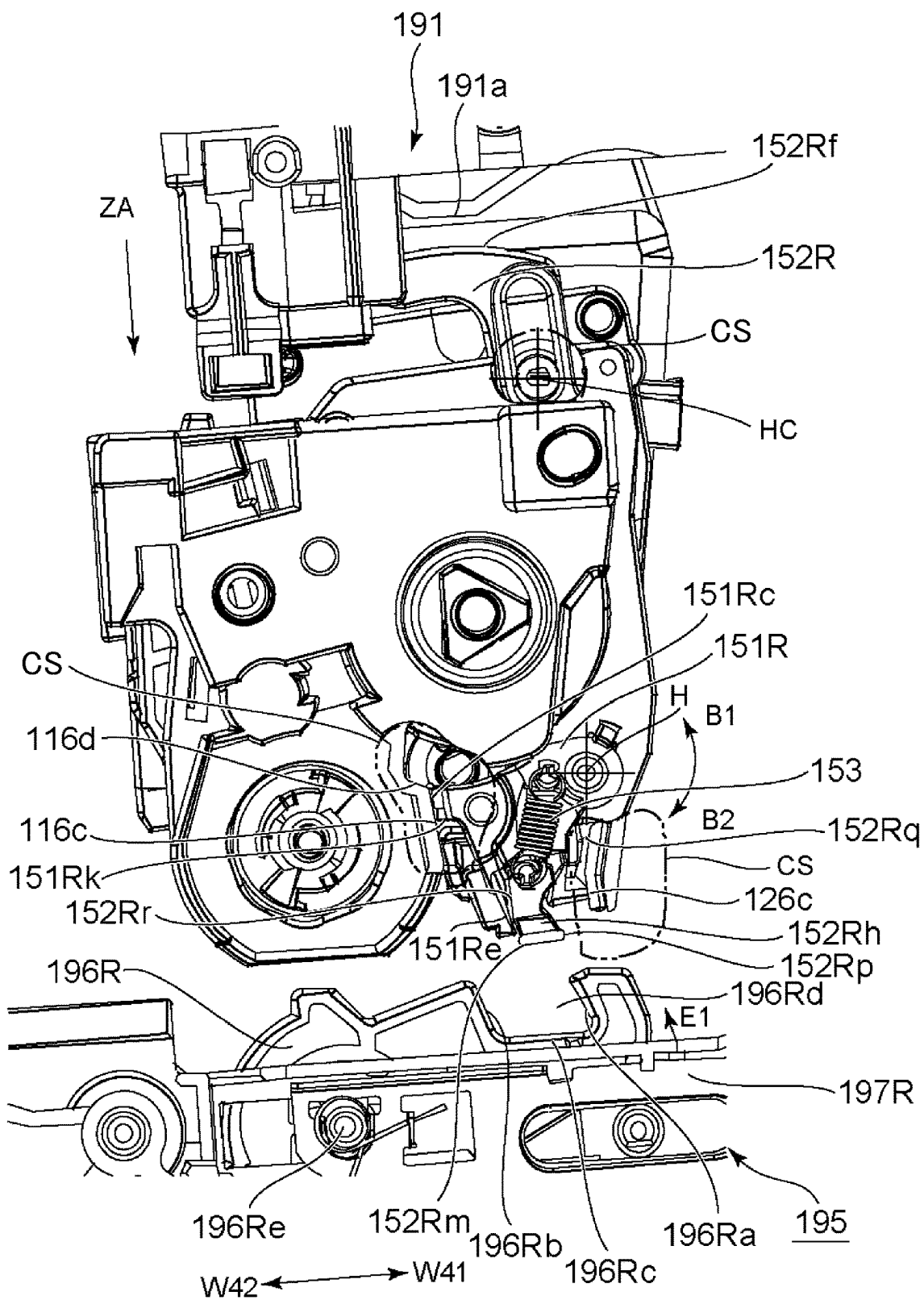


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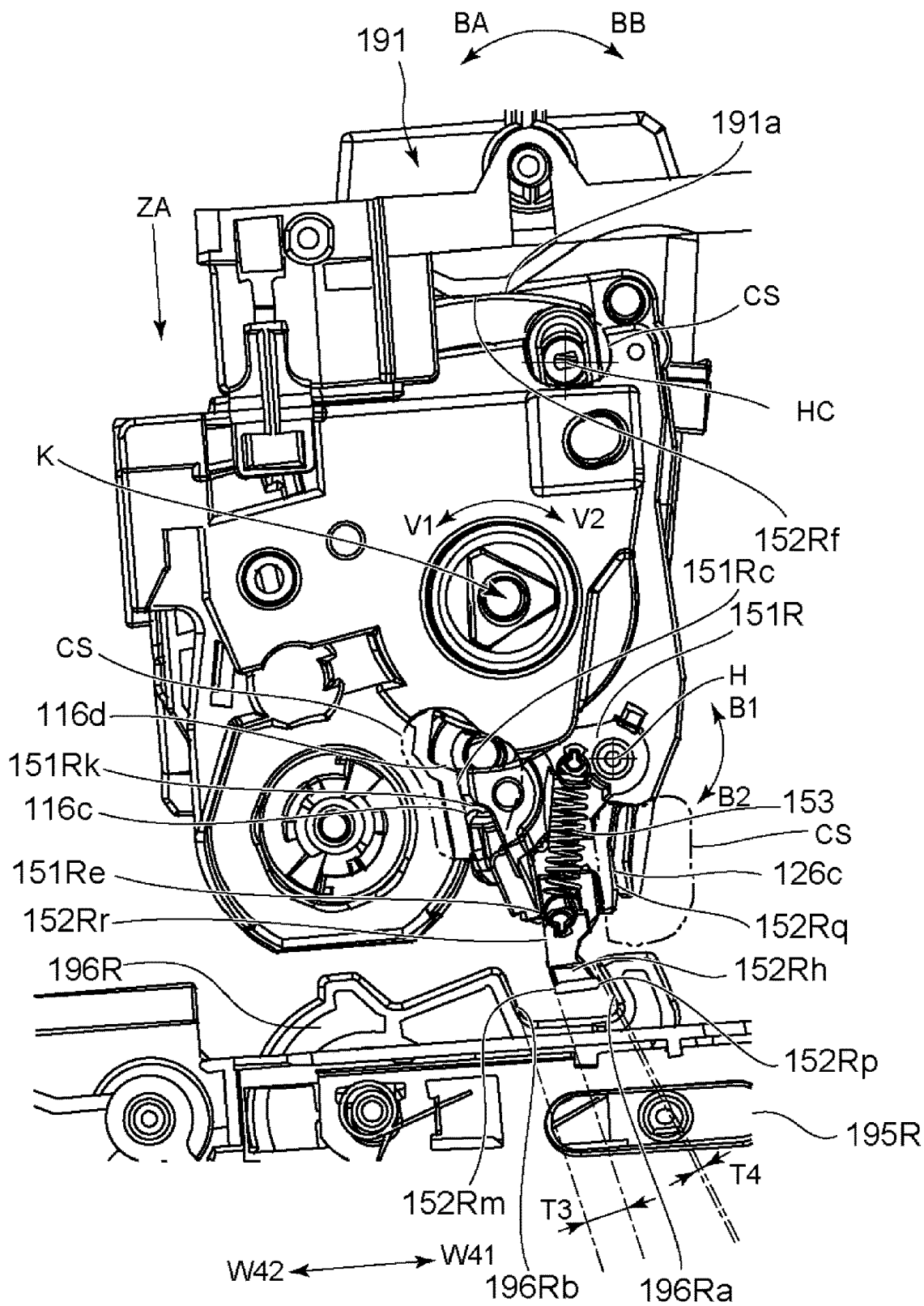


Fig. 24

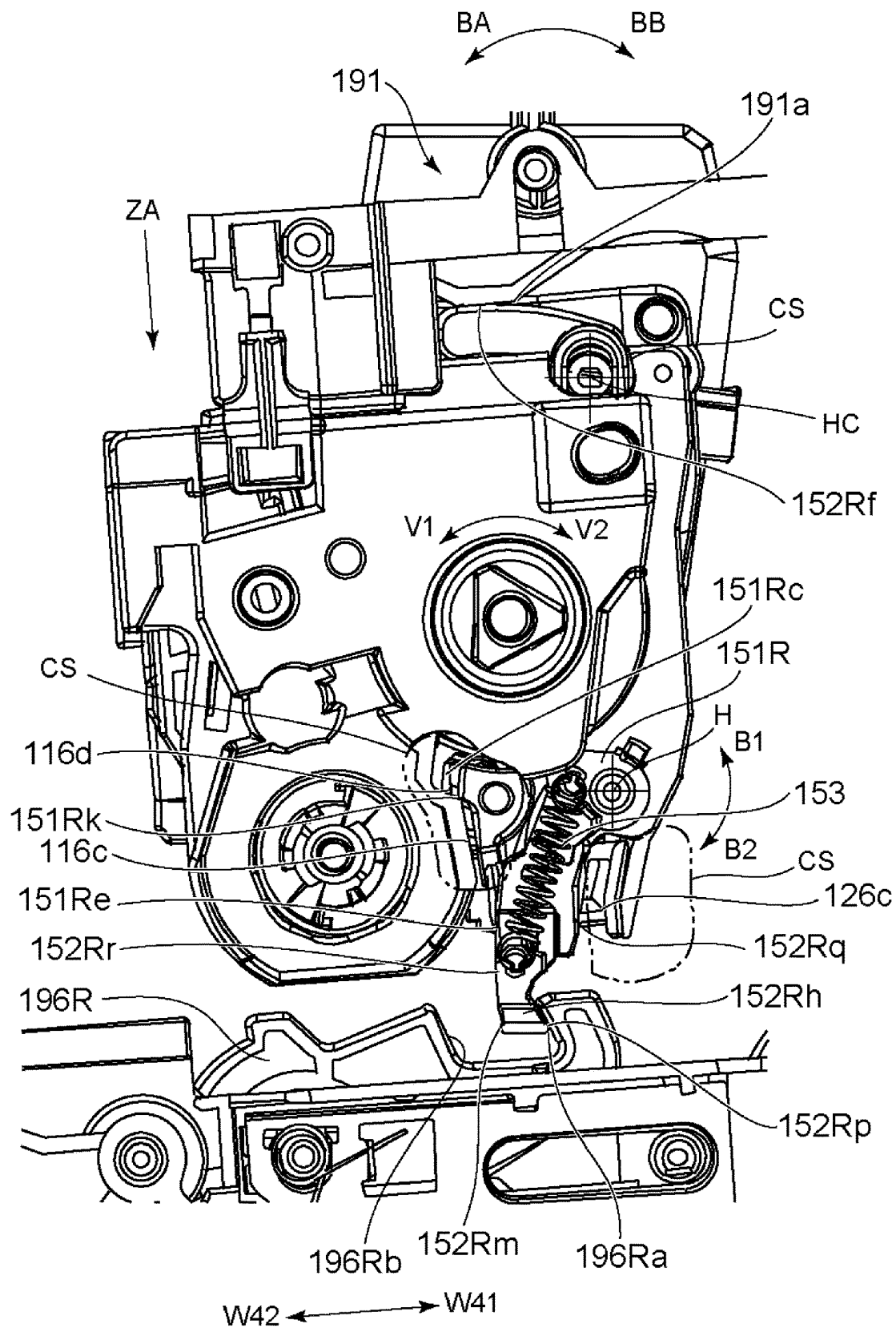


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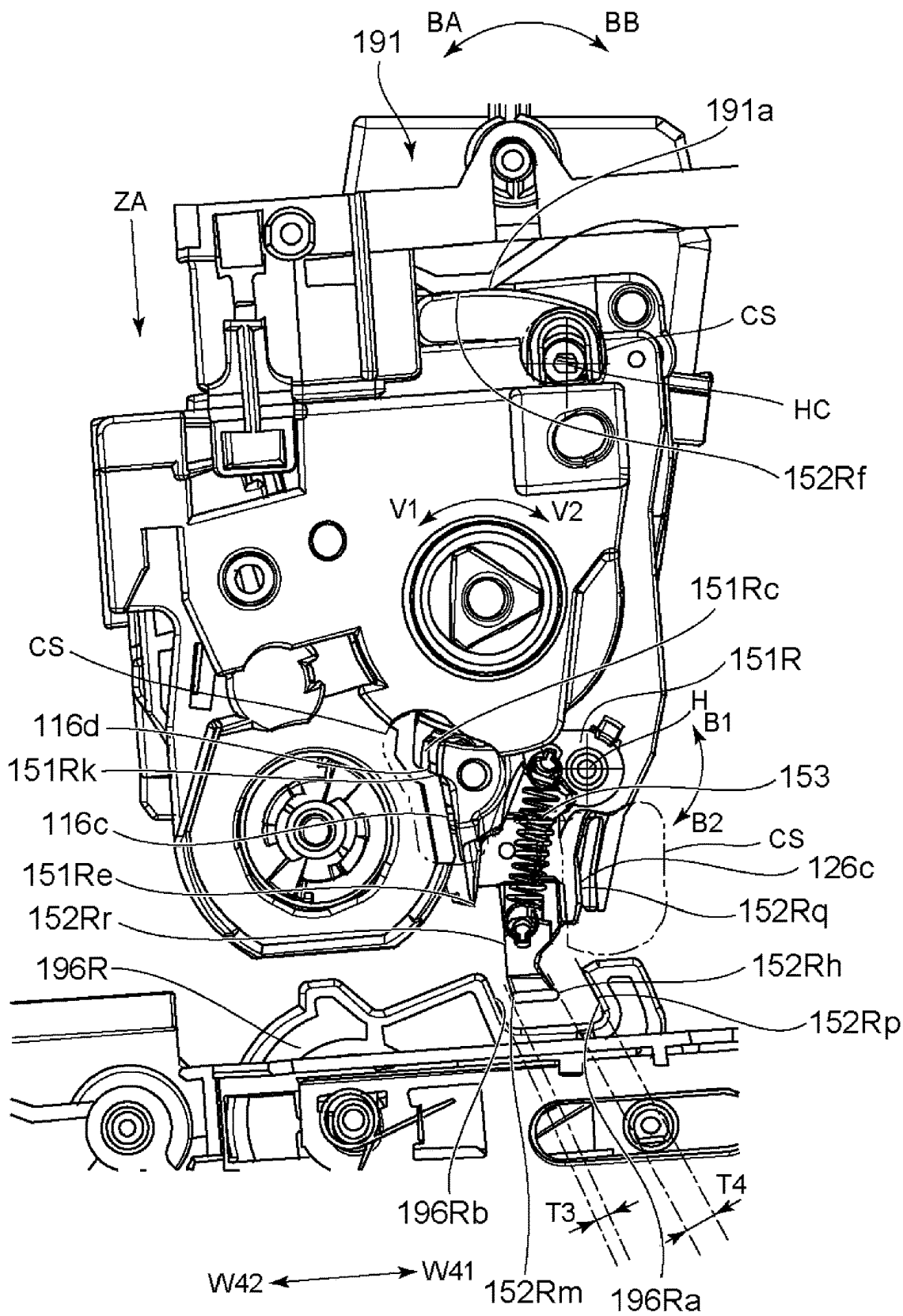


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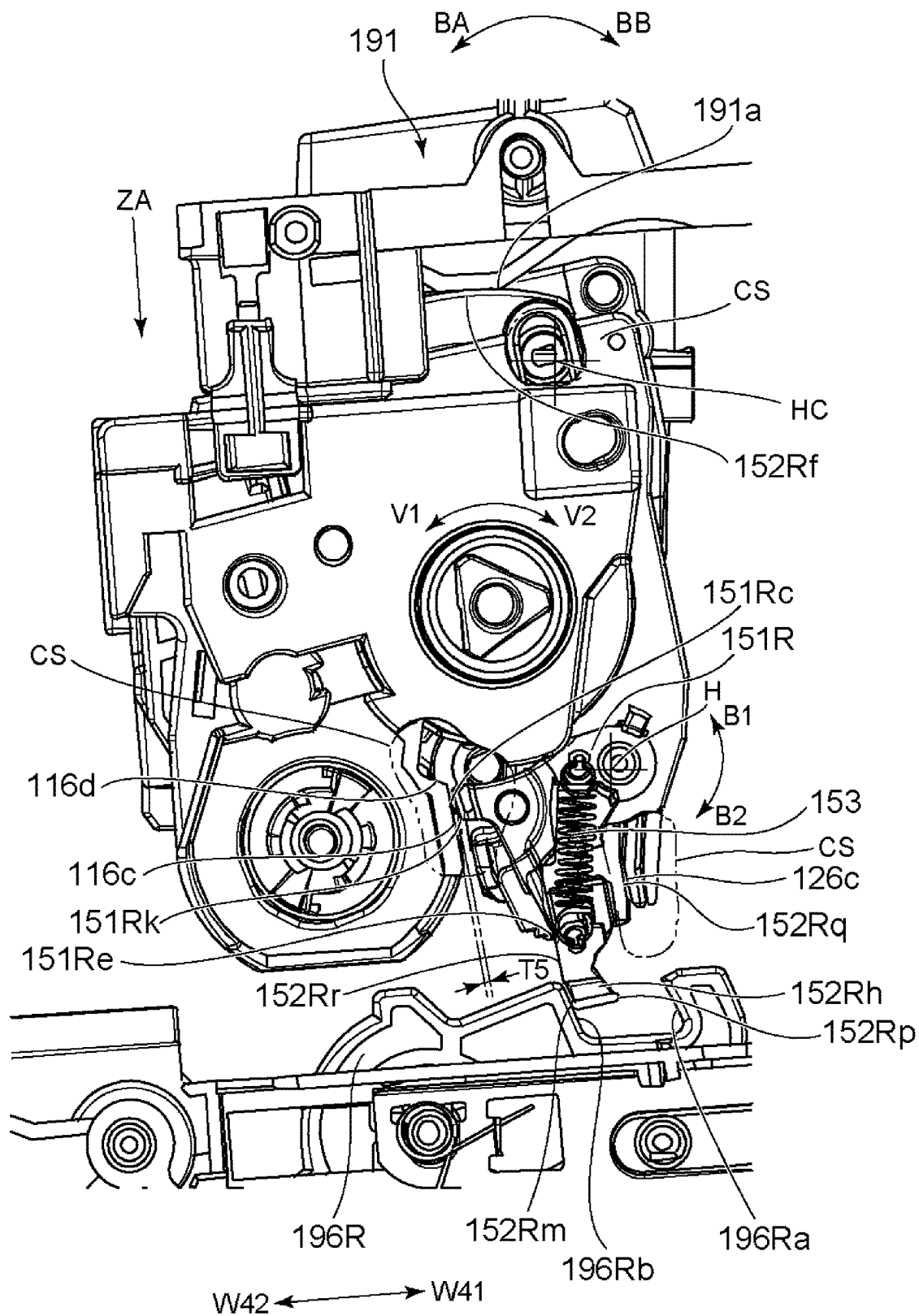


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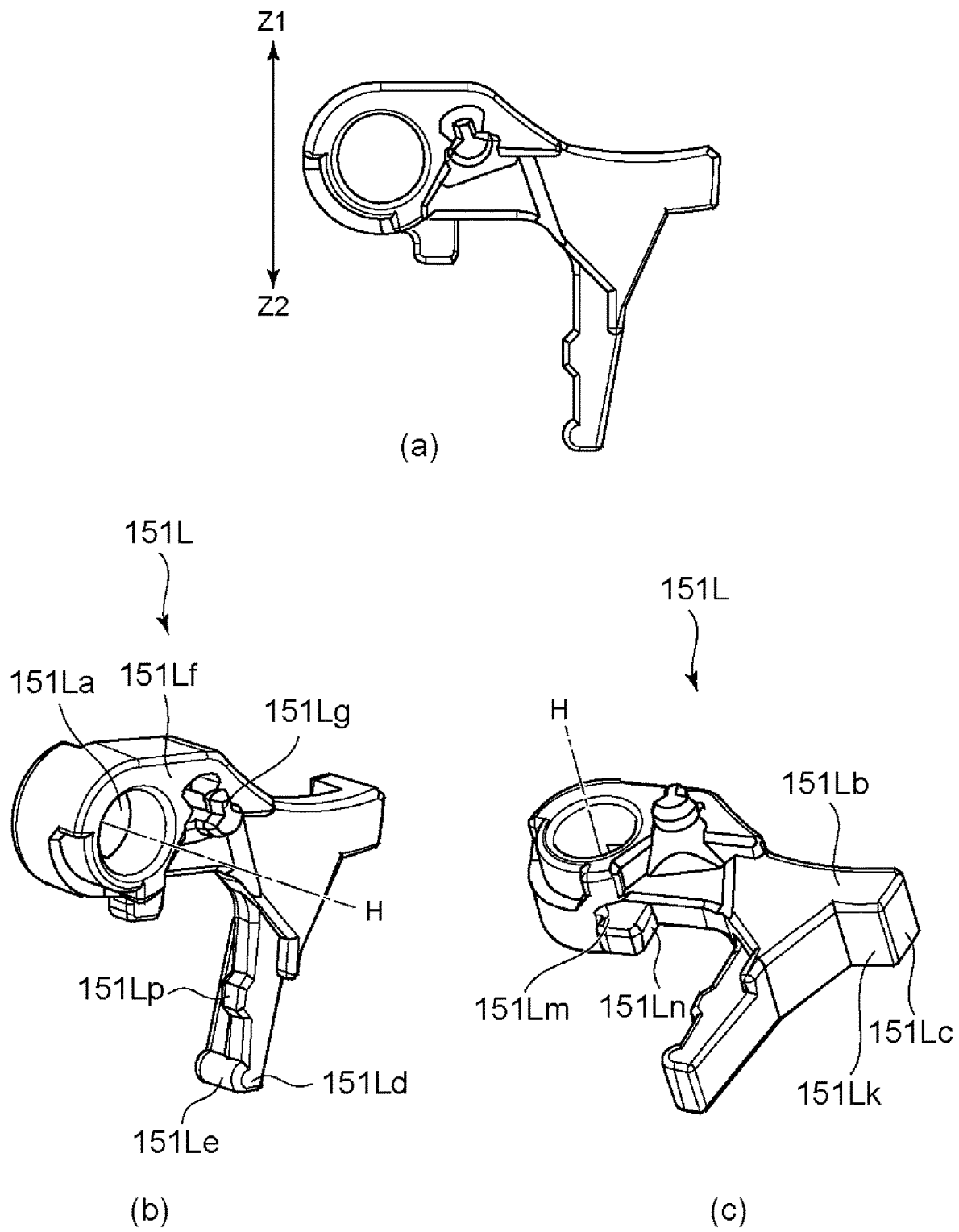


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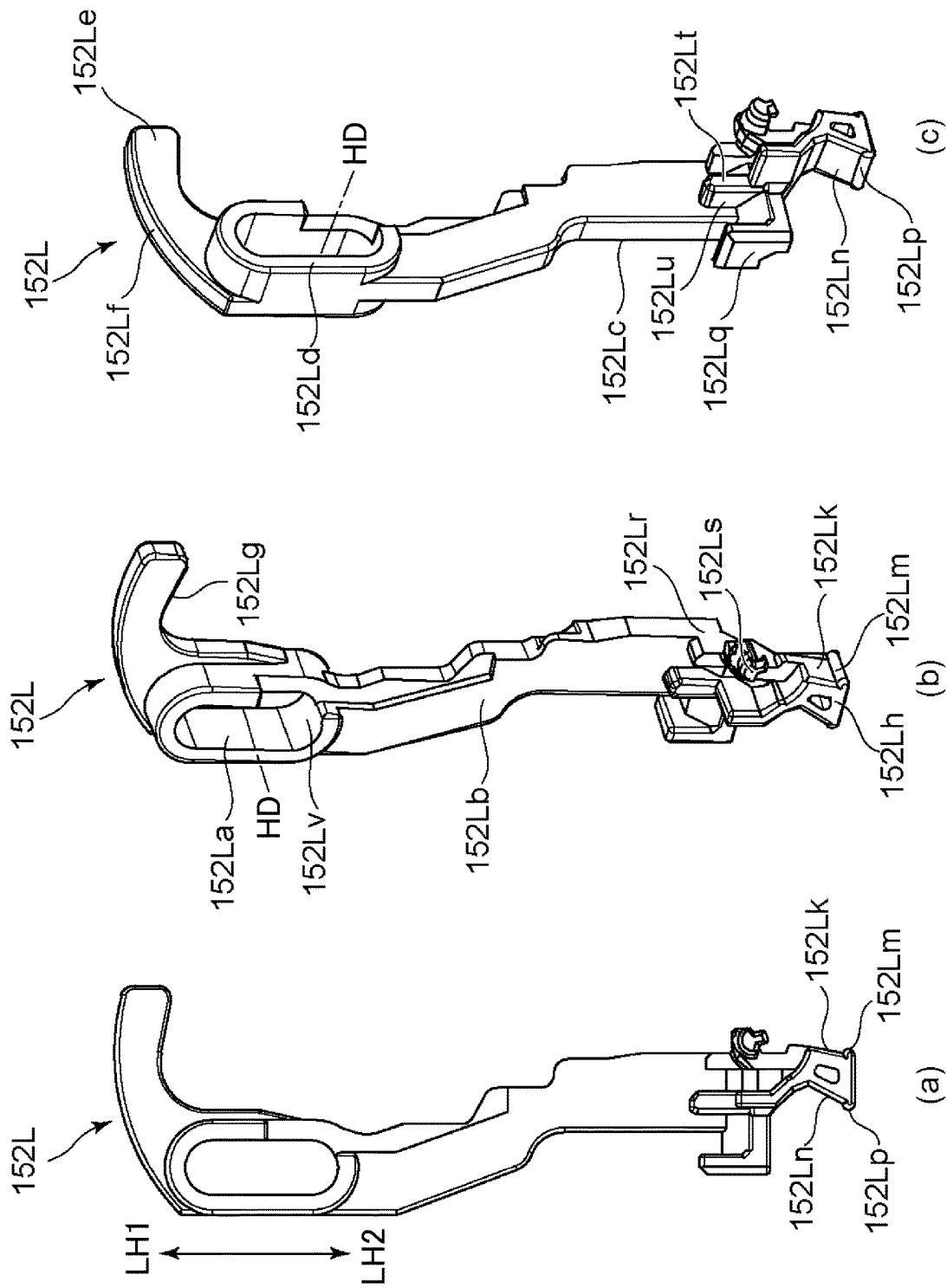


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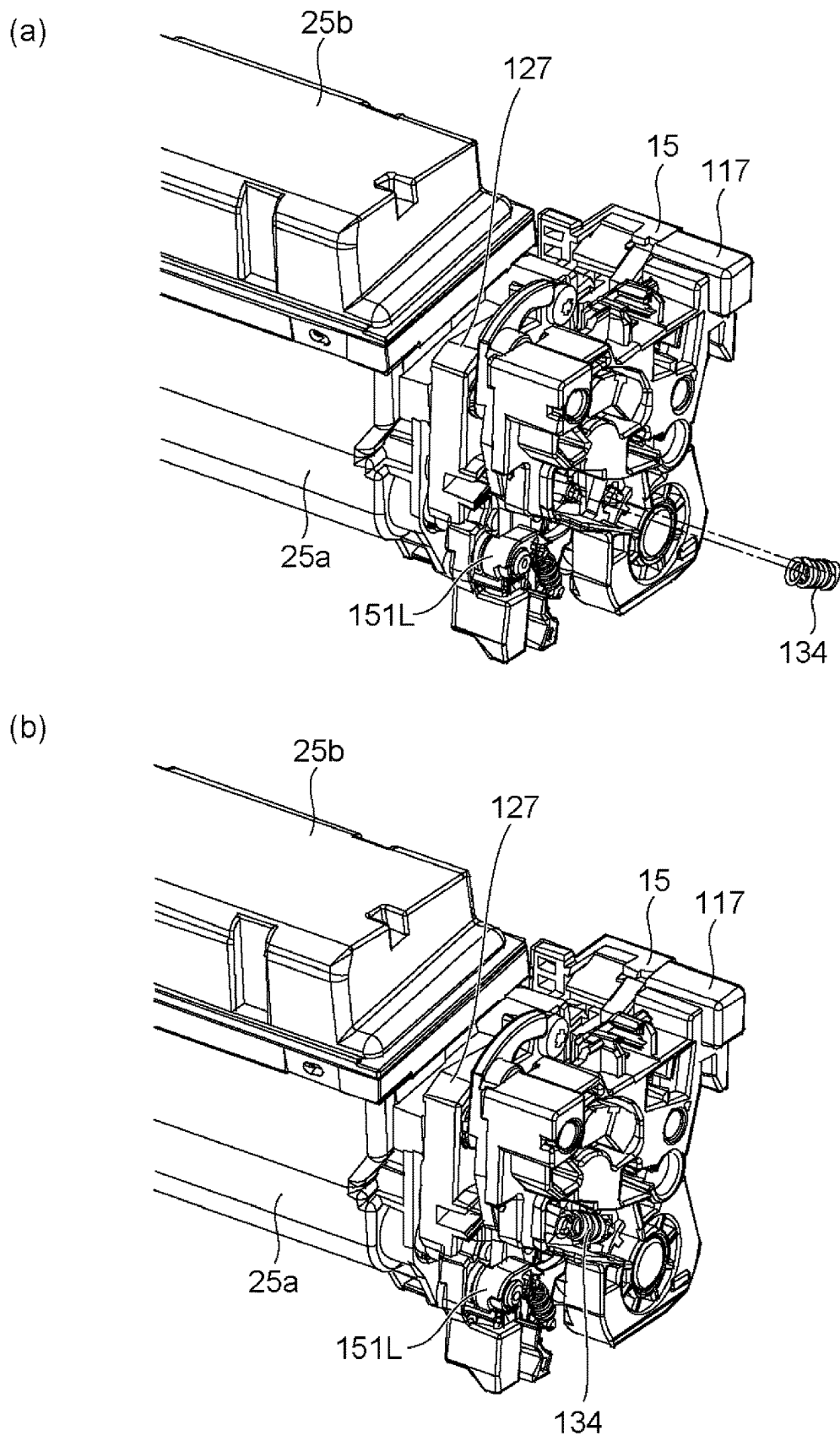


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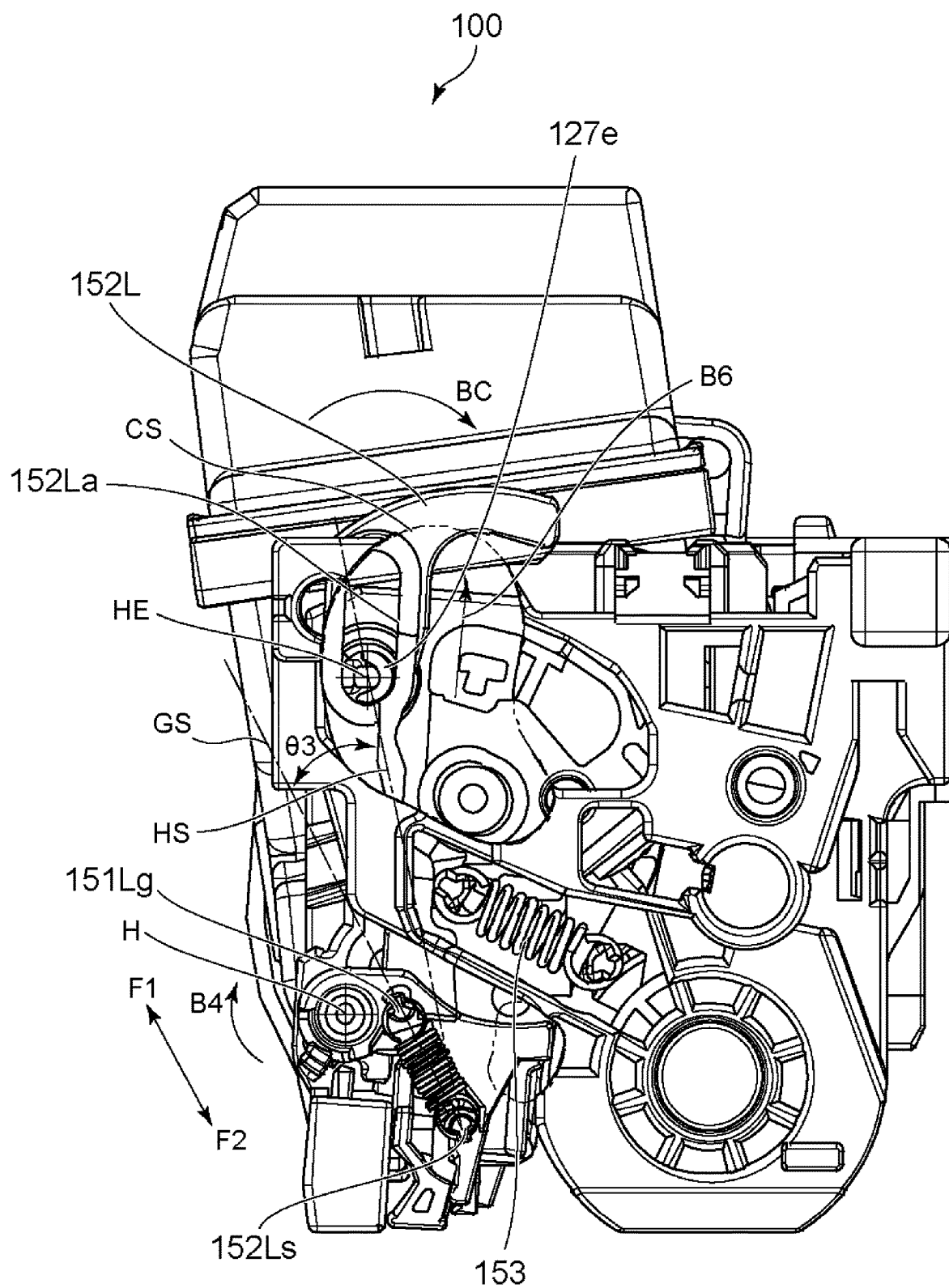


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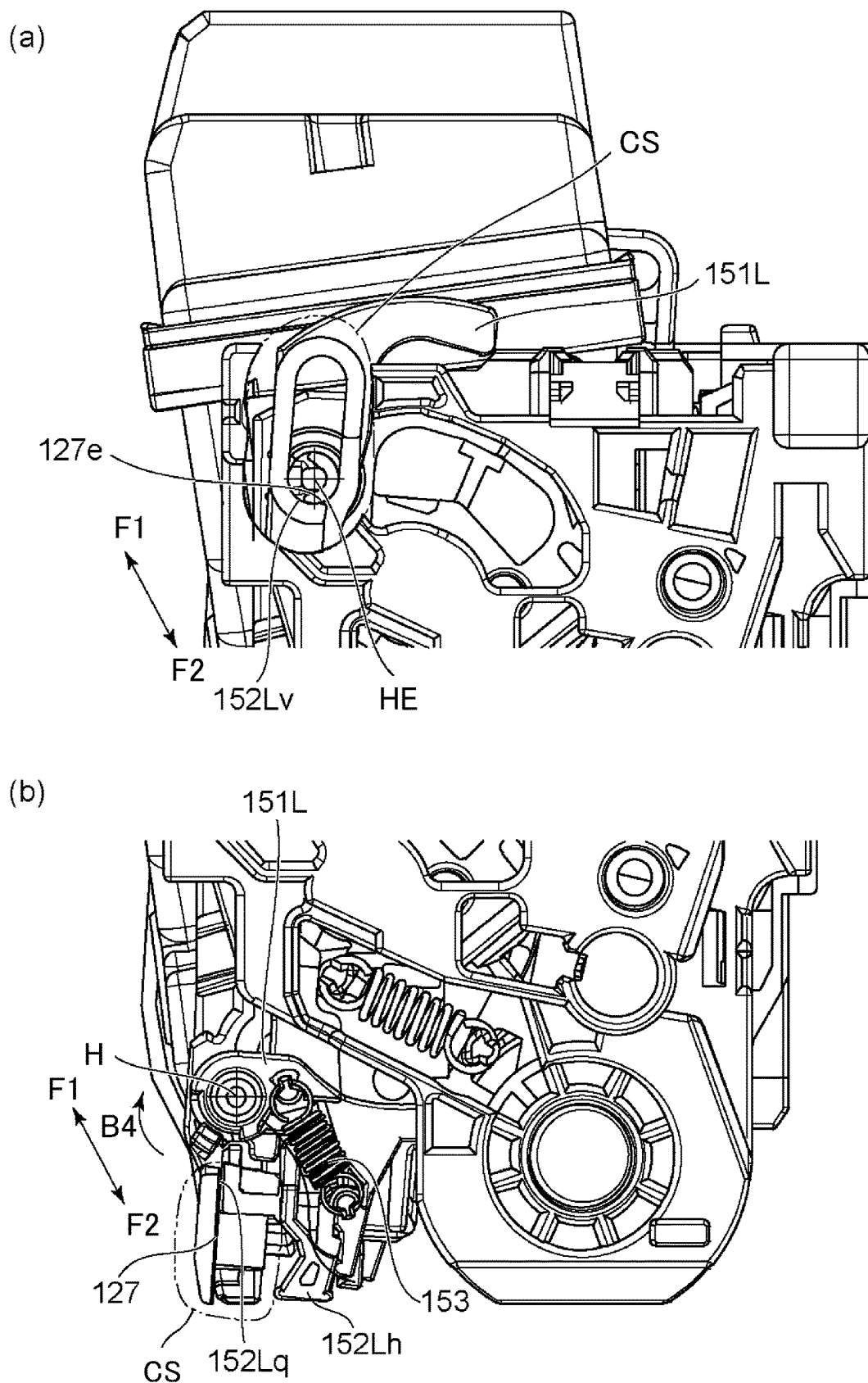


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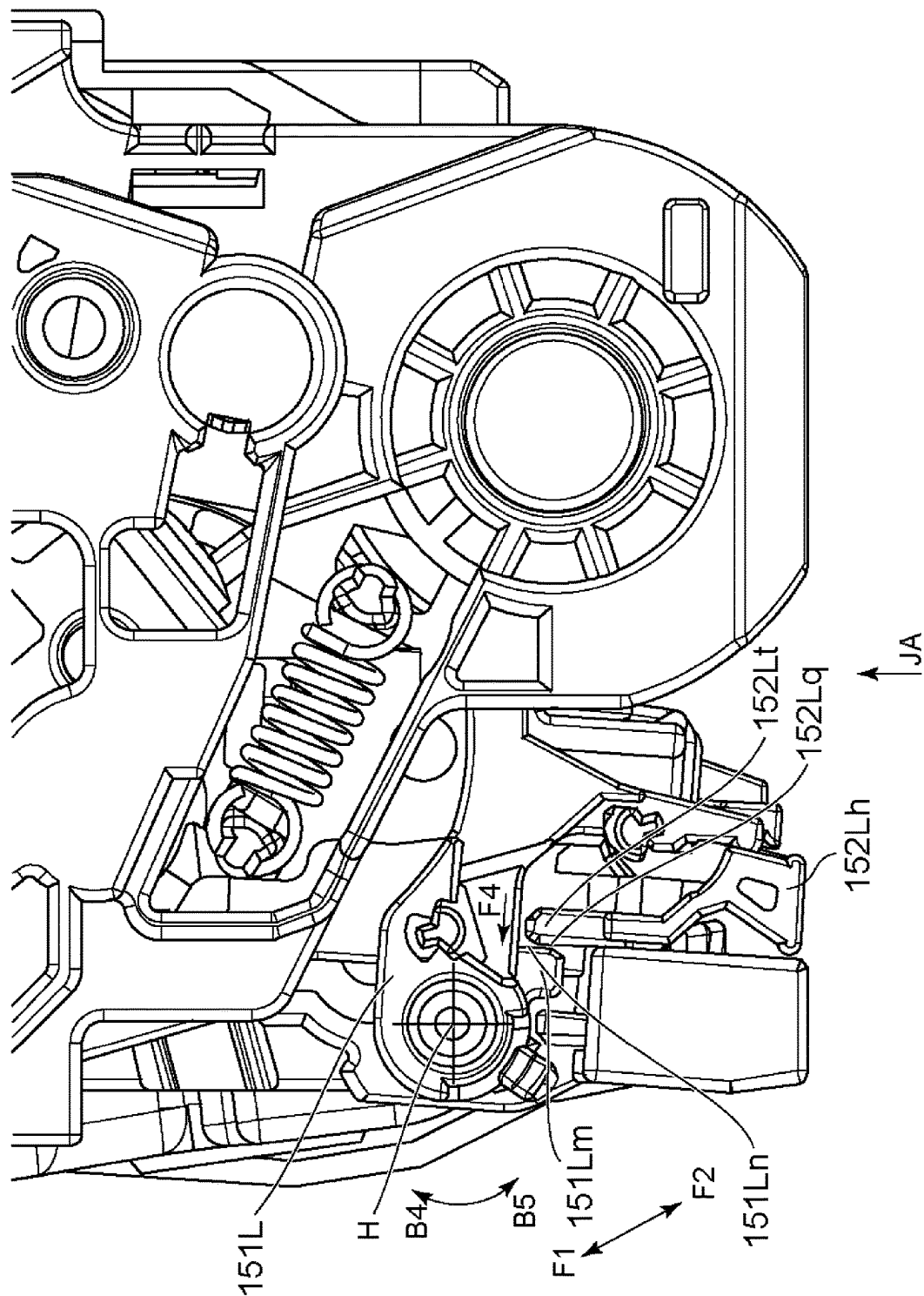
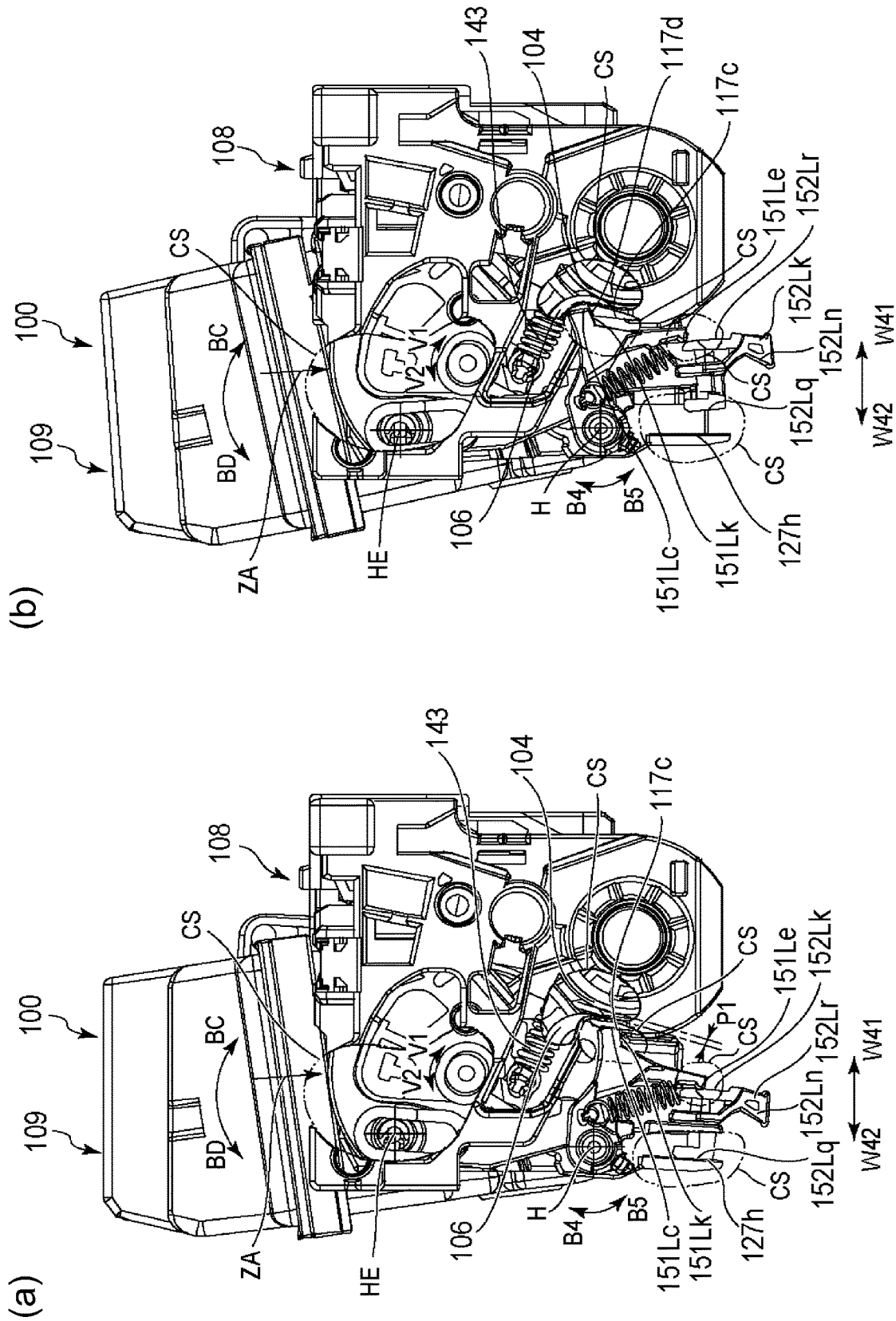


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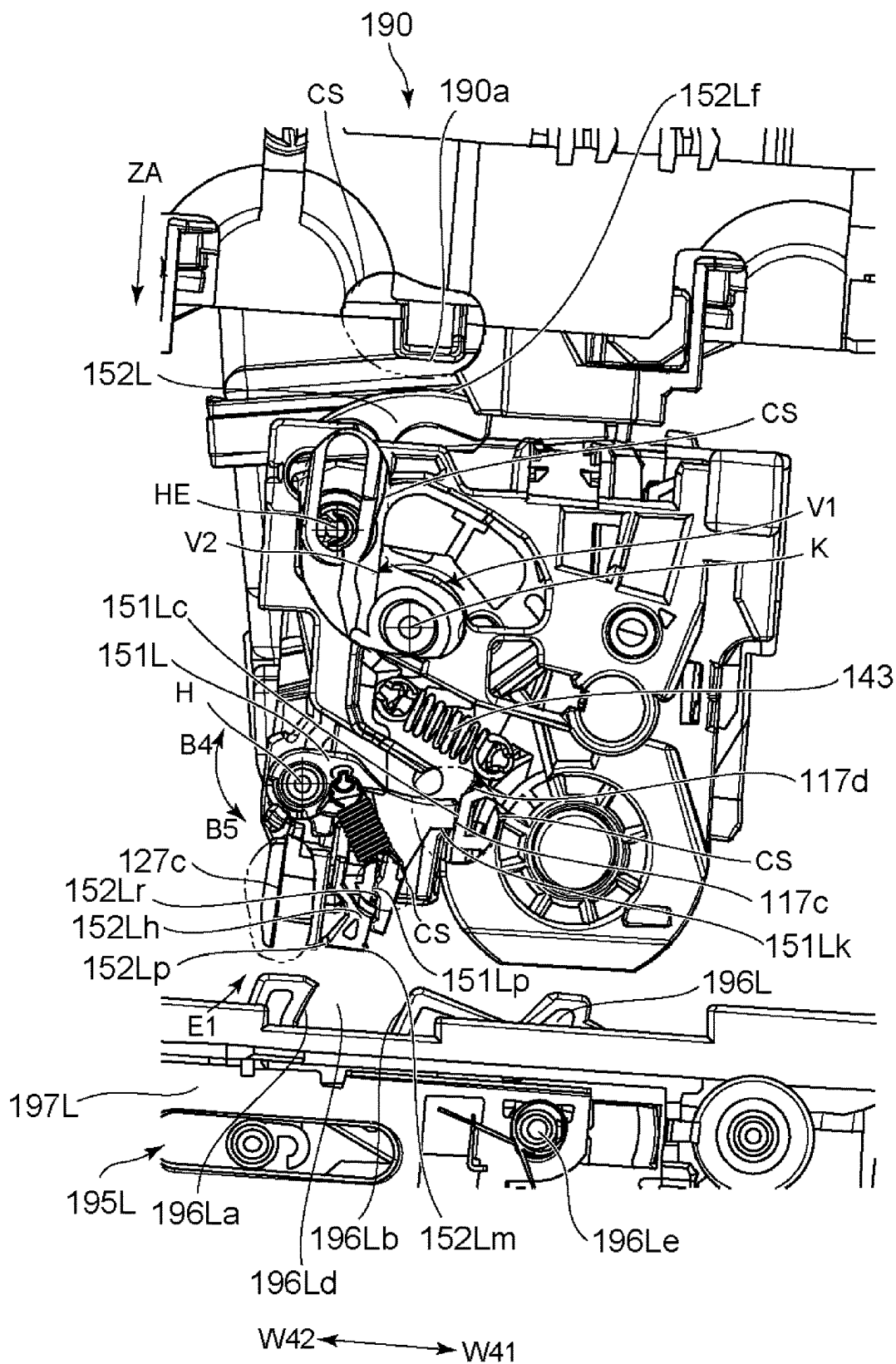


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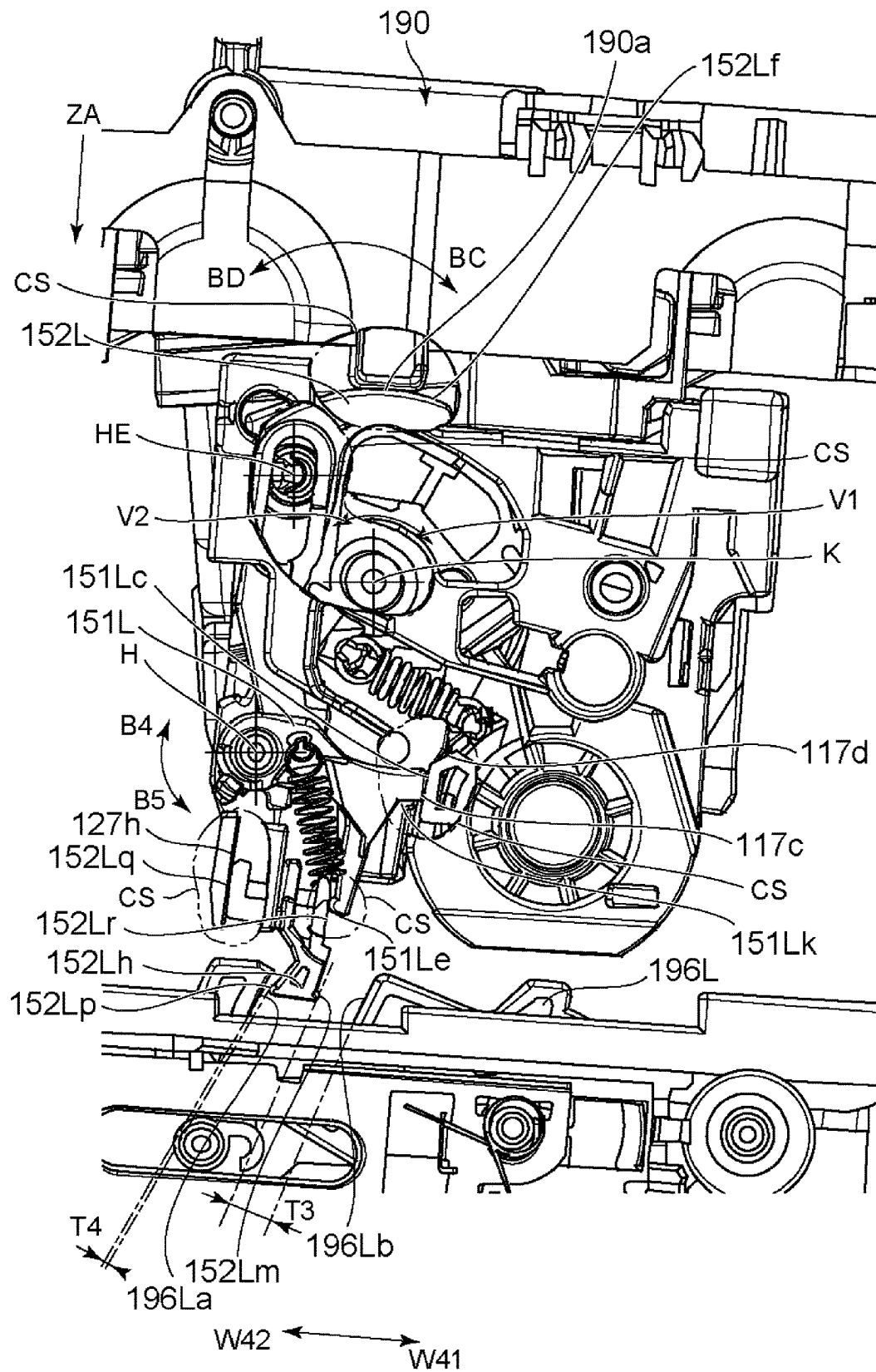


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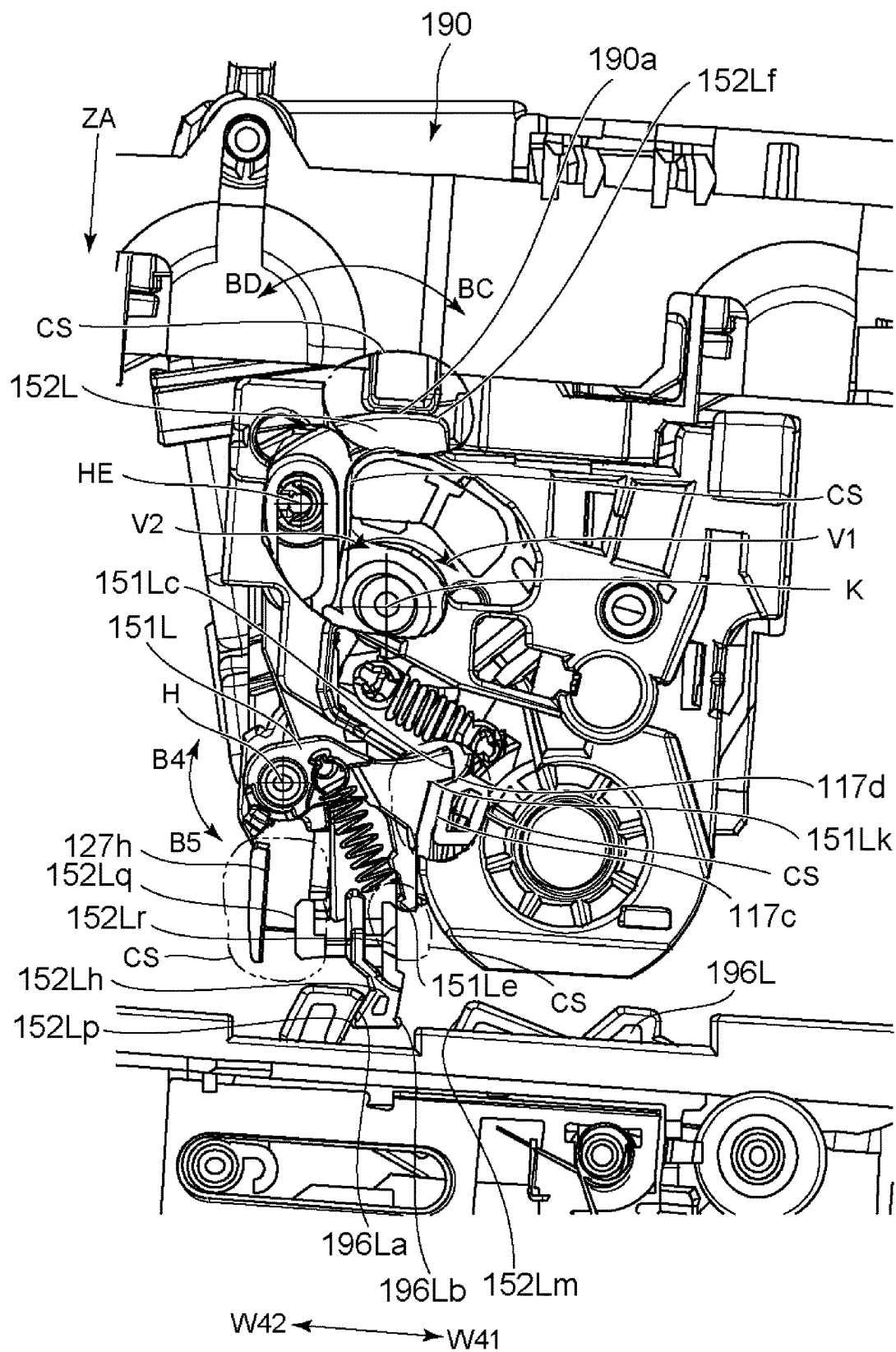


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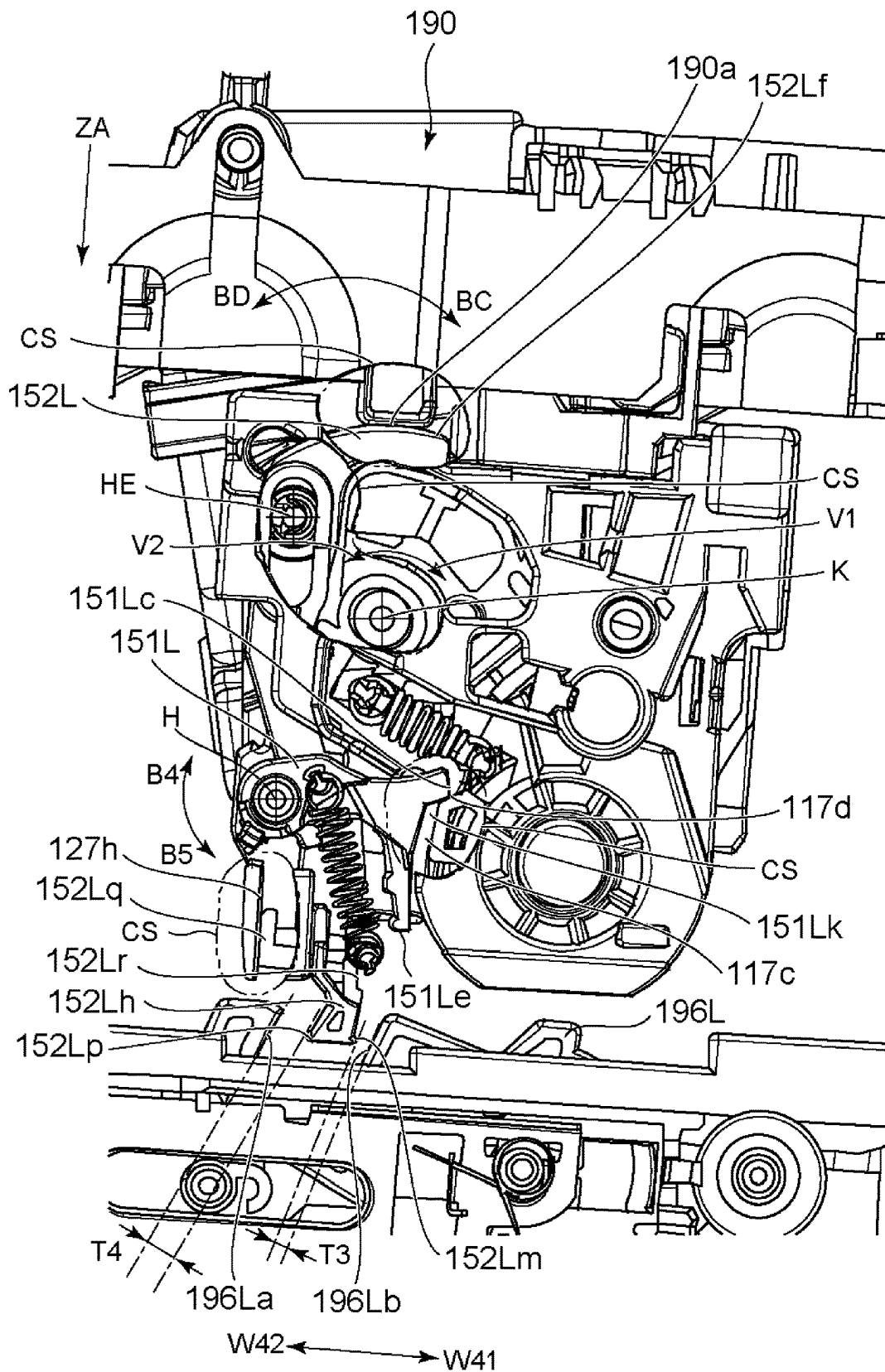


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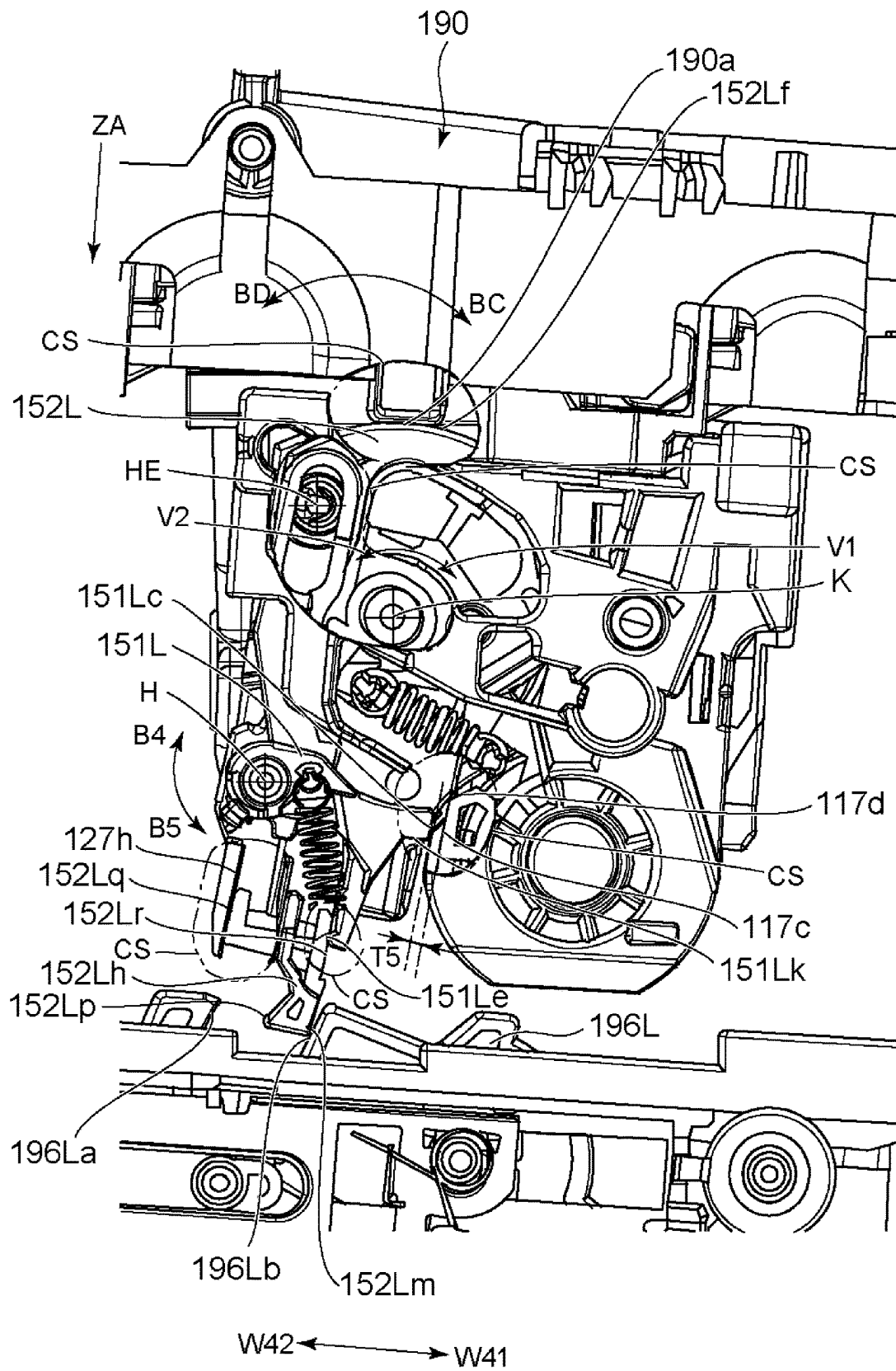


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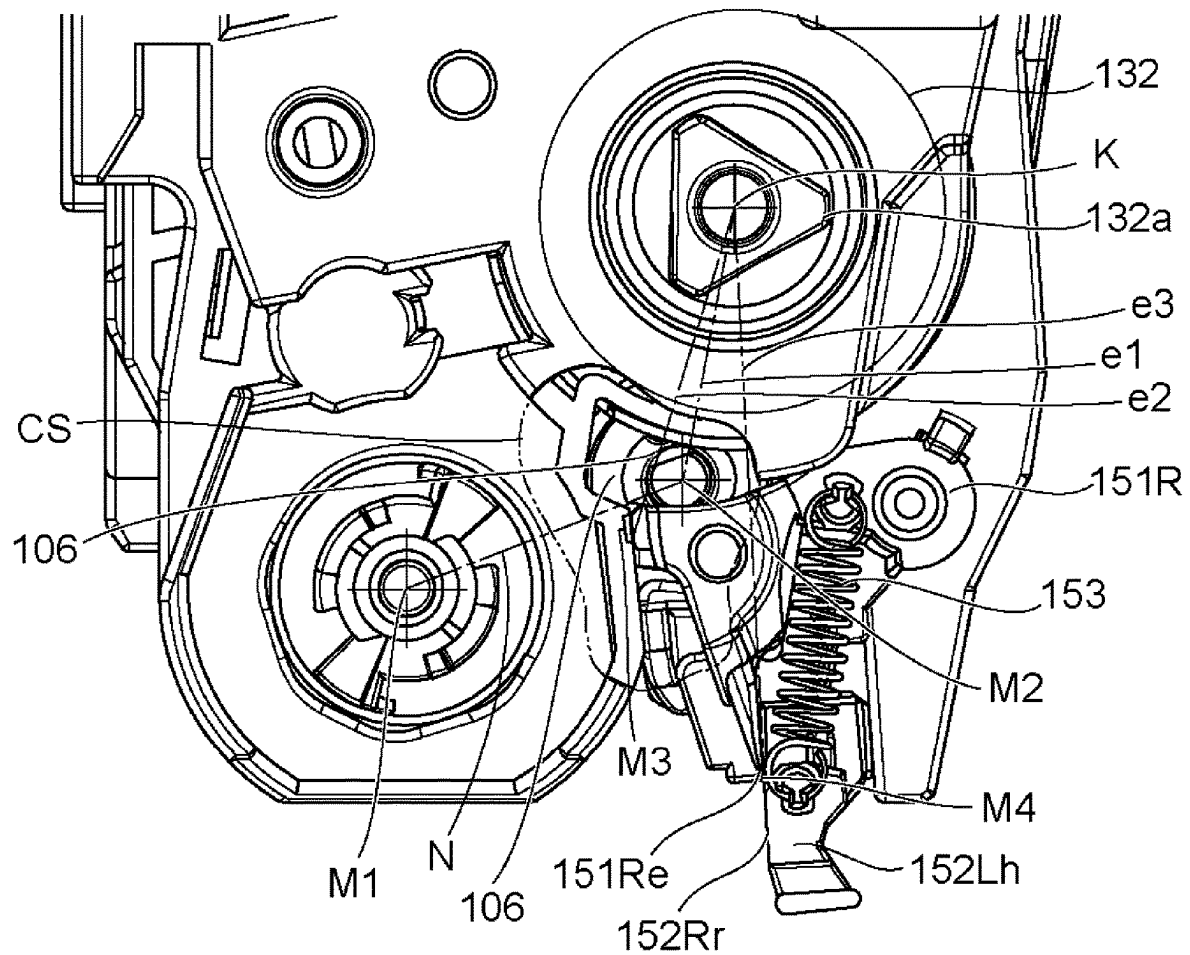


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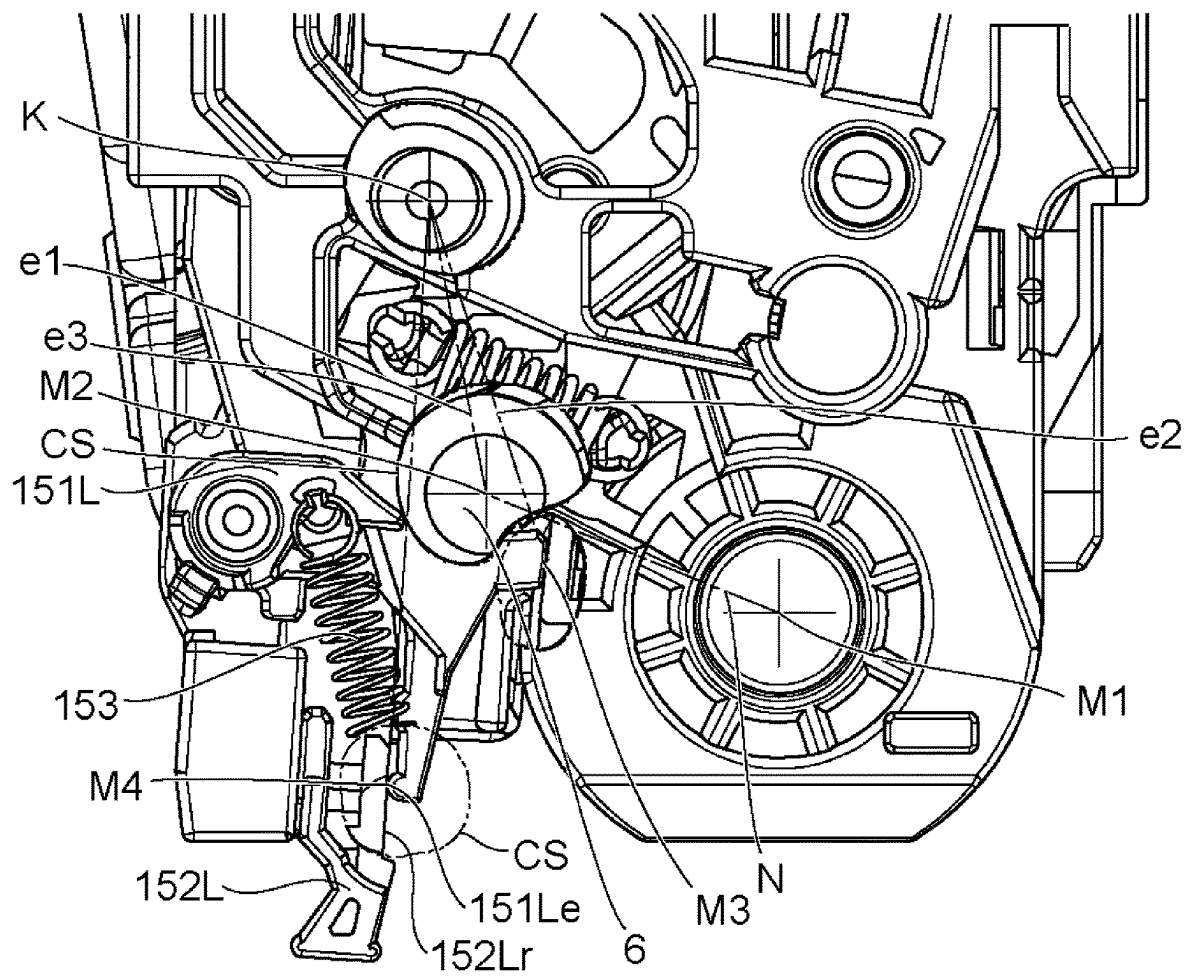


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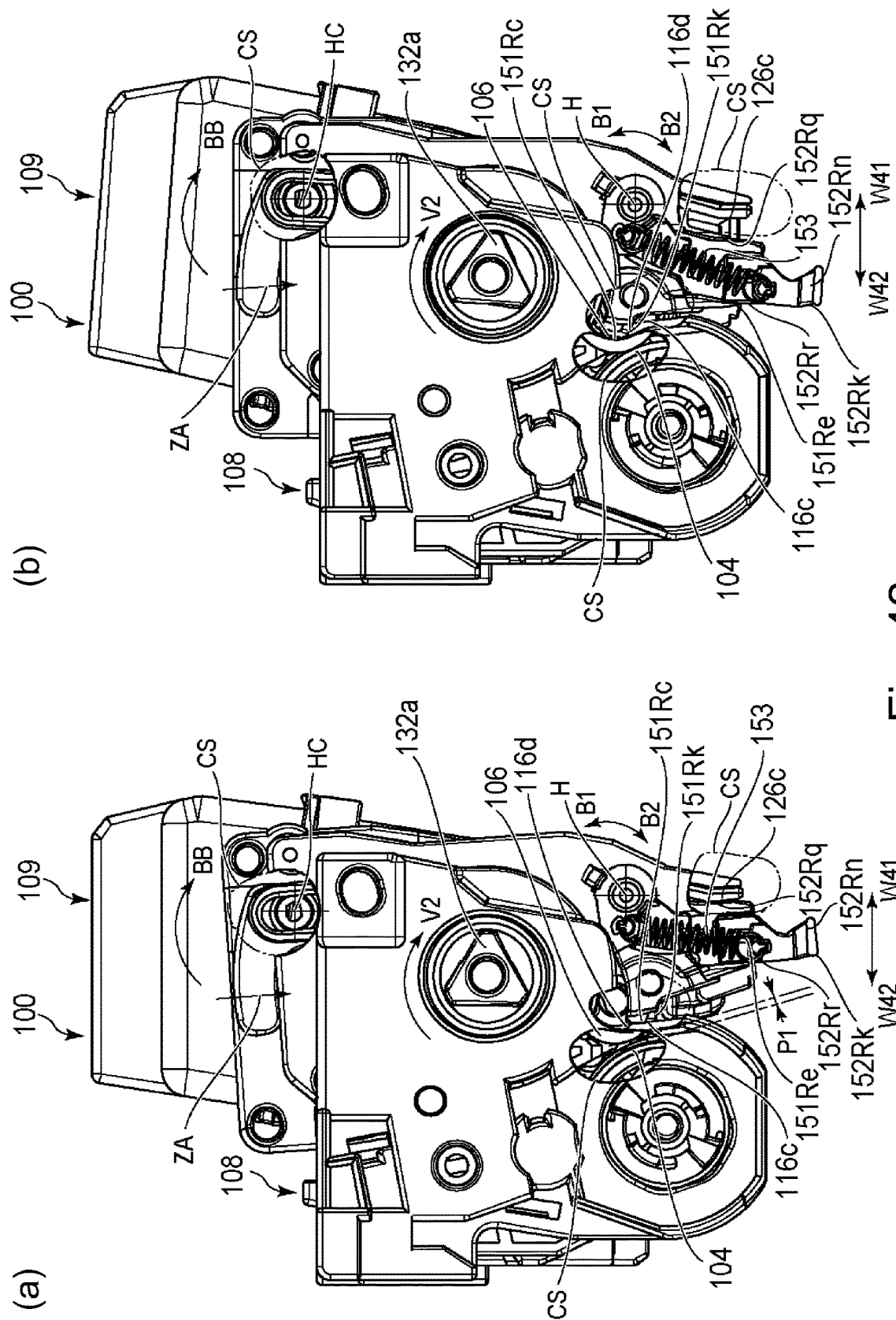


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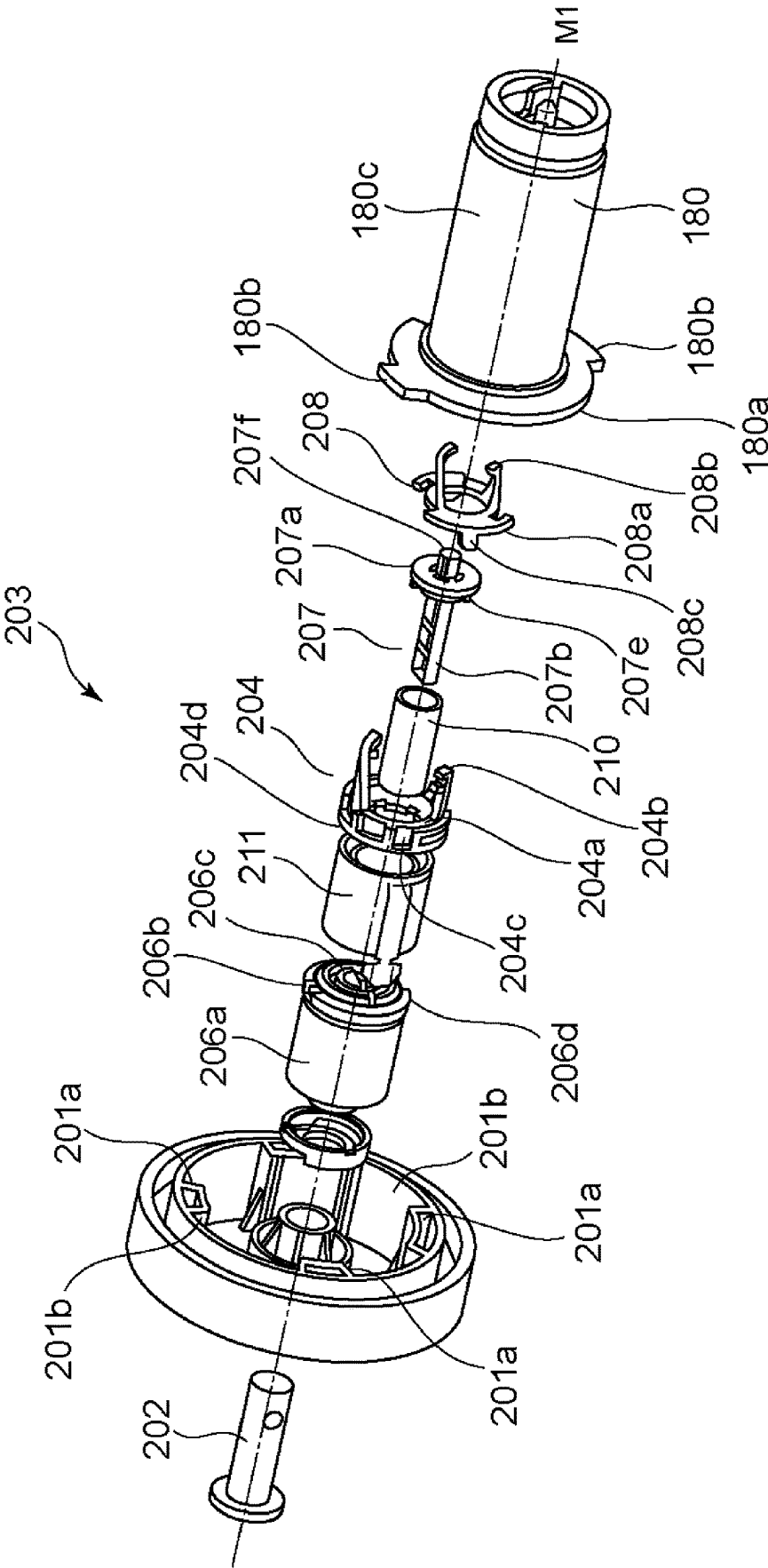


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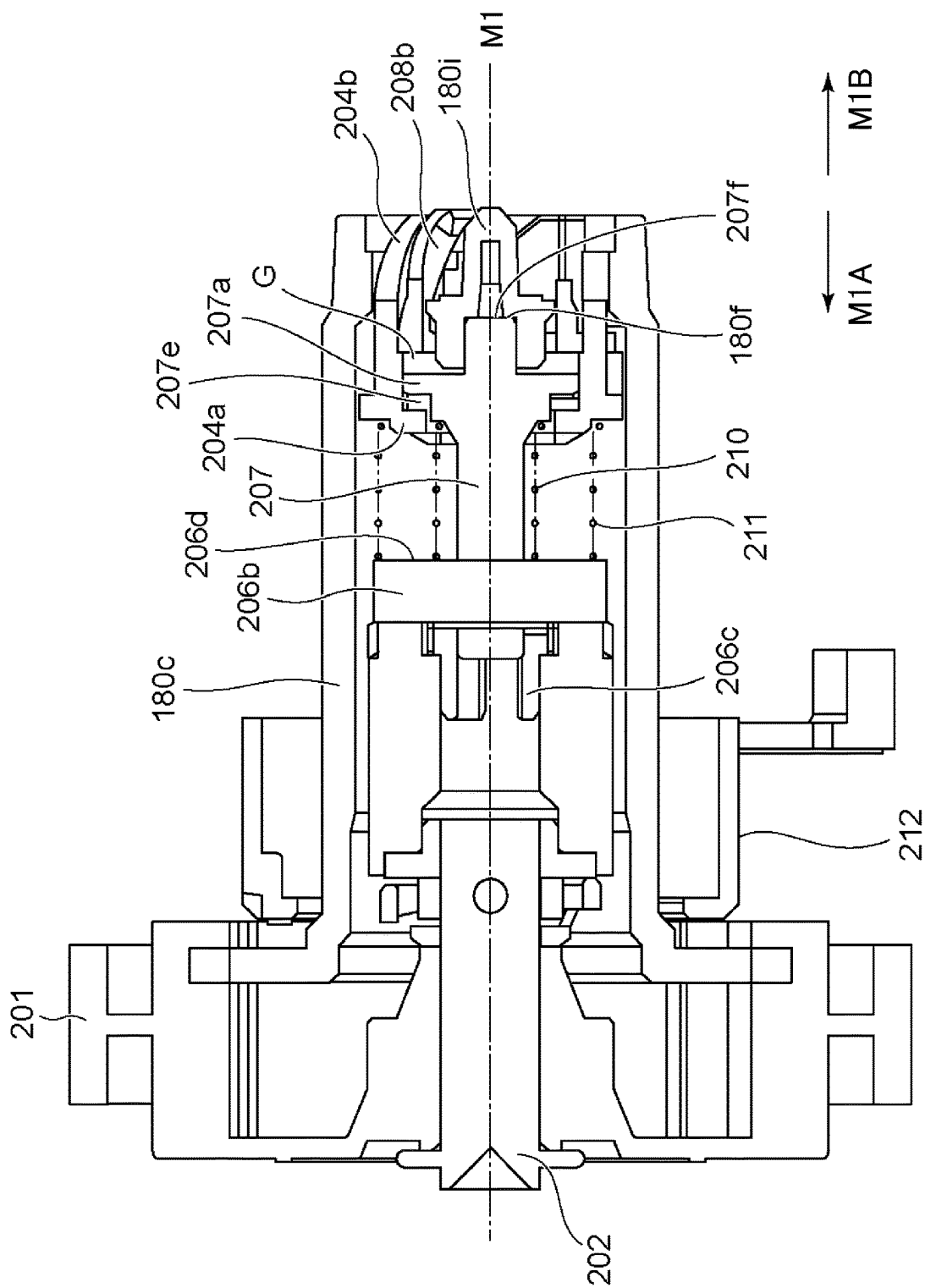


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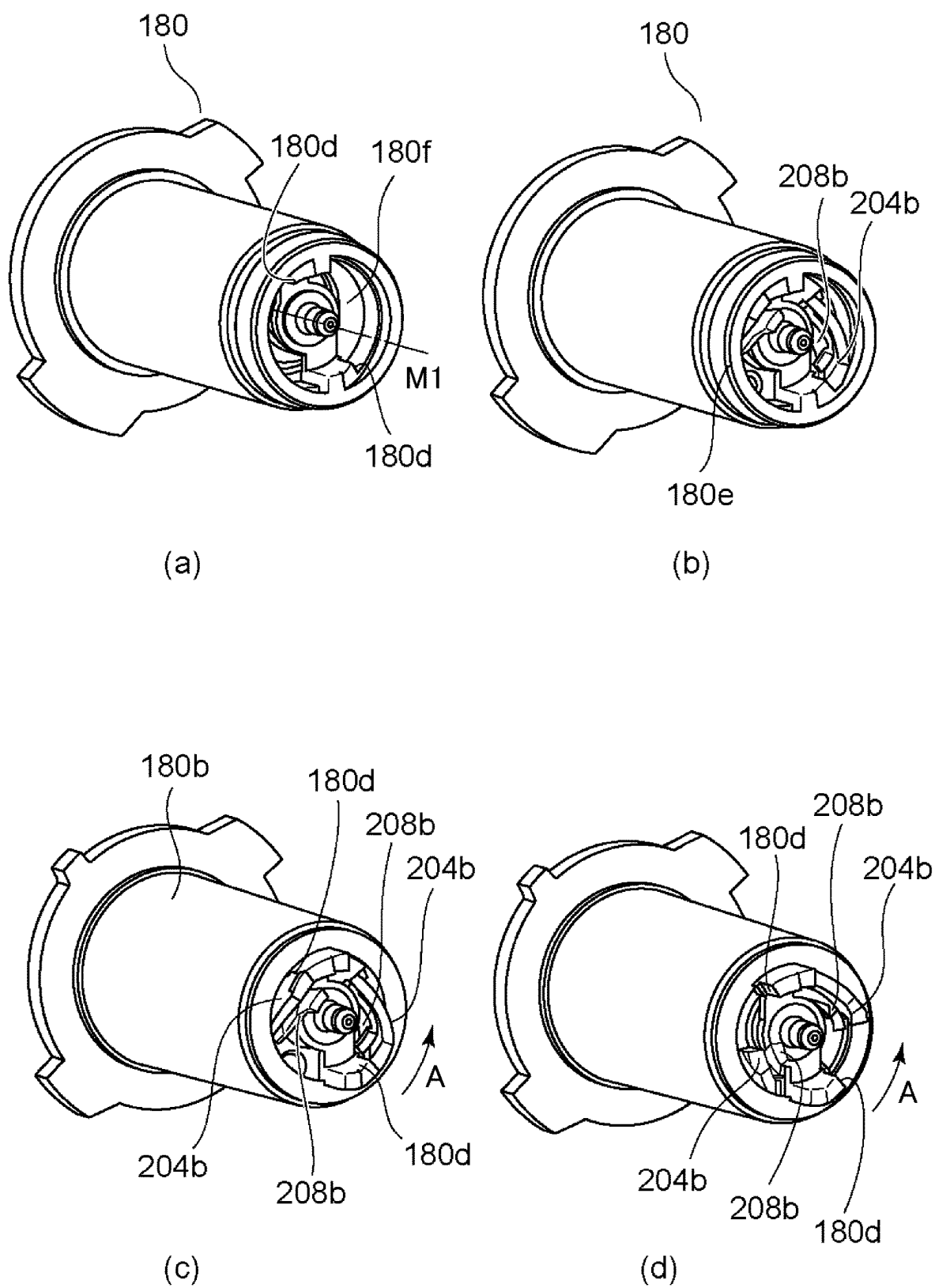


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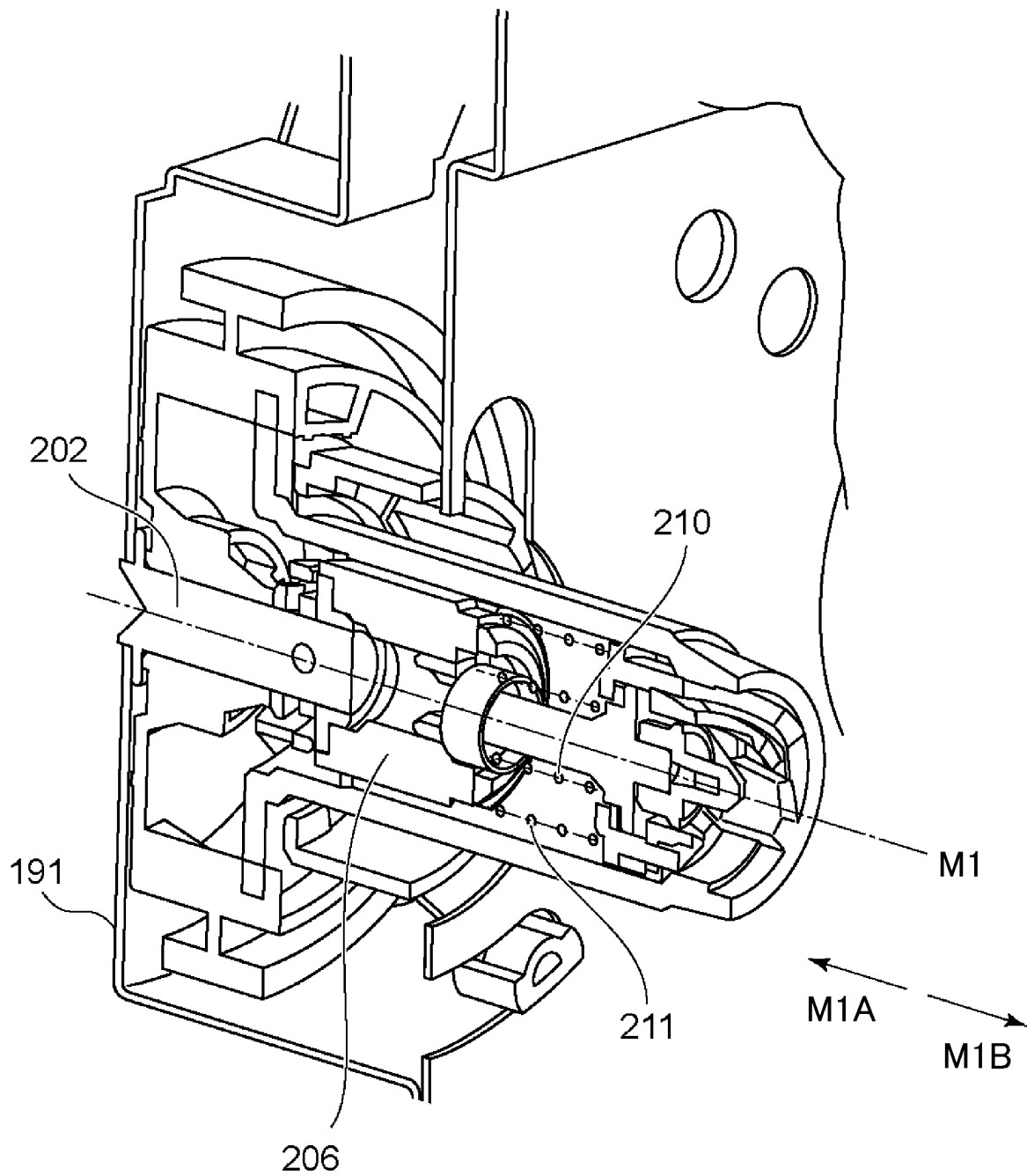
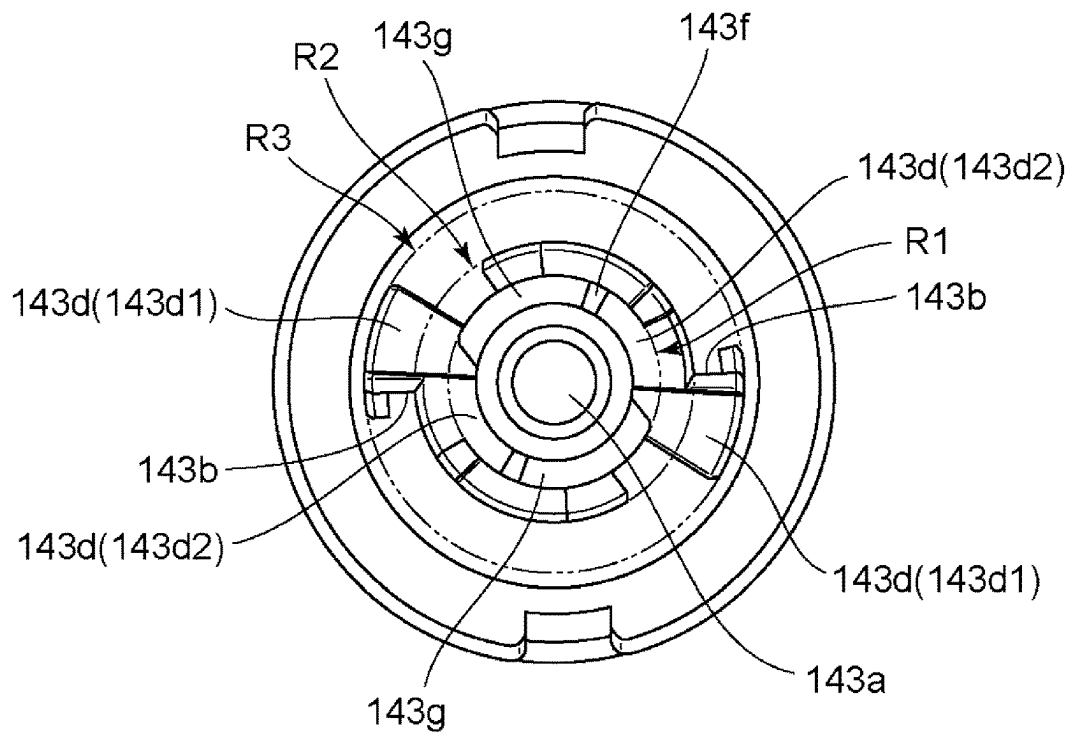
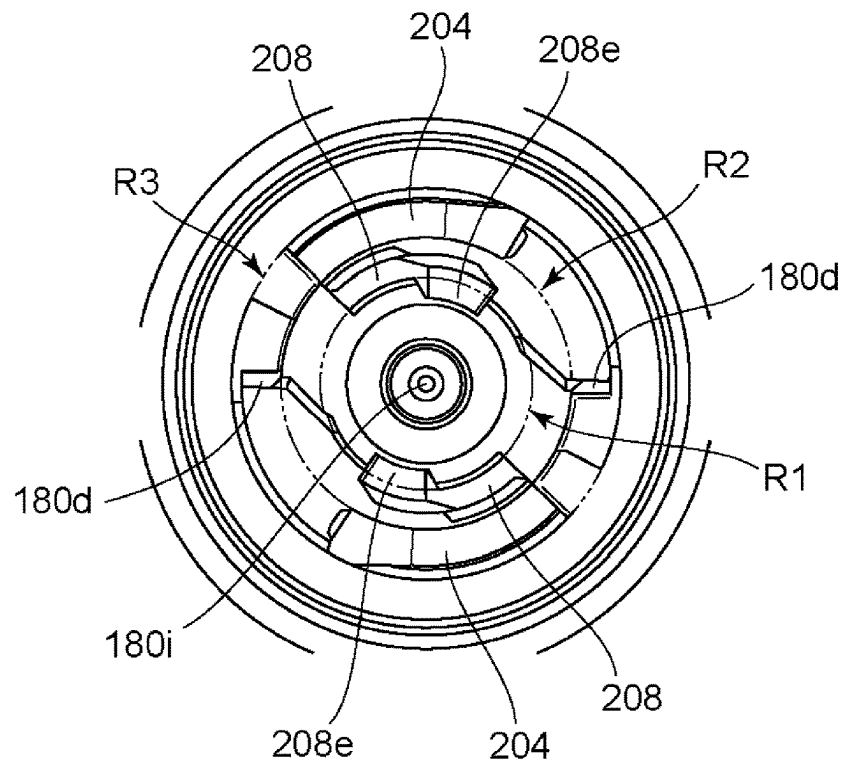


Fig. 46



(a)



(b)

Fig. 47

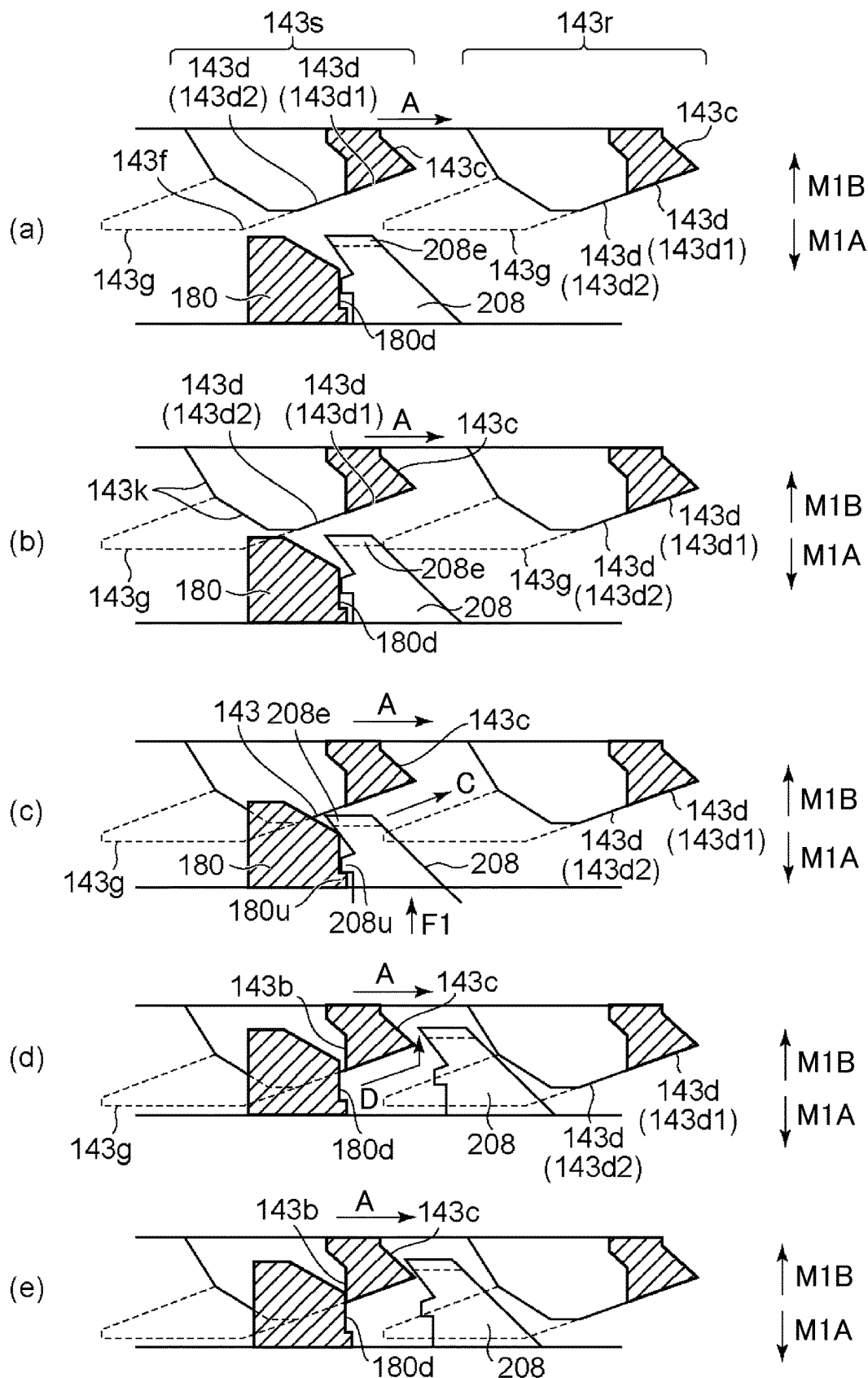


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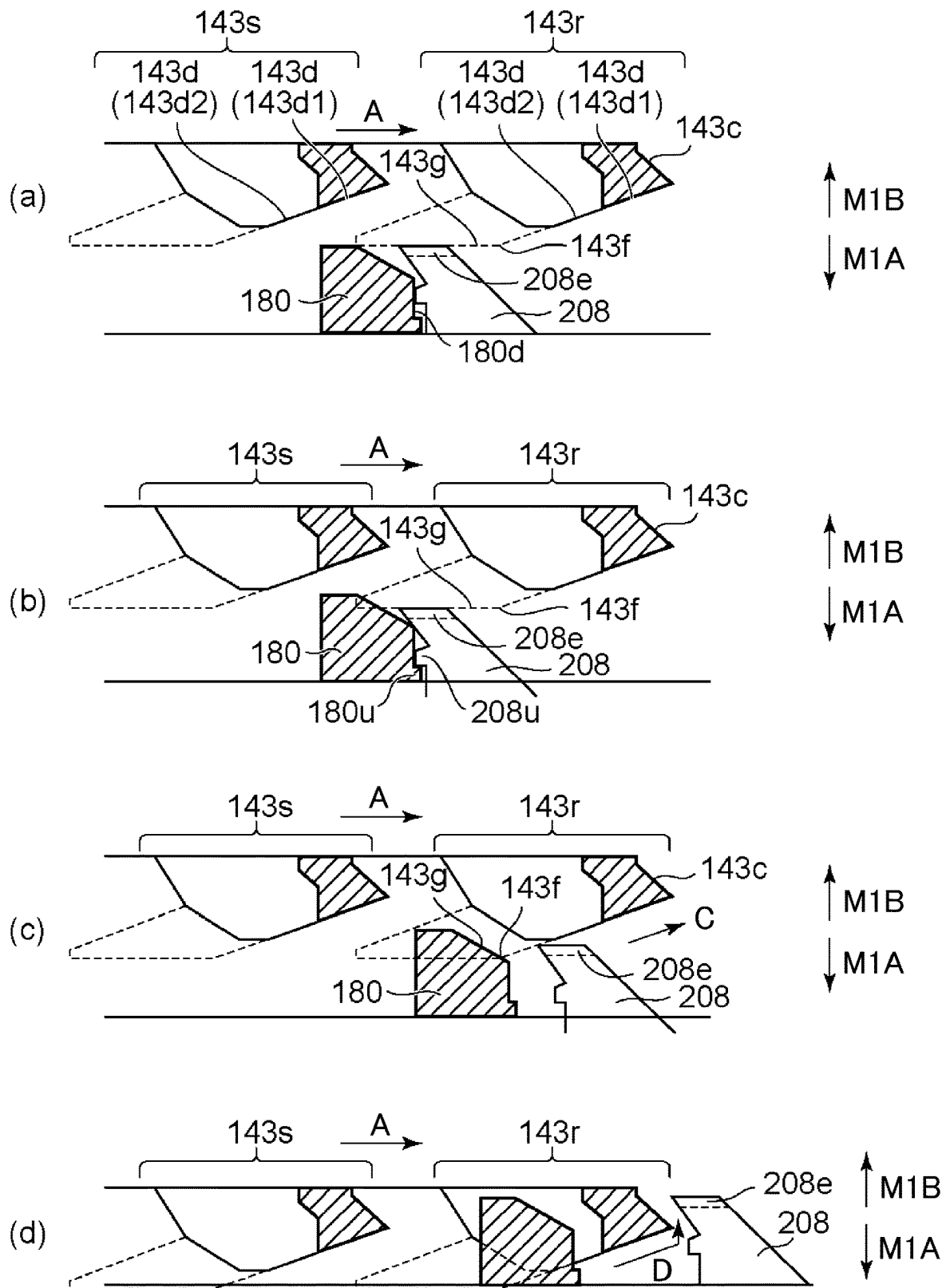


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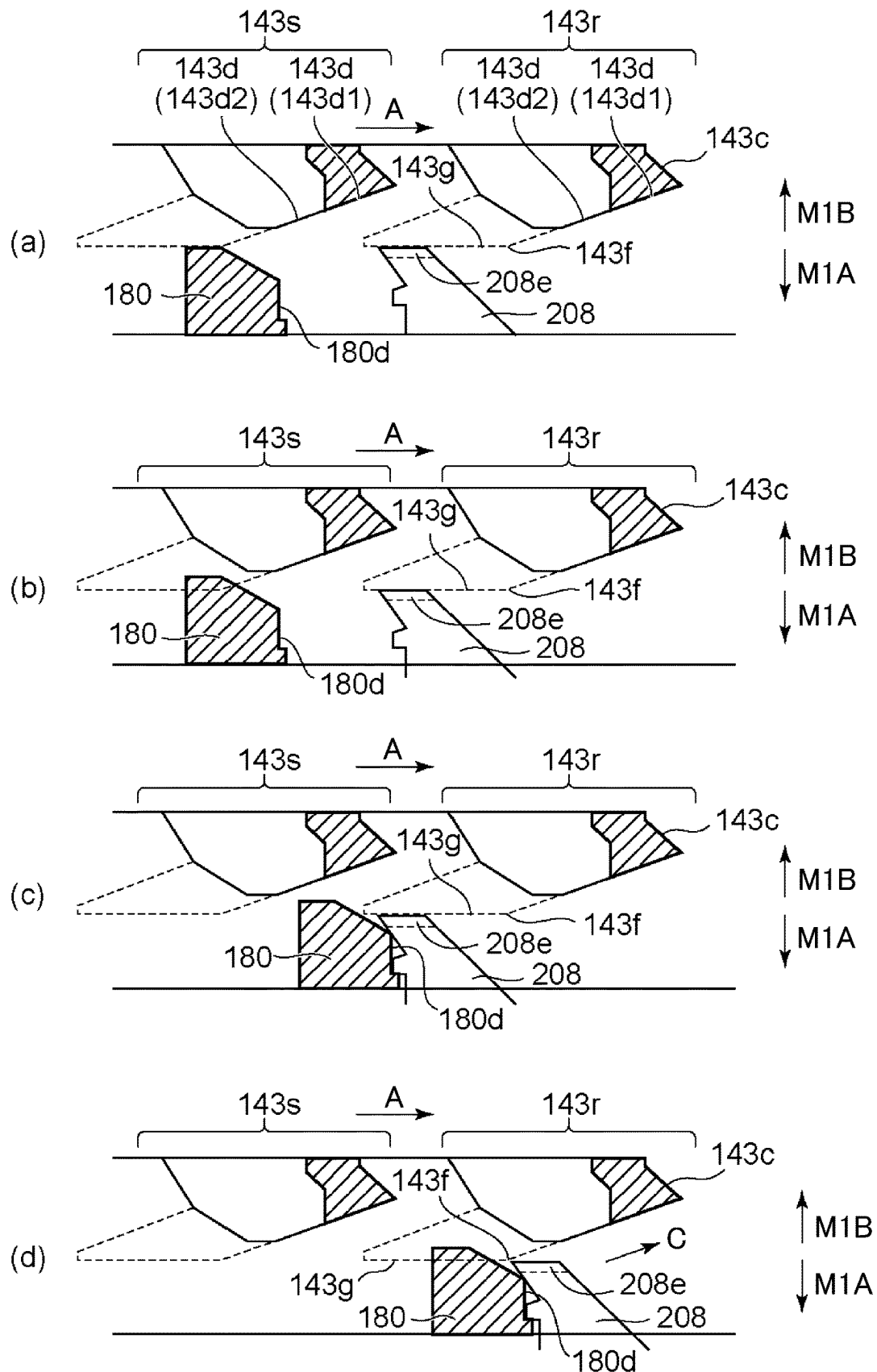


Fig. 50

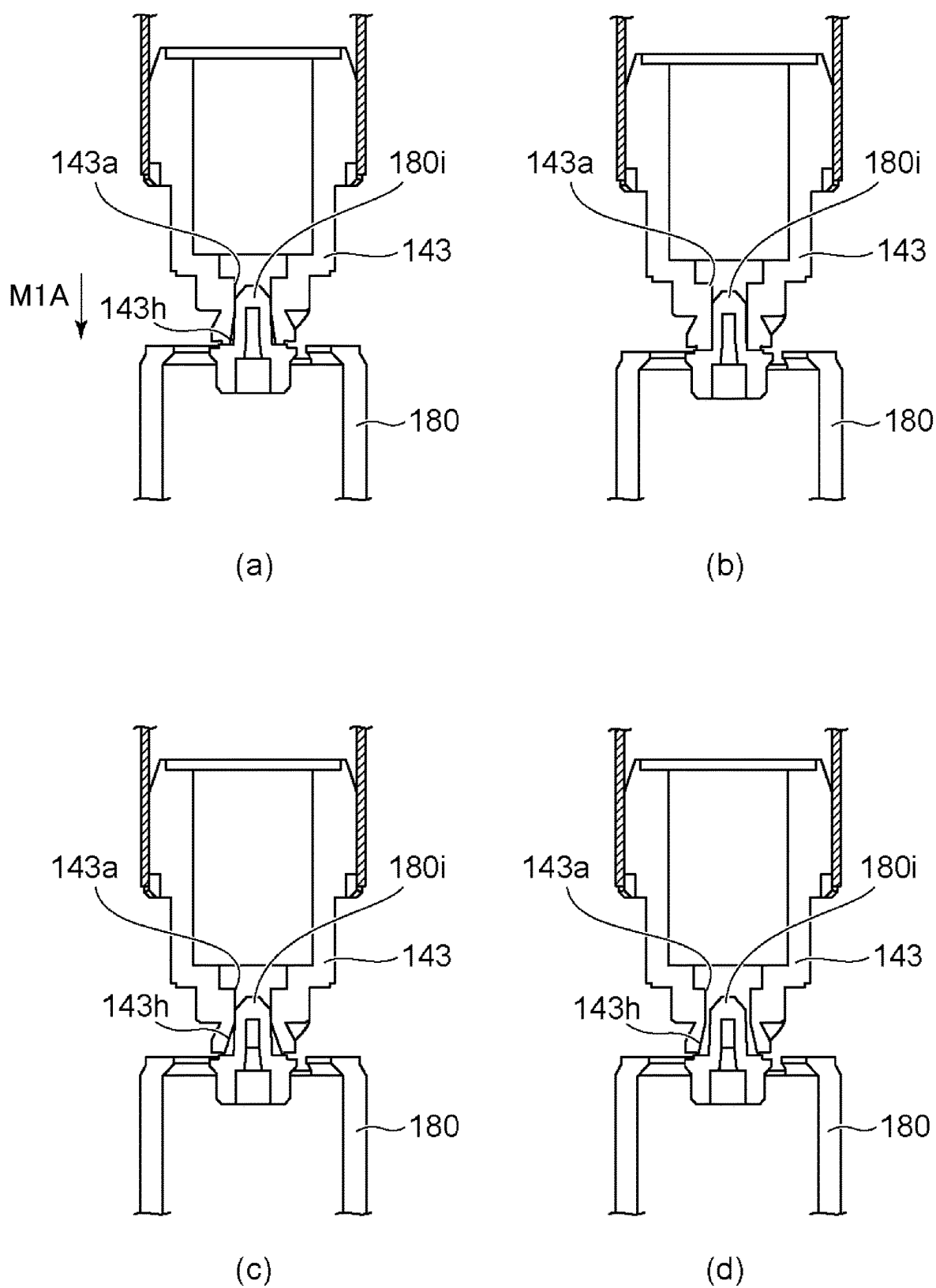


Fig. 51

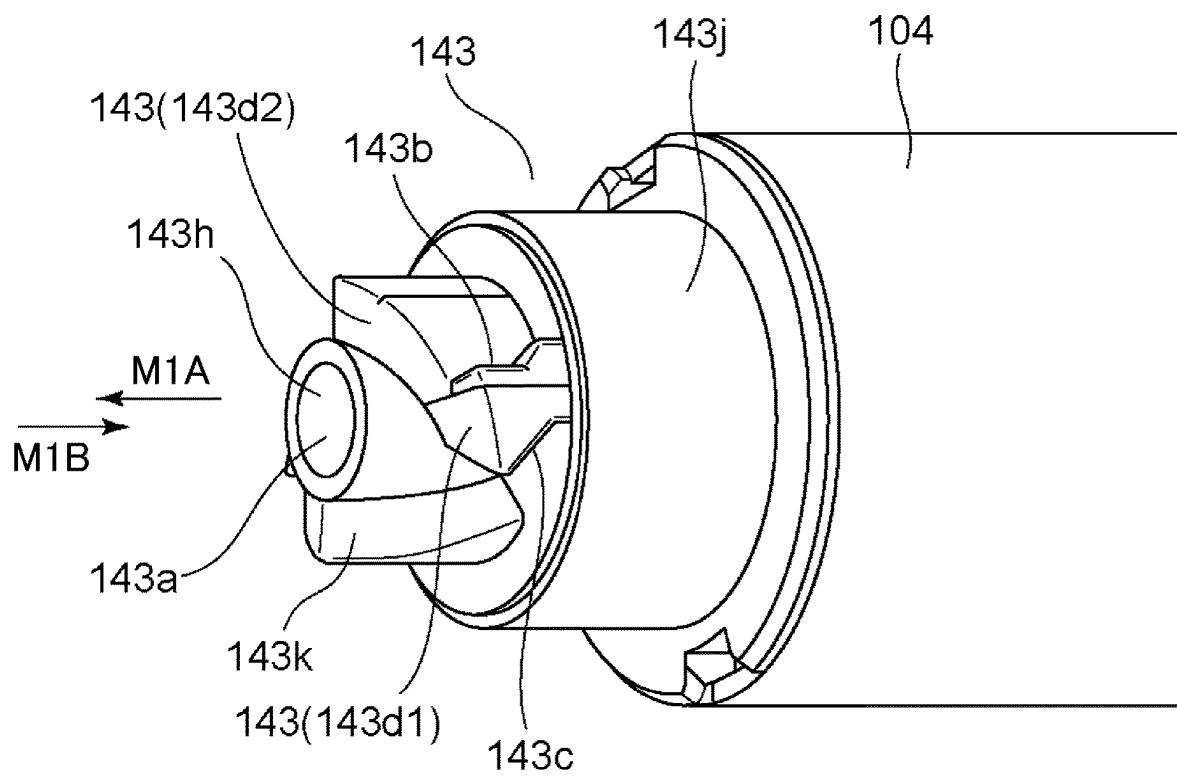


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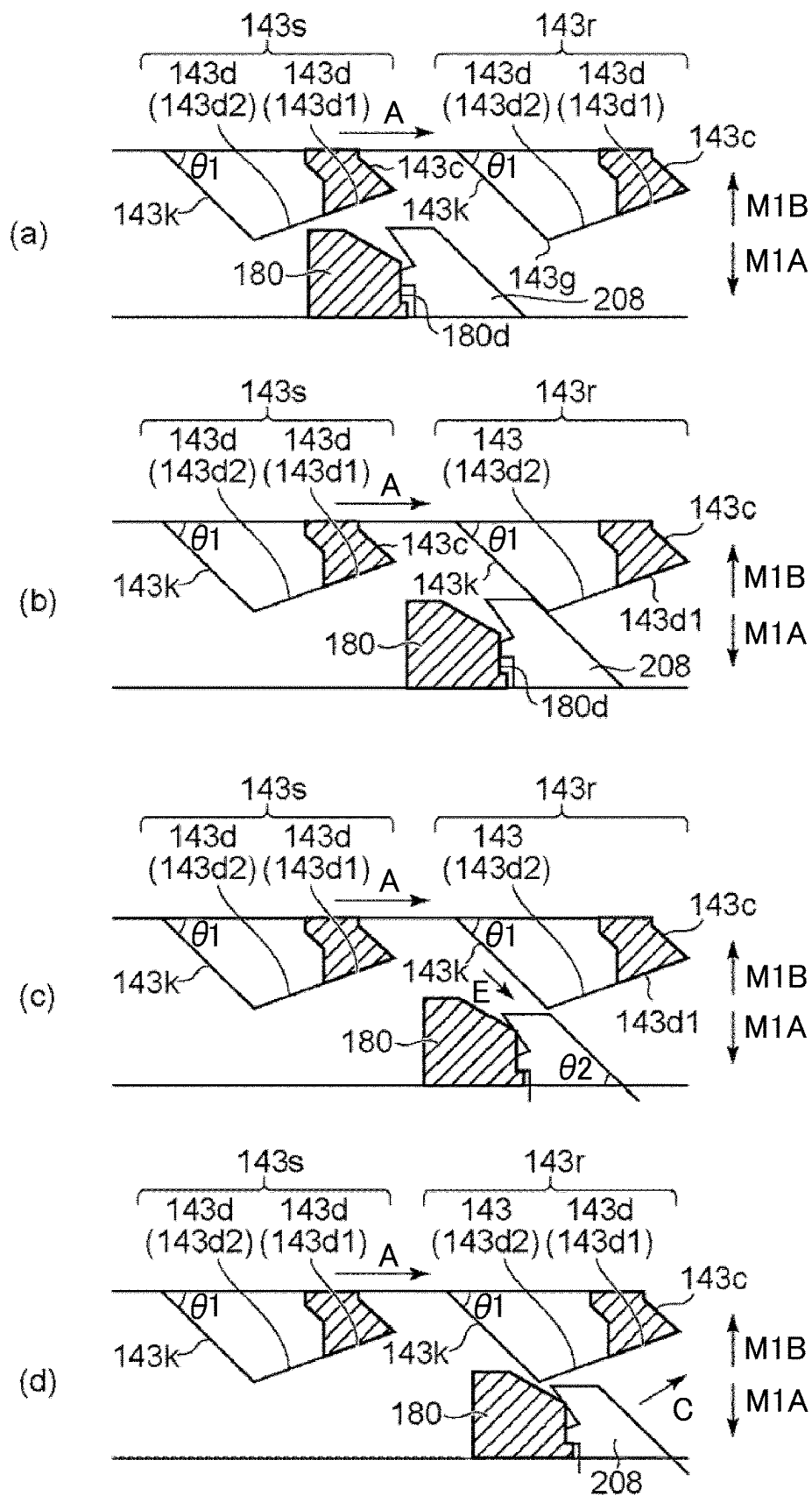


Fig. 53

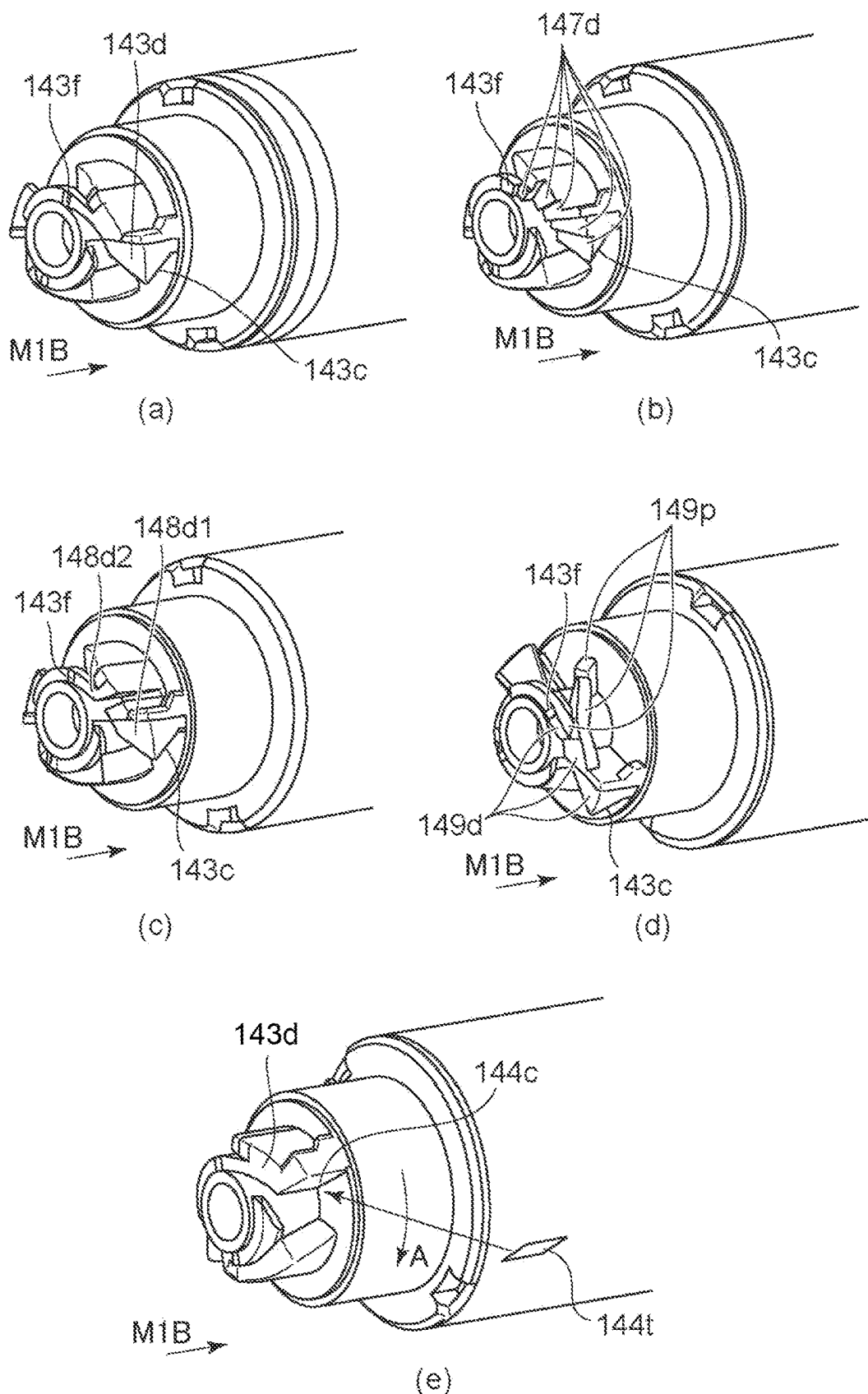
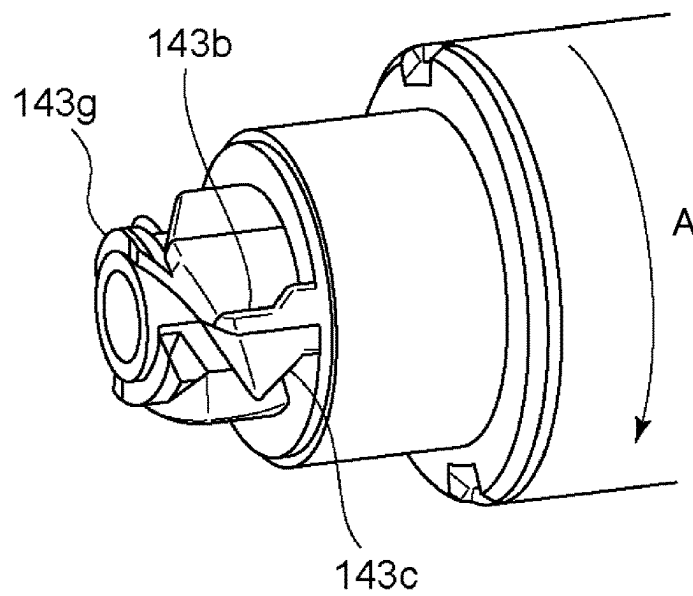
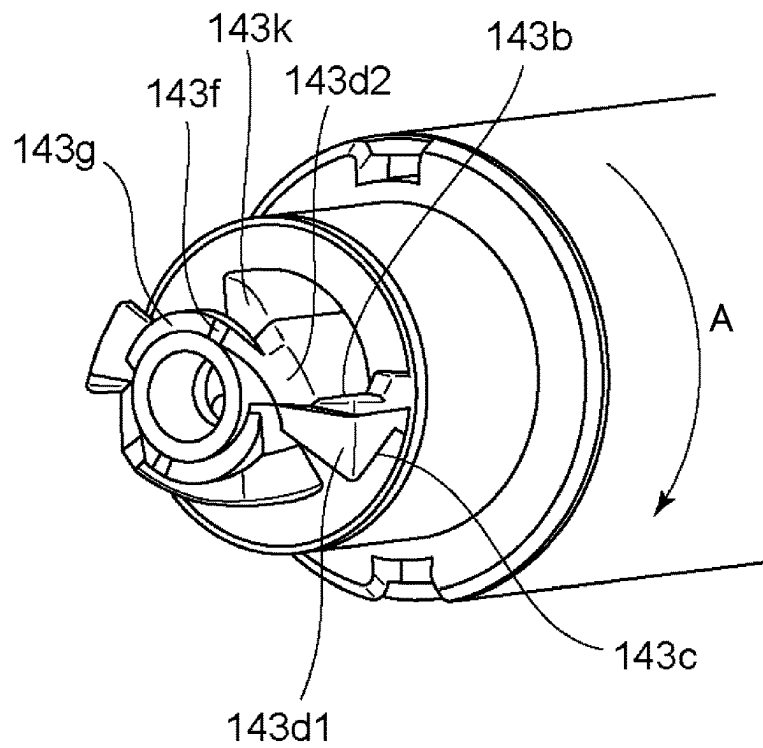


Fig. 54



(a)



(b)

Fig. 55

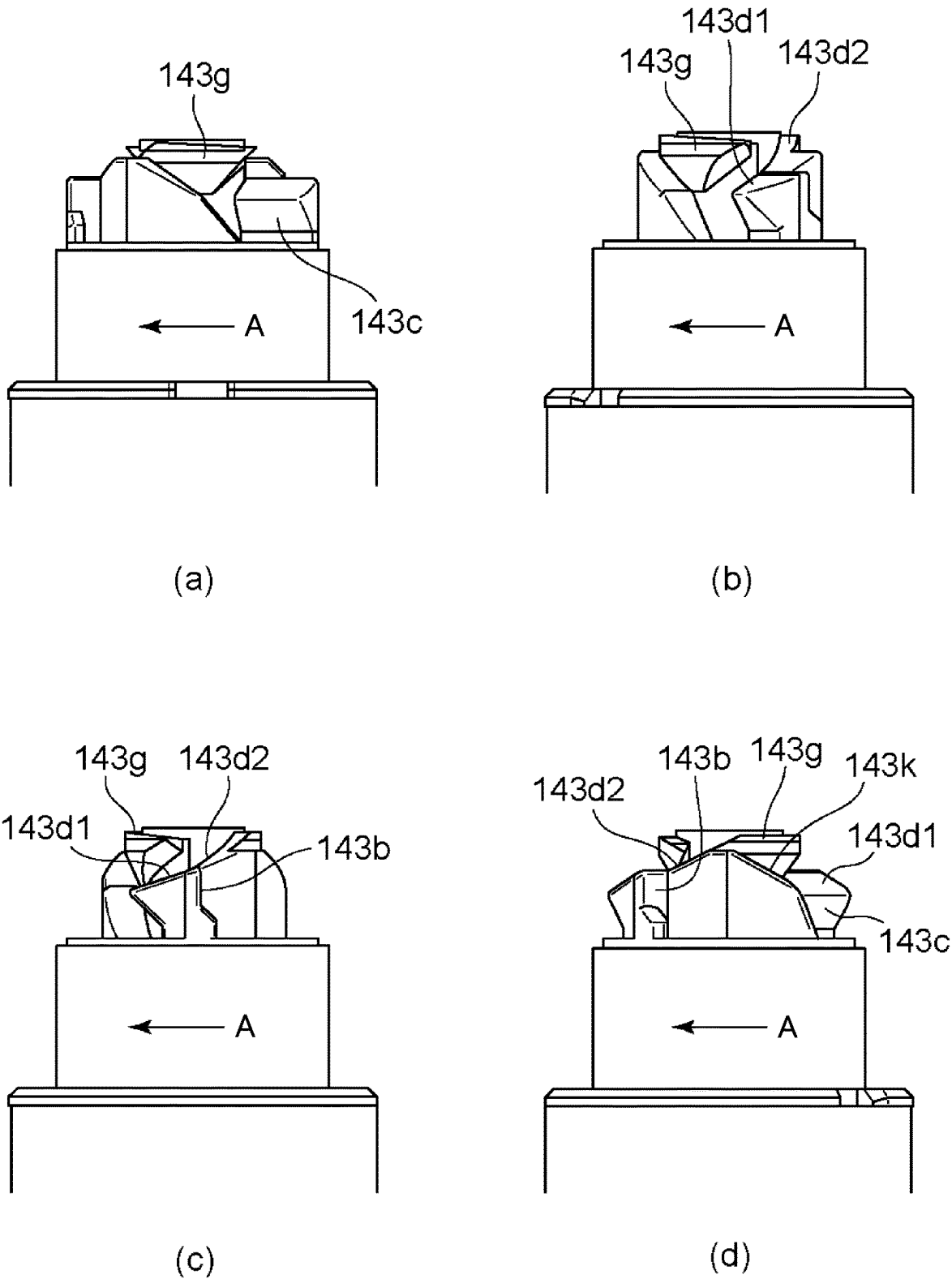


Fig. 56

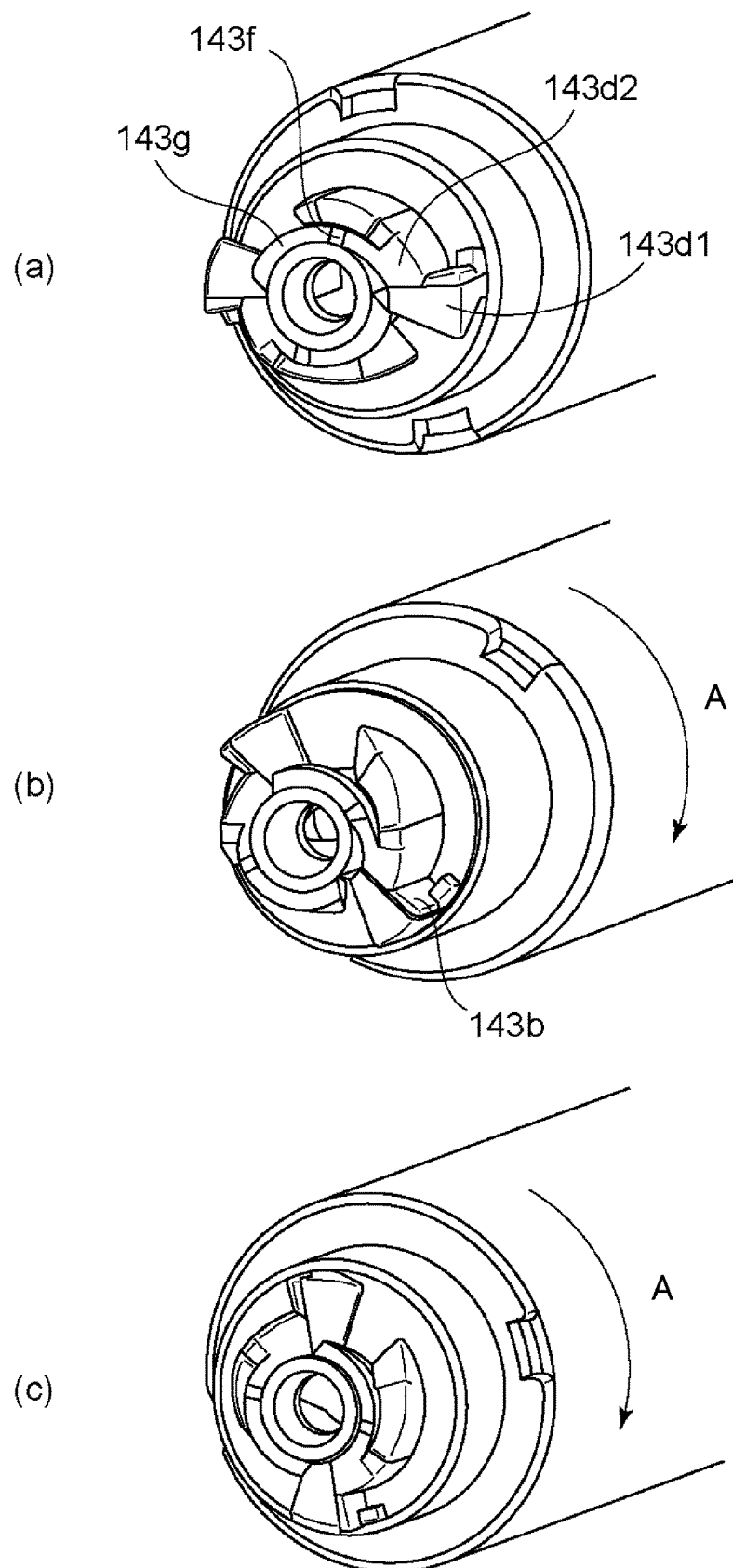


Fig. 57

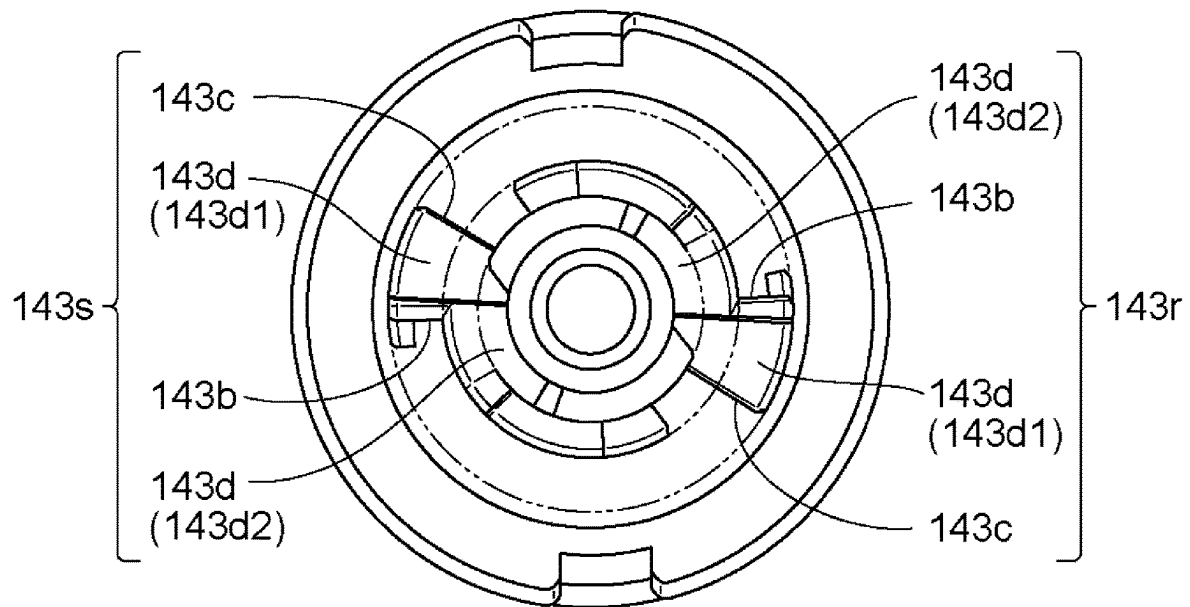


Fig. 58

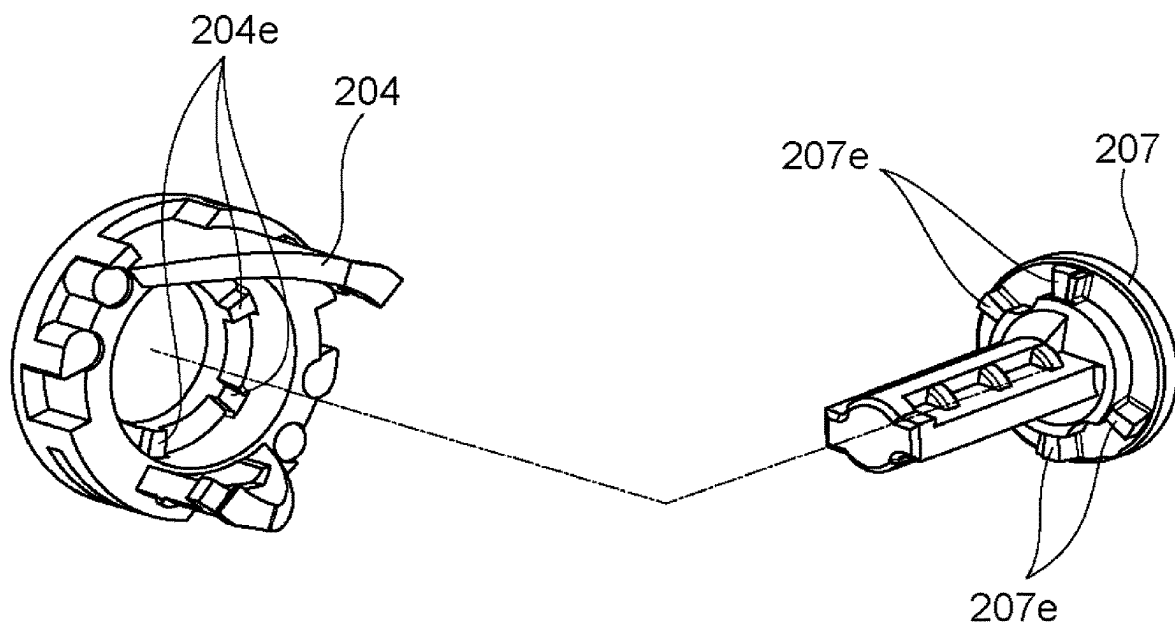


Fig. 59

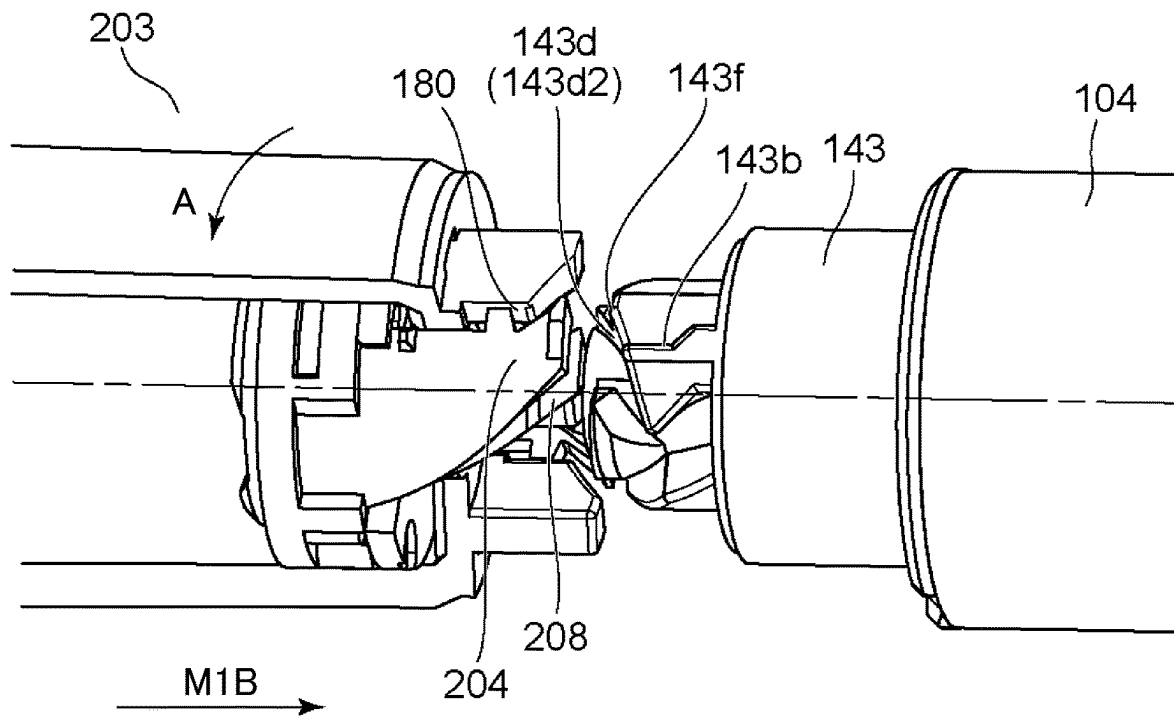


Fig. 60

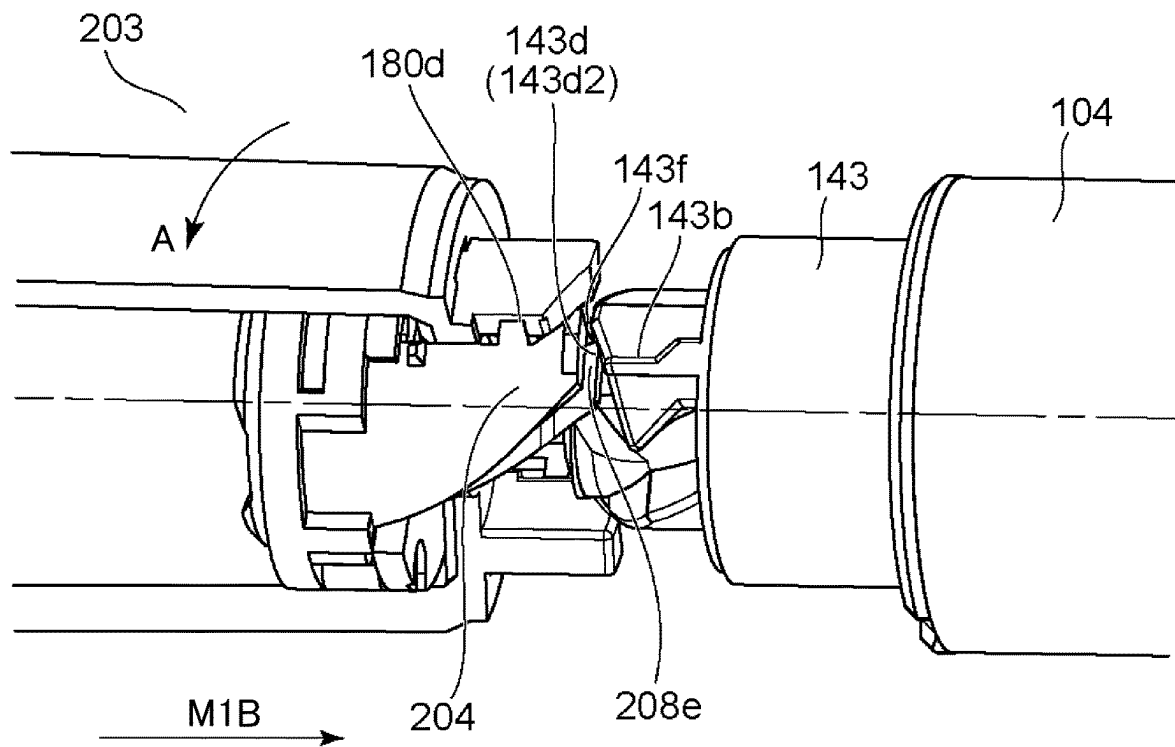


Fig. 61

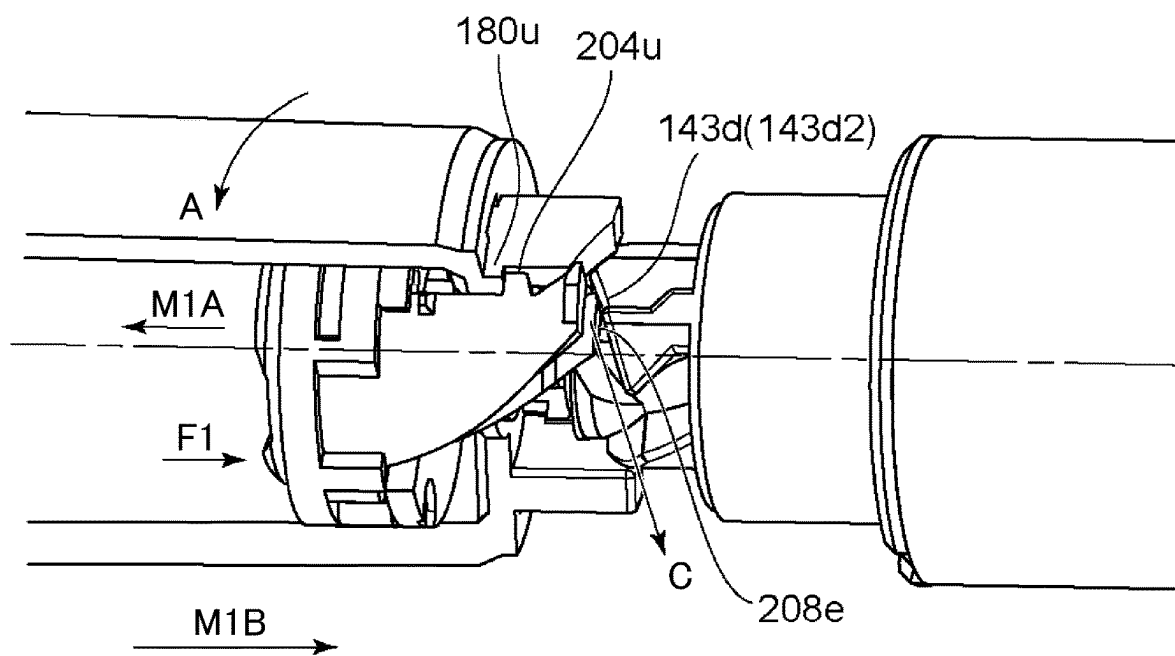


Fig. 62

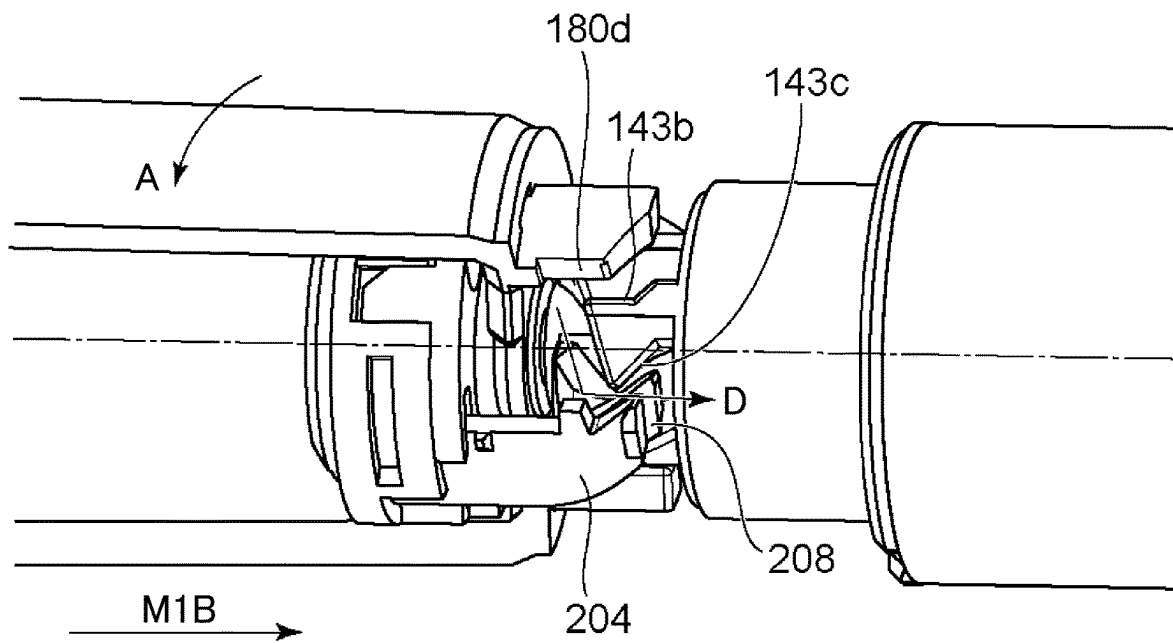


Fig. 63

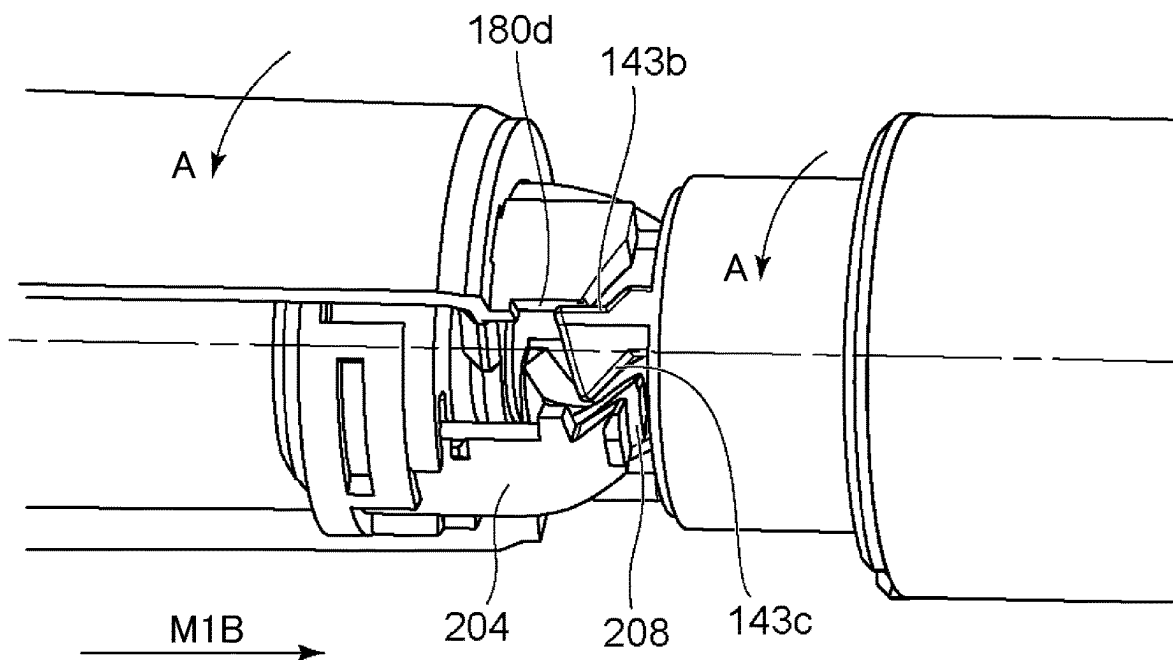


Fig. 64

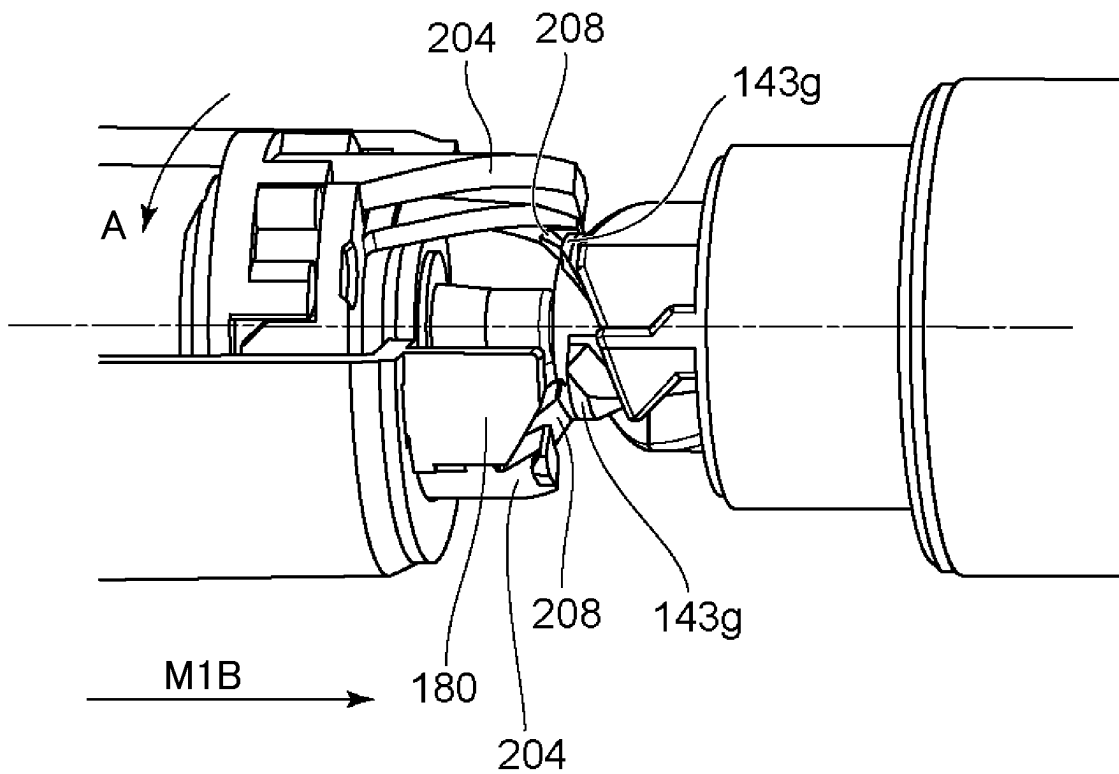


Fig. 65

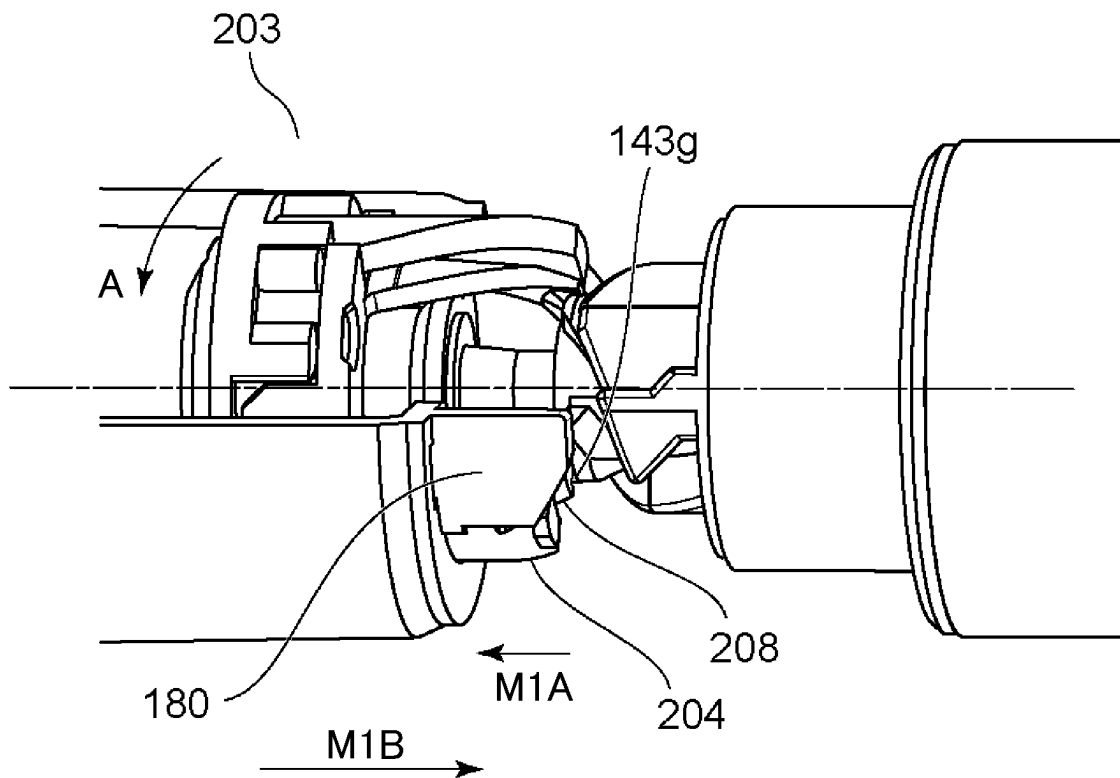


Fig. 66

Fig. 67

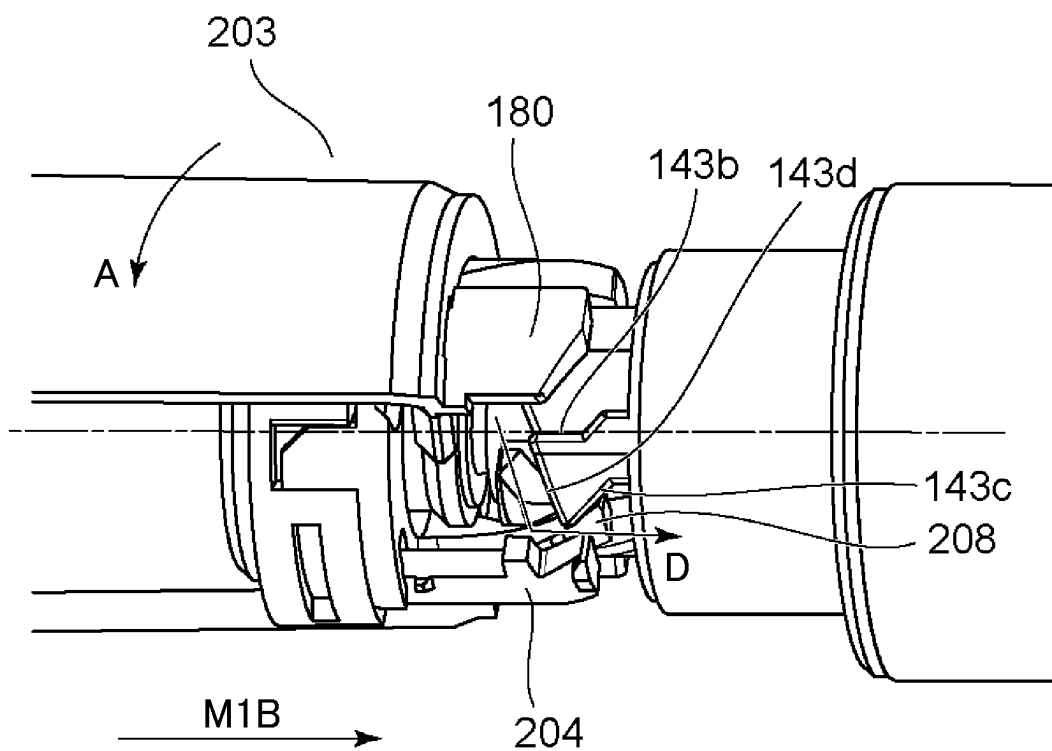


Fig. 68

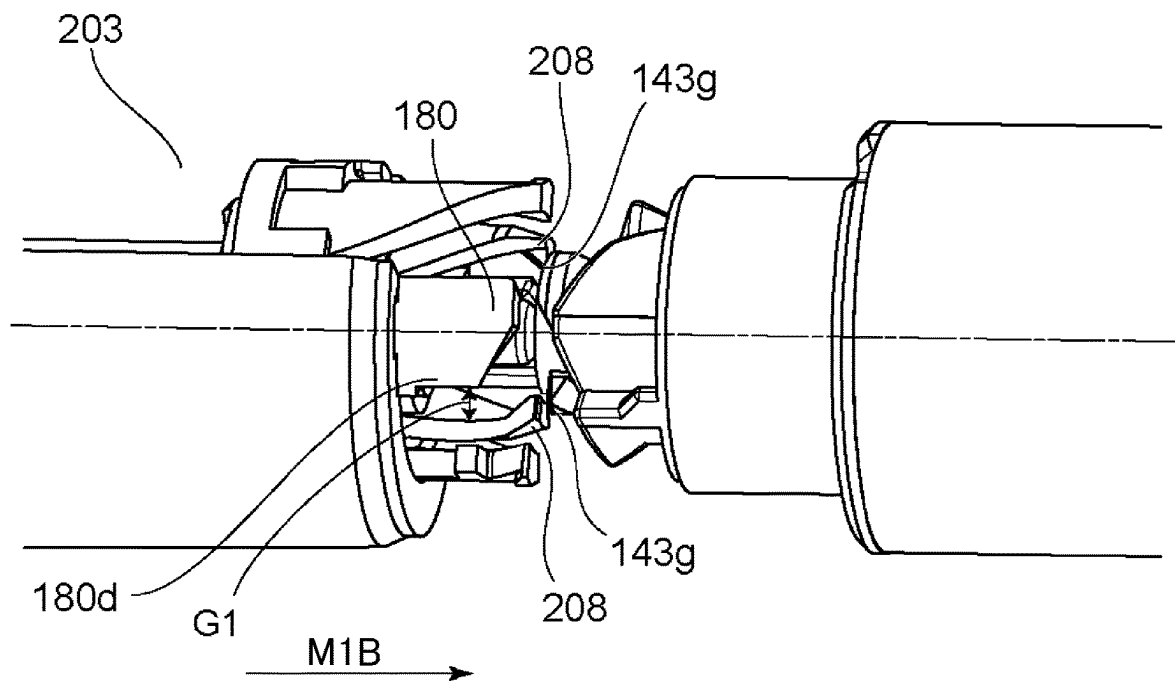


Fig. 69

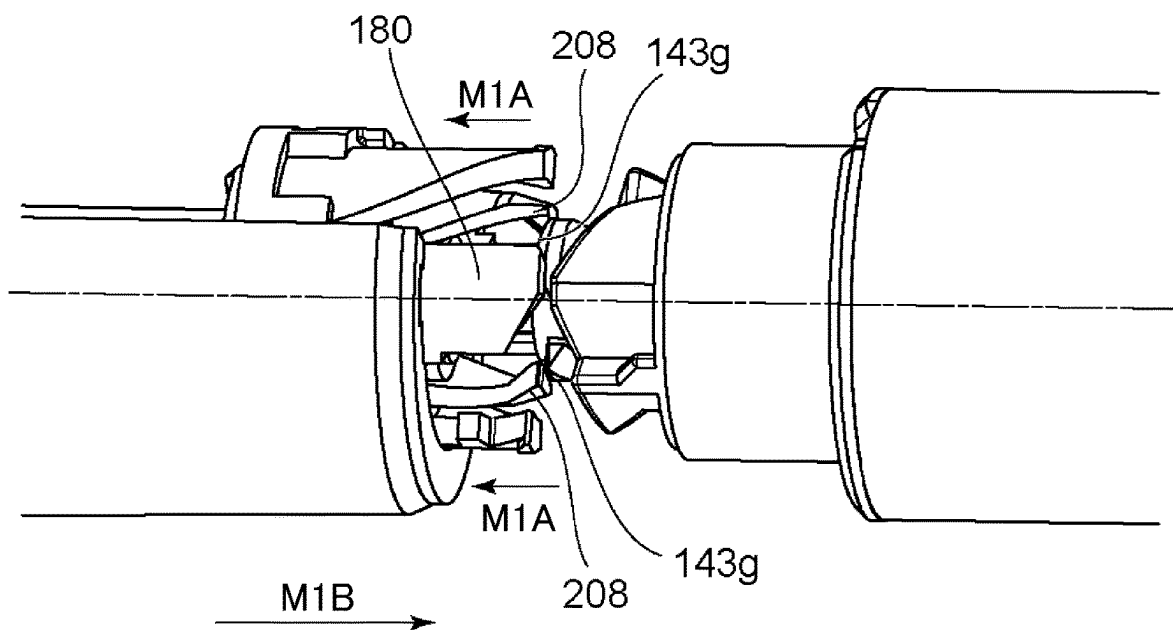


Fig. 70

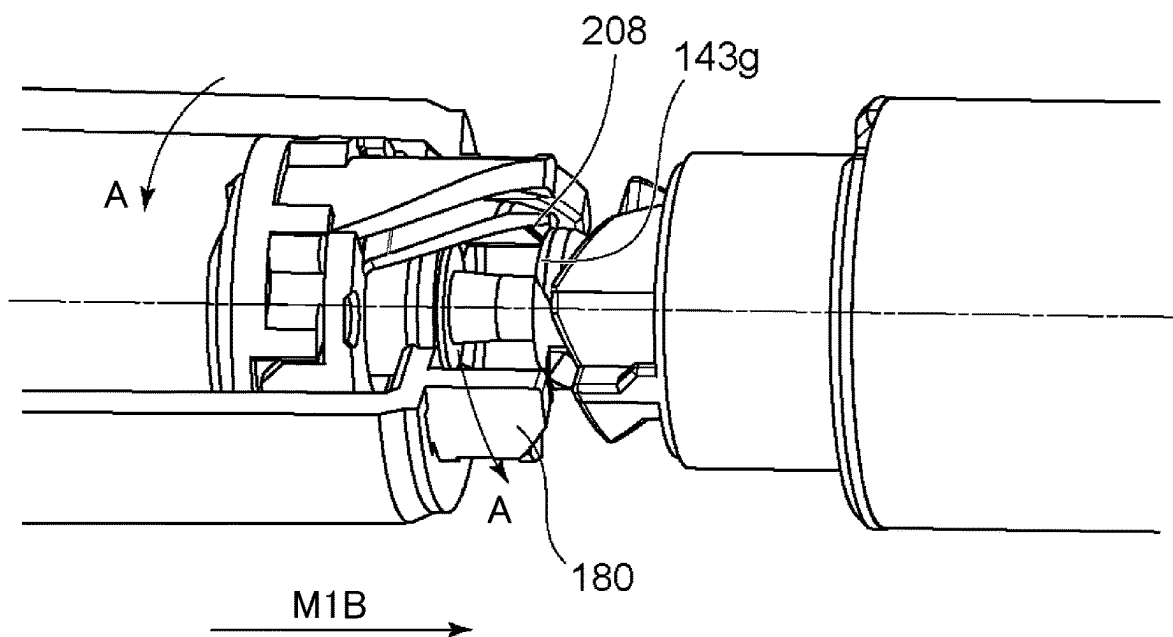


Fig. 71

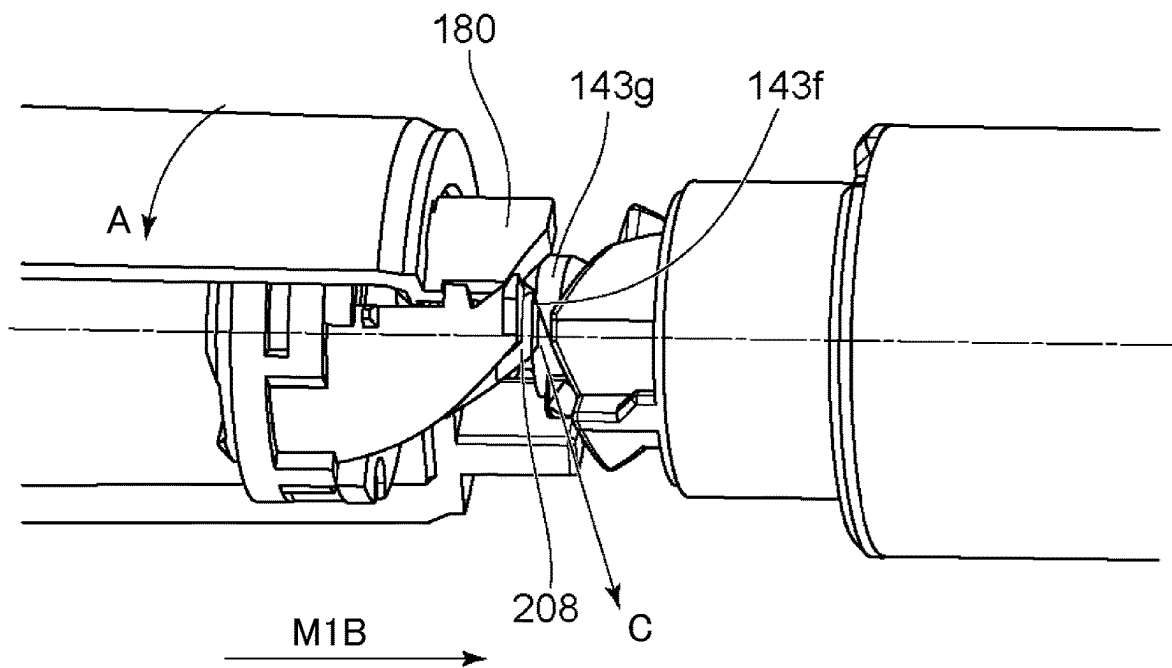


Fig. 72

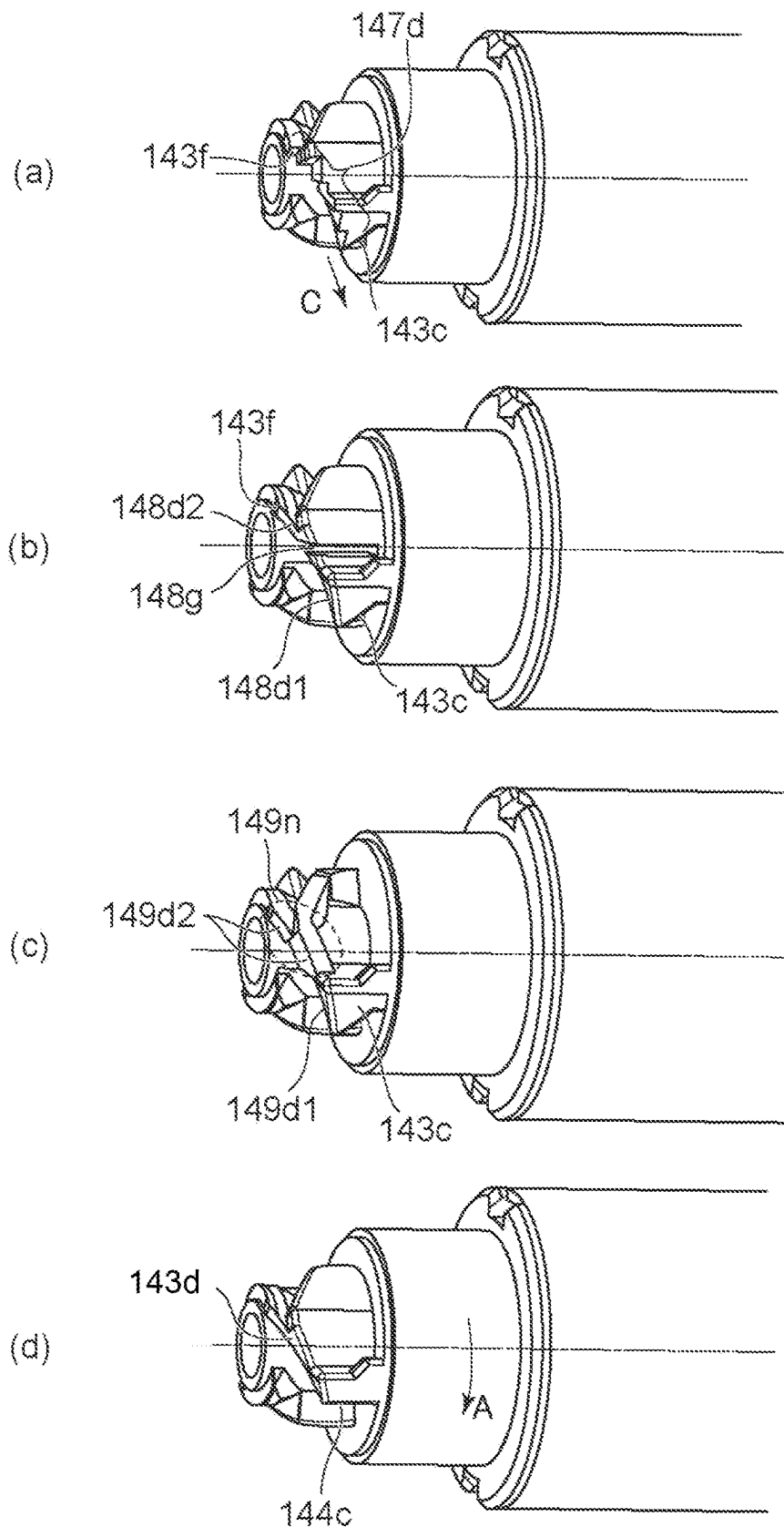


Fig. 73

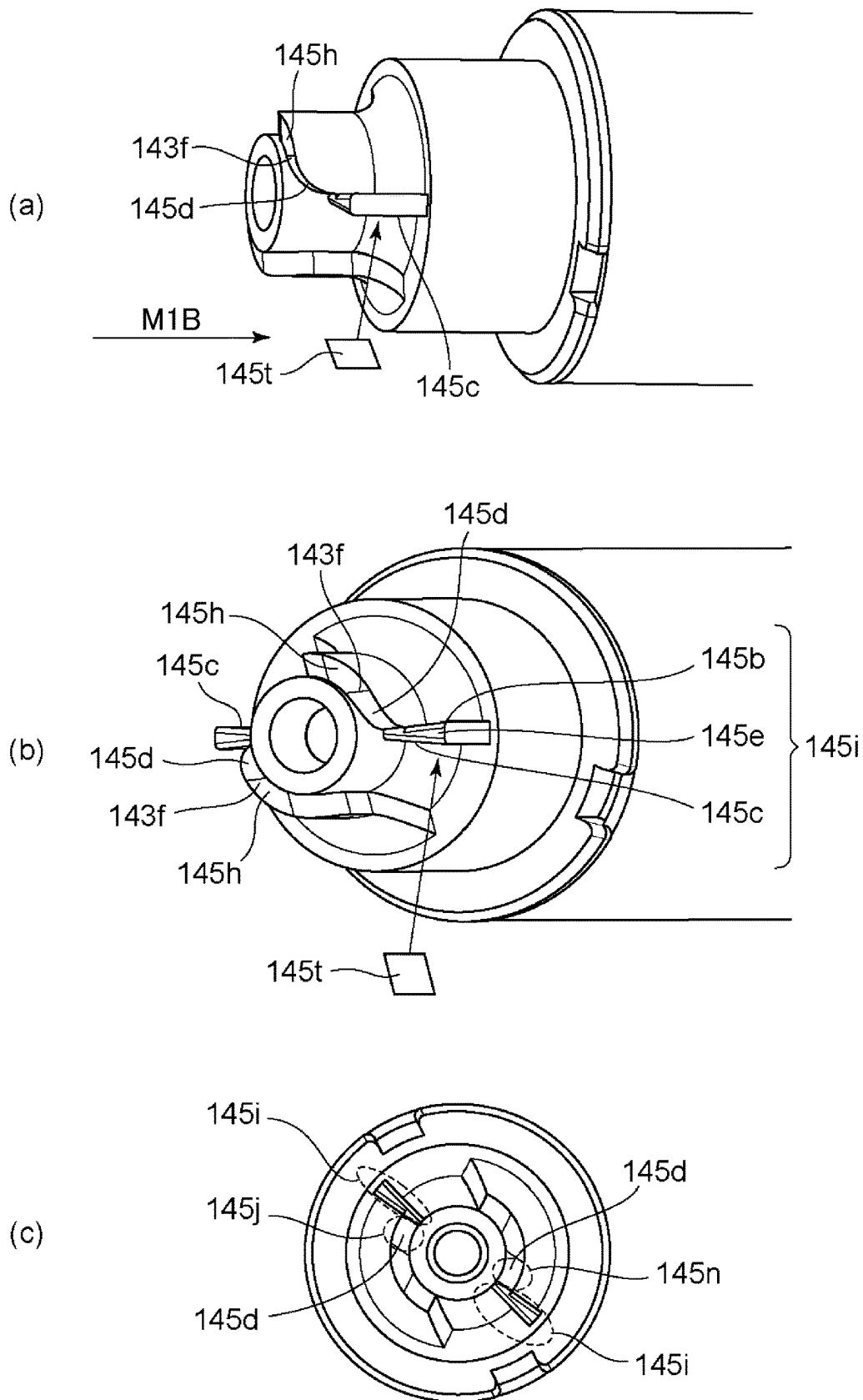


Fig. 74

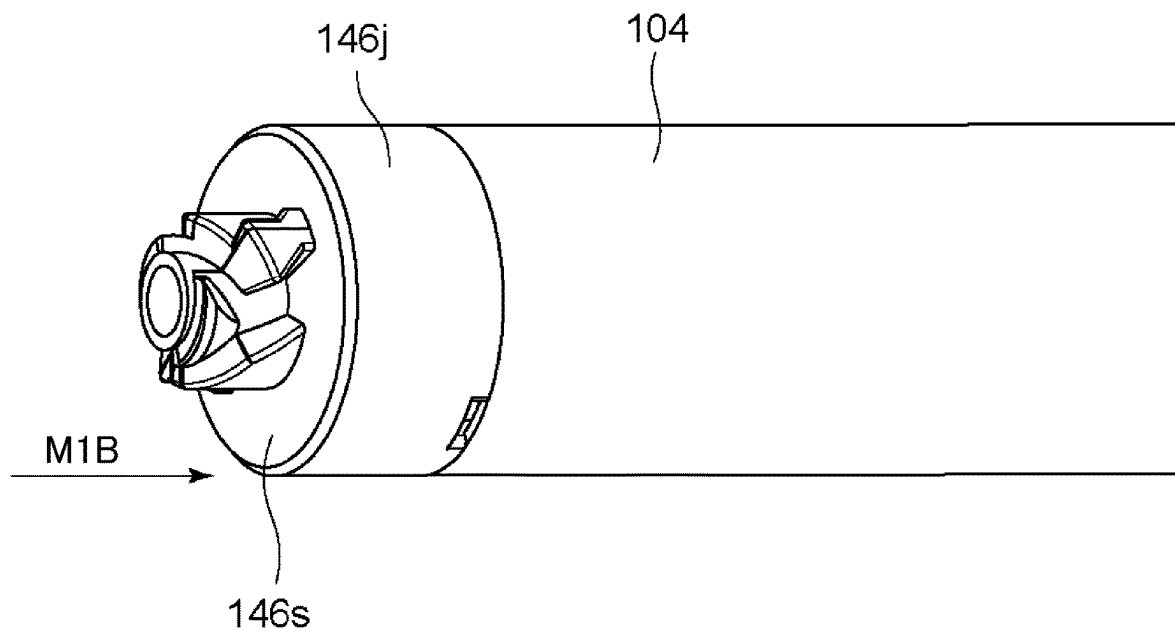


Fig. 75

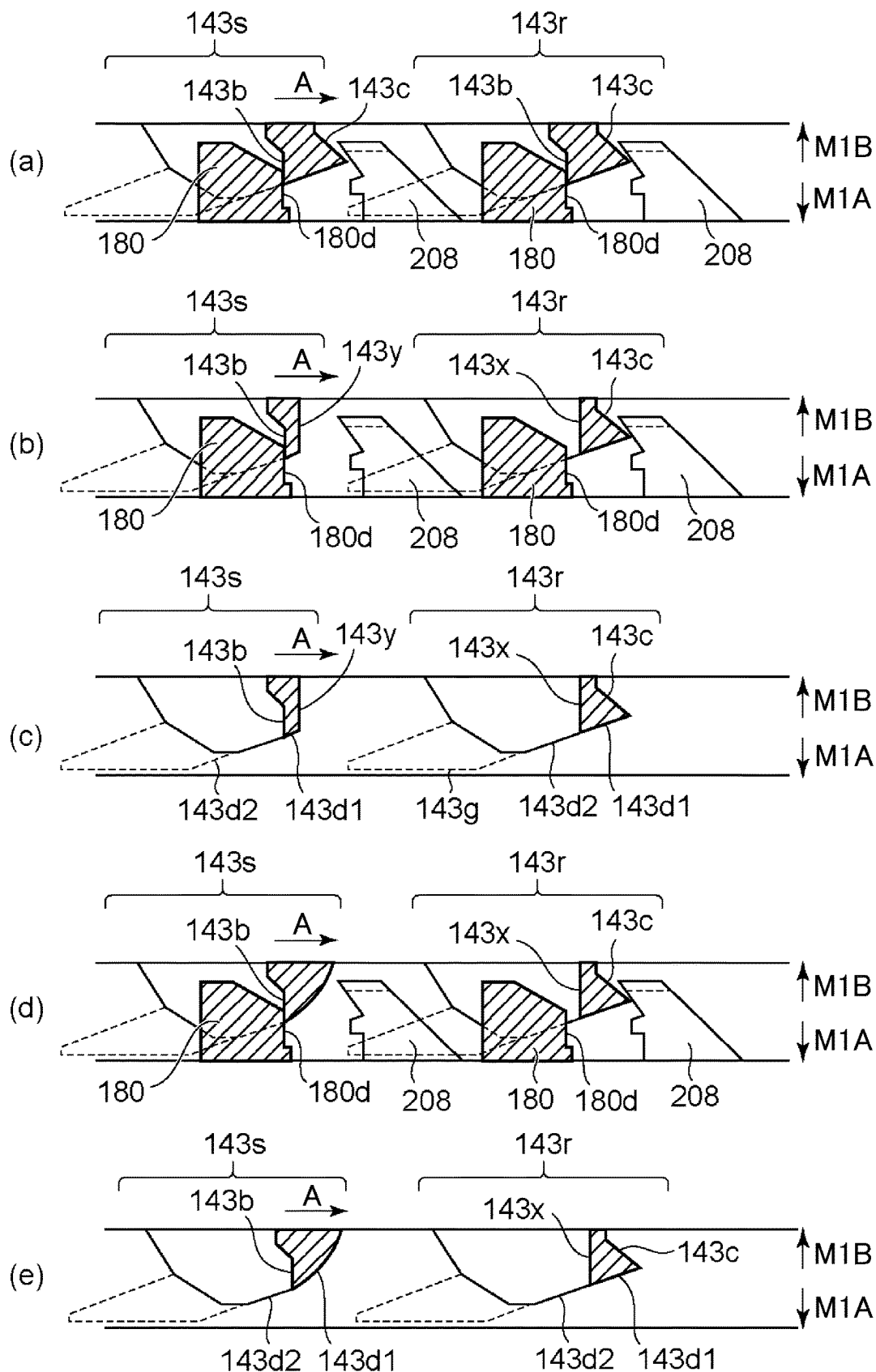


Fig. 76

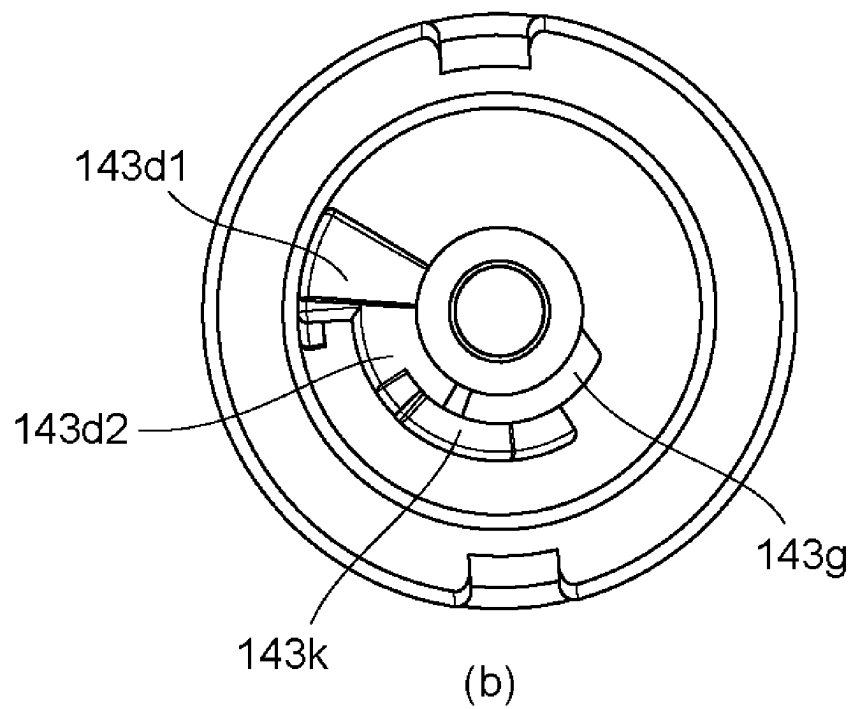
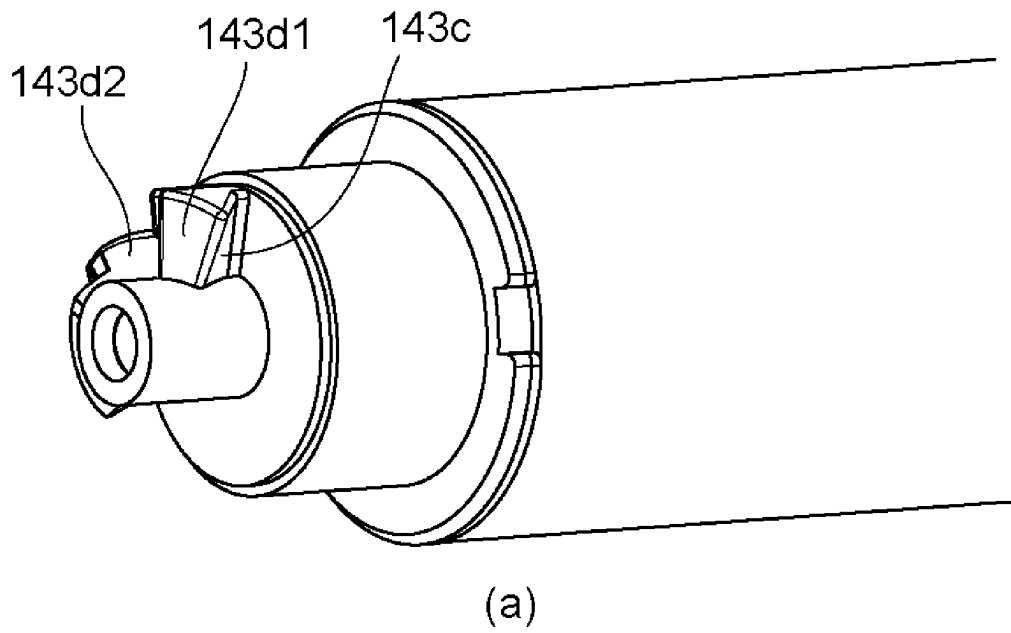


Fig. 77

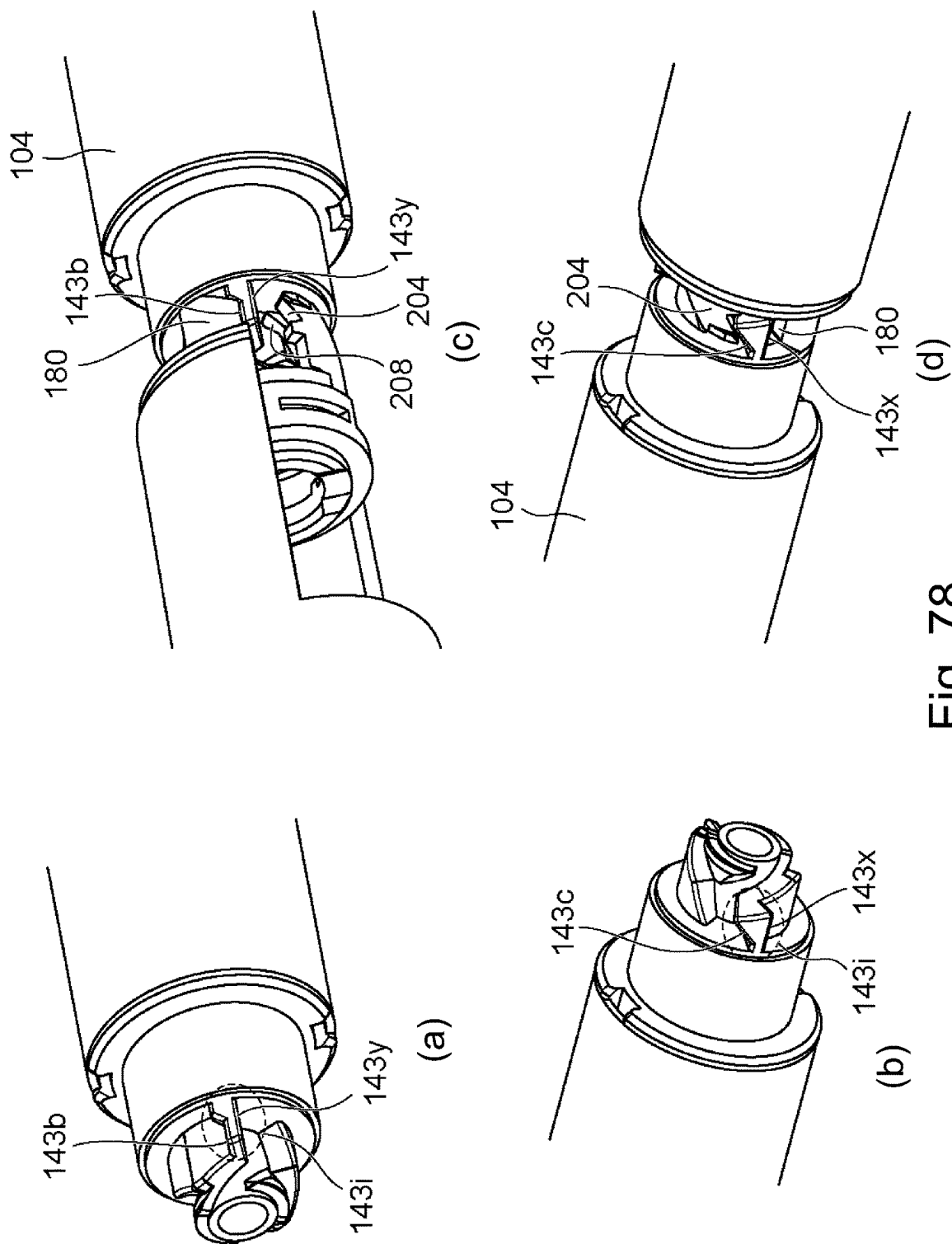


Fig. 78

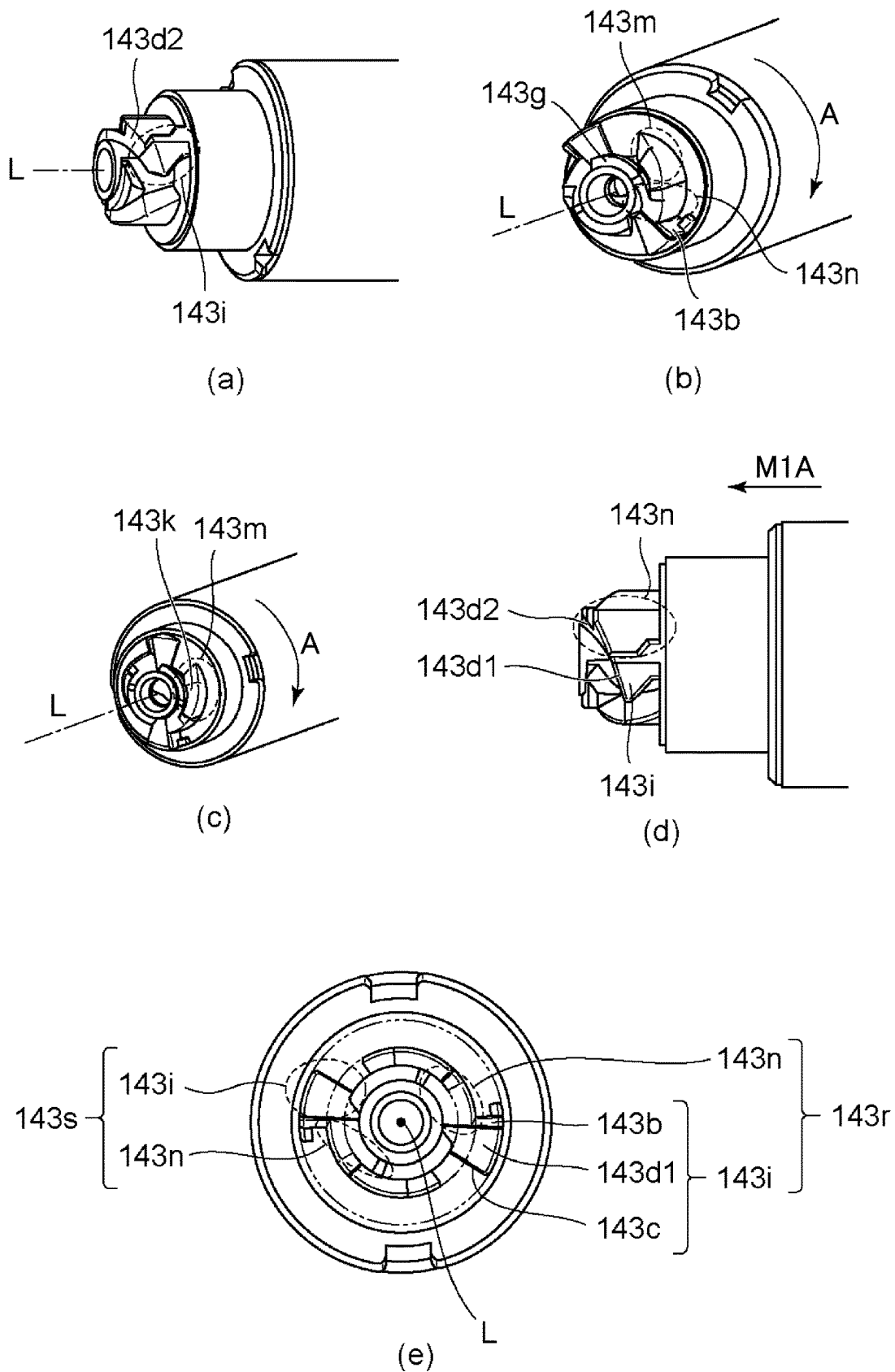


Fig. 79

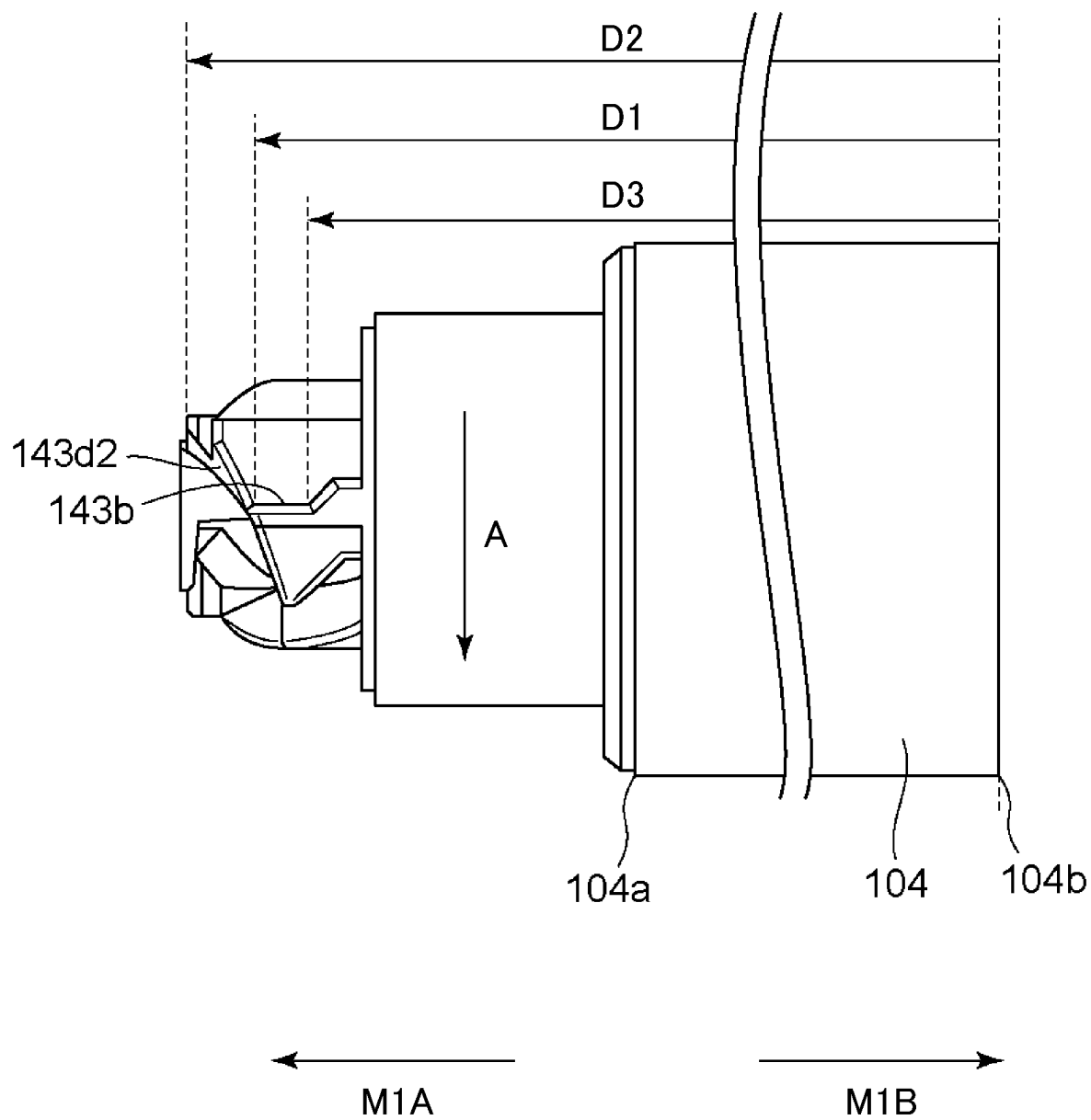


Fig. 80

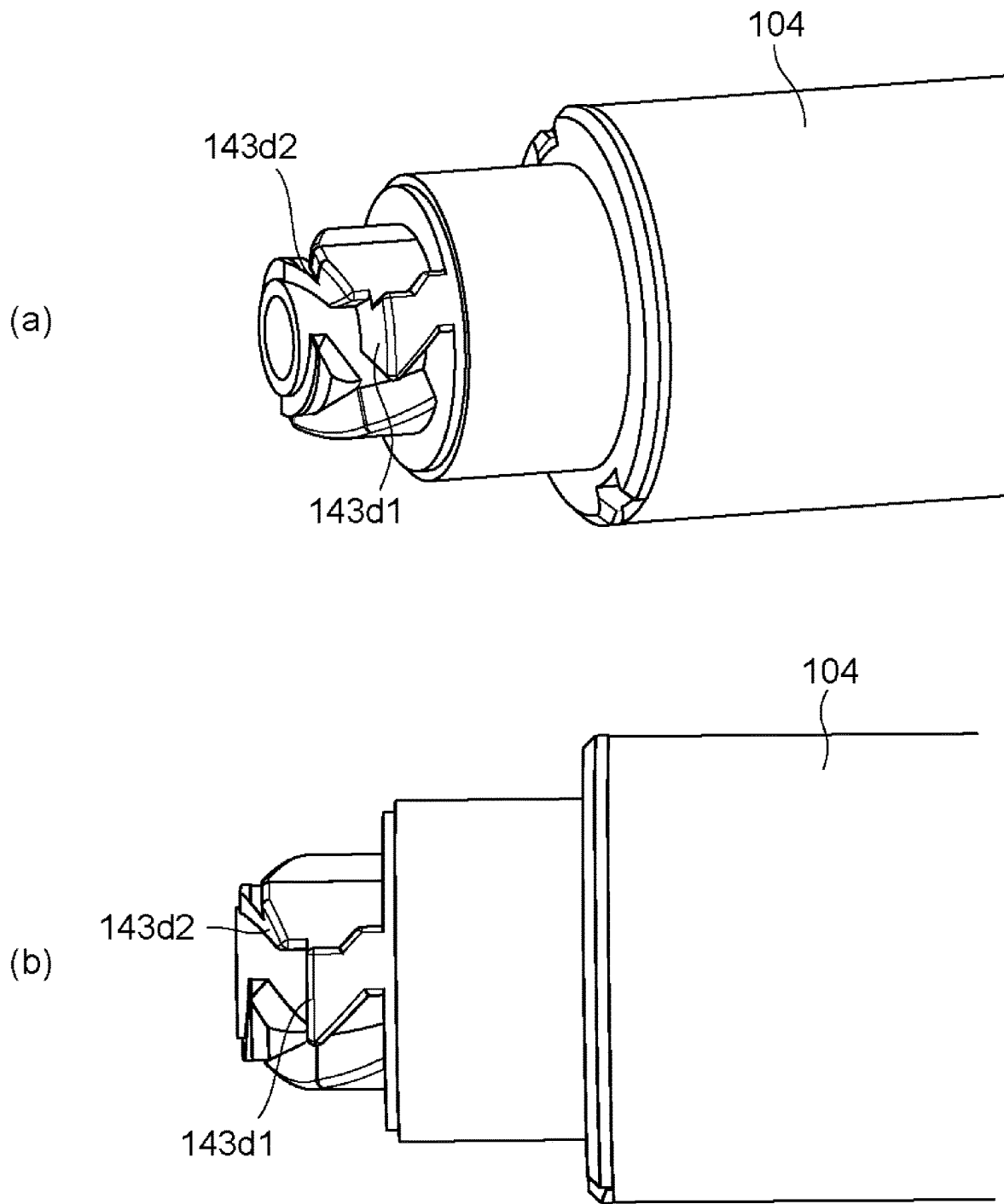
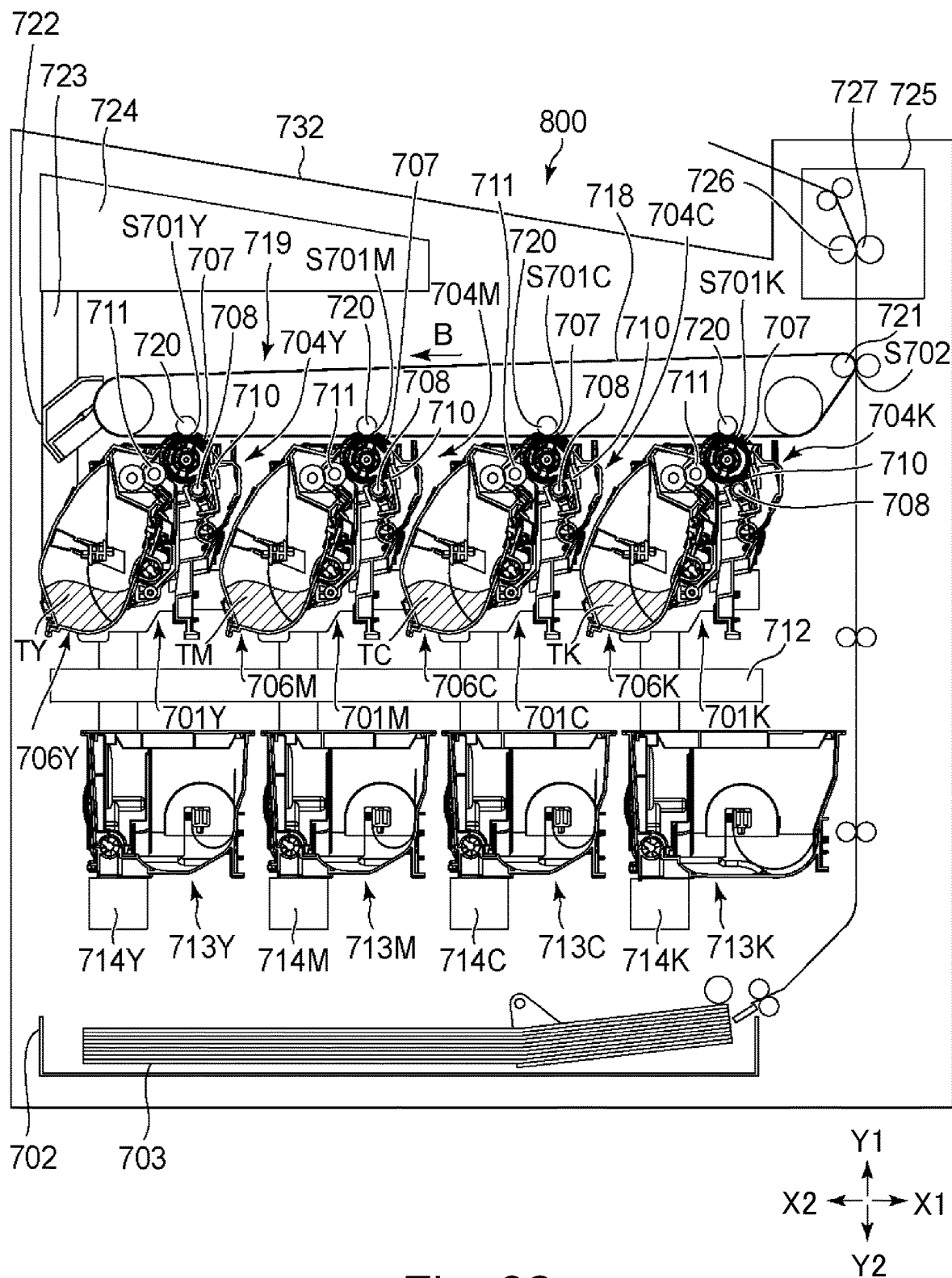


Fig. 81



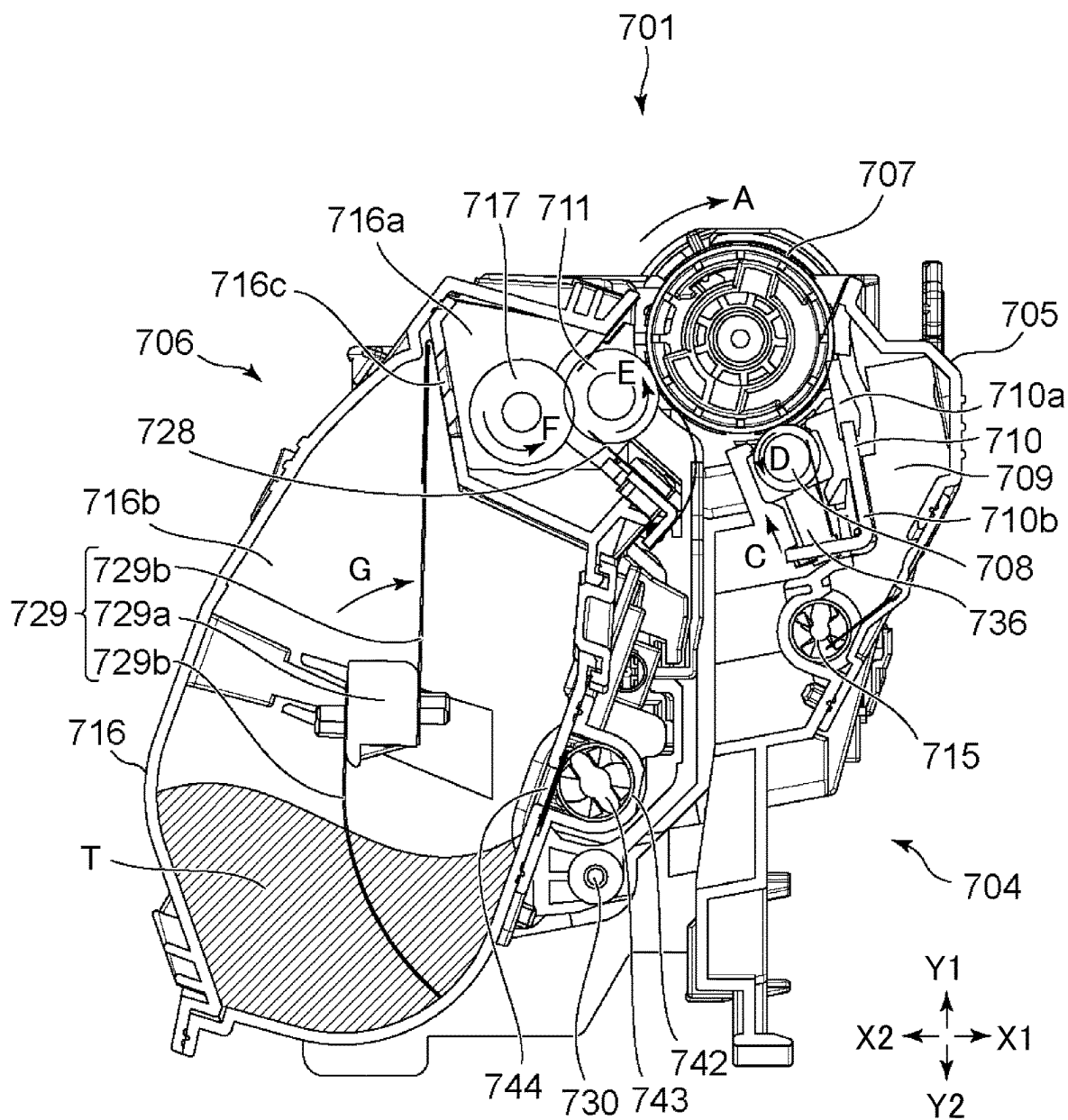


Fig. 83

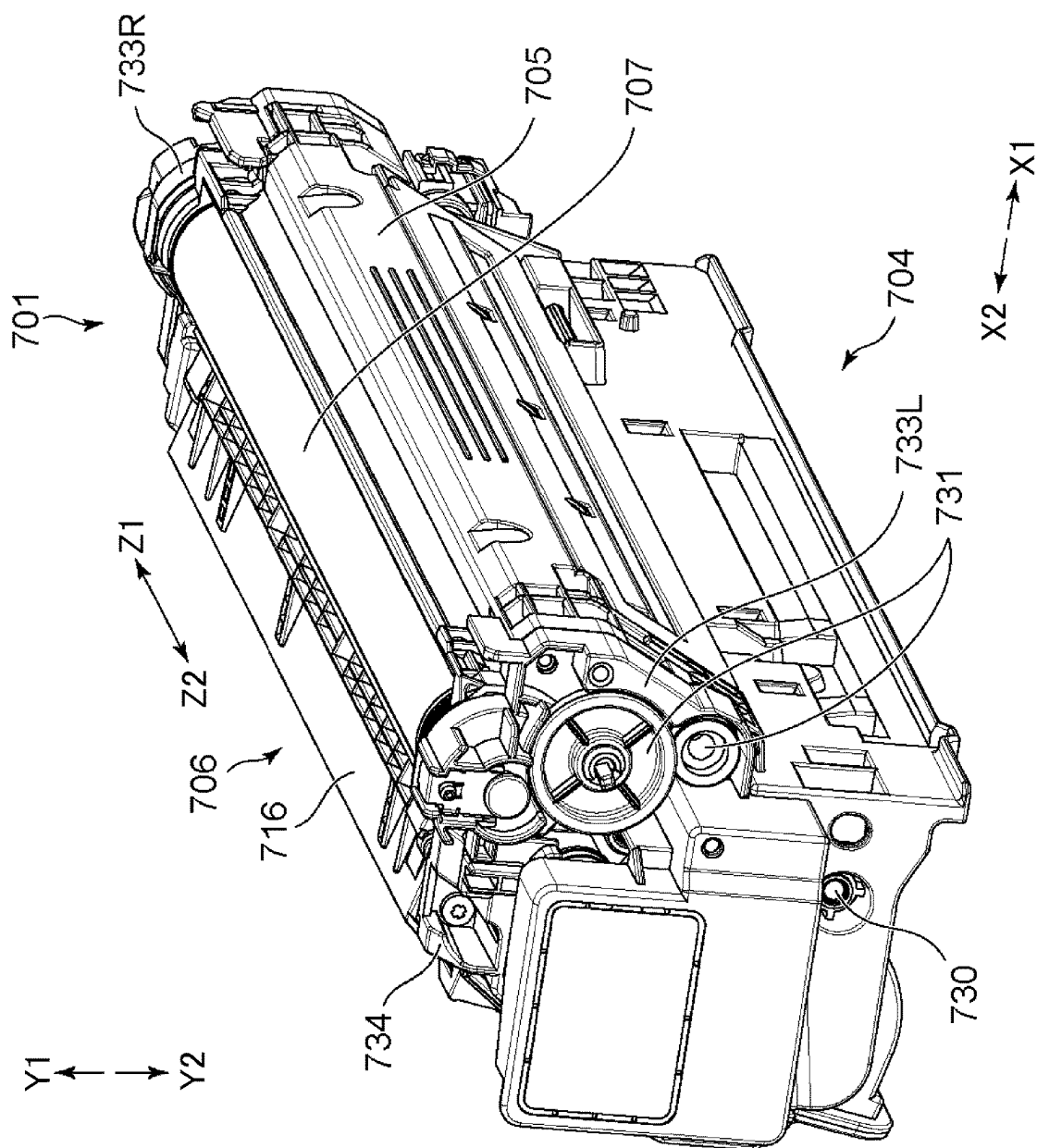


Fig. 84

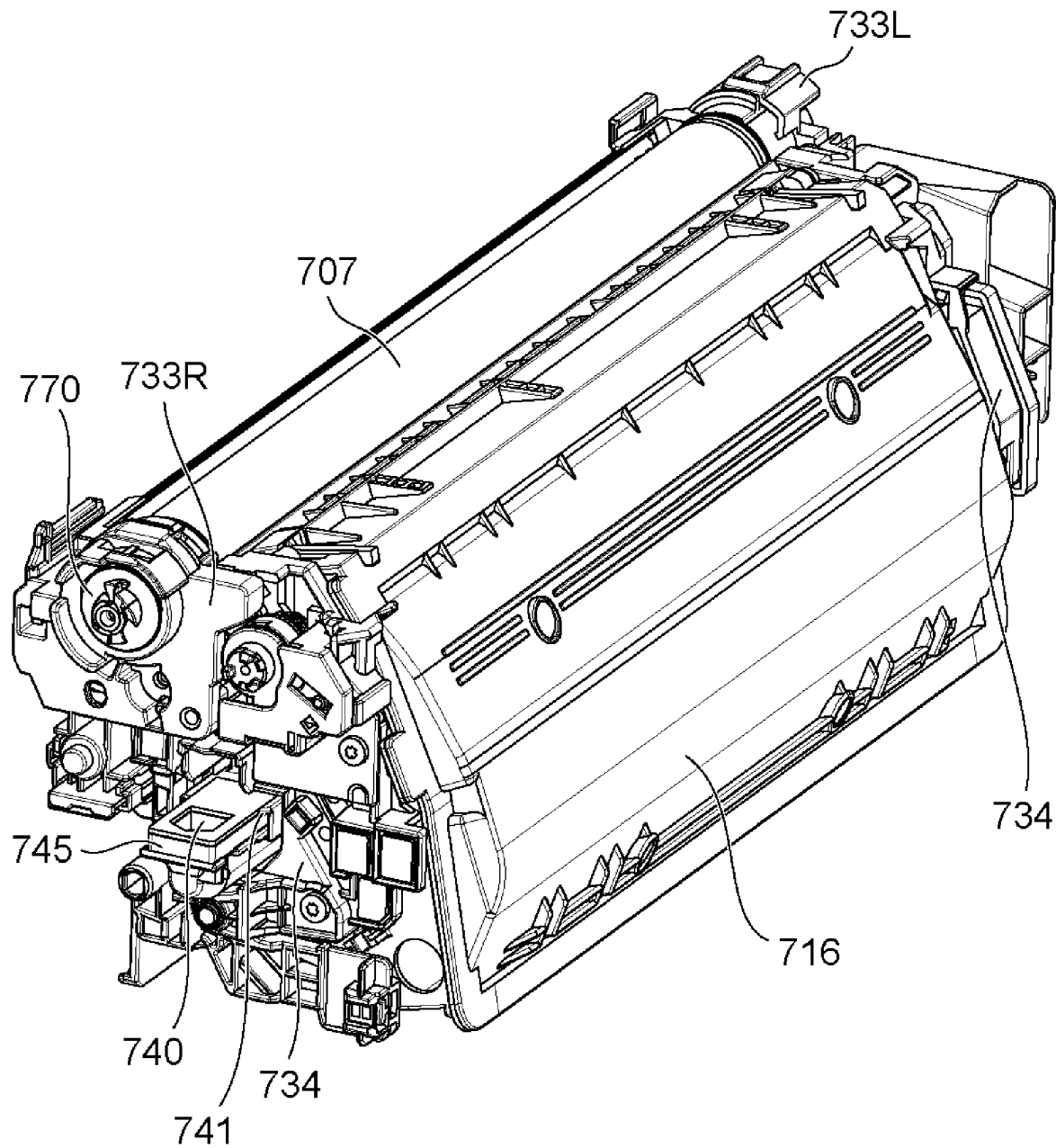


Fig. 85

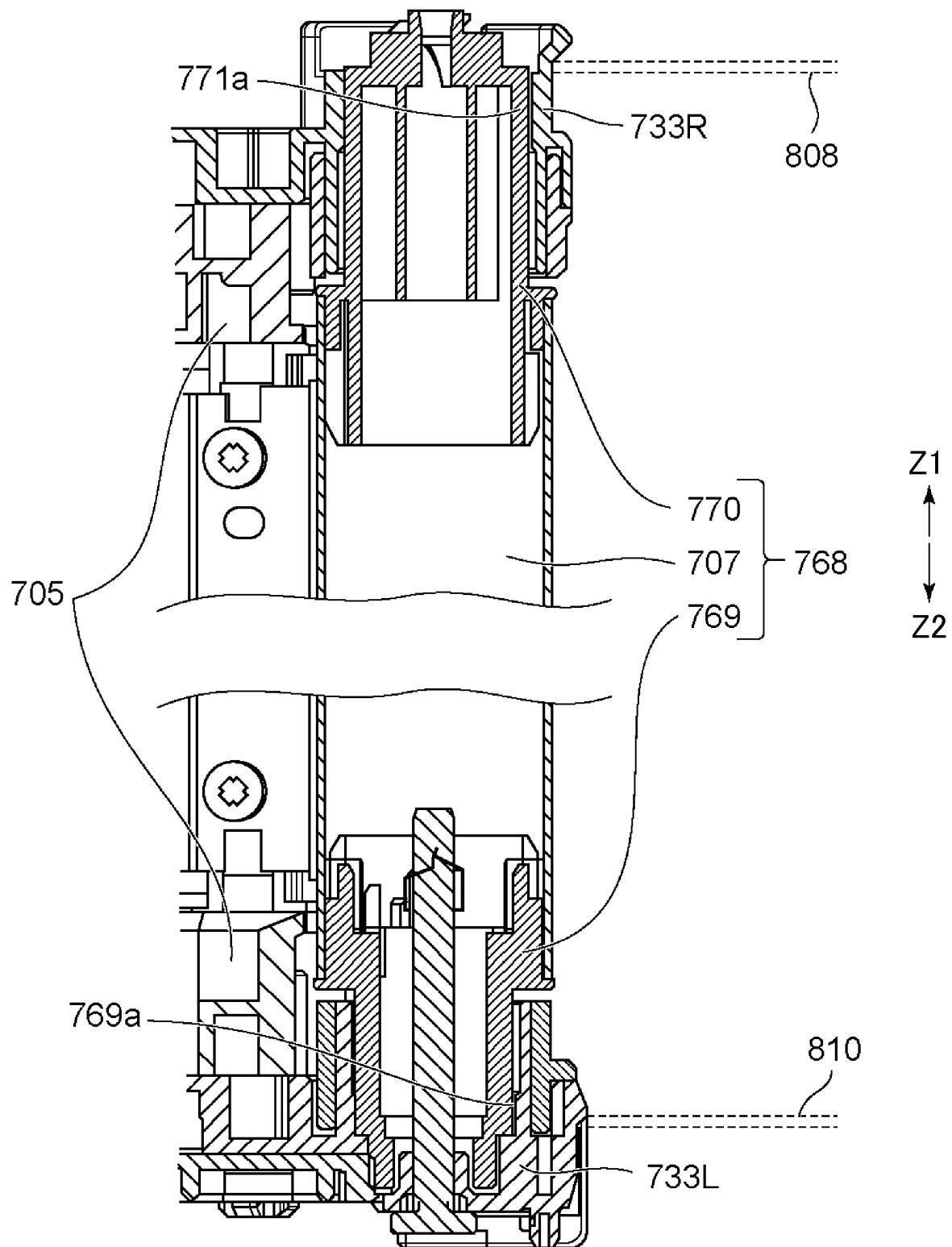


Fig. 86

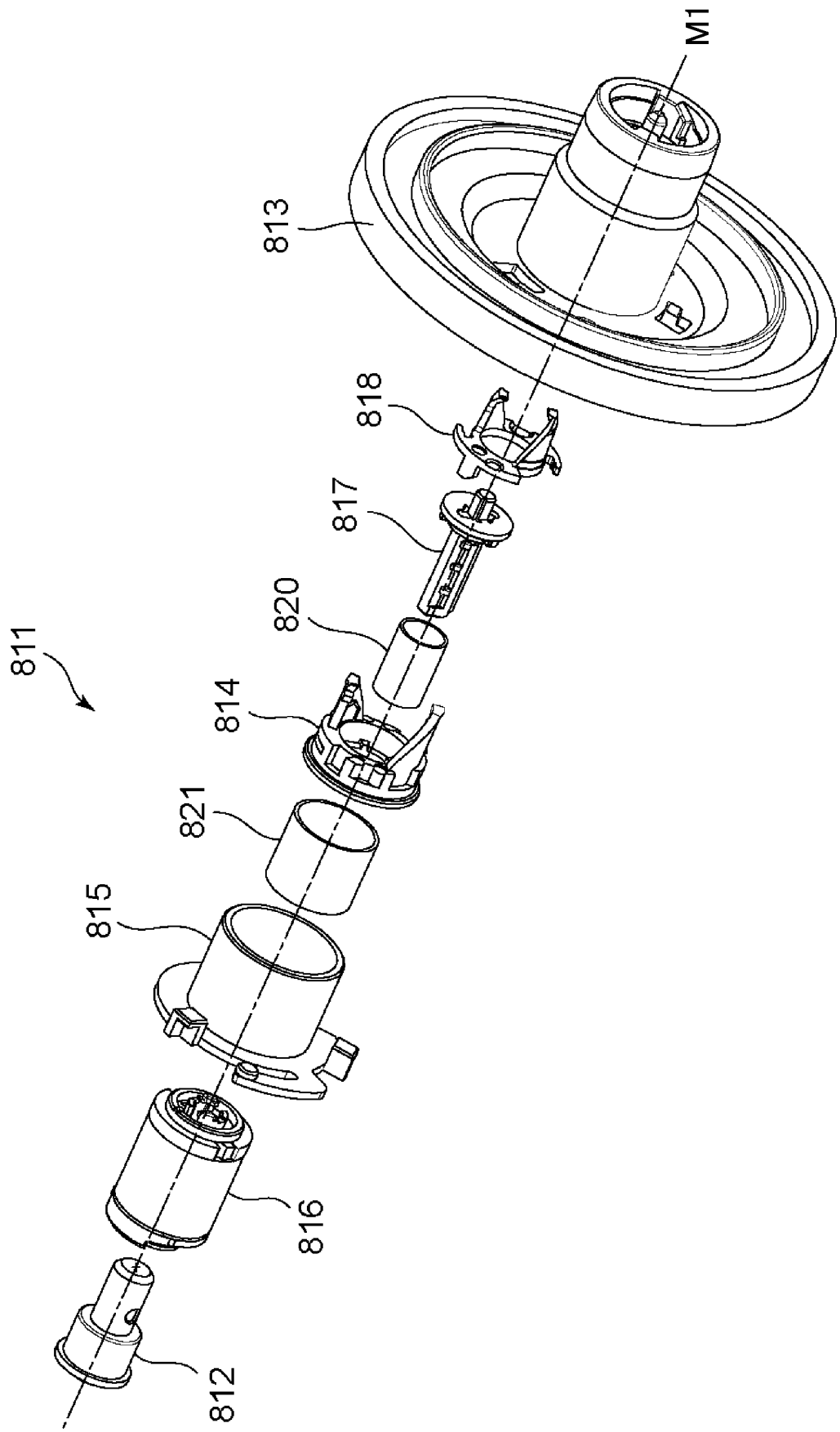


Fig. 87

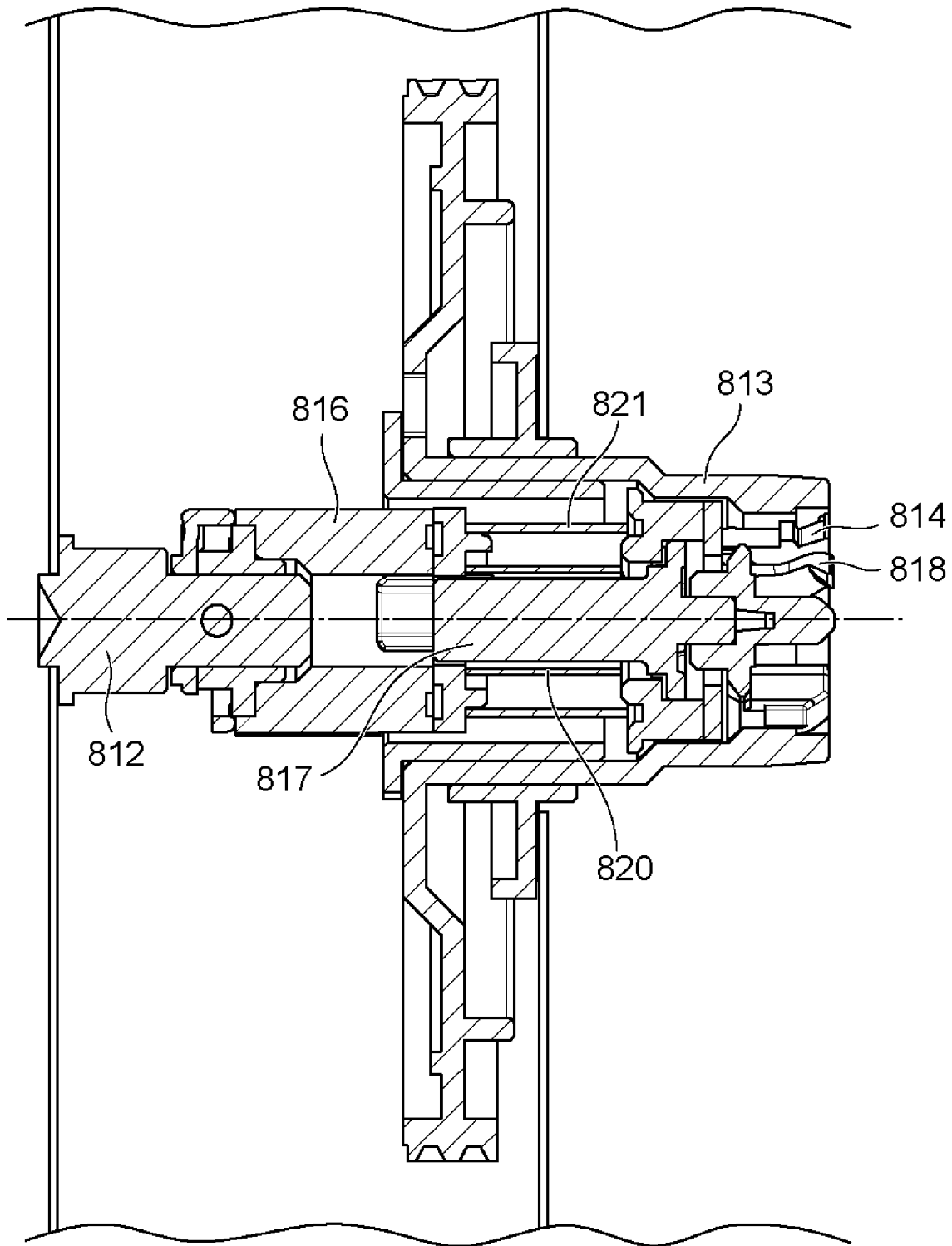


Fig. 88

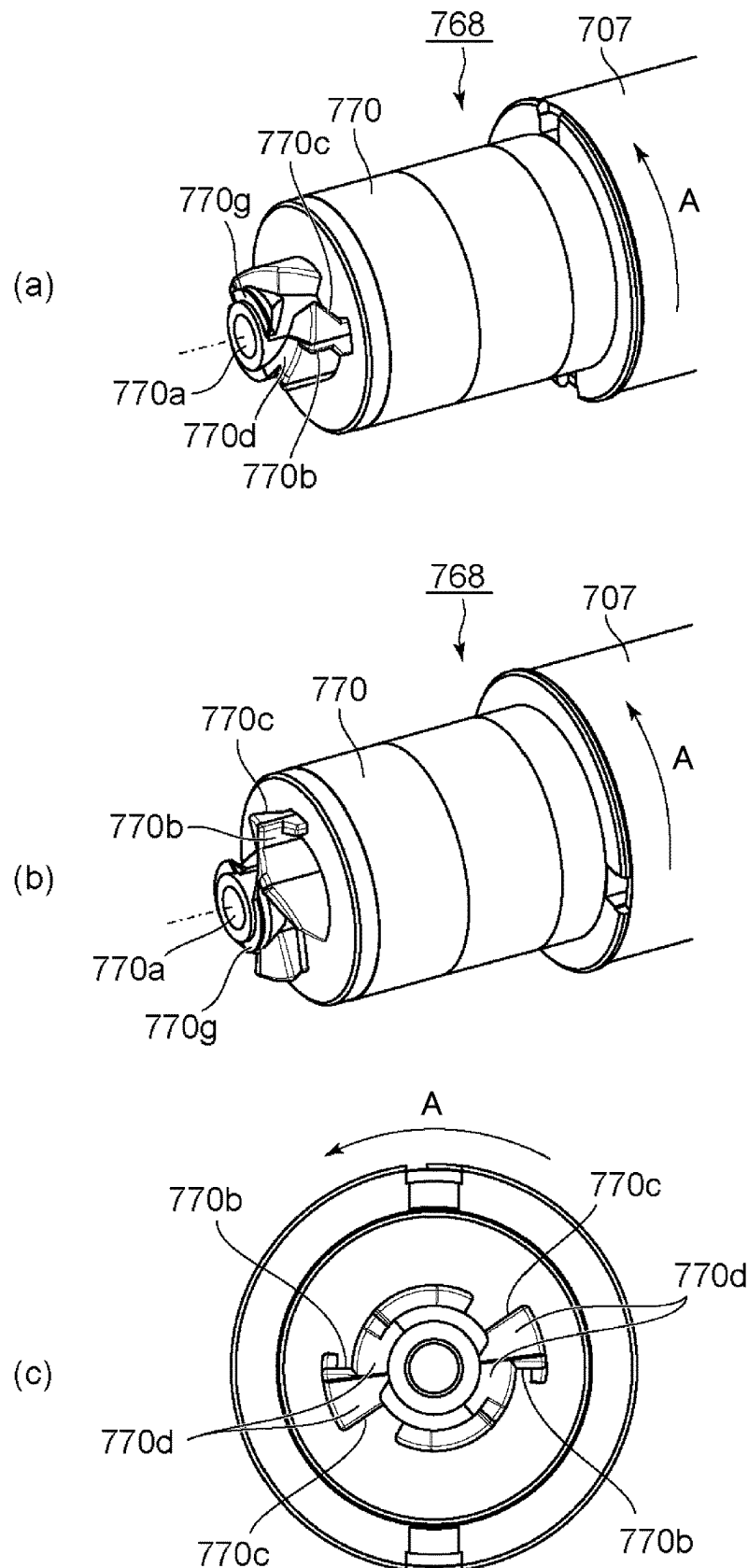


Fig. 89

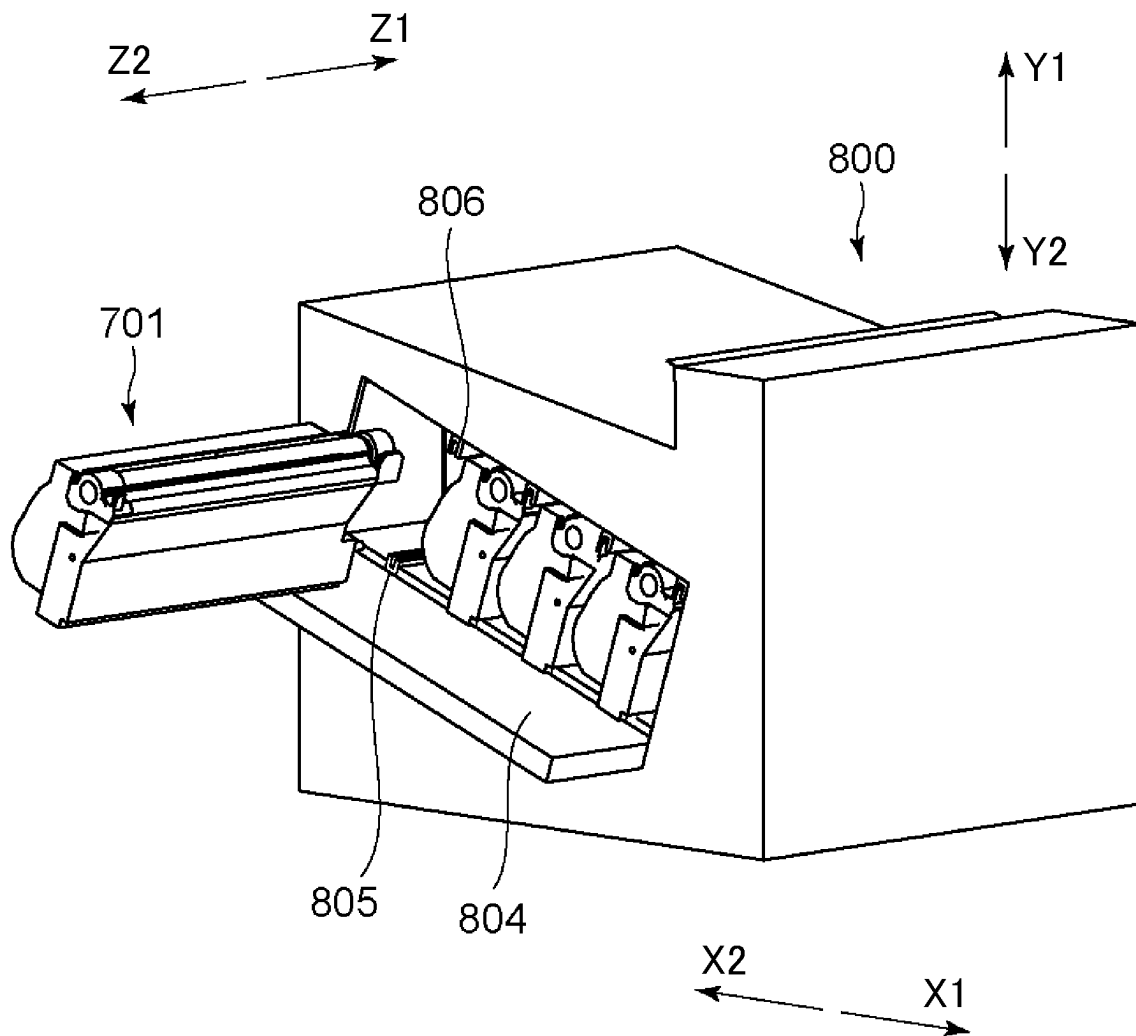


Fig. 90

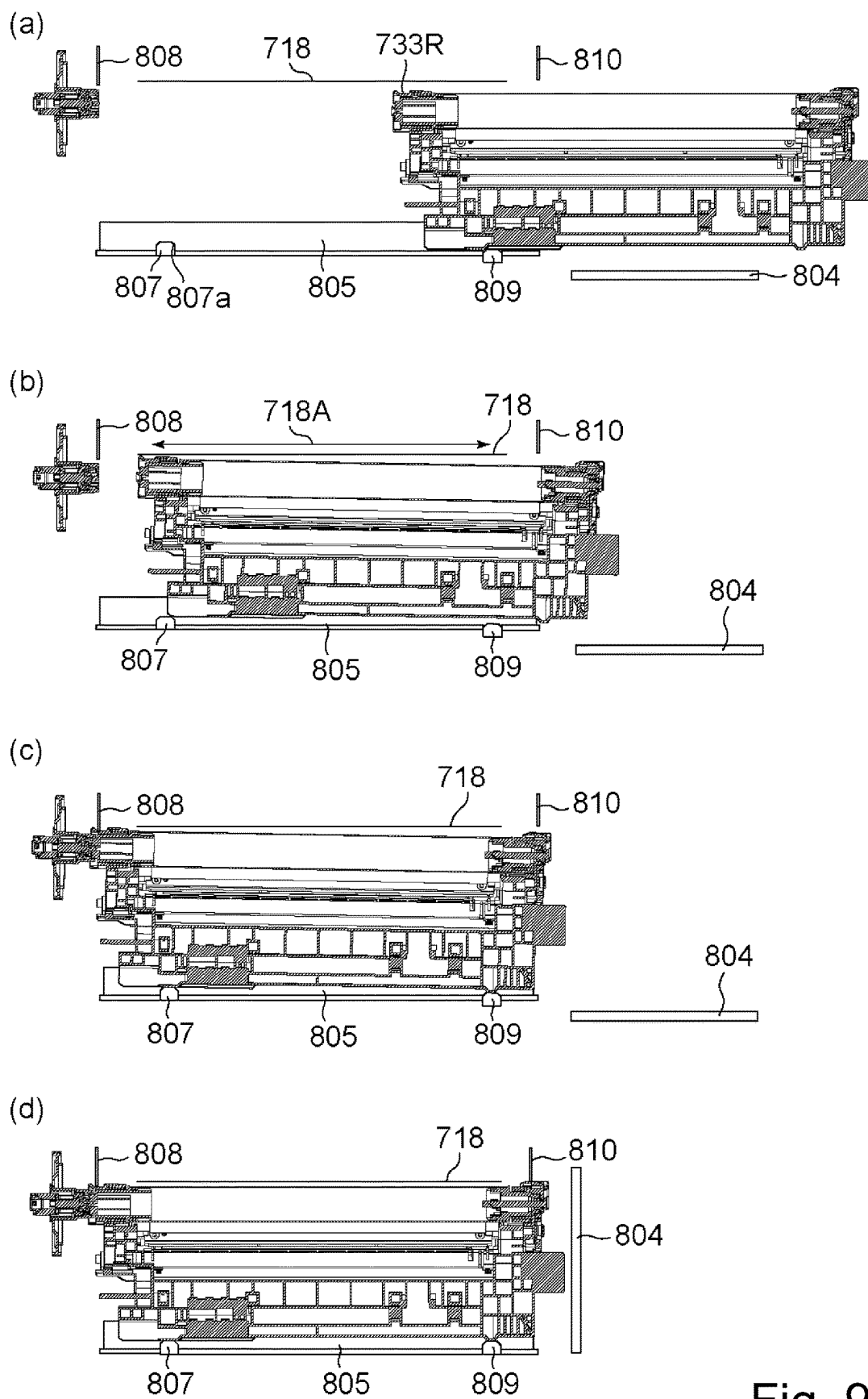
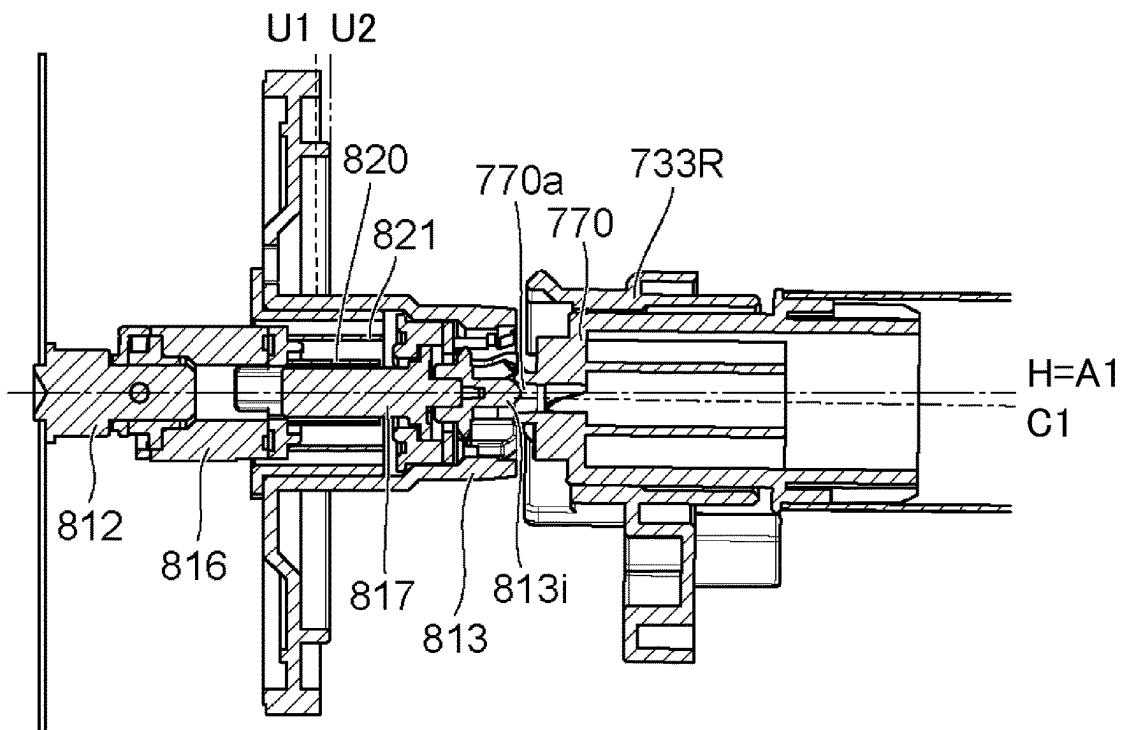


Fig. 91

(a)



(b)

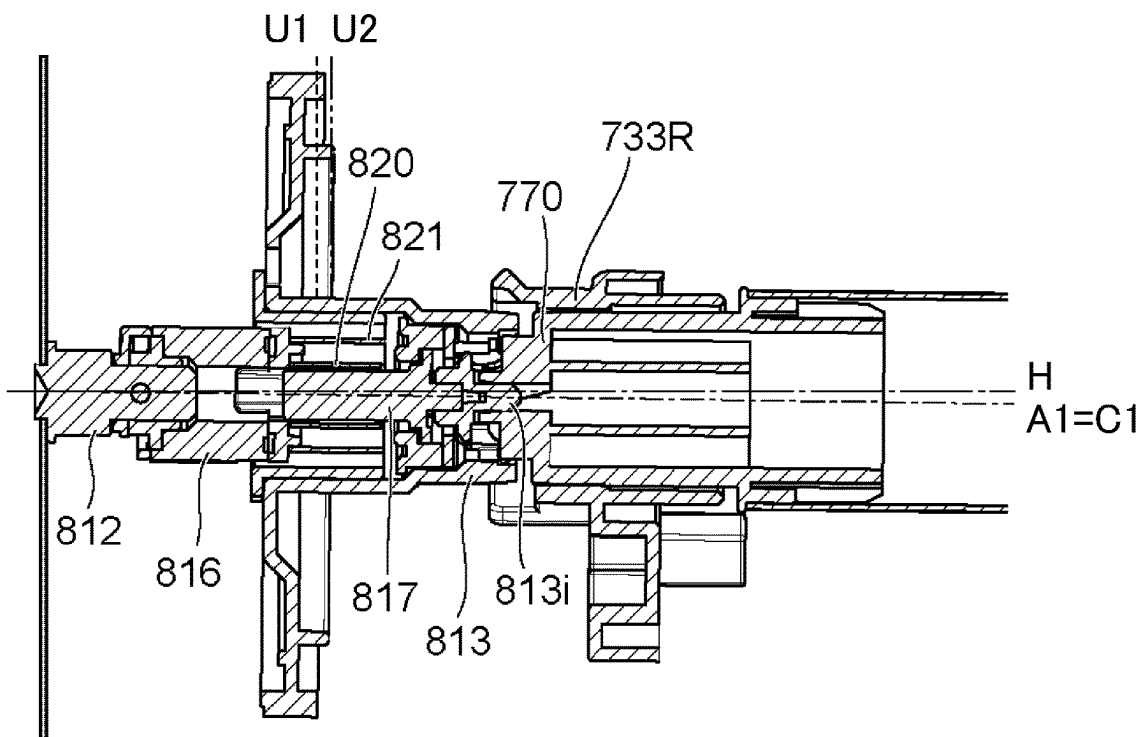
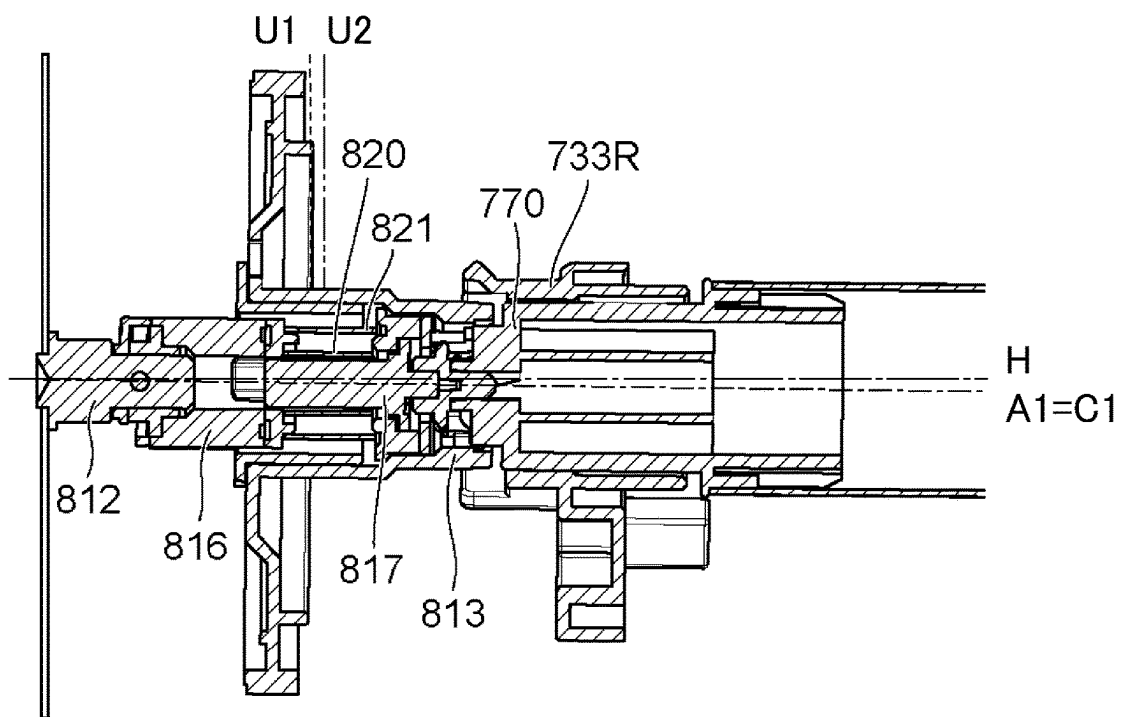


Fig. 92

(a)



(b)

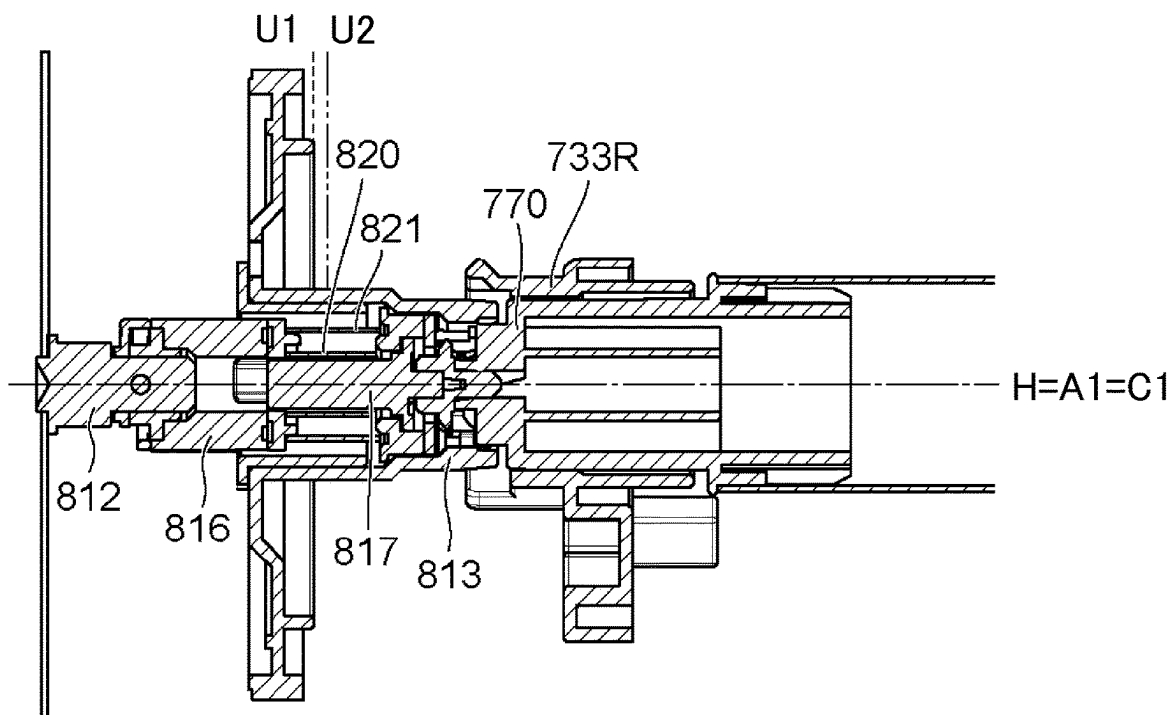


Fig. 93

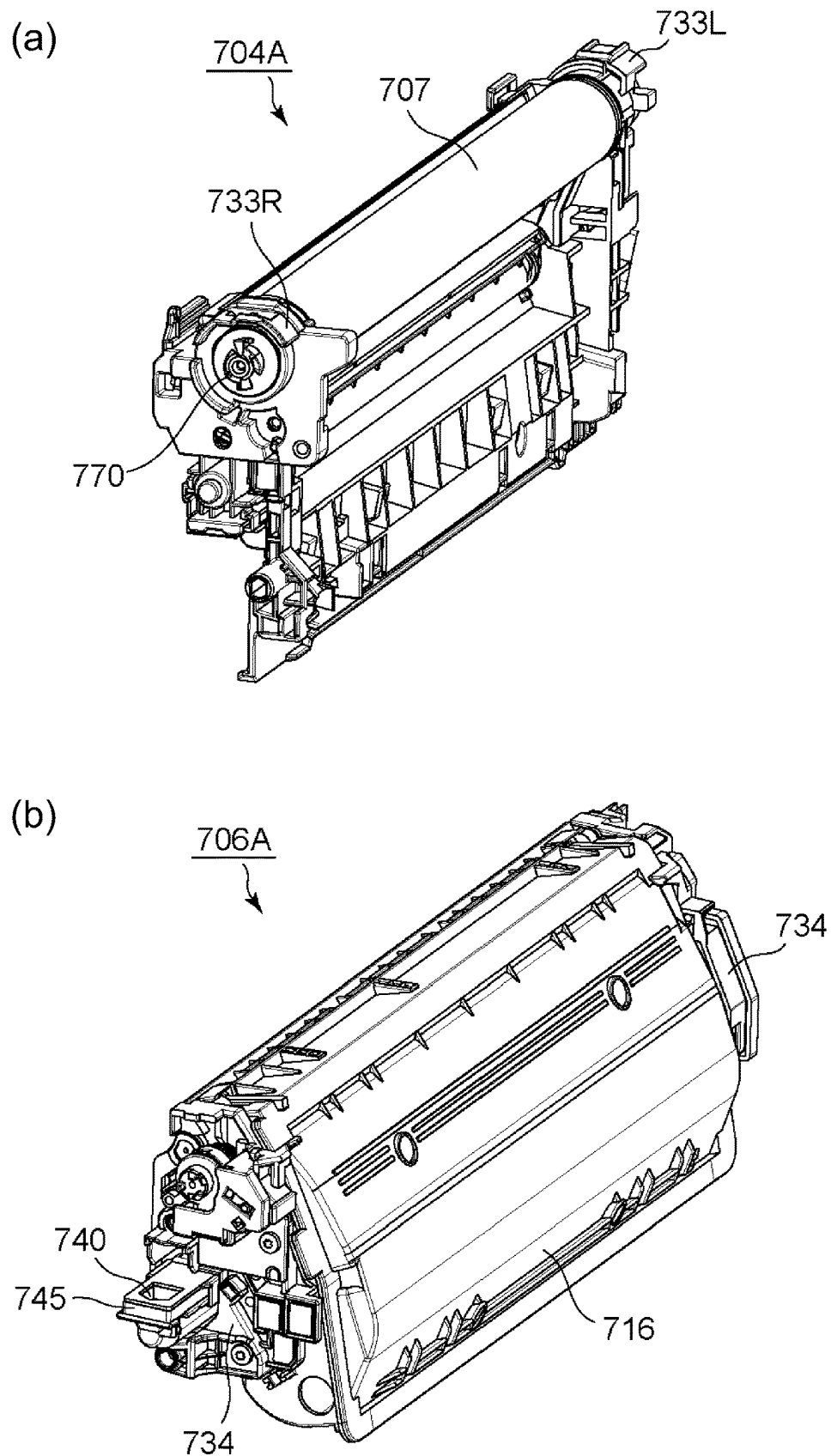


Fig. 94

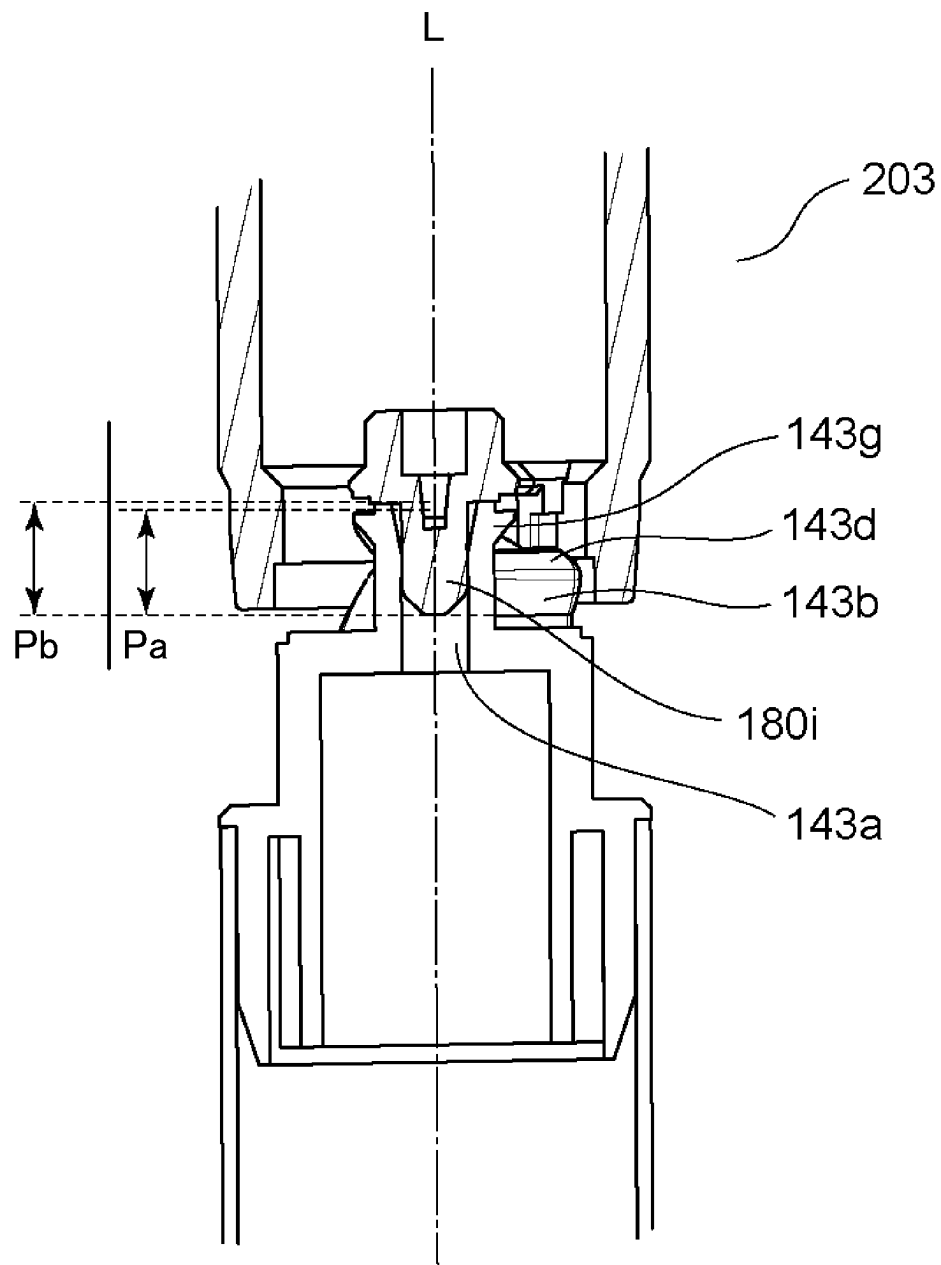


Fig. 95

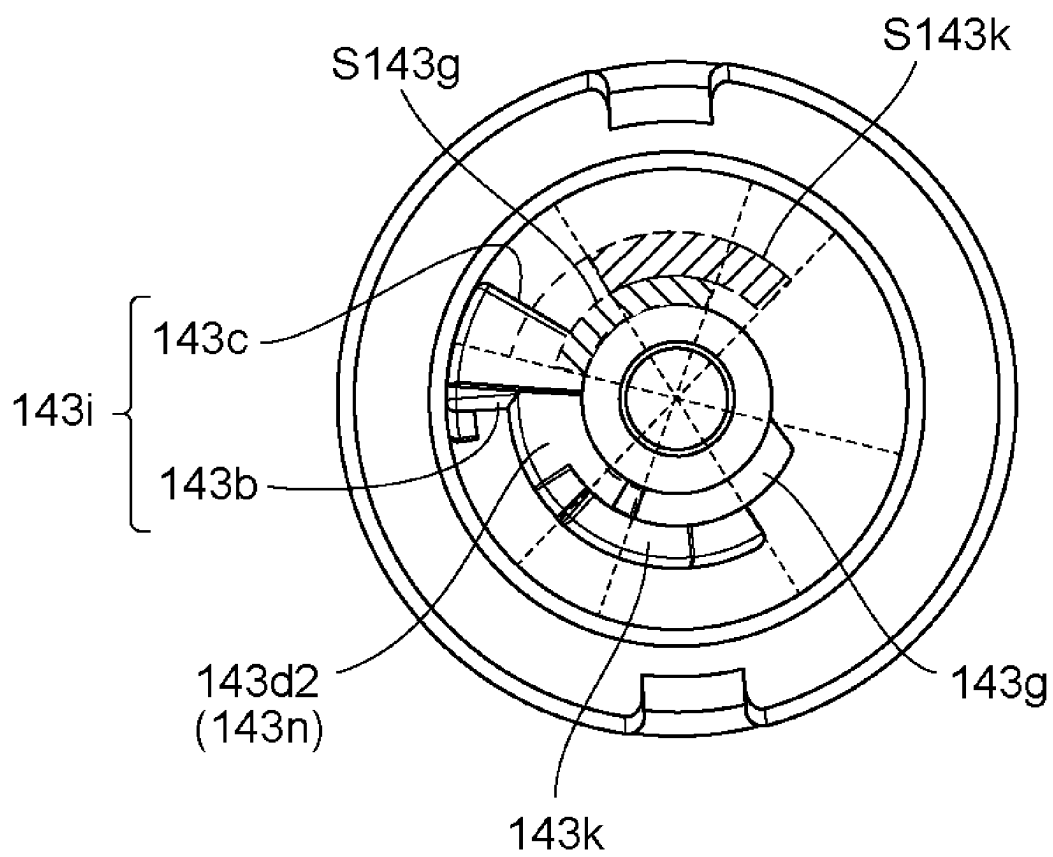
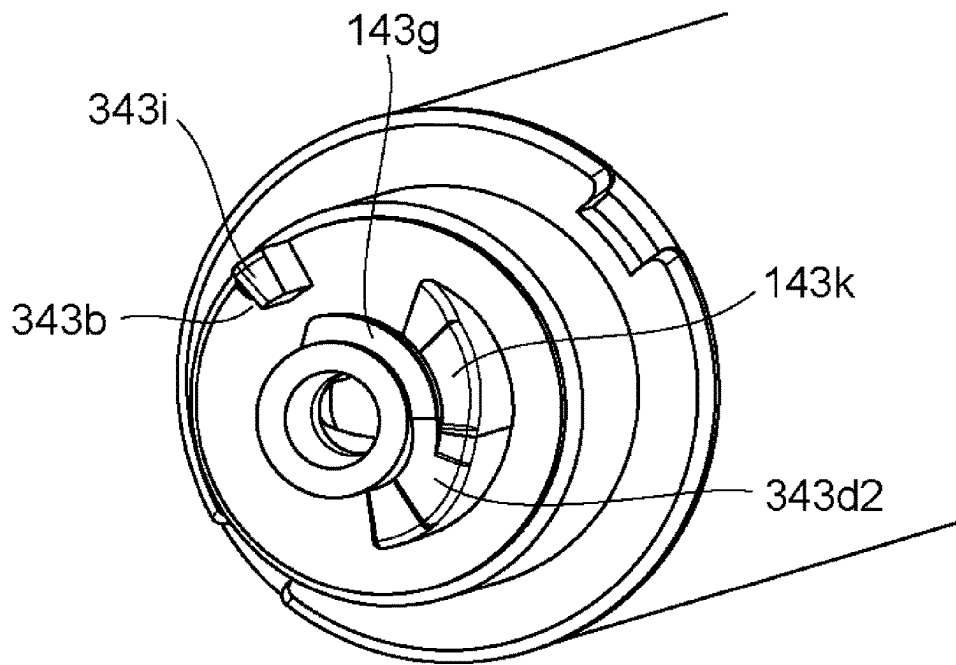
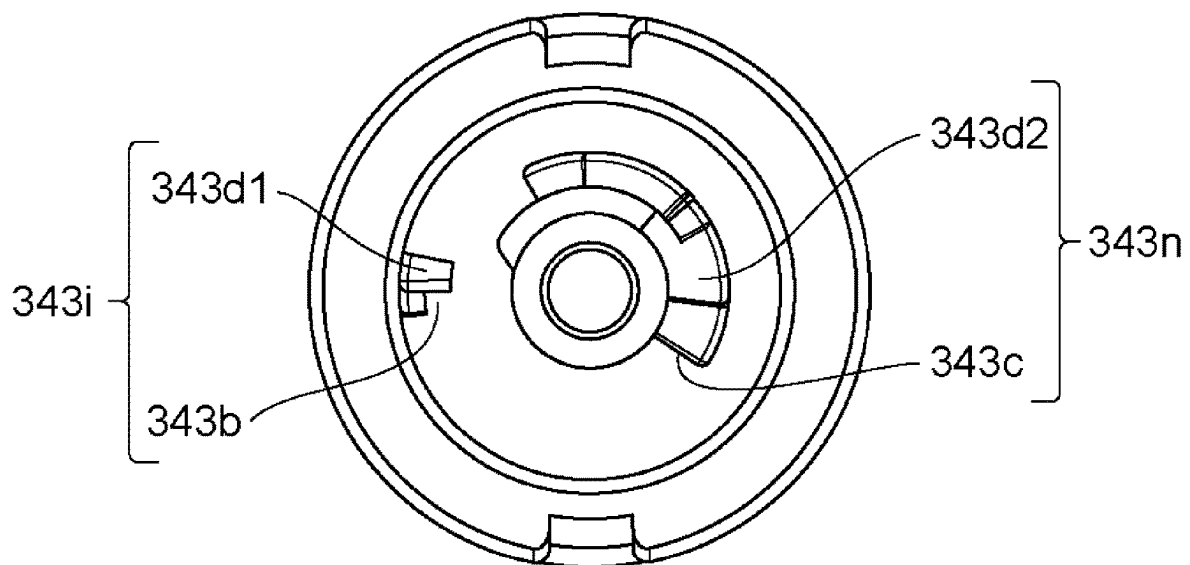


Fig. 96



(a)



(b)

Fig. 97

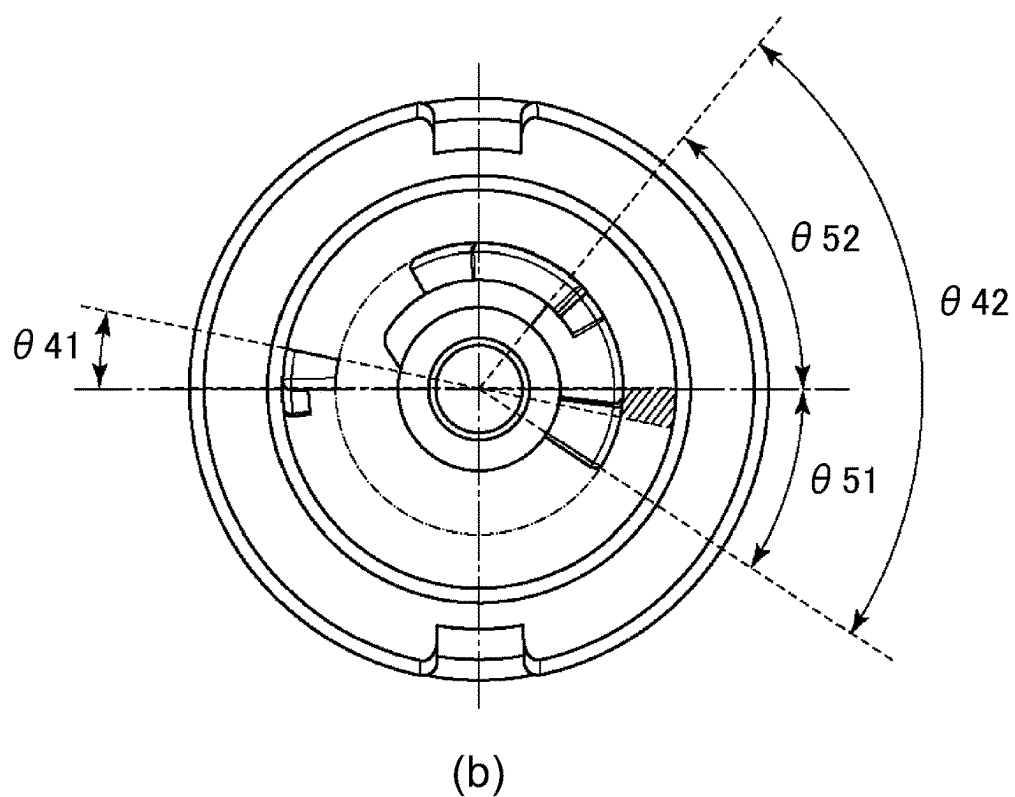
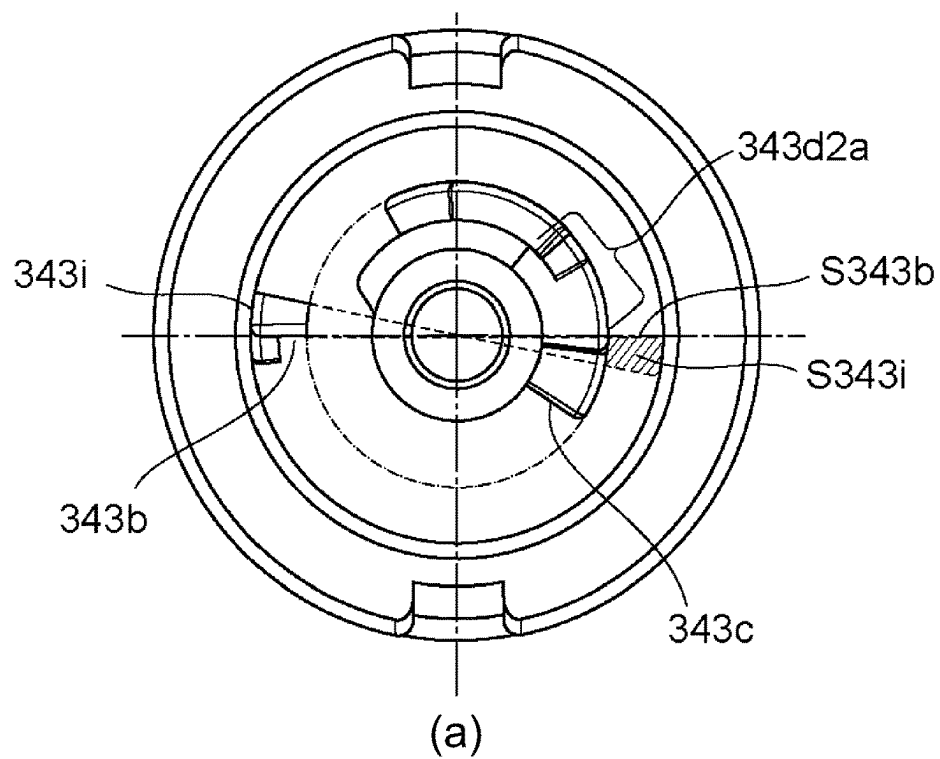


Fig. 98

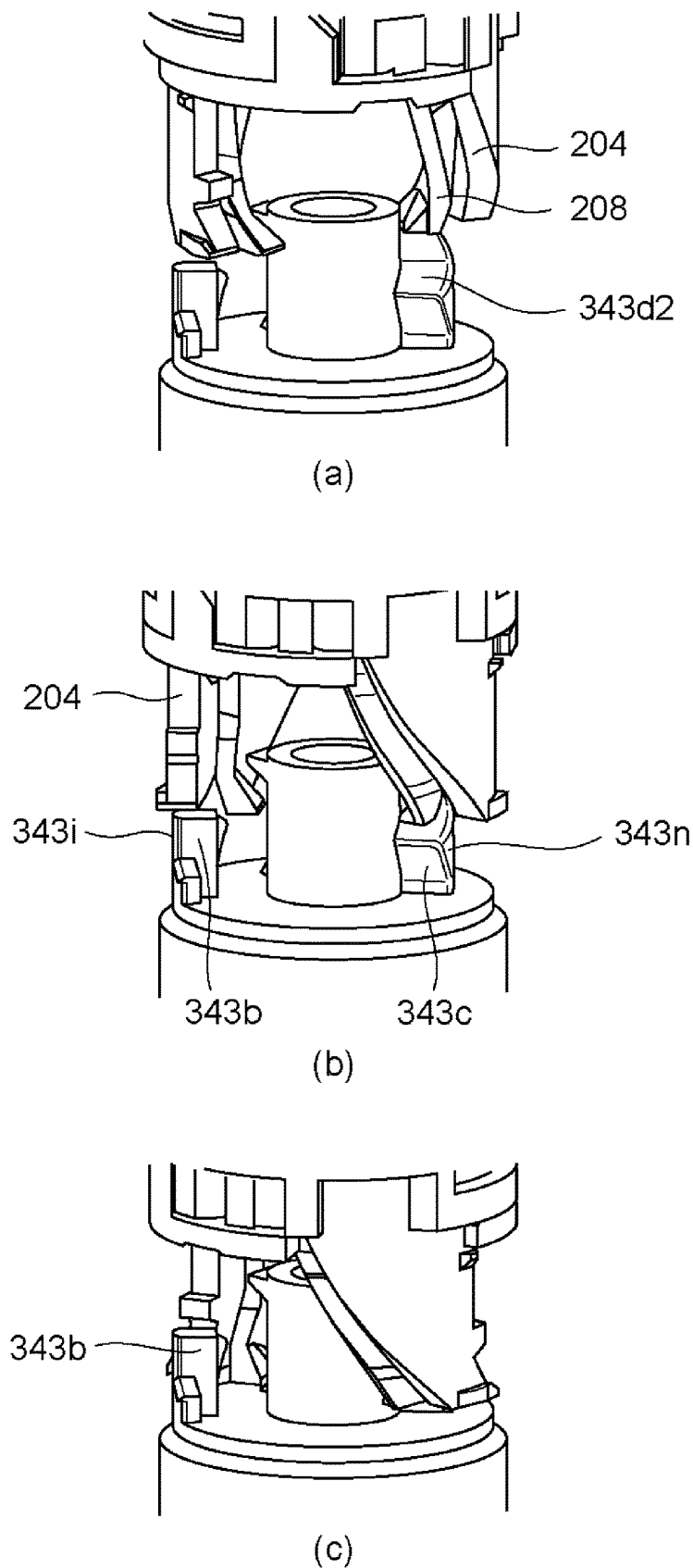


Fig. 99

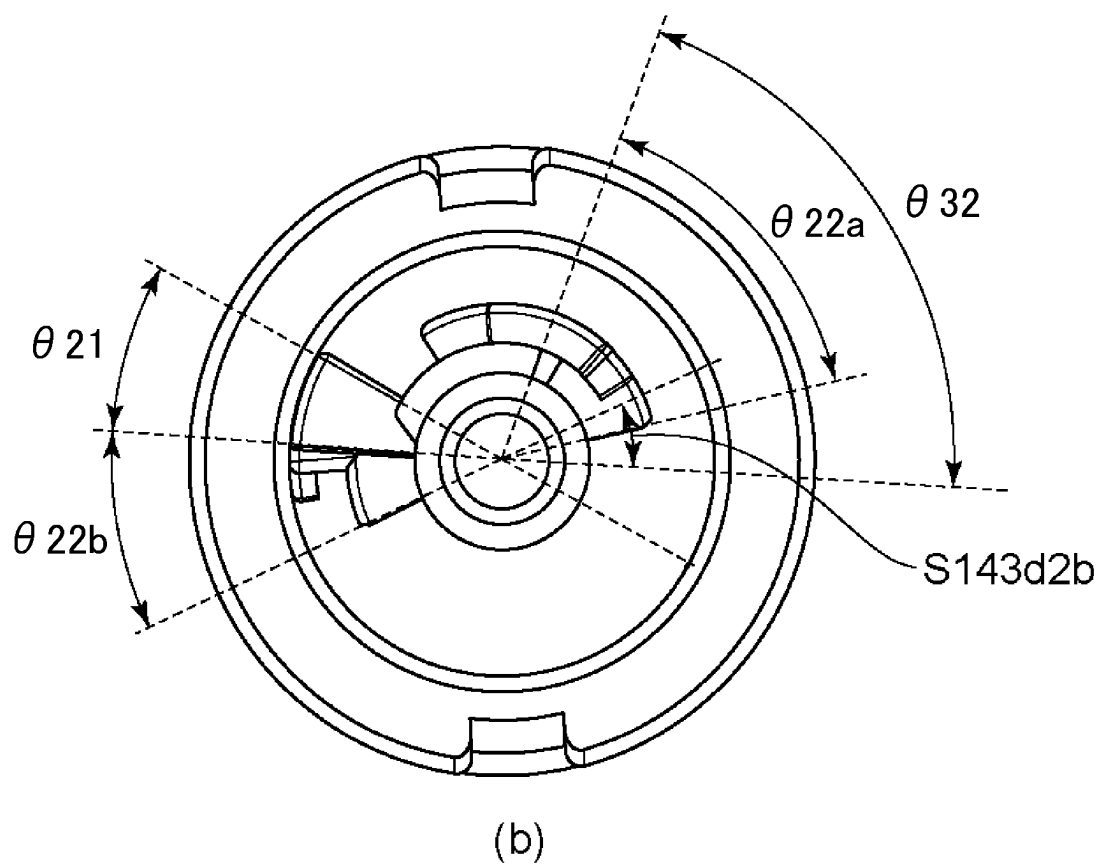
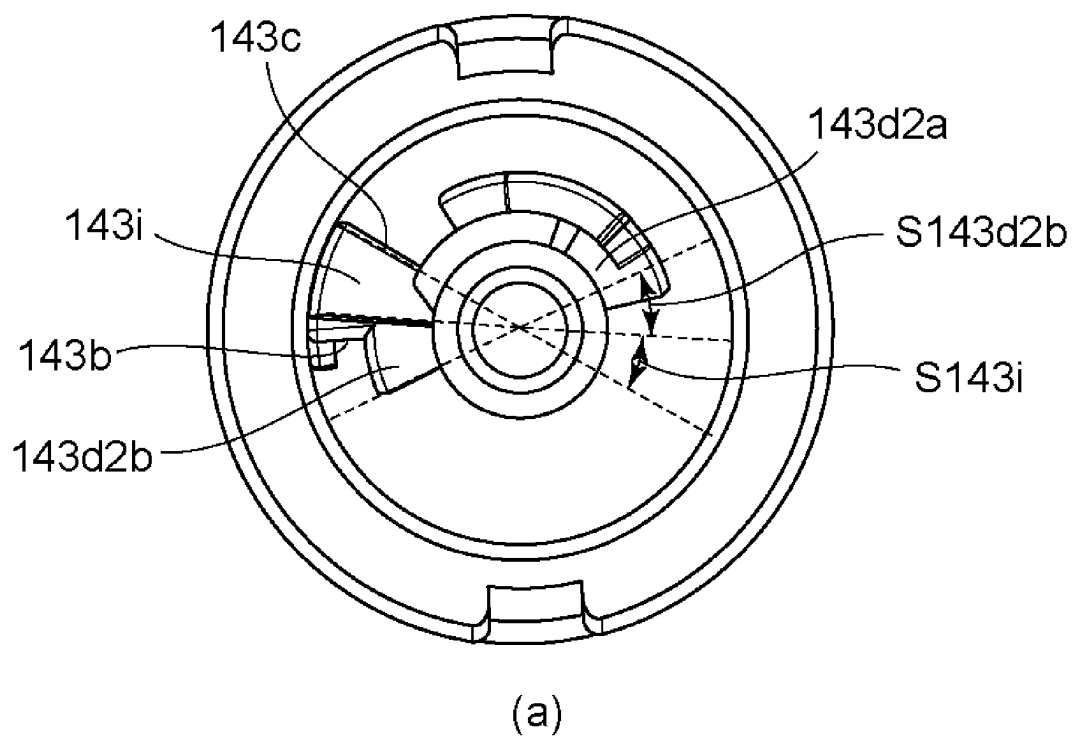


Fig. 100

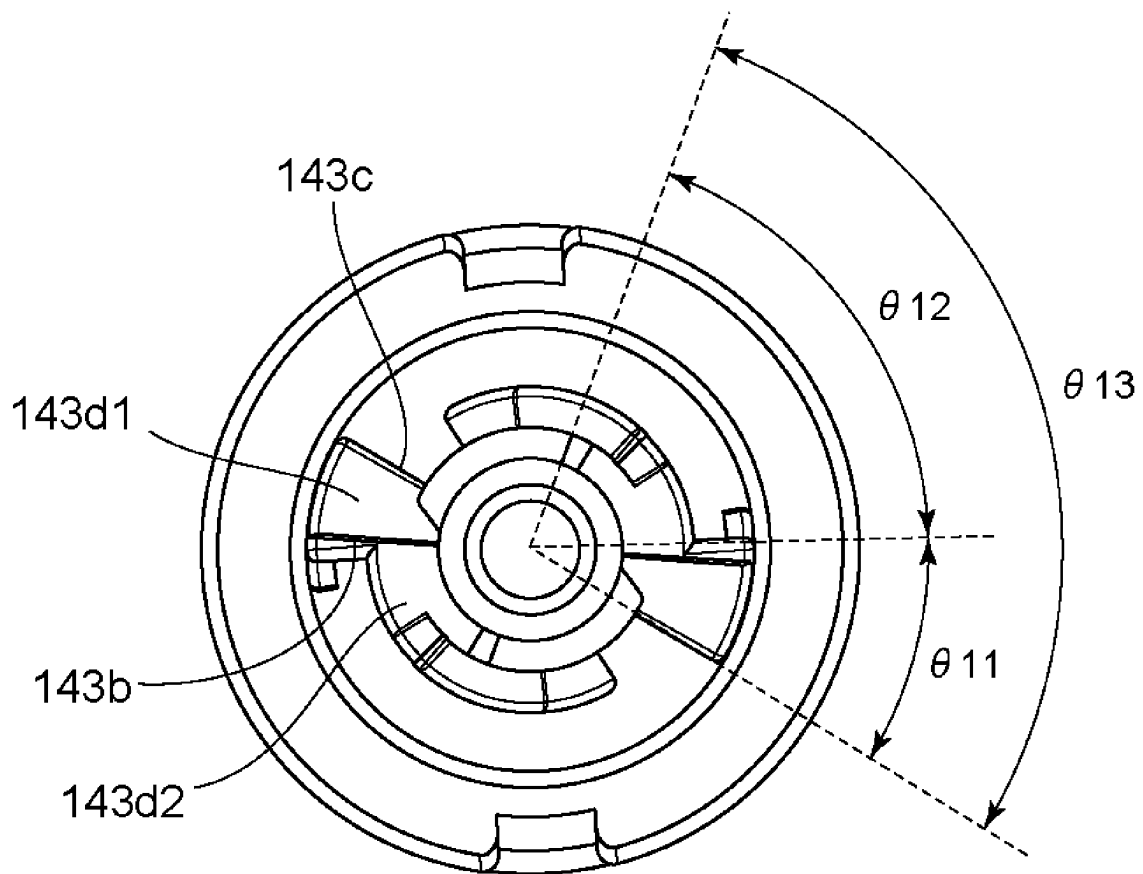


Fig. 101

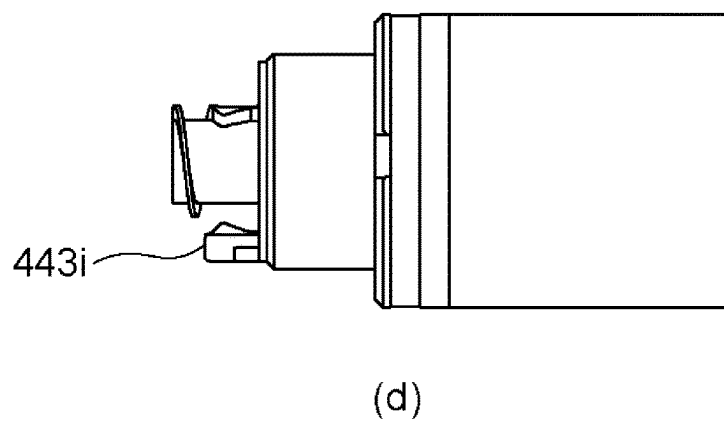
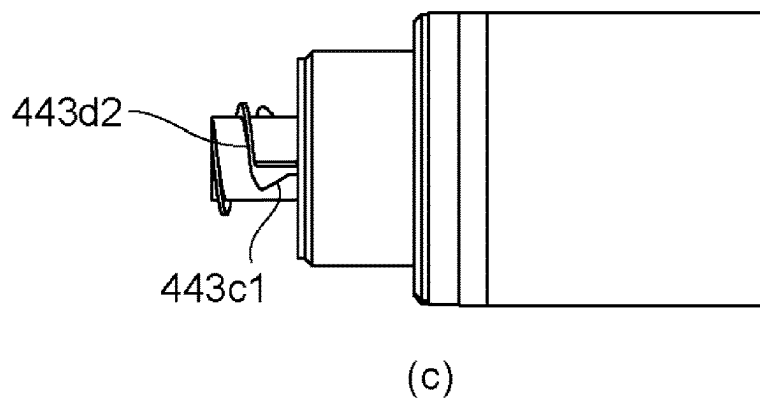
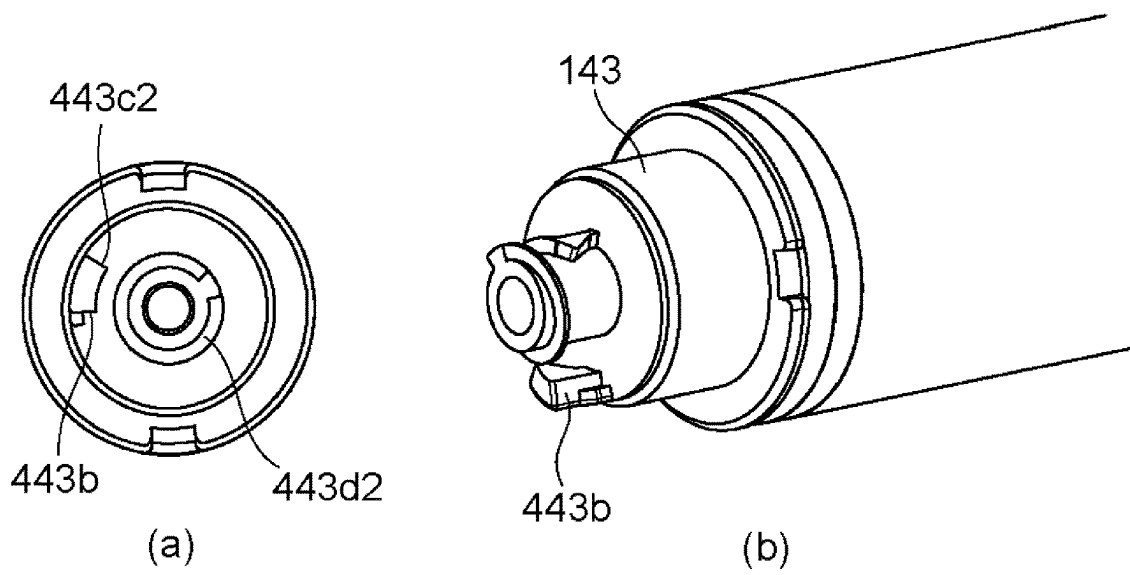


Fig. 102

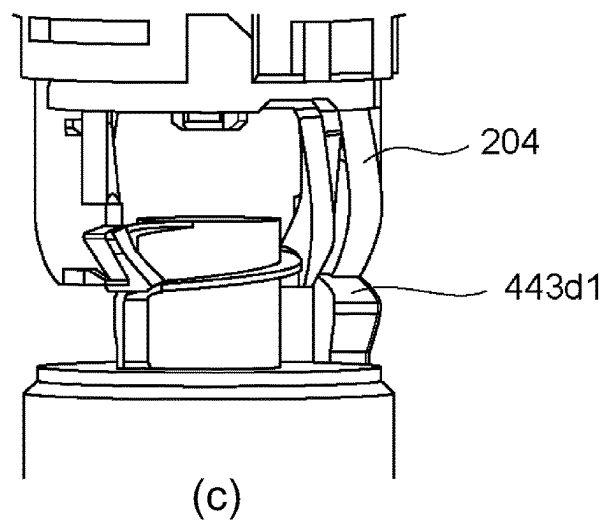
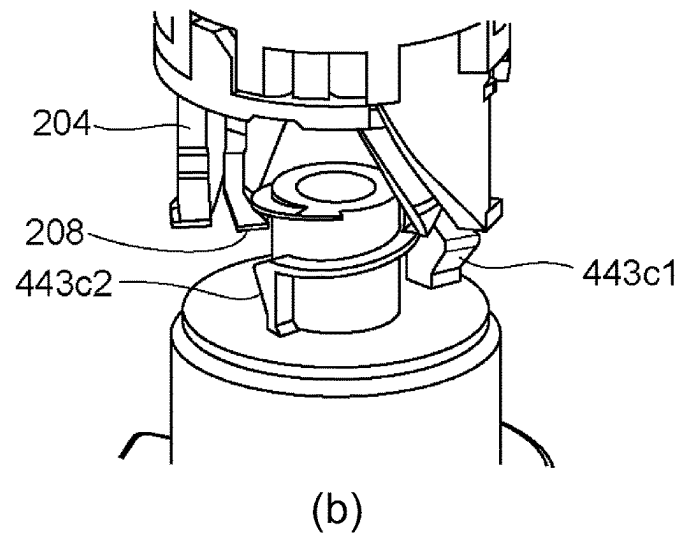
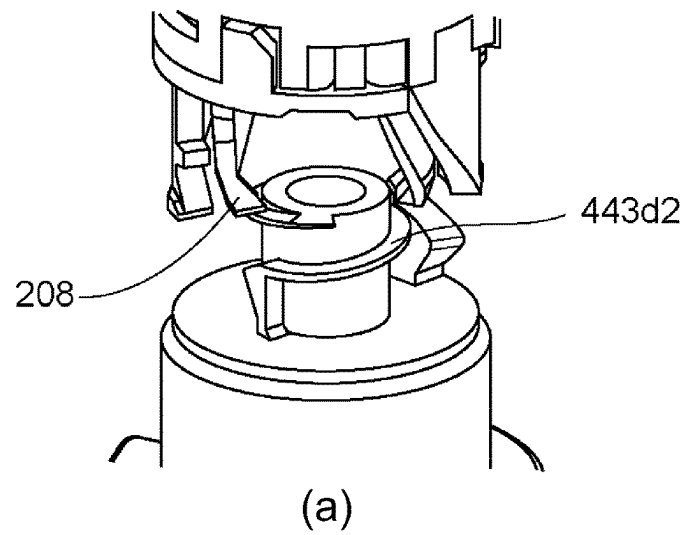


Fig. 103

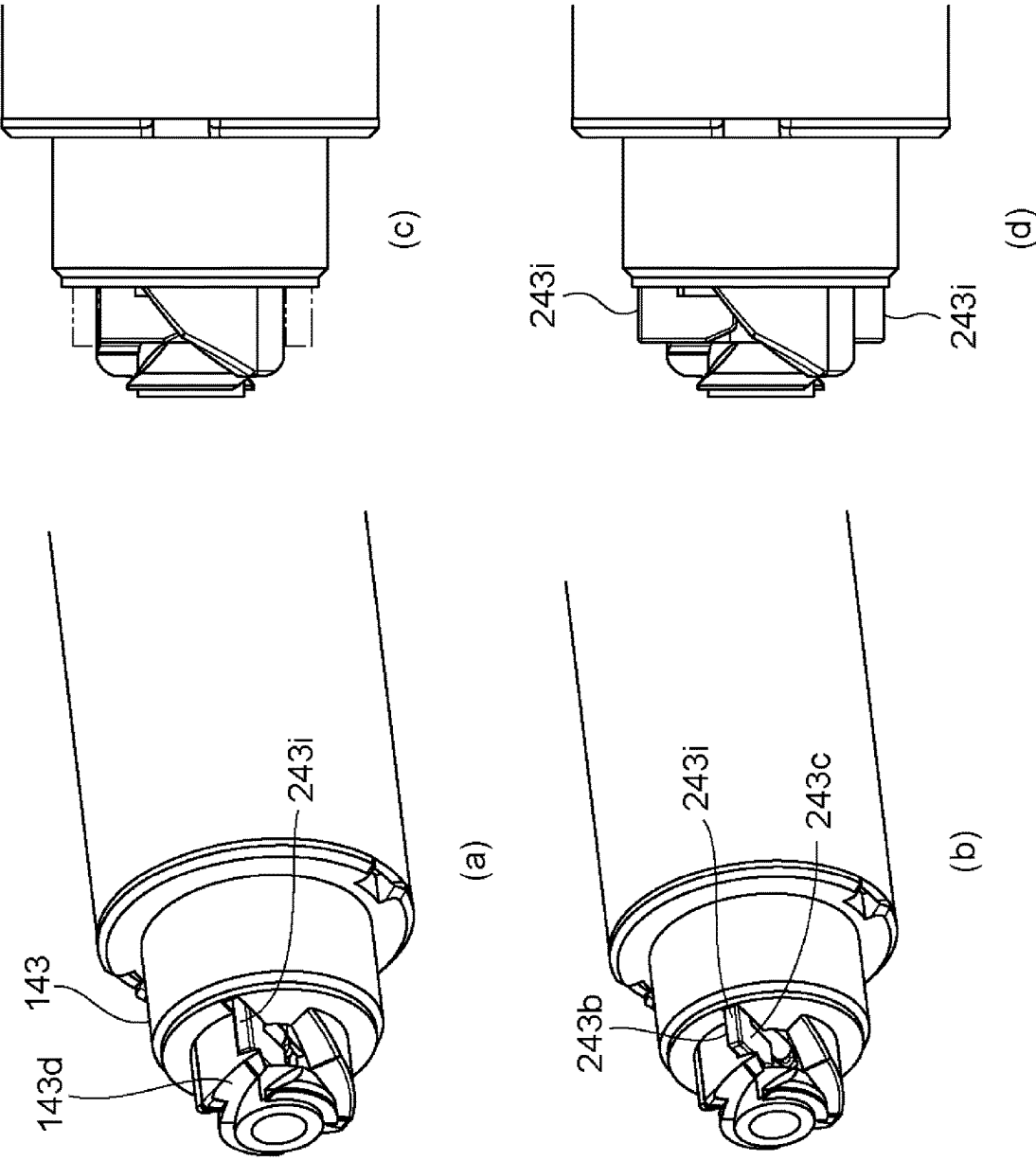


Fig. 104

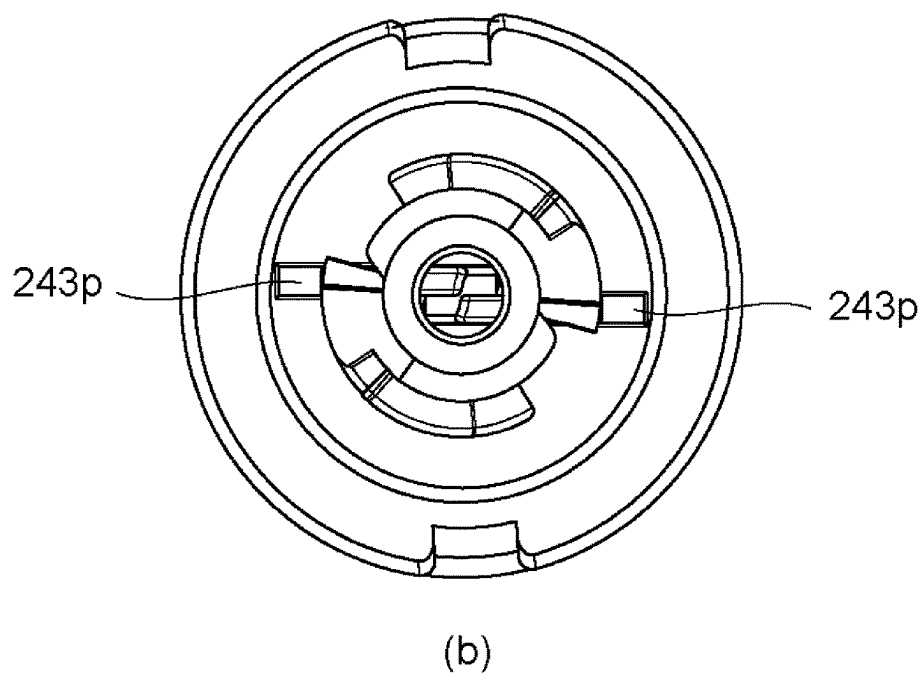
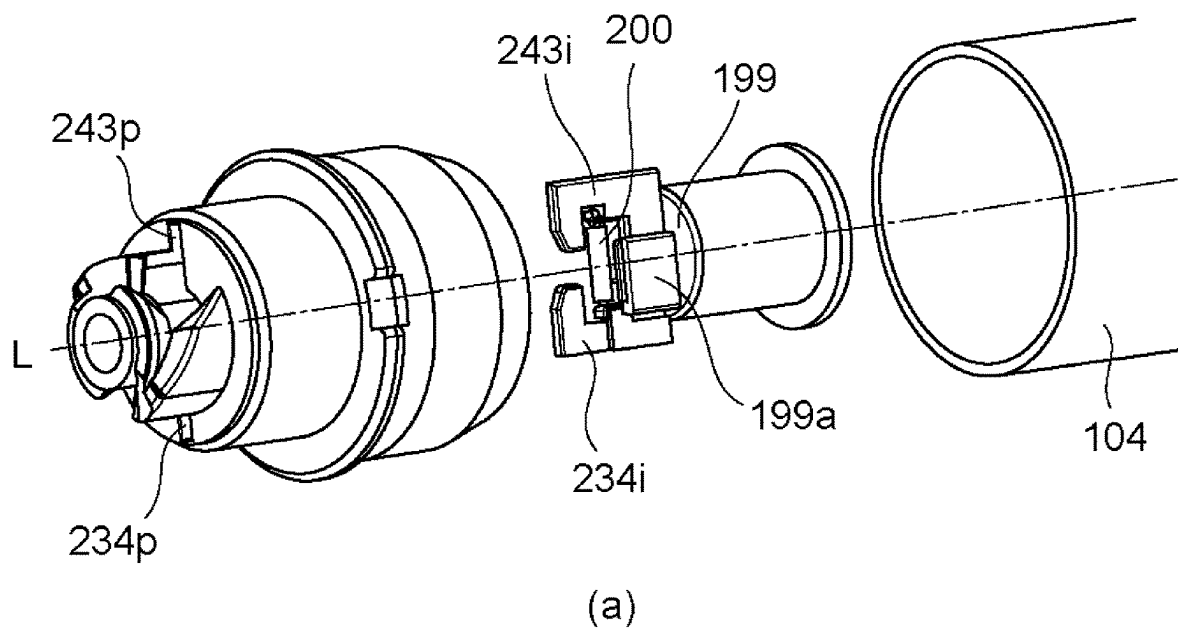
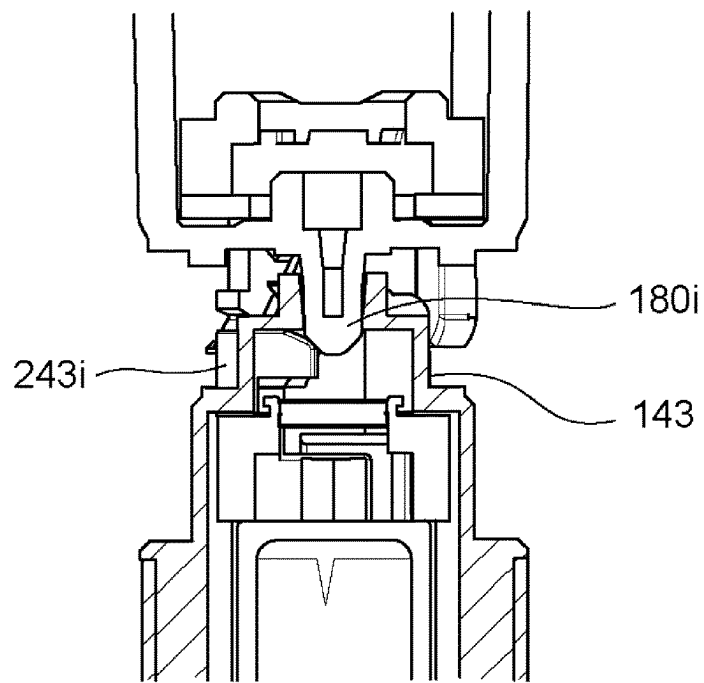
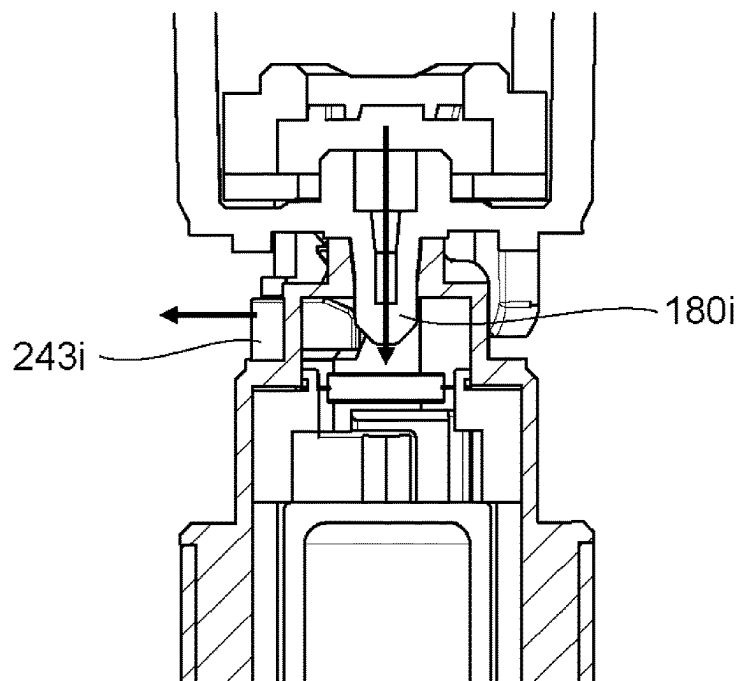


Fig. 105



(a)



(b)

Fig. 106

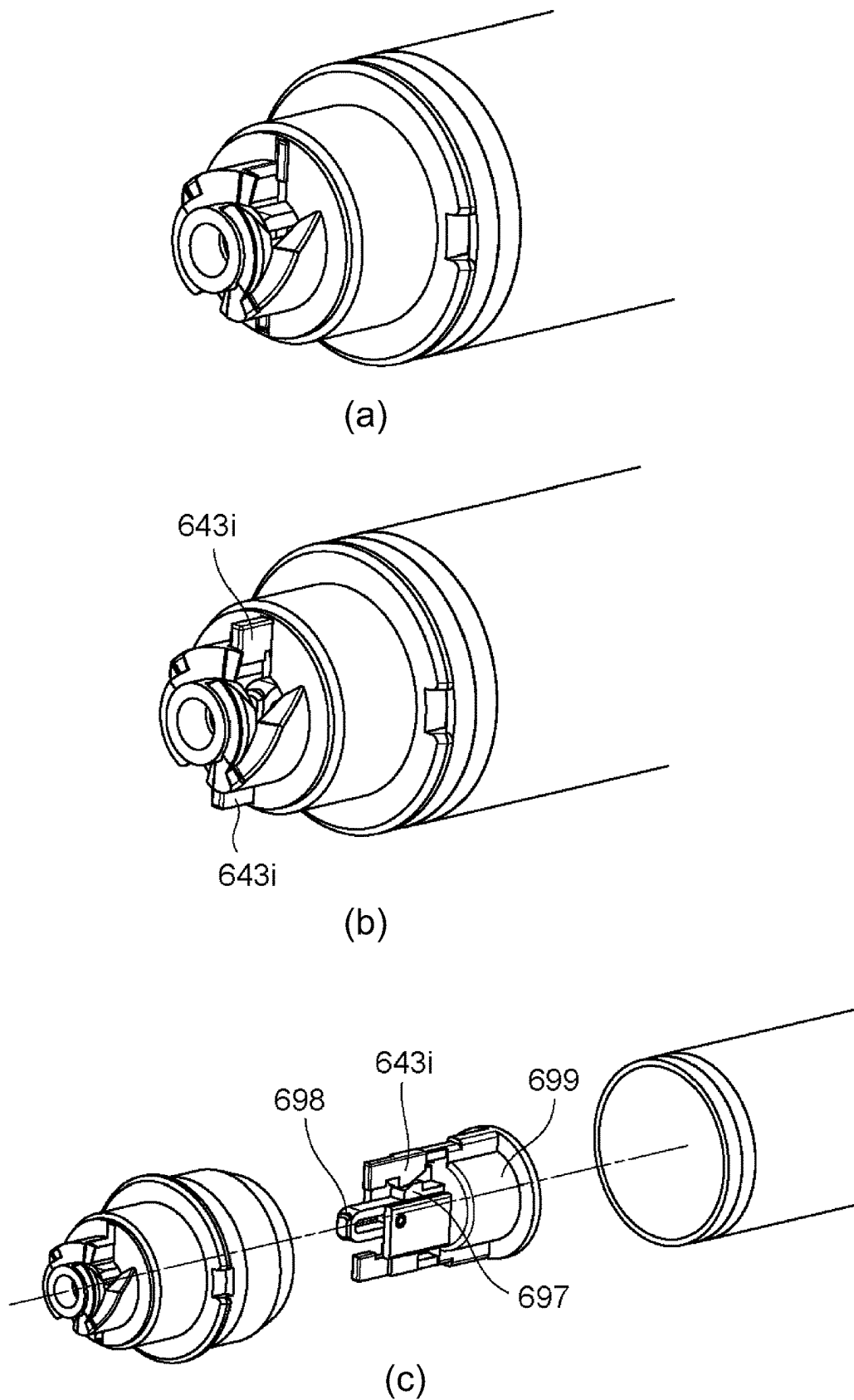
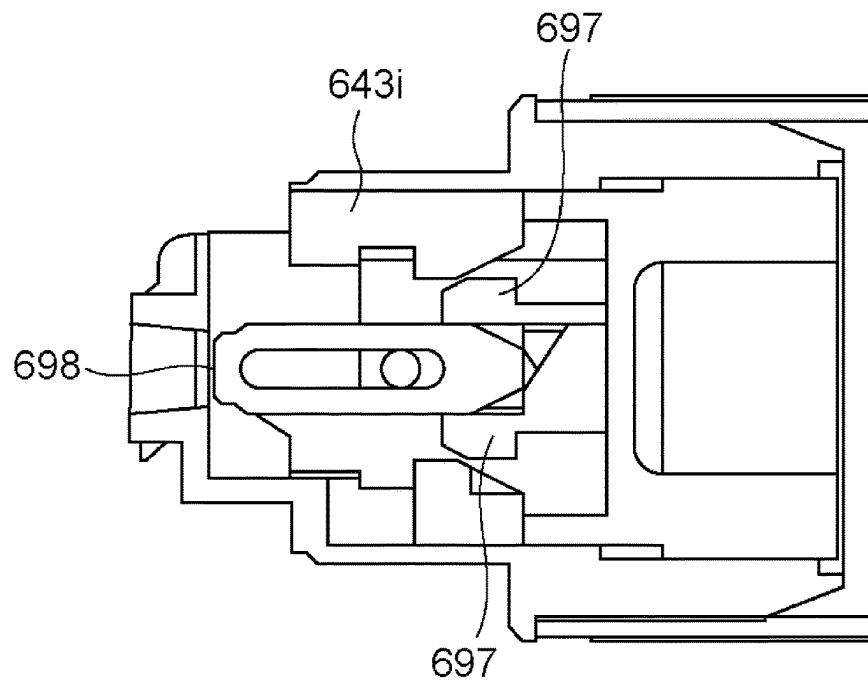
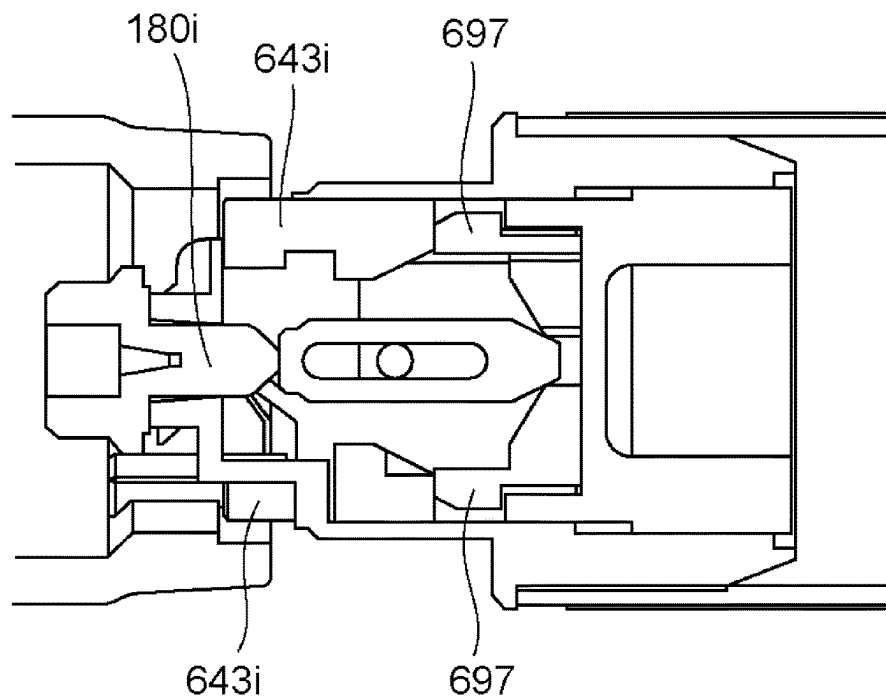


Fig. 107



(a)



(b)

Fig. 108

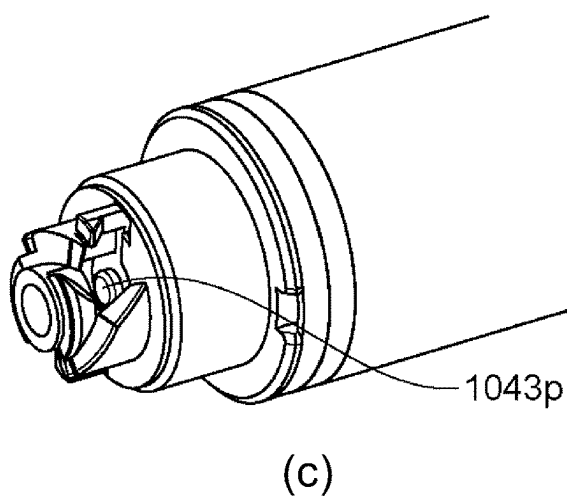
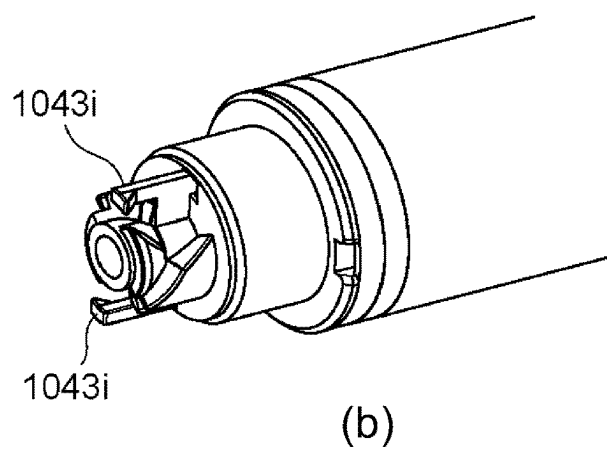
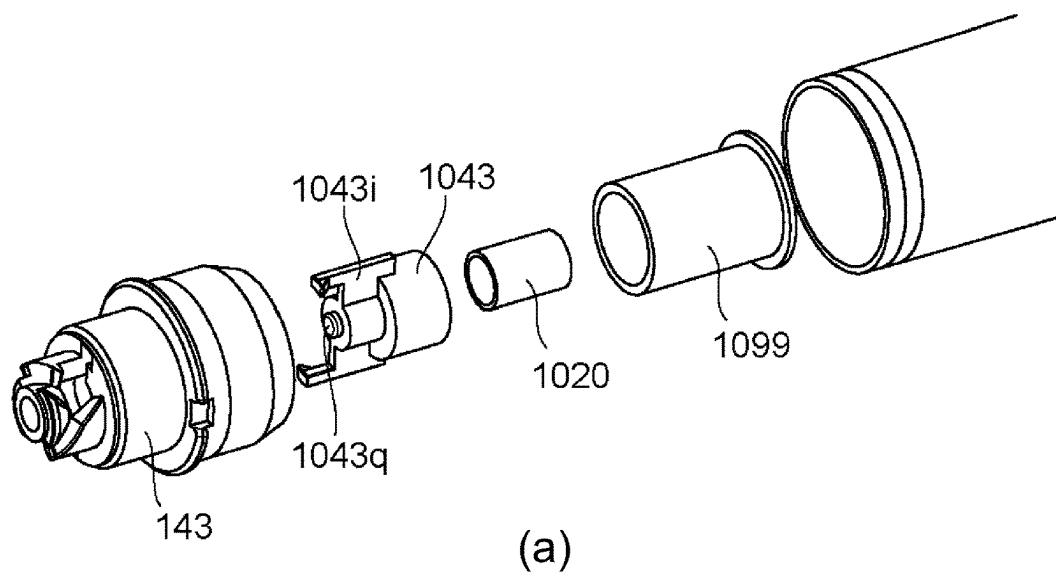


Fig. 109

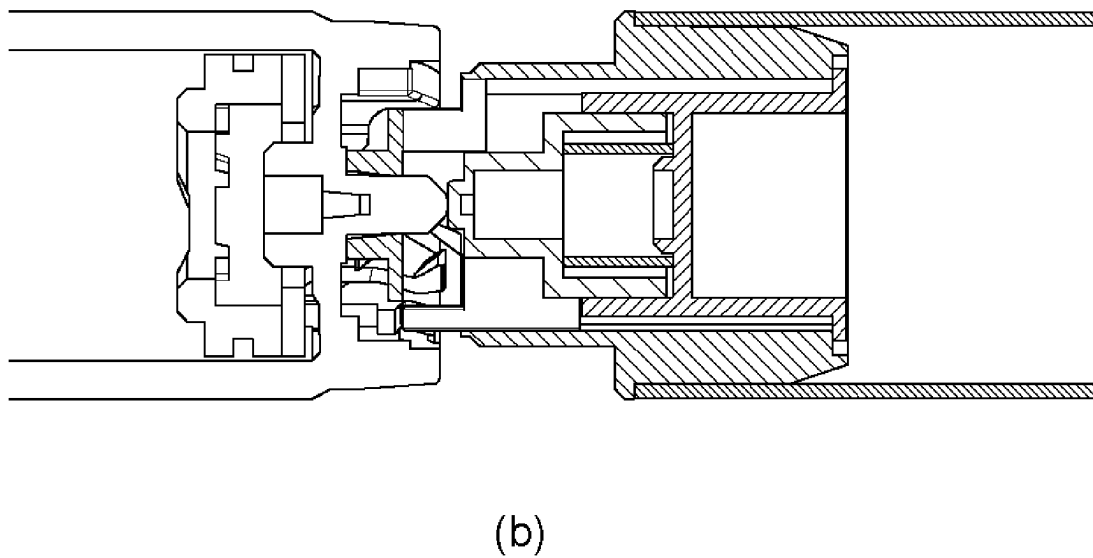
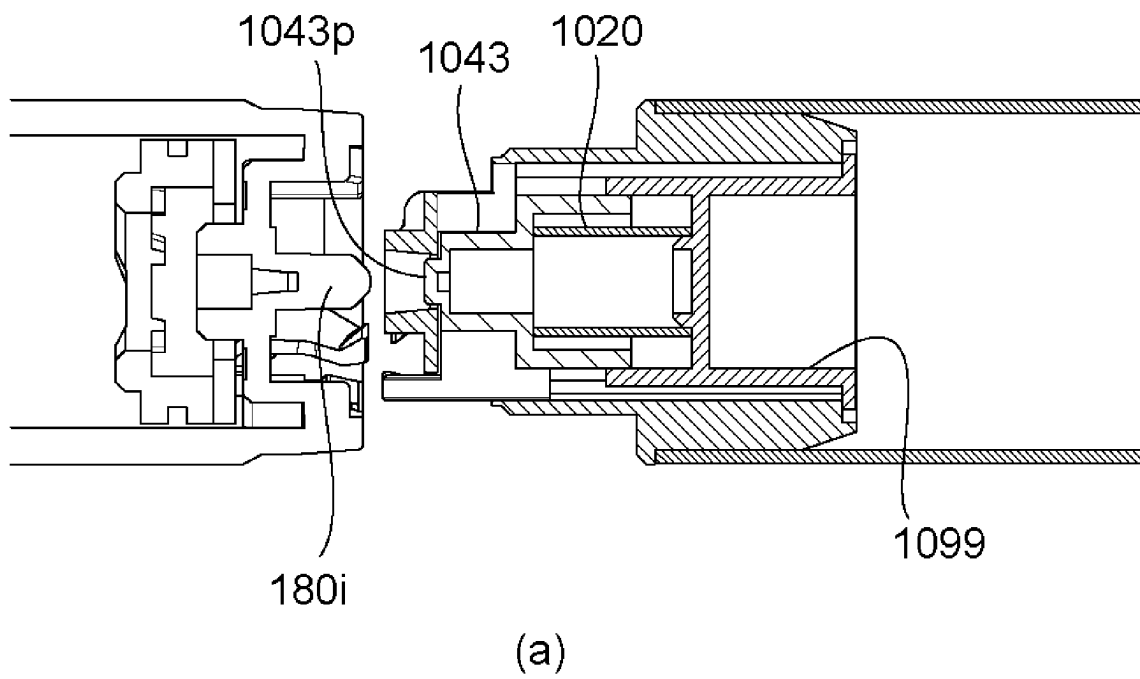


Fig. 110

1

ELECTROPHOTOGRAPHIC IMAGE FORMING APPARATUS, CARTRIDGE AND DRUM UNIT

FIELD OF THE INVENTION

The present invention relates to an electrophotographic image forming apparatus such as a copying machine or a printer which employs an electrophotographic method, and a cartridge usable with the electrophotographic image forming apparatus. The present invention also relates to a drum unit usable with the electrophotographic image forming apparatus and the cartridge.

Here, the electrophotographic image forming apparatus (hereinafter, also referred to as an "image forming apparatus") is an apparatus which forms an image on a recording material by using the electrophotographic image forming method. Examples of the image forming apparatus include a copying machine, a facsimile machine, a printer (laser beam printer, LED printer, and so on), a multifunction printer of them, and the like.

The cartridge is dismountable from the main assembly of the image forming apparatus (apparatus main assembly). Examples of the cartridge include a process cartridge in which a photosensitive member and at least one of the process means acting on the photosensitive member is integrally formed into a cartridge.

The drum unit is a unit including a photosensitive drum, and is used for the cartridge or the image forming apparatus.

BACKGROUND ART

Conventionally, in the field of the image forming apparatus using the electrophotographic forming process, it is known that an electrophotographic photosensitive member (hereinafter referred to as a photosensitive drum) and a process means acting on the photosensitive drum are integrally formed into a cartridge. Such a cartridge is dismountable from the main assembly of the image forming apparatus.

According to this cartridge method, the maintenance of the image forming apparatus can be performed by the user himself/herself without relying on a service person, so that the maintainability can be remarkably improved. Therefore, this cartridge type is widely used in an image forming apparatus.

In a structure in which the cartridge can be mounted to and dismounted from the image forming apparatus main assembly (device main assembly), there is a structure in which the main assembly and the cartridge are connected by using a coupling to input a driving force from the device main assembly to the cartridge (JP H8-328449).

The amount of torque required to drive the cartridge varies depending on the structure of the cartridge.

JP 2002-202690 proposes a structure of a cartridge including a load generating member which applies a load to the rotation of the photosensitive drum. The load generating member stabilizes the rotation of the photosensitive drum by increasing the torque of the photosensitive drum (JP 2002-202690).

SUMMARY OF THE INVENTION

The object of the present invention is to further develop the above-mentioned conventional technology.

An example of the cartridge according to the present application is a cartridge detachably mountable to a main

2

assembly of an electrophotographic image forming apparatus, the main assembly including a driving force application member and a braking force application member, the cartridge comprising:

- 5 a casing;
- a photosensitive drum rotatably supported by the casing;
- a coupling connected with the photosensitive drum so as to be capable of drive transmission,
- wherein the coupling including,
- 10 a driving force receiving portion for receiving a driving force for rotating the coupling by engagement with the driving force application member, and
- a braking force receiving portion for receiving a braking force for applying a load against rotation of the coupling, by engagement with the braking force application member, and
- 15 a guide for moving the braking force application member relative to the driving force application member.

An example of the drum unit according to the present application is a drum unit detachably mountable to a main assembly of an image forming apparatus, the main assembly including a driving force application member and a braking force application member, the drum unit comprising:

- a photosensitive drum;
- 25 a coupling connected with the photosensitive drum so as to be capable of drive transmission,
- wherein the coupling including,
- a driving force receiving portion for receiving driving force for rotating the coupling by engagement with the driving force application member, and
- 30 a braking force receiving portion for receiving a braking force for applying a load against rotation of the coupling, by engagement with the braking force application member, and
- a guide for moving the braking force application member relative to the driving force application member.

Another example of the cartridge according to the present application is a cartridge comprising:

- a casing having a first end portion and a second end portion opposite from the first end portion;
- 40 a photosensitive drum rotatably supported by the first end portion and the second end portion of the casing; and
- a coupling connected with the photosensitive drum so as to be capable of drive transmission, the coupling being provided adjacent to the first end portion of the casing,
- wherein the coupling includes a first shaped portion and a second shaped portion,

the first shaped portion has a portion at a position which is more remote from the second end portion of the casing than the second shaped portion,

a distance measured from the second end portion of the casing to the remote portion of the first shaped portion along an axis direction of the coupling decreases toward downstream in a rotational moving direction of the coupling,

the second shaped portion has a first side portion at a position upstream in the rotational moving direction and the second side portion at a position downstream in the rotational moving direction, and

at least a part of the second shaped portion is more remote from an axis of the coupling than the remote portion of the first shaped portion in a radial direction of the coupling.

Another example of the drum unit according to the present application is usable with a cartridge a drum unit comprising,

- 65 a photosensitive drum rotatably supported by the first end portion and the second end portion of the casing, and

3

a coupling connected with the photosensitive drum so as to be capable of drive transmission, the coupling being provided adjacent to the first end portion of the photosensitive drum,

wherein the coupling includes a first shaped portion and a second shaped portion,

the first shaped portion has a portion at a position which is more remote from the second end portion of the photosensitive drum than the second shaped portion,

a distance measured from the second end portion of the photosensitive drum to the remote portion of the first shaped portion along an axis direction of the coupling decreases toward downstream in a predetermined circumferential direction of the coupling,

the second shaped portion has a first side portion at a position upstream in the circumferential direction and the second side portion at a position downstream in the circumferential directing direction, and

at least a part of the second shaped portion is more remote from an axis of the coupling than the remote portion of the first shaped portion in a radial direction of the coupling.

Another example of the cartridge according to the present application is a cartridge comprising:

a casing having a first end portion and a second end portion opposite from the first end portion;

a photosensitive drum rotatably supported by the first end portion and the second end portion of the casing;

a coupling provided adjacent to the first end portion of the casing, the coupling being connected with the photosensitive drum so as to be capable of drive transmission,

wherein the coupling including,

a first side portion facing upstream in a rotational moving direction of the coupling;

a second side portion facing downstream in the rotational moving direction; and

a guide extending so as to be closer to the second end portion of the casing toward downstream in the rotational moving direction of the coupling, the guide having a portion more remote from the second end portion of the photosensitive drum than the first side portion, in an axial direction of the coupling,

wherein at least a part of the first side portion is more remote from an axis of the drum unit than the remote portion of the guide, in a radial direction of the coupling.

Another example of the drum unit according to the present application is a drum unit comprising:

a photosensitive drum having a first end portion and a second end portion opposite from the first end portion; and

a coupling provided adjacent to the first end portion of the photosensitive drum, the coupling being connected with the photosensitive drum so as to be capable of drive transmission,

wherein the coupling including,

a first side portion facing the upstream in a predetermined circumferential direction of the coupling,

a second side portion facing downstream in the circumferential direction, and

a guide extending so as to be closer to the second end portion of the casing toward a downstream in the circumferential direction, the guide having a portion more remote from the second end portion of the photosensitive drum in an axial direction of the coupling than the first side portion,

wherein at least a part of the first side portion is more remote from an axis of the coupling than the remote portion of the guide, in a radial direction of the coupling.

Another example of the cartridge according to the present application is a cartridge detachably mountable to a main

4

assembly of an electrophotographic image forming apparatus, the main assembly including a driving force application member and a braking force application member movable relative to the driving force application member, the cartridge comprising:

a casing;

a photosensitive drum rotatably supported by the casing; and

a coupling connected with the photosensitive drum so as to be capable of drive transmission,

wherein the coupling including,

a driving force receiving portion for receiving a driving force for rotating the coupling by engagement with the driving force application member, and

a braking force receiving portion for receiving a braking force for applying a load against rotation of the coupling, by engagement with the braking force application member.

Another example of the drum unit according to the present application is a drum unit detachably mountable to a main assembly of an electrophotographic image forming apparatus, the main assembly including a driving force application member and a braking force application member movable relative to the driving force application member, the drum unit comprising:

a photosensitive drum rotatably supported by the casing; and

a coupling connected with the photosensitive drum so as to be capable of drive transmission,

wherein the coupling including,

a driving force receiving portion for receiving a driving force for rotating the coupling by engagement with the driving force application member, and

a braking force receiving portion for receiving a braking force for applying a load against rotation of the coupling, by engagement with the braking force application member.

Further, another example of the cartridge according to the present application includes one of the above-mentioned drum units and a casing which supports the drum unit.

Furthermore, an example of the electrophotographic image forming apparatus according to the present application includes any of the above-mentioned cartridges and the main assembly of the electrophotographic image forming apparatus.

Effect of the Invention

Conventional technology can be developed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a drum coupling **143**.

FIG. 2 is a schematic sectional view of an image forming apparatus.

FIG. 3 is a sectional view of a process cartridge.

FIG. 4 is a sectional view of the image forming apparatus.

FIG. 5 is a sectional view of the image forming apparatus.

FIG. 6 is a sectional view of the image forming apparatus.

FIG. 7 is a partial detailed view of the tray.

FIG. 8 is a perspective view of the storing element pressing unit and the cartridge pressing unit.

FIG. 9 is a partial perspective view of the image forming apparatus.

FIG. 10 is a side view (partial sectional view) of the process cartridge.

FIG. 11 is a sectional view of the image forming apparatus.

5

FIG. 12 is a perspective view of a development separation control unit.

FIG. 13 is an assembly perspective view of the process cartridge.

FIG. 14 is a perspective view of the process cartridge.

FIG. 15 is an assembly perspective view of the process cartridge.

FIG. 16 is an assembly perspective view of the process cartridge.

FIG. 17 is a view of a separation holding member R per se.

FIG. 18 is a view of a force applying member R per se.

FIG. 19 is a partial sectional view of the separation holding member R after assembly.

FIG. 20 is an enlarged view of the periphery of the separation holding member R.

FIG. 21 is an enlarged view of the periphery of the separation holding member R.

FIG. 22 is a bottom view of a driving side of the process cartridge.

FIG. 23 is an illustration showing operation of a developing unit in the main assembly of the image forming apparatus.

FIG. 24 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 25 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 26 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 27 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 28 is a view of the separation holding member L per se.

FIG. 29 is a view of the force applying member L per se.

FIG. 30 is an assembly perspective view after assembling the development pressure spring and assembling the separation holding member L.

FIG. 31 is a partial sectional view of the separation holding member L after assembly.

FIG. 32 is an enlarged view of the peripheries of the separation holding member L and the force applying member L.

FIG. 33 is an enlarged view of the periphery of the separation holding member.

FIG. 34 is a side view as viewed from the driving side with the process cartridge mounted inside the image forming apparatus main assembly.

FIG. 35 is an illustration showing a process cartridge in the main assembly of the image forming apparatus.

FIG. 36 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 37 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 38 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

FIG. 39 is an illustration showing the operation of the developing unit in the main assembly of the image forming apparatus.

6

FIG. 40 is an illustration showing the arrangement of the separation holding member R and the force applying member.

FIG. 41 is an illustration showing the arrangement of the separation holding member and the force applying member.

FIG. 42 is a side view as viewed from the driving side with the process cartridge 100 mounted inside the image forming apparatus main assembly.

FIG. 43 is an exploded perspective view of the drive transmission unit 203.

FIG. 44 is a sectional view of the drive transmission unit 203.

FIG. 45 is a perspective view of the drive transmission unit 203.

FIG. 46 is a sectional perspective view of the main assembly of the device including the drive transmission unit 203.

FIG. 47 is a front view of the drive transmission unit 203 and the drum coupling 143.

FIG. 48 is a developed view illustrating engagement of the drum coupling.

FIG. 49 is a developed view illustrating the engagement of the drum coupling.

FIG. 50 is a developed view illustrating the engagement of the drum coupling.

FIG. 51 is a sectional view illustrating the engagement of the drum coupling.

FIG. 52 is a perspective view illustrating a modified example of the drum coupling.

FIG. 53 is a developed view illustrating the engagement of the drum coupling.

FIG. 54 is a development view illustrating the engagement of the drum coupling.

FIG. 55 is a perspective view of the drum unit showing the drum coupling.

FIG. 56 is an illustration of a drum unit showing a drum coupling.

FIG. 57 is a perspective view of the drum unit showing the drum coupling.

FIG. 58 is a top view of the drum coupling.

FIG. 59 is a perspective view illustrating parts of the drive transmission unit.

FIG. 60 is a perspective view of the drive transmission unit and the drum unit.

FIG. 61 is a perspective view of the drive transmission unit and the drum unit.

FIG. 62 is a perspective view of the drive transmission unit and the drum unit.

FIG. 63 is a perspective view of the drive transmission unit and the drum unit.

FIG. 64 is a perspective view of the drive transmission unit and the drum unit.

FIG. 65 is a perspective view of the drive transmission unit and the drum unit.

FIG. 66 is a perspective view of the drive transmission unit and the drum unit.

FIG. 67 is a perspective view of the drive transmission unit and the drum unit.

FIG. 68 is a perspective view of the drive transmission unit and the drum unit.

FIG. 69 is a perspective view of the drive transmission unit and the drum unit.

FIG. 70 is a perspective view of the drive transmission unit and the drum unit.

FIG. 71 is a perspective view of the drive transmission unit and the drum unit.

7

FIG. 72 is a perspective view of the drive transmission unit and the drum unit.

FIG. 73 is a perspective view illustrating a modified example of the drum coupling.

FIG. 74 is a perspective view and a front view illustrating a modified example of the drum coupling.

FIG. 75 is a perspective view of the drum unit.

FIG. 76 is a developed view illustrating the engagement of the drum coupling.

FIG. 77 is a perspective view of the drum unit and a front view of the coupling.

FIG. 78 is a perspective view of the drum unit and the drive transmission unit.

FIG. 79 is a side view, a perspective view, and a front view of the coupling.

FIG. 80 is a side view of the coupling.

FIG. 81 is a side view and a perspective view of the coupling.

FIG. 82 is a schematic sectional view of the image forming apparatus.

FIG. 83 is a schematic sectional view of the process cartridge.

FIG. 84 is a schematic perspective view of the process cartridge.

FIG. 85 is a schematic perspective view of the process cartridge.

FIG. 86 is a schematic sectional view of the process cartridge taken along a rotational axis of the photosensitive drum.

FIG. 87 is an exploded perspective view of a drive transmission unit 811.

FIG. 88 is a sectional view taken along the rotation axis of the drive transmission unit 811 mounted to the main assembly of the image forming apparatus.

FIG. 89 is a schematic perspective view of another form of the drum coupling 770.

FIG. 90 is a schematic perspective view illustrating mounting of the cartridge 701 to the image forming apparatus main assembly 800.

FIG. 91 is a schematic sectional view illustrating the mounting operation of the cartridge 701 to the image forming apparatus main assembly 800.

FIG. 92 is a schematic sectional view illustrating the mounting operation of the drum coupling 770 to the main assembly drive transmission unit 811.

FIG. 93 is a schematic sectional view illustrating the mounting operation of the drum coupling 770 to the main assembly drive transmission unit 811.

FIG. 94 is a perspective view illustrating another form of the process cartridge.

FIG. 95 is a sectional view of the drum unit.

FIG. 96 is a front view of the coupling.

In FIG. 97, part (a) is a perspective view of the coupling, and part (b) is a front view.

FIG. 98 is a front view of the coupling.

FIG. 99 is a perspective view illustrating an engaged state of the coupling and the braking engagement member.

FIG. 100 is a front view of the coupling.

FIG. 101 is a front view of the coupling.

FIG. 102 is a front view, a perspective view, and a side view of the coupling.

FIG. 103 is a perspective view illustrating an engaged state of the coupling and the braking engagement member.

FIG. 104 is a perspective view and a side view of the drum unit.

FIG. 105 is a perspective view of the drum unit and a front view of the coupling.

8

FIG. 106 is a sectional view of the drum unit.

FIG. 107 is a perspective view of the drum unit.

FIG. 108 is a sectional view of the coupling.

FIG. 109 is a perspective view of the drum unit.

FIG. 110 is a sectional view of the drum unit and the drive transmission unit.

DESCRIPTION OF THE EMBODIMENTS

Embodiment 1

Hereinafter, a mode for carrying out the present invention will be described in detail exemplarily with reference to the drawings and examples. However, the functions, materials, shapes, relative arrangements, and the like of the components described in this embodiment are not intended to limit the scope of the present invention to those, unless otherwise specified.

Hereinafter, the Embodiment 1 will be described with reference to the drawings.

In the following embodiment, as the image forming apparatus, an image forming apparatus which four process cartridges can be mounted to and dismounted from is illustrated.

The number of process cartridges mounted on the image forming apparatus is not limited to this example. It is selected appropriately as needed.

Further, in the embodiment described below, a laser beam printer is exemplified as one aspect of the image forming apparatus.

[Outline Structure of Image Forming Apparatus]

FIG. 2 is a schematic sectional view of the image forming apparatus M. Further, FIG. 3 is a sectional view of the process cartridge 100.

The image forming apparatus M is a four-color full-color laser printer using an electrophotographic process, and forms a color image on the recording material S. The image forming apparatus M is a process cartridge type, and a process cartridge is dismountably mounted to the image forming apparatus main assembly (apparatus main assembly) 170 to form a color image on the recording material S.

Here, regarding the image forming apparatus M, the side where the front door 11 is provided is the front surface (front surface), and the surface opposite to the front surface is the back surface (rear surface). Further, the right side of the image forming apparatus M as viewed from the front is referred to as a driving side, and the left side is referred to as a non-driving side.

Further, as the image forming apparatus M is viewed from the front side, the upper side is the upper surface and the lower side is the lower surface. FIG. 2 is a sectional view of the image forming apparatus M as viewed from the non-driving side; the front side of the sheet of the drawing is the non-driving side of the image forming apparatus M; the right side of the sheet of the drawing is the front side; and the rear side of the sheet of the drawing is the driving side of the image forming apparatus.

The driving side of the process cartridge 100 is the side on which the drum coupling (photosensitive member coupling) which will be described hereinafter is disposed in the axial direction of the photosensitive drum. Further, the driving side of the process cartridge 100 is also the side on which the development coupling described hereinafter is arranged in the axial direction of the developing roller (developing member).

The axial direction of the photosensitive drum is a direction parallel to the rotation axis of the photosensitive drum, which will be described hereinafter. Similarly, the axial direction of the developing roller is a direction parallel to the rotation axis of the developing roller, which will be described hereinafter. In this embodiment, the axis of the photosensitive drum and the axis of the developing roller are substantially parallel, and therefore, the axial direction of the photosensitive drum and the axial direction of the developing roller are considered to be substantially the same.

The image forming apparatus main assembly 170 has four process cartridges 100 (100Y, 100M, 100C, 100K), namely a first process cartridge 100Y, a second process cartridge 100M, a third process cartridge 100C, and a fourth process cartridge 100K, which are arranged almost horizontally.

Each of the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) has the same electrophotographic process mechanism, and the colors of the developer (hereinafter referred to as toner) are different. Rotational driving force is transmitted to the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) from a drive output portion (details will be described hereinafter) of the image forming apparatus main assembly 170.

Further, bias voltages (charging bias, development bias, and so on) are supplied from the image forming apparatus main assembly 170 to each of the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) (not shown).

As shown in FIG. 3, each of the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) of this embodiment includes a photosensitive drum 104 and a drum holding unit 108 which is provided with charging means functioning as a process means acting on the photosensitive drum 104. Further, each of the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) includes a developing unit 109 provided with a developing means for developing an electrostatic latent image on the photosensitive drum 104.

The drum holding unit 108 and the developing unit 109 are coupled to each other. A more specific structure of the process cartridge 100 will be described hereinafter.

The first process cartridge 100Y contains yellow (Y) toner in a development frame 125, and forms a yellow-color toner image on the surface of the photosensitive drum 104.

The second process cartridge 100M contains magenta (M) toner in a development frame 125, and forms a magenta-color toner image on the surface of the photosensitive drum 104.

The third process cartridge 100C contains cyan (C) toner in a development frame 125, and forms a cyan-color toner image on the surface of the photosensitive drum 104.

The fourth process cartridge 100K contains black (K) toner in a development frame 125, and forms a black toner image on the surface of the photosensitive drum 104. A laser scanner unit 14 as an exposure means is provided above the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K). The laser scanner unit 14 outputs a laser beam U corresponding to the image information. The laser beam U passes through the exposure window 110 of the process cartridge 100 and scans so that the surface of the photosensitive drum 104 is exposed to the laser beam U.

Below the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K), an intermediary transfer unit 12 as a transfer member is provided. The intermediary transfer unit 12 includes a drive roller 12e, a turn roller 12c, and a tension roller 12b, and a flexible transfer belt 12a is extended around these rollers.

The lower surface of the photosensitive drum 104 of each of the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) is in contact with the upper surface of the transfer belt 12a. The contact portion is the primary transfer portion. Inside the transfer belt 12a, a primary transfer roller 12d is provided so as to oppose the photosensitive drum 104.

The secondary transfer roller 6 is brought into contact with the turn roller 12c by way of the transfer belt 12a. The contact portion between the transfer belt 12a and the secondary transfer roller 6 is the secondary transfer portion.

A feeding unit 4 is provided below the intermediary transfer unit 12. The feeding unit 4 includes a sheet feed tray 4a on which the recording material S is loaded and accommodated, and a sheet feeding roller 4b.

A fixing device 7 and a paper discharge ion device 8 are provided on the upper left side of the image forming apparatus main assembly 170 in FIG. 2. The upper surface of the image forming apparatus main assembly 170 functions as a paper discharge tray 13.

The toner image is fixed on the recording material S by a fixing means provided in the fixing device 7, and the recording material is discharged to the paper discharge tray 13.

[Image Forming Operation]

The operation for forming a full-color image is as follows.

The photosensitive drum 104 of each of the first to fourth process cartridges 100 (100Y, 100M, 100C, 100K) is rotationally driven at a predetermined speed (in the direction of arrow A in FIG. 3).

The transfer belt 12a is also rotationally driven in the forward direction (direction of arrow C in FIG. 2) codirectionally with the rotation of the photosensitive drum at a speed corresponding to the speed of the photosensitive drum 104.

The laser scanner unit 14 is also driven. In synchronization with the drive of the laser scanner unit 14, the charging roller 105 uniformly charges the surface of the photosensitive drum 104 to a predetermined polarity and potential in each process cartridge. The laser scanner unit 14 scans and exposes the surface of each photosensitive drum 104 with laser beam U in accordance with the image signals of each color.

By this, an electrostatic latent image corresponding to the image signal of the corresponding color is formed on the surface of each photosensitive drum 104. The formed electrostatic latent image is developed by a developing roller 106 which is rotationally driven at a predetermined speed. More specifically, the developing roller 106 is in contact with the photosensitive drum 104, and the toner moves from the developing roller 106 to the latent image of the photosensitive drum 104, so that the latent image is developed into a toner image. In this embodiment, the contact developing method is employed, and the developing roller 106 and the photosensitive drum 104 are in contact with each other. However, there a non-contact development method may be employed in which toner jumps from the developing roller 106 to the photosensitive drum 104 through a small gap between the developing roller 106 and the photosensitive drum 104.

Through the electrophotographic image forming process operation as described above, a yellow toner image corresponding to the yellow component of the full-color image is formed on the photosensitive drum 104 of the first process cartridge 100Y. Then, the toner image is primary-transferred onto the transfer belt 12a. A part of the photosensitive drum 104 is exposed to the outside of the cartridge and is in contact with the transfer belt 12a. At this contact portion, the

11

toner image on the surface of the photosensitive drum **104** transferred onto the transfer belt **12a**.

Similarly, a magenta color toner image corresponding to the magenta component of the full color image is formed on the photosensitive drum **104** of the second process cartridge **100M**. Then, the toner image is superimposedly transferred onto the yellow toner image already transferred on the transfer belt **12a**.

Similarly, a cyan toner image corresponding to the cyan component of the full-color image is formed on the photosensitive drum **104** of the third process cartridge **100C**. Then, the toner image is superimposedly primary-transferred onto the yellow-colored and magenta-colored toner images already transferred on the transfer belt **12a**.

Similarly, a black toner image corresponding to the black component of the full-color image is formed on the photosensitive drum **104** of the fourth process cartridge **100K**. Then, the toner image is superimposedly primary-transferred onto the yellow, magenta, and cyan toner images already transferred on the transfer belt **12a**.

In this manner, a four-color full-color unfixed toner image of yellow, magenta, cyan, and black is formed on the transfer belt **12a**.

On the other hand, the recording materials **S** are separated and fed one by one at a predetermined controlled timing. The recording material **S** is introduced then into the secondary transfer portion, which is the contact portion between the secondary transfer roller **6** and the transfer belt **12a**, at a predetermined control timing.

By this, in the process of feeding the recording material **S** to the secondary transfer unit, the four-color superimposed toner images on the transfer belt **12a** are sequentially and collectively transferred onto the surface of the recording material **S**.

In more detail, the structure of the image forming apparatus main assembly will be described below.

[Outline of Process Cartridge Mounting/Dismounting Structure]

Referring to FIGS. **42** and **4** to **7**, the tray **171** which supports the process cartridge will be described in more detail. FIG. **4** is a sectional view of the image forming apparatus **M** in which the tray **171** is located inside the image forming apparatus main assembly **170** with the front door **11** open. FIG. **5** is a sectional view of the image forming apparatus **M** in a state in which the tray **171** is located outside the image forming apparatus main assembly **170** with the front door **11** open and the process cartridges **100** accommodated in the tray. FIG. **6** is a sectional view of the image forming apparatus **M** in a state in which the tray **171** is located outside the image forming apparatus main assembly **170** with the front door **11** open and the process cartridge **100** having been removed from the tray. Part (a) of FIG. **7** is a partial detailed view of the tray **171** as viewed from the driving side in the state shown in FIG. **4**. Part (b) of FIG. **7** is a partial detailed view of the tray **171** as viewed from the non-driving side in the state of FIG. **4**.

As shown in FIGS. **4** and **5**, the tray **171** can be moved in the arrow **X1** direction (pushing direction) and the arrow **X2** direction (pulling direction) relative to the image forming apparatus main assembly **170**. That is, the tray **171** is provided so as to be retractable from and insert able into the image forming apparatus main assembly **170**, and the tray **171** is structured to be movable in a substantially horizontal direction in a state where the image forming apparatus main assembly **170** is installed on a horizontal floor. Here, the state in which the tray **171** is located outside the image forming apparatus main assembly **170** (the state shown in

12

FIG. **5**) is referred to as an outside position. Further, a state in which the tray is placed inside the image forming apparatus main assembly **170** with the front door **11** open and the photosensitive drum **104** and the transfer belt **12a** are separated from each other (state in FIG. **4**) is referred to as an inner position.

Further, the tray **171** has a mounting portion **171a** in which the process cartridges **100** can be dismountably mounted as shown in FIG. **6** in the outer position. Then, each process cartridge **100** mounted on the mounting portion **171a** in the outer position of the tray **171** is supported by the tray **171** by the driving side cartridge cover member **116** and the immovable side cartridge cover member **117** as shown in FIG. **7**. Then, the process cartridge moves inside the image forming apparatus main assembly **170** with the movement of the tray **171** in a state of being placed in the mounting portion **171a**. At this time, in the movement, a gap is kept between the transfer belt **12a** and the photosensitive drum **104**. The tray **171** can carry the process cartridge **100** into the image forming apparatus main assembly **170** without the photosensitive drum **104** contacting with the transfer belt **12a** (details will be described hereinafter).

As described above, by using the tray **171**, a plurality of process cartridges **100** can be collectively moved to a position where image formation is possible inside the image forming apparatus main assembly **170**, and is collectively moved to the outside of the image forming apparatus main assembly **170**.

[Positioning of Process Cartridge Relative to Electrophotographic Image Forming Apparatus Main Assembly]

Referring to FIG. **7**, the positioning of the process cartridge **100** relative to the image forming apparatus main assembly **170** will be described more specifically.

As shown in FIG. **7**, the tray **171** is provided with positioning portions **171VR** and **171VL** for holding the cartridge **100**. The positioning portion **171VR** has straight portions **171VR1** and **171VR2**, respectively. The center of the photosensitive drum is determined by the arc portions **116VR1** and **116VR2** of the cartridge cover member **116** shown in FIG. **7** contacting with the straight portions **171VR1** and **171VR2**.

Further, the tray **171** shown in FIG. **7** is provided with a rotation-determining projection **171KR**. The attitude of the process cartridge **100** is determined relative to the apparatus main assembly by fitting it with the rotation determining recess **116KR** of the cartridge cover member **116** shown in FIG. **7**.

The positioning portion **171VL** and the rotation determining projection **171KL** are arranged at positions (non-driving side) so as to oppose each other across the intermediary transfer belt **12a** in the longitudinal direction of the positioning portion **171VR** and the process cartridge **100**. That is, on the non-driving side as well, the position of the process cartridge is determined by engagement of the arc portions **117VL1** and **117VL2** of the cartridge cover member **117** with the positioning portion **171VL** and engagement of the rotation determining recess **117KL** with the rotation determining projection **171KL**.

By doing so, the position of the process cartridge **100** relative to the tray **171** is correctly determined.

Then, as shown in FIG. **5**, the process cartridge **100** integrated with the tray **171** is moved in the direction of the arrow **X1** and inserted to the position shown in FIG. **5**.

Then, by closing the front door **11** in the direction of the arrow **R**, the process carriage **100** is pressed by a cartridge pressing mechanism (not shown) which will be described hereinafter, and is fixed to the image forming apparatus main

13

assembly 170 together with the tray 171. Further, the transfer belt 12a comes into contact with the photosensitive member 104 in interrelation with the operation of the cartridge pressing mechanism. In this state, an image formation is enabled (FIG. 2).

In this embodiment, the positioning portion 171VR and the positioning portion 171V also serve as reinforcements for maintaining the rigidity in the pull-out operation of the tray 171, and for this reason, the use is made with metal sheet, but the present invention is not limited to this.

[Cartridge Pressing Mechanism]

Next, referring to FIG. 8, the details of the cartridge pressing mechanism will be described.

Part (a) of FIG. 8 shows only the process cartridge 100, the tray 171, the cartridge pressing mechanisms 190 and 191 and the intermediary transfer unit 12 in the state of FIG. 4. Part (b) of FIG. 8 shows only the process cartridge 100, the tray 171, the cartridge pressing mechanisms and 191 and the intermediary transfer unit 12 in the state of FIG. 2.

The process cartridge 100 receives a driving force during image formation, and further receives a reaction force from the primary transfer roller 12d (FIG. 2) in the direction of arrow Z1. Therefore, it is necessary to press the process cartridge in the Z2 direction in order to maintain a stable attitude without the process cartridge spacing from the positioning portions 171VR and 171VL during the image forming operation.

In order to achieve these, in this embodiment, the image forming apparatus main assembly 170 is provided with cartridge pressing mechanisms (190, 191).

As for the cartridge pressing mechanism (190, 191), the storing element pressing unit 190 works for the non-driving side, and the cartridge pressing unit 191 works for the driving side. This will be described in more detail below.

By closing the front door 11 shown in FIG. 4, the storing element pressing unit 190 and the cartridge pressing unit 191 shown in FIG. 8 lowers in the direction of arrow Z2.

The storing element pressing unit 190 is provided with a main assembly side electric contact (not shown) which mainly contacts with the electric contact of the storing element (not shown) provided in the process cartridge 100. By interlocking with the front door 11 by a link mechanism (not shown), the storing element 140 and the electric contact on the main assembly side can be brought into and out of contact with each other.

That is, the contacts are brought into contact with each other by closing the front door 11, and the contacts are separated by opening the front door 11.

By such a structure, when the process cartridge 100 moves inside the image forming apparatus main assembly together with the tray 171, the electric contacts are not rubbed and the contacts are retracted from the insertion/removal locus of the process cartridge 100, whereby insertion and removal operations of the tray 171 are not hindered.

The storing element pressing unit 190 also functions to press the process cartridge against the positioning portion 171VR described above.

Further, similarly to the storing element pressing unit 190, the cartridge pressing unit 121 also lowers in the direction of arrow Z2 in interrelation with the operation of closing the front door 11 and functions to press the process cartridge 100 against the above-mentioned positioning portion 171VL.

Further, although the details will be described hereinafter, the cartridge pressing mechanism (190, 191) also functions to press down the force applying members 152L and 152R of the process cartridge 100 as will be described hereinafter.

14

[Drive Transmission Mechanism]

Next, referring to FIGS. 9 and 10 (for better illustration, the tray 171 is omitted), the drive transmission mechanism of the main assembly in this embodiment will be described.

Part (a) of FIG. 9 is a perspective view in which the process cartridge 100 and the tray 171 are omitted in the state of FIG. 4 or FIG. 5. FIG. 9B is a perspective view in which the process cartridge 100, the front door 11 and the tray 171 are omitted.

FIG. 10 is a side view of the process cartridge 100 as viewed from the driving side.

As shown in FIG. 10, the process cartridge in this embodiment includes a development coupling portion 32a and a drum coupling (photosensitive member coupling) 143.

The structure is such that by closing the front door 11 (state of part (b) of FIG. 9, the main assembly side drum drive coupling and the main assembly side development drive coupling 185 which drive and transmit the driving forces to the process cartridge 100 are projected in the arrow Y1 direction by a link mechanism (not shown).

Further, by opening the front door 11 (state of part (a) of FIG. 9, the drum drive coupling 180 and the development drive coupling 185 are retracted in the direction of arrow Y2.

By retracting each coupling from the insertion/removal locus of the process cartridge (X1 direction, X2 direction), the insertion/removal of the tray 171 is not hindered.

By closing the front door 11 and starting the driving of the image forming apparatus main assembly, the drum drive coupling 180 described above engages with the drum coupling (coupling member, cartridge side coupling) 143. Along with this, the development drive coupling 185 on the main assembly side engages with the development coupling portion 32a. As a result, the drive is transmitted to the process cartridge 100. The drive transmission to the process cartridge 100 is not limited to the structure described above, and a mechanism which inputs the drive only to the drum coupling and transmits the drive to the developing roller may be provided.

[Intermediary Transfer Unit Structure]

Next, referring to FIG. 9, the intermediary transfer unit 12 of the image forming apparatus main assembly in this embodiment will be described.

In this embodiment, the structure is such that the intermediary transfer unit 12 is raised in the direction of arrow R2 by a link mechanism (not shown) by closing the front door 11, and moves to the position for the image forming operation (photosensitive drum 104 and intermediary transfer belt 12a are in contact with each other).

Further, by opening the front door 11, the intermediary transfer unit 12 lowers in the direction of arrow R1, and the photosensitive drum 2 and the intermediary transfer belt 12a are separated from each other.

That is, in a state in which the process cartridge 100 is set in the tray 171, the photosensitive drum 104 and the intermediary transfer belt 12a come into and out of contact with each other depending on the opening/closing operation of the front door 11.

The structure is such that in the contact/separation operation, the intermediary transfer unit rises and falls while drawing a rotation locus about the center point PV1 shown in FIG. 4.

The intermediary transfer belt 12a is driven by receiving a force from a gear (not shown) provided coaxially with the PVI. Therefore, by setting the above-mentioned position PV1 as the rotation center, the intermediary transfer unit 12 can be raised and lowered without moving the gear center.

15

By doing so, it is not necessary to move the center of the gear, and the position of the gear can be maintained with high accuracy.

With the above-described structure, in the state that the process cartridge **100** is set in the tray **171**, when the tray **11** is inserted or removed, the photosensitive drum **104** and the intermediary transfer belt **12a** do not rub relative to each of, and therefore, damage of the photosensitive drum **104** and deterioration of the image by charge memory are prevented. [Development Separation Control Unit]

Next, referring to FIGS. **8**, **11** and **12**, the separation mechanism of the image forming apparatus main assembly in this embodiment will be described.

FIG. **11** is a sectional view of the image forming apparatus **M** taken along the driving side end of the process cartridge **100**. FIG. **12** is a perspective view of the development separation control unit as viewed obliquely from above.

In this embodiment, the development separation control unit **195** controls the separation contact operation of the developing unit **109** relative to the photosensitive drum **104** by engaging with a portion of the developing unit **109**. The development separation control unit **195** is disposed in a lower portion the image forming apparatus main assembly **170** as shown in FIG. **8**.

Specifically, the development separation control unit **195** is placed below the development input coupling portion **32a** and the drum coupling **143** in the vertical direction (downward in the arrow **Z2** direction).

Further, the development separation control unit **195** is placed in the longitudinal direction (**Y1**, **Y2** direction) of the photosensitive drum **104** of the intermediary transfer belt **12**. That is, the development separation control unit **195** includes a development separation control unit **195R** on the driving side and a development separation control unit **195L** on the non-driving side.

By disposing the development separation control unit **195** in the dead space of the image forming apparatus main assembly **170** as described above, the main assembly can be downsized.

The development separation control unit **195R** has four separation control members **196R** corresponding to the process cartridges **100** (**100Y**, **100M**, **100C**, **100K**), respectively. The four separation control members have substantially the same shape. The development separation control unit **195R** is always fixed to the image forming apparatus main assembly. However, the separation control member **196R** is structured to be movable in the **W41** and **W42** directions by a control mechanism (not shown). The detailed structure will be described hereinafter.

The development separation control unit **195L** has four separation control members **196L** corresponding to the process cartridge **100** (**100Y**, **100M**, **100C**, **100K**). The four separation control members have substantially the same shape. The development separation control unit **195L** is always fixed to the image forming apparatus main assembly. However, the separation control member **196L** is structured to be movable in the **W41** and **W42** directions by a control mechanism (not shown). The detailed structure will be described hereinafter.

Further, in order for the development separation control unit **195** to engage with a portion of the developing unit **109** and control the separation contact operation of the developing unit **109**, a portion of the development control unit **196** and a portion of the developing unit are required to overlap in the vertical direction (**Z1**, **Z2** direction).

Therefore, for the overlapping in the vertical direction (**Z1** and **Z2** directions) as described above after the developing

16

unit **109** of the process cartridge **100** is inserted in the **X1** direction, a part of the developing unit (in the case of this embodiment, the force applying member **152**) is required to project. Details will be described hereinafter.

In the case that the development separation control unit **195** itself is raised in the same manner as in the case of the intermediary transfer unit **12** for the engagement, there are problems such as an increase in the operating force of the interlocked front door **11** and complication of the drive train.

In this embodiment, a method is employed in which the development separation control unit **195** is fixed to the image forming apparatus main assembly **170**, and a part of the developing unit **109** (force applying member **152**) is projected downward (**Z2**) in the image forming apparatus main assembly **170**, and one of the reasons for this arrangement is to address this problem. Further, the mechanism for projecting the force applying member **152** utilized the mechanisms of the storing element pressing unit **190** and the cartridge pressing unit described above, and therefore, there is no above-described problem and an increase in the cost of the device main assembly can be suppressed.

The entire unit of the development separation control unit **195** is fixed to the image forming apparatus main assembly **170**. However, as will be described hereinafter, a part of the developing unit is movable in order to engage with the force applying member **152** to cause an operation so that the developing unit **109** is in a separated state and a contacted state relative to the photosensitive drum **104**. Details will be described hereinafter.

[Overall Structure of Process Cartridge]

Referring to FIGS. **3**, **13** and **14**, the structure of the process cartridge will be described.

FIG. **13** is an assembly perspective view of the process cartridge **100** as viewed from the driving side, which is one side in the axial direction of the photosensitive drum **104**. FIG. **14** is a perspective view of the process cartridge **100** as viewed from the driving side.

In this embodiment, the first to fourth process cartridges **100** (**100Y**, **100M**, **100C**, **100K**) have the same electrophotographic process mechanism, but the color of the contained toner and the filling amount of the toner are different from each other.

The process cartridge **100** includes a photosensitive drum **104** (**4Y**, **4M**, **4C**, **4K**) and process means which act on the photosensitive drum **104**. The cartridge **100** includes a charging roller **105** as a process means, which is a charging means (charging member) for charging the photosensitive drum **104**. Further, the cartridge **100** includes a developing roller **106** which is a developing means (developing member) for developing the latent image formed on the photosensitive drum **104** as another process means.

In addition, as an example of the process means, there is a cleaning means, (for example, a cleaning blade or the like) for removing residual toner remaining on the surface of the photosensitive drum **104** can be considered. However, the image forming apparatus of this embodiment employs a structure in which the cleaning means contacting the photosensitive drum **104** is not provided.

The process cartridge **100** is divided into a drum holding unit **108** (**108Y**, **108M**, **108C**, **108K**) and a developing unit **109** (**109Y**, **109M**, **109C**, **109K**).

[Drum Holding Unit Structure]

As shown in FIGS. **3** and **13**, the drum holding unit **108** comprises a photosensitive drum **104**, a charging roller **105**, and a drum frame **115** which is a first frame, and so on. The photosensitive drum **104** unified together with the coupling

17

143 and the drum flange **142** to provide the drum unit **103** (see part (a) of FIG. 1, the details will be described hereinafter).

The drum unit **103** is rotatably supported by a driving side cartridge cover member **116** and a non-driving side cartridge cover member **117** provided at the opposite ends in the longitudinal direction of the process cartridge **100**.

The driving side cartridge cover member **116** and the non-driving side cartridge cover member **117** will be described hereinafter.

Further, as shown in FIGS. **13** and **14**, a drum coupling **143** for transmitting a driving force to the photosensitive drum **104** is provided in the neighborhood of one end in the longitudinal direction of the photosensitive drum **104**. As described above, the coupling **143** engages with the main assembly side drum drive coupling **180** (see FIG. **9**) as the drum drive output unit of the image forming apparatus main assembly **170**. The driving force of the driving motor (not shown) of the image forming apparatus main assembly **170** is transmitted to the photosensitive drum **104** to rotate it in the direction of arrow A. Further, the photosensitive drum **104** is provided with a drum flange **142** in the neighborhood of the other end (second end portion) in the longitudinal direction.

The shaft portion **143j** (see FIG. **1**) of the coupling **143** is supported by the driving side cartridge cover **116**, and the drum flange **142** is supported by the shaft fixed to the non-driving side cartridge cover **117**. By this, the drum unit **103** is rotatably supported in the cartridge. That is, the ends of the photosensitive drum **104** are rotatably supported by the ends of the casing of the cartridge (that is, the cartridge covers **116** and **117**) by way of the coupling **143** and the drum flange **142**.

The charging roller **105** is supported by the drum frame **115** in contact with the photosensitive drum **104** so that it can be rotationally driven by the photosensitive drum **104**.

Of the opposite sides of the drum unit **103** in the longitudinal direction (axial direction), the side on which the coupling **143** is provided is the driving side, and the side on which the drum flange **142** is placed is the non-driving side. That is, of the opposite ends of the photosensitive drum **104** in the axial direction, the coupling **143** is fixed in the neighborhood of the end on the driving side, and the drum flange **142** is fixed in the neighborhood of the end on the opposite side to the driving side. Of opposite ends of the photosensitive drum **104**, one may be referred to as a first end and the other may be referred to as a second end. FIG. **80** shows the end portion **104a** on the drum driving side and the end portion **104b** on the non-driving side of the photosensitive drum.

Similarly to the drum unit **103**, of the opposite sides of the cartridge **100**, the side on which the coupling **143** is placed is referred to as the driving side, and the side opposite to the driving side is referred to as the non-driving side. For example, FIGS. **10** and **19** are illustrations showing the driving side of the cartridge. Further, FIG. **16** is an illustration showing the non-driving side of the cartridge.

As shown in FIGS. **13** and **14**, the driving side cartridge cover **116** is a component provided at the driving side end of the casing of the cartridge **100**, and the non-driving side cartridge cover is a component provided at the non-driving side end of the casing. The drum coupling **143** supported by the driving side cartridge cover **116** can be considered to be located in the neighborhood of the non-driving side end of the casing of the cartridge **100**. Of the opposite ends of the cartridge **100**, one may be referred to as a first end and the other may be referred to as a second end.

18

[Development Unit Structure]

As shown in FIGS. **3** and **13**, the developing unit **109** includes a developing roller **106**, a toner feeding roller (toner supply roller) **107**, a developing blade **130**, a developing unit frame **125**, and the like. The developing unit frame **125** comprises a lower frame **125a** and a lid member **125b**. The lower frame **125a** and the lid member **125b** are connected by ultrasonic welding or the like.

The development frame **125**, which is the second frame (second casing), includes a toner accommodating portion **129** for accommodating toner to be supplied to the developing roller **106**. Further, the development frame **125** rotatably supports the developing roller **106** and the toner feeding roller **107** by way of the driving side bearing **126** and the non-driving side bearing **127**, which will be described hereinafter, and holds the developing blade **130** for regulating a layer thickness of the toner on the peripheral surface of the developing roller **106**.

The developing blade **130** is formed by mounting an elastic member **130b**, which is a sheet-like metal having a thickness of about 0.1 mm, on a support member **130a**, which is a metal material having an L-shaped cross-section, by welding or the like. The developing blade **130** is mounted to the development frame **125** with fixing screws **130c** at two locations, one in the neighborhood of one end and the other in the neighborhood of the other end in the longitudinal direction. The developing roller **106** comprises a core metal **106c** and a rubber portion **106d**.

The developing roller **106** is rotatably supported by a driving side bearing **126** and a non-driving side bearing **127** mounted to the opposite ends in the longitudinal direction of the development frame **125**, respectively. The development frame **125**, the driving side bearing **126**, and the non-driving side bearing **127** are a part of the frame (casing) of the cartridge. In a broad sense, the bearings **126** and **127** may be regarded as a part of the development frame **125**, and the bearings **126** and **127** and the development frame **125** may be collectively referred to as a development frame.

The toner feeding roller **107** conveys and supplies the toner contained in the toner accommodating portion **129** toward the developing roller **106** to develop the latent image on the photosensitive drum **104**. The toner feeding roller **107** is in contact with the developing roller **106**.

Further, as shown in FIGS. **13** and **14**, a development input coupling portion (development coupling) **32a** for transmitting a driving force to the developing unit **109** is provided on one side of the developing unit **109** in the longitudinal direction. The development input coupling portion **32a** engages with the development drive coupling **185** (see FIG. **9**) on the main assembly side as the development drive output portion of the image forming apparatus main assembly **170**, and the driving force of the drive motor (not shown) of the image forming apparatus main assembly **170** is input to the developing unit **109**.

The driving force inputted to the developing unit **109** is transmitted by a driving train (not shown) provided in the developing unit **109**, so that the developing roller **106** can be rotated in the direction of arrow D in FIG. **3**. Similarly, the driving force received by the development input coupling portion **32a** also rotates the toner feeding roller **107** to supply toner to the developing roller **106**.

On one side of the developing unit **109** in the longitudinal direction, a development cover member **128** which supports and covers a developing input coupling portion **32a** and a drive train (not shown) is provided. The outer diameter of the developing roller **106** is selected to be smaller than the outer diameter of the photosensitive drum **104**. The outer

diameter of the photosensitive drum **104** of this embodiment is selected to be in the range of $\Phi 18$ to $\Phi 22$ (mm), and the outer diameter of the developing roller **106** is selected to be in the range of $\Phi 8$ to $\Phi 14$. By the selections of such outer diameters, efficient arrangement is possible.

[Assembling of Drum Holding Unit and Developing Unit]

Referring to Figure, the assembly of the drum holding unit **108** and the developing unit **109** will be described. The drum holding unit **108** and the developing unit **109** are connected by a driving side cartridge cover member **116** and a non-driving side cartridge cover member **117** provided at respective ends in the longitudinal direction of the process cartridge **100**.

The driving side cartridge cover member **116** provided on one side (driving side) of the process cartridge **100** in the longitudinal direction is provided with a developing unit support hole **116a** for supporting the developing unit so as to be swingable (movable). Similarly, the non-driving side cartridge cover member **117** provided on the other side (non-driving side) of the process cartridge **100** in the longitudinal direction is provided with a developing unit support hole **117a** for swingably supporting the developing unit **109**.

Further, the driving side cartridge cover member **116** and the non-driving side cartridge cover member **117** are provided with drum support holes **116b** and **117b** for rotatably supporting the photosensitive drum **104**. Here, on the driving side, the outer diameter portion of the cylindrical portion **128b** of the development cover member **128** is fitted into the developing unit support hole **116a** of the driving side cartridge cover member **116**. On the non-driving side, the outer diameter portion of the cylindrical portion (not shown) of the non-driving side bearing **127** is fitted into the developing unit support hole **117a** of the non-moving side cartridge cover member **117**.

Further, the opposite ends of the photosensitive drum **104** in the longitudinal direction are fitted into the drum support holes **116b** of the driving side cartridge cover member **116** and the drum support holes **117b** of the non-driving side cartridge cover member **117**, respectively. Then, the driving side cartridge cover member **116** and the non-driving side cartridge cover member are fixed to the drum frame **115** of the drum holding unit **108** with screws or adhesives (not shown). By this, the developing unit **109** is rotatably supported by the driving side cartridge cover member **116** and the non-driving side cartridge cover member **117**. The developing unit **109** can be moved (rotated) relative to the drum holding unit **108**, and the developing roller **106** can be moved with respect to the photosensitive drum by this movement. At the time of image formation, the developing roller **106** can be placed at the position acting on the photosensitive drum **104**.

The drum frame **115** and the cover members **116** and **117** are a part of the cartridge frame (casing). More specifically, they are frames of the drum holding unit **108**. Further, since the cover members **116** and **117** are fixed to one end and the other end of the drum frame **115**, respectively, the cover members **116** and **117** may be regarded as a part of the drum frame **115**. Or, the cover members **116** and **117** and the drum frame **115** may be collectively referred to as a drum frame.

Further, one of the frame (**115**, **116**, **117**) of the drum holding unit **108** and the frame (**125**, **126**, **127**) of the developing unit may be called a first frame (first casing), and the other may be called a second frame (second casing) or the like. Further, the frame (**115**, **116**, **117**) of the drum holding unit **108** and the frame (**125**, **126**, **127**) of the

developing unit may be collectively referred to as a frame of the cartridge (casing of the cartridge), without particular distinction between them.

FIG. **14** shows a state in which the drum holding unit **108** and the developing unit **109** are assembled by the above-described steps to provide an integral process cartridge **100**.

The axis connecting the center of the developing unit support hole **116a** of the driving side cartridge cover member **116** and the center of the developing unit support hole **117a** of the non-moving side cartridge cover member **117** is referred to as a swing axis K. Here, the cylindrical portion **128b** of the development cover member **128** on the driving side is coaxial with the development input coupling **74**. That is, the developing unit **109** has a structure in which a driving force is transmitted from the image forming apparatus main assembly **170** on the swing axis K. Further, the developing unit **109** is rotatably supported about the swing axis K.

[Structure of Separation/Contact Mechanism]

The structure in which the photosensitive drum **104** of the process cartridge **100** and the developing roller **106** of the developing unit **109** are separated from and contacted with each other in this embodiment will be described in detail. The process cartridge includes a separation contact mechanism **150R** on the driving side and a separation contact mechanism **150L** on the non-driving side. FIG. **15** shows an assembly perspective view of the driving side of the developing unit **109** including the separation contact mechanism **150R**. FIG. **16** shows an assembly perspective view of the developing unit including the separation contact mechanism **150L** on the non-driving side. Regarding the separation contact mechanism, the details of the separation contact mechanism **150R** on the driving side will first be described, and then the separation contact mechanism **150L** on the non-driving side will be described.

Since the separation contact mechanisms on the driving side and the non-driving side have almost the same functions, the same reference numerals are used for both sides with the exception that R is added at the end for the driving side, and L is added for the non-driving side.

The separation contact mechanism **150R** includes a separation holding member **151R** which is a restriction member, a force applying member **152R** which is a pressing member, and a tension spring **153**.

The separation contact mechanism **150L** includes a separation holding member **151L** which is a restriction member, a force applying member **152L** which is a pressing member, and a tension spring **153**.

[Detailed Description of Separation Holding Member R]

Referring to FIG. **17**, the separation holding member **151R** will be described in detail.

Part (a) of FIG. **17** is a front view of the separation holding member **151R** per se of the process cartridge **100** as viewed from the driving side longitudinal direction. Parts (b) and (c) of FIG. **17** are perspective views of the separation holding member **151R** per se. Part (d) of FIG. **17** is a view of the separation holding member **151R** as viewed in the direction of arrow Z2 in part (a) of FIG. **17** (vertically upward in the image forming state). The separation holding member **151R** includes an annular support receiving portion **151Ra**, and includes a separation holding portion **151Rb** projecting from the support receiving portion **151Ra** in the radial direction of the support receiving portion **151Ra**. The free end of the separation holding portion **151Rb** has a separation holding surface **151Rc** having an arc shape having a center on the separation holding member swing axis H and inclined by an angle $\theta 1$ with respect to the line

21

HA parallel to the separation holding member swing axis H. The angle $\theta 1$ is selected so as to satisfy the equation (1).

$$0 \leq \theta 1 \leq 45^\circ \quad (1)$$

Further, the separation holding member **151R** has a second restricted surface **151Rk** adjacent to the separation holding surface **151Rc**. Further, the separation holding member **151R** is provided with a second pressed portion **151Rd** projecting in the **Z2** beyond the support receiving portion **151Ra**, and an arc-shaped second pressed surface **151Re** projecting from the second pressed portion **151Rd** in the direction of the separation holding member swing axis H of the support receiving portion **151Ra**.

Furthermore, the separation holding member **151R** includes a main body portion **151Rf** connected to the support receiving portion **151Ra**, and the main body portion **151Rf** is provided with a spring hooked portion **151Rg** projecting in the direction of the separation holding member swing axis H of the support receiving portion **151Ra**. Further, the main body portion **151Rf** is provided with a rotation (on its own axis) prevention portion **151Rm** projecting in the **Z2** direction, and the rotation prevention surface **151Rn** is provided in a direction facing the second pressed surface **151Re**.

[Detailed Description of Force Applying Member R]

Referring to FIG. **18**, the force applying member **152R** will be described in detail.

Part (a) of FIG. **18** is a front view of the force applying member **152R** per se as viewed from the longitudinal direction of the process cartridge **100**, and FIGS. **18B** and **18C** are perspective views of the force applying member **152R** per se.

The force applying member **152R** is provided with an oblong-shaped oblong support receiving portion **152Ra**. Here, the longitudinal direction of the oblong shape of the oblong support receiving portion **152Ra** is indicated by an arrow LH, the upward direction is indicated by an arrow LH1, and the downward direction is indicated by an arrow LH2. Further, the direction in which the oblong support receiving portion **152Ra** is formed is indicated by as HB. The force applying member **152R** has a projecting portion **152Rh** formed on the downstream side in the arrow LH2 direction of the oblong support receiving portion **152Ra**. The oblong support receiving portion **152Ra** and the projecting portion **152Rh** are connected by a main body portion **152Rb**. On the other hand, the force applying member **152R** includes a pressed portion **152Re** projecting in the arrow LH1 direction and substantially perpendicular to the arrow LH1 direction, and has an arc-shaped pressed surface **152Rf** on the downstream side in the arrow LH1 direction and has a pushing restriction surface **152Rg** on the upstream side. Further, the force applying member **152R** has a first at-accommodation restriction surface **152Rv** extending from the main body portion **152Rb** on the upstream side in the arrow LH2 direction, and a second at-accommodation restricting surface **152Rw** which is adjacent to the first at-accommodation restriction surface **152Rv** and which is substantially parallel to the first pressing surface **152Rq**.

The projecting portion **152Rh** includes a first force receiving portion **152Rk** and a second force receiving portion **152Rn** which are arranged so as to be opposite from each other in a direction substantially perpendicular to the arrow LH2 direction at an end portion in the arrow LH2 direction. The first force receiving portion **152Rk** and the second force receiving portion **152Rn** have a first force receiving surface **152Rm** and a second force receiving surface **152Rp** extending in the HB direction and having arc shapes, respectively.

22

Further, the projecting portion **152Rh** has a spring hooked portion **152Rs** projecting in the HL direction and a locking portion **152Rt**, and the locking portion **152Rt** has a locking surface **152Ru** facing in the same direction as the first force receiving surface **152Rp**.

Further, the force applying member **152R** is a part of the main body portion **152Rb**, and is arranged on the upstream side of the second force receiving portion **152Rn** in the arrow LH2 direction, and has a first pressing surface **152Rq** facing in the same direction as the second force receiving surface **152Rp**. Further, the force applying member **152R** has a second pressing surface **152Rr** which is perpendicular to the first at-accommodation restriction surface **152Rv** and which is opposite from the first pressing surface **152Rq**.

When the process cartridge **100** is mounted on the image forming apparatus main assembly **170**, the LH1 direction is substantially the same as the Z1 direction, and the LH2 direction is substantially the same as the Z2 direction. Further, the HB direction is substantially the same as the longitudinal direction of the process cartridge **100**.

[Assembling of Separation/Contact Mechanism R]

Next, referring to FIGS. **10** and **15** to **19**, the assembly of the separation contact mechanism will be described. FIG. **19** is a perspective view of the process cartridge **100** after being assembled with the separation holding member **151R**, as viewed from the driving side.

As shown in FIG. **15** described above, in the developing unit **109**, the outer diameter portion of the cylindrical portion **128b** of the development cover member **128** is fitted into the developing unit support hole portion **116a** of the driving side cartridge cover member **116**. By this, the developing unit **109** is rotatably supported relative to the photosensitive drum **104** about the swing axis K. Further, the development cover member **128** includes a cylindrical first support portion **128c** and a second support portion **128k** projecting in the direction of the swing axis K.

The outer diameter of the first support portion **128c** fits with the inner diameter of the support receiving portion **151Ra** of the separation holding member **151R**, to rotatably support the separation holding member **151R**. Here, the swing center of the separation holding member **151R** assembled to the development cover member **128** is the separation holding member swing axis H. The development cover member includes a first retaining portion **128d** which projects in the direction of the separation holding member swing axis H. As shown in FIG. **15**, the movement of the separation holding member **151R** assembled to the development cover member **128** in the swing axis H direction is restricted by abutment of the first retaining portion **128d** to the separation holding member **151R**.

Further, the outer diameter of the second support portion **128k** fits with the inner wall of the oblong support receiving portion **152Ra** of the force applying member **152R**, to support the force applying member **152R** so as to be rotatable and movable in the oblong direction. Here, the swing center of the force applying member **152R** assembled to the development cover member **128** is a force applying member swing axis HC. As shown in FIG. **15**, the movement of the force applying member **152R** assembled to the development cover member **128** in the swing axis HC direction is restricted by abutment of the second retaining portion **128m** to the separation holding member **151R**.

FIG. **10** is a sectional view taken along a line CS with a part of the driving side cartridge cover member **116** and a part of the development cover member **128** omitted such that the fitting portion between the oblong support receiving portion **151Ra** of the force applying member **152R** and the

cylindrical portion **128b** of the development cover member **128** can be seen. The separation contact mechanism **150R** is provided with a tension spring **153**, as an urging means, for urging the separation holding member **151R** to rotate in the direction of arrow **B1** in the drawing about the separation holding member swing axis **H** and for urging the force applying member **152R** in the direction of arrow **B3**.

The arrow **B3** direction is a direction substantially parallel to the oblong direction **LH2** (see FIG. **18**) of the oblong support receiving portion **152Ra** of the force applying member **152R**. The tension spring **153** is assembled between the spring hooked portion **151Rg** provided on the separation holding member **151R** and the spring hooked portion **152Rs** provided on the force applying member **152R**. The tension spring **153** applies a force to the spring hooked portion **151Rg** of the separation holding member **151R** in the direction of arrow **F2** in FIG. **10** to apply an urging force for rotating the separation holding member **151R** in the direction of arrow **B1**. Further, the tension spring **153** applies a force to the spring hooked portion **152Rs** of the force applying member **152R** in the direction of the arrow **F1** to apply an urging force for moving the force applying member **152R** in the direction of the arrow **B3**.

The line connecting the spring hooked portion **151Rg** of the separation holding member **151R** and the spring hooked portion **152Rs** of the force holding member **152R** is **GS**. The line connecting the spring hooked portion **152Rs** of the force applying member **152R** and the force applying member swing axis **HC** is **HS**. Here, an angle $\theta 2$ formed by the line **GS** and the line **HS** is selected to satisfy the following equation (2) with the clockwise direction about the spring hooked portion **152Rs** of the force applying member **152R** being positive. By this, the force applying member **152R** is urged to rotate in the direction of arrow **BA** about the force applying member swing axis **HC**.

$$0^\circ \leq \theta 2 \leq 90^\circ \quad (2)$$

As shown in FIG. **15**, in the development drive input gear **132**, the inner diameter portion of the cylindrical portion **128b** of the development cover member **128** and the outer diameter portion of the cylindrical portion **32b** of the development drive input gear **132** are fitted, and in addition, the support portion **126a** of the driving side bearing **126** is fitted and the cylindrical portion (not shown) of the development drive input gear are fitted. By this, the driving force can be transmitted to the developing roller gear **131**, the toner feeding roller gear **133**, and other gears.

In this embodiment, the mounting positions of the separation holding member **151R** and the force applying member **152R** are as follows. As shown in FIG. **15**, in the direction of the swing axis **K**, the separation holding member **151R** is disposed on the side (outside in the longitudinal direction) where the driving side cartridge cover member **116** is provided, with the development cover member **128** interposed therebetween. The force applying member **152R** is disposed on the side (inside in the longitudinal direction) where the development drive input gear **13** is arranged. However, the position thereof is not limited to this, and the positions of the separation holding member **151R** and the force applying member **152R** may be interchanged, and the separation holding member **151R** and the force applying member **152R** may be disposed in one side in the swing axis **K** direction with respect to the development cover member **128**. Further, the arrangement order of the separation holding member **151R** and the force applying member **152R** may be exchanged.

The development cover member **128** is fixed to the development frame **125** by way of the driving side bearing **126** to form the developing unit **109**. As shown in FIG. **15**, the fixing method in this embodiment uses a fixing screw **145** and an adhesive (not shown), but the fixing method is not limited to this example, and welding such as welding by heating or pouring and hardening of resin material, for example, may be used.

Here, FIG. **20** is a sectional view in which the periphery of the separation holding portion **151R** in FIG. **10** is enlarged and a part of the tension spring **153** and the separation holding member **151R** is partially omitted by the partial sectional line **CS4** for the sake of illustration. In the force applying member **152R**, the first restriction surface **152Rv** of the force applying member **152R** comes into contact with the first restriction surface **128h** of the development cover member **128** by the urging force of the tension spring **153** in the **F1** direction in the drawing, as described above. Further, the second restriction surface **152Rw** of the force applying member **152R** comes into contact with the second restriction surface **128g** of the development cover member **128** and is positioned thereby. This position is referred to as an accommodation position (reference position) of the force applying member **152R**. Further, the separation holding member **151R** is rotated in the **B1** direction about the swing axis **H** of the separation holding member by the urging force of the tension spring **153** in the **F2** direction, and the second pressed portion **151Rd** of the separation holding member **151R** comes into contact with the second pressing surface **152Rr** of the force applying member **152R**, by which the rotation is stopped. This position is referred to as a separation holding position (restriction position) of the separation holding member **151R**.

Further, FIG. **21** is an illustration in which the periphery of the separation holding portion **151R** in FIG. **10** is enlarged, and the tension spring **153** is omitted, for the sake of illustration. Here, the case is considered in which the process cartridge **100** including the separation contact mechanism **150R** according to this embodiment is dropped in the **JA** direction of FIG. **21** when the process cartridge **100** is transported. At this time, the separation holding member **151R** receives a force of rotating in the direction of arrow **B2** by its own weight about the separation holding swing axis **H**. For this reason, when the rotation in the **B2** direction occurs starts, the rotation prevention surface **151Rn** of the separation holding member **151R** comes into contact with the locking surface **152Ru** of the force applying member **152R**, and the separation holding member **151R** receives the force in the **F3** direction in the drawing so as to suppress the rotation in the **B2** direction. By this, it is possible to prevent the separation holding member **151R** from rotating in the **B2** direction during transportation, and it is possible to prevent the state of separation between the photosensitive drum **104** and the developing unit **109** from being impaired.

In this embodiment, the tension spring **153** is mentioned as an urging means for urging the separation holding member **151R** to the separation holding position and for urging the force applying member **152R** to the accommodating position, but the urging means is not limited to this example. For example, a torsion coil spring, a leaf spring, or the like may be used as an urging means to urge the force applying member **152R** to the accommodating position and to urge the separation holding member **151R** to the separation holding position. Further, the material of the urging means may be metal, a mold, or the like, which has elasticity and can urge the separation holding member **151R** and the force applying member **152R**.

25

As described above, the developing unit 109 provided with the separation contact mechanism 150R is integrally coupled with the drum holding unit 108 by the driving side cartridge cover member 116 as described above (state in FIG. 19).

FIG. 22 is a view as seen in the direction of arrow J in part (a) of FIG. 19s shown in FIG. 15, the driving side cartridge cover 116 of this embodiment has a contact surface 116c. As shown in FIG. 22, the contact surface 116c is slanted with an inclination of an angle $\theta 3$ relative to the swing axis K. It is desirable that the angle $\theta 3$ is the same as the angle $\theta 1$ forming the separation holding surface 151Rc of the separation holding member 151R, but the angle $\theta 3$ is not limited to this example. Further, as shown in FIGS. 15 and 19 when the driving side cartridge cover member 116 is assembled to the developing unit 109 and the drum holding unit 108, the contact surface 116c faces the separation holding surface 151Rc of the separation holding member 151R placed at a separation holding position. The contact surface 116c contacts the separation holding surface 151Rc by the urging force of the development pressure spring 134 which will be described hereinafter. The structure is such that when the engaging surface 116Rc and the separation holding surface 151Rc contact each other, the attitude of the developing unit 109 is positioned so that the developing roller 106 of the developing unit 109 and the photosensitive drum 104 are separated by a gap P1. The state in which the developing roller 106 (developing member) is separated from the photosensitive drum 104 by the gap P1 by the separation holding member 151R is referred to as a separation position (retraction position) of the developing unit 109 (see part (a) of FIG. 42).

Here, referring to FIG. 42, the separated state and the contact state of the process cartridge 100 will be described in detail.

FIG. 42 is a side view of the process cartridge 100 as viewed from the driving side with the process cartridge 100 mounted inside the image forming apparatus main assembly 170. Part (a) of FIG. 42 shows a state in which the developing unit 109 is separated from the photosensitive drum 104. Part (b) of FIG. 42 shows a state in which the developing unit 109 is in contact with the photosensitive drum 104.

First, in a state where the separation holding member 151R is placed at the separation holding position and the developing unit 109 is located at the separation position, the pressed portion 152Re of the force applying member 152R is pushed in the ZA direction. By this, the projecting portion 152Rh of the force applying member 152R projects from the process cartridge 100. The second pressed surface 151Re of the separation holding member 151R is in contact with the second pressing surface 152Rr of the force applying member 152R by the tension spring 153 as described above. Therefore, when the second force receiving portion 152Rn is pressed in the direction of the arrow W42, the force applying member 152R rotates in the direction of the arrow BB about the force applying member swing axis HC to rotate the separation holding member 151R in the direction of the arrow B2. When the separation holding member 151R rotates in the direction of arrow B2, the separation holding surface 151Rc separates from the contact surface 116c, by which the developing unit 109 can rotate from the separation position in the direction of arrow V2 about the swing axis K. That is, the developing unit 109 rotates in the V2 direction from the separated position, and the developing roller 106 of the developing unit 109 comes into contact with the photosensitive drum 104. Here, the position of the developing unit

26

109 in which the developing roller 106 and the photosensitive drum 104 contact each other is referred to as a contact position (development position) (state of part (b) of FIG. 42). The position where the separation holding surface 151Rc of the separation holding member 151R is separated from the contact surface 116c is referred to as a separation permission position (permission position). When the developing unit 109 is located at the contact position, the second restriction surface 151Rk of the separation holding member 151R contacts the second restriction surface 116d of the driving side cartridge cover 116, so that the separation holding member 151R is maintained at the separation release position.

Further, the driving side bearing 126 has a first pressed surface 126c which is a surface perpendicular to the swing axis K. Since the driving side bearing 126 is fixed to the developing unit 109, the developing unit 109 presses the first force receiving portion 152Rk of the force applying member 152R in the direction of the arrow 41 in the state that the developing unit is in the contact position. Then, by the first pressing surface 152Rq being brought into contact with the first pressed surface 126c, the developing unit 109 rotates about the swing axis K in the direction of arrow V1 to move to a separated position (state shown in part (a) of FIG. 42). Here, the direction in which the first force receiving surface 126c moves when the developing unit 109 moves from the contact position to the separated position is shown by arrows W41 in part (a) of FIGS. 42 and 42(b). Further, the direction opposite to the arrow W41 is depicted by an arrow W42, and the arrow W41 direction and the arrow W42 direction are substantially horizontal (X1, X2 directions). The second force receiving surface 152Rp of the force applying member 152R assembled to the developing unit 109 as described above is on the upstream side of the first force receiving surface 126c of the driving side bearing 126 in the direction of the arrow W41. Further, the first force receiving surface 126c and the second force receiving surface 151Re of the separation holding member 151R are disposed at positions where they overlap at least partly in the W1 and W2 direction.

The detailed description of the operation of the separation contact mechanism 150R in the image forming apparatus main assembly 170 will be made below.

[Mounting of Process Cartridge to Image Forming Apparatus Main Assembly]

Next, referring to FIGS. 12, 23, and 24 the description will be made as to the engaging operation of 195 between the separation contact mechanism 150R of the process cartridge 100 and the development separation control unit of the image forming apparatus main assembly 170 when the process cartridge 100 is mounted to the image forming apparatus main assembly 170. For the sake of illustration, these Figures are sectional views in which a part of the development cover member 128 and a part of the driving side cartridge cover member 116 are omitted along the partial sectional lines CS1 and CS2, respectively.

FIG. 23 is a view as seen from the driving side of the process cartridge 100 when the process cartridge 100 is mounted on the cartridge tray 171 (not shown) of the image forming apparatus M and the cartridge tray 171 is inserted into the first mounting position. In this Figure, except for the process cartridge 100, the cartridge pressing unit 121, and the separation control member 196R are omitted.

As described above, the image forming apparatus main assembly 170 of this embodiment includes the separation control member 196R corresponding to each process cartridge 100 as described above. The separation control mem-

bers 196R are arranged on the lower side of the image forming apparatus main assembly 170 below the separation holding member 151R when the process cartridge 100 is placed at the first inner position and the second inner position. The separation control member 196R has a first force applying surface 196Ra and a second force applying surface 196Rb which project toward the process cartridge 100 and face each other across the space 196Rd. The first force applying surface 196Ra and the second force applying surface 196Rb are connected with each other by way of a connecting portion 196Rc in the lower side of the image forming apparatus main assembly 170. Further, the separation control member 196R is supported by the control sheet metal 197 rotatably about a rotation center 196Re. The separating member 196R is normally urged in an E1 direction by an urging spring. Further, the control sheet metal 197 is structured to be movable in the W41 and W42 directions by a control mechanism (not shown), so that the separation control member 196R is structured to be movable in the W41 and W42 directions.

As described above, in interrelation with the transition of the front door 11 of the image forming apparatus main assembly 170 from the open state to the closed state, the cartridge pressing unit 121 lowers in the direction of arrow ZA, and the first force applying portion 121a is brought into contact with the pressed surface 152Rf of the force applying member 152R. After that, when the cartridge pressing unit 121 is lowered to a predetermined position which is the second mounting position, the projecting portion 152Rh of the force applying member 152R projects downward in the Z2 direction of the process cartridge 100 (state in FIG. 24). This position is referred to as a projecting position of the force applying member 152R. When this operation is completed, as shown in FIG. 24, a gap T4 is formed between the first force applying surface 196Ra of the separation control member 196R and the first force receiving surface 152Rp of the force applying member 152R, and a gap T3 is formed between the second force applying surface 196Rb and the second force receiving surface 152Rp. Then, it is placed at the second mounting position where the separation control member 196R does not act on the force applying member 152R. This position of the separation control member 196R is referred to as a home position. The arrangement is such that at this time, the first force receiving surface 152Rp of the force applying member 152R and the first force applying surface 196Ra of the separation control member 196R are partly overlapped in the W1 and W2 direction. Similarly, the arrangement is such that the second force receiving surface 152Rp of the force applying member 152R and the second force applying surface 196Rb of the separation control member 196R are partly overlapped in the W1 and W2 direction.

[Contact Operation of Developing Unit]

Next, referring to FIGS. 24 to 26, the detailed description will be made as to the operation of contacting between the photosensitive drum 104 and the developing roller 106 by the separation contact mechanism 150R. For the sake of illustration, these Figures are sectional views of a part of the development cover member 128, a part of the driving side cartridge cover member 116, and a part of the driving side bearing 126, taken along lines CS1, CS2 and CS3, respectively.

In the structure of this embodiment, the development input coupling 32 receives a driving force from the image forming apparatus main assembly 170 in the direction of arrow V2 in FIG. 24, so that the developing roller 106 rotates. That is, the developing unit 109 including the

developing input coupling 32 receives torque in the arrow V2 direction about the swing axis K from the image forming apparatus main assembly 170. As shown in FIG. 24, when the developing unit 109 is in the separated position and the separation holding member 151R is in the separation holding position, the developing unit 109 receives this torque and an urging force by the development pressure spring 134 as will be described hereinafter. Even in this case, the separation holding surface 151Rc of the separation holding member 151R contacts the contact surface 116c of the driving side cartridge cover member 116, and therefore, the attitude of the developing unit 109 is maintained at the separation position.

The separation control member 196R of this embodiment is structured to be movable in the direction of arrow W42 in FIG. 24 from the home position. When the separation control member 196R moves in the W42 direction, the second force applying surface 196Rb of the separation control member 196R and the second force receiving surface 152Rp of the force applying member 152R come into contact with each other, so that the force applying member 152R rotates about the swing axis HC of the force applying member 152R in the BB direction. Further, as the force applying member 152R rotates further, the separation holding member 151R is rotated in the B2 direction, while the second pressing surface 152Rr of the force applying member 152R contacts the second pressed surface 151Re of the separation holding member 151R. Then, the separation holding member 151R is rotated by the force applying member 152R to the separation permission position where the separation holding surface 151Rc and the contact surface 116c are separated from each other. Here, the position of the separation control member 196R for moving the separation holding member 151R to the separation permission position shown in FIG. 25 is referred to as a first position.

In this manner, the separation control member 196R moves the separation holding member 151R to the separation permission position. Then, the developing unit 109 is rotated in the V2 direction by the torque received from the image forming apparatus main assembly 170 and the development pressure spring 134 which will be described hereinafter, and moves to the contact position where the developing roller 106 and the photosensitive drum 104 are in contact with each other (state shown in FIG. 25). At this time, the separation holding member 151R urged in the direction of arrow B1 by the tension spring 153 is maintained at the separation permission position by the second restricted surface 151Rk coming into contact with the second restriction surface 116d of the driving side cartridge cover member 116. Thereafter, the separation control member 196R moves in the direction of W41 and returns to the home position. At this time, the force applying member 152R is rotated in the BA direction by the tension spring 153, and the first pressing surface 152Rq of the force applying member 152R and the first pressing surface 126c of the driving side bearing 126 become in contact with each other (state shown in FIG. 26).

By this, the above-mentioned gaps T3 and T4 are formed again, and are placed at positions where the separation control member 196R does not act on the force applying member 152R. The transition from the state of FIG. 25 to the state of FIG. 26 is performed without a delay.

As described above, in the structure of this embodiment, by the separation control member 196R moving from the home position to the first position, the force applying member 152R can be rotated and the separation holding member 151R is moved from the separation holding position

29

to the separation permission position. By this, the developing unit **109** can move from the separated position to the contacting position where the developing roller **9** and the photosensitive drum **104** are in contact with each other. The position of the separation control member **196R** in FIG. **26** is the same as that in FIG. **24**.

[Separation Operation of Developing Unit]

Next, referring to FIGS. **26** and **27**, the operation of moving the developing unit **109** from the contact position to the distance position by the separation contact mechanism **150R** will be described in detail. For the sake of better illustration, these Figures are cross-sectional views taken along the line CS, in which a part of the development cover member **128**, a part of the driving side cartridge cover member **116**, and a part of the driving side bearing **126** are partially omitted.

The separation control member **196R** in this embodiment is structured to be movable from the home position in the direction of arrow **W41** in FIG. **26**. When the separation control member **196R** moves in the **W41** direction, the first force applying surface **196Rb** and the first force receiving surface **152Rm** of the force applying member **152R** are brought into contact with each other, and the force applying member **152R** rotates about the force applying member swing axis **HC** in the direction indicated by the arrow **BB**. Rotate in the direction. Then, the developing unit **109** rotates from the contact position in the direction of the arrow **V1** about the swing axis **K**, by the first pressing surface **152Rq** of the force applying member **152R** being brought into contact with the first pressed surface **126c** of the driving side bearing **126** (State shown in FIG. **27**). Here, the pressed surface **152Rf** of the force applying member **152R** has the arc shape, and the center of the arc is placed so as to coincide with the swing axis **K**. By this, when the developing unit **109** moves from the contact position to the separated position, the force received by the pressed surface **152Rf** of the force applying member **152R** from the cartridge pressing unit **121** is directed in the swing axis **K** direction. Therefore, the developing unit **109** can be operated so as not to hinder the rotation in the arrow **V1** direction. In the separation holding member **151R**, the second restricted surface **151Rk** of the separation holding member **151R** and the second restriction surface **116d** of the driving side cartridge cover member **116** are separated from each other, and the separation holding member **151R** is rotated in the arrow **B1** direction by the urging force of the tension spring **153**. By this, the separation holding member **151R** rotates until the second pressed surface **151Re** comes into contact with the second pressing surface **152Rr** of the force applying member **152R**, and by the contacts, the separation holding member **151R** shifts to the separation holding position. When the developing unit **109** is moved from the contact position to the separation position by the separation control member **196R** and the separation holding member **151R** is in the separation holding position, the gap **T5** is formed between the separation holding surface **151Rc** and the contact surface **116c** as shown in FIG. **27**. Here, the position shown in FIG. **27** in which the developing unit **109** is rotated from the contact position toward the separation position and the separation holding member **151** can move to the separation holding position is referred to as a second position of the separation control member **196R**.

Thereafter, the separation control member **196R** moves in the direction of the arrow **W42** and returns from the second position to the home position. Then, while the separation holding member **151R** is maintained in the separation holding position, the developing unit is rotated in the arrow **V2**

30

direction by the torque received from the image forming apparatus main assembly **170** and the development pressure spring **134** which will be described hereinafter, and the separation holding surface **151Rc** is contacted to the contact surface **116c**. That is, the developing unit **109** is in a state where the separation position is maintained by the separation holding member **151R**, and the developing roller **106** and the photosensitive drum **104** are in a state where they are separated by a gap **P1** (states shown in FIG. **24** and part (a) of FIG. **42**. By this, the above-mentioned gaps **T3** and **T4** are formed again, and the separation control member **196R** is placed at a position not acting on the force applying member **152R** (state in FIG. **24**). The transition from the state of FIG. **27** to the state of FIG. **24** is executed without a delay.

As described above, in this embodiment, the separation control member **196R** moves from the home position to the second position, so that the separation holding member **151R** moves from the separation permission position to the separation holding position. Then, by the separation control member **196R** returning from the second position to the home position, the developing unit **109** becomes in a state of maintaining the separation position by the separation holding member **151R**.

[Detailed Description of Separation Holding Member L]

Here, referring to FIG. **28**, the separation holding member **151L** will be described in detail.

Part (a) of FIG. **28** is a front view of the process cartridge **100** per se of the separation holding member **151L** as viewed in the longitudinal direction of the driving side, and FIGS. **28B** and **28C** are perspective views of the separation holding member **151L** per se. The separation holding member **151L** includes an annular support receiving portion **151La**, and includes a separation holding portion **151Lb** projecting from the support receiving portion **151La** in the radial direction of the support receiving portion **151La**. The free end of the separation holding portion **151Lb** has an arc-shaped separation holding surface **151Lc** extending about the separation holding member swing axis **H**.

Further, the separation holding member **151L** has a second regulated surface **151Lk** adjacent to the separation holding surface **151Lc**. Further, the separation holding member **151L** includes a second pressed portion **151Ld** projecting from the support receiving portion **151La** in the **Z2** direction, and includes a arc-shaped second pressed surface **151Le** projecting from the second pressed portion **151Ld** in the direction of the separation holding member swing axis **H** of the support receiving portion **151La**.

Further, the separation holding member **151L** is provided with a main body portion **151Lf** connected with the support receiving portion **151La**, and the main body portion **151Lf** is provided with a spring hooked portion **151Lg** projecting in the direction of the separation holding member swing axis **H** of the support receiving portion **151La**. Further, the main body portion **151Lf** is provided with a rotation prevention portion **151lm** projecting in the **Z2** direction, and a rotation prevention surface **151Ln** is provided in a direction facing the second pressed surface **151Le**.

[Detailed Description of Force Applying Member L]

Referring to FIG. **29**, the force applying member **152L** will be described in detail.

Part (a) of FIG. **29** is a front view of the force applying member **152L** as viewed in the longitudinal direction of the process cartridge **100**, and parts (b) and (c) of FIG. **29** are perspective views of the force applying member **152L**.

The force applying member **152L** is provided with an oblong-shaped oblong support receiving portion **152La**. Here, the longitudinal direction of the oblong shape of the

31

oblong support receiving portion **152La** is depicted by an arrow LH, the upward direction is depicted by an arrow LH1, and the downward direction is depicted by an arrow LH2. Further, the direction in which the oblong support receiving portion **152La** is extended is depicted by HD. The force applying member **152L** is provided with a projecting portion **152Lh** formed on the downstream side in the arrow LH2 direction of the oblong support receiving portion **152La**. The oblong support receiving portion **152La** and the projecting portion **152Lh** are connected by a main body portion **152Lb** with each other. On the other hand, the force applying member **152L** includes a pushed portion **152Le** projecting in the direction of arrow LH1 and in the direction substantially perpendicular to the direction of arrow LH1, and is provided with an arc-shaped pressed surface **152Lf** on the downstream side in the arrow LH1 direction and is further provided with a pushing restriction surface of **152Lg** on the upstream side. Further, the force applying member **152L** has a first at-accommodation restriction surface **152Lv** which is a part of the oblong support receiving portion **152La** and which is provided on the downstream side in the arrow LH2 direction.

The projecting portion **152Lh** includes a first force receiving portion **152Lk** and a second force receiving portion **152Ln** which are arranged so as to oppose each other in a direction substantially perpendicular to the arrow LH2 direction and a terminal portion in the arrow LH2 direction. The first force receiving portion **152Lk** and the second force receiving portion **152Ln** have a first force receiving surface **152Lm** and a second force receiving surface **152Lp** extending in the HD direction and having an arc shape, respectively. In addition, the projecting portion **152Lh** is provided with a spring hooked portion **152Ls** and a locking portion **152Lt** projecting in the HB direction, and the locking portion **152Lt** is provided with a locking surface **152Lu** facing in the same direction as the second force receiving surface **152Lp**.

Further, the force applying member **152L** is a part of the main body portion **152Lb**, is placed on the upstream side of the second force receiving portion **152Ln** in the arrow LH2 direction, and has a first pressing surface **152Lq** facing in the same direction as the second force receiving surface **152Lp**. Further, the force applying member **152L** is a part of the main body portion **152Lb**, is placed on upstream side of the first force receiving portion **152Lk** in the arrow LH2 direction, and has a first pressing surface **152Lr** facing in the same direction as the first force receiving surface **152Lm**.

In the state that the process cartridge **100** is mounted to the image forming apparatus main assembly **170**, the LH1 direction is substantially the same as the Z1 direction, and the LH2 direction is substantially the same as the Z2 direction. Further, the HB direction is substantially the same as the longitudinal direction of the process cartridge **100**. [Assembling of Separation/Contact Mechanism L]

Next, referring to FIGS. **16** and **29** to **35**, the assembly of the separation mechanism will be described. FIG. **30** is a perspective view of the process cartridge **100** after assembling the separation holding member therewith, as viewed from the driving side. As described above, as shown in FIG. **16**, in the developing unit **109**, the outer diameter portion of the cylindrical portion **127a** of the non-driving side bearing **127** is fitted into the developing unit support hole portion **117a** of the non-driving side cartridge cover member **117**. By this, the developing unit **109** is supported so as to be rotatable relative to the photosensitive drum **104** about the swing axis K. Further, the non-driving side bearing **127**

32

includes a cylindrical first support portion **127b** and a second support portion **127e** projecting in the direction of the swing axis K.

The outer diameter of the first support portion **127b** fits with the inner diameter of the support receiving portion **151La** of the separation holding member **151L**, to rotatably support the separation holding member **151L**. Here, the swing center of the separation holding member **151L** assembled to the non-driving side bearing **127** is the separation holding member swing axis H. The non-driving side bearing **127** includes a first retaining portion **127c** projecting in the direction of the separation holding member swing axis H. As shown in FIG. **16**, the movement of the separation holding member **151L** assembled to the non-driving side bearing **127** in the swing axis H direction is restricted by the first retaining portion **127c** coming into contact with the separation holding member **151L**.

Further, the outer diameter of the second support portion **127e** fits with the inner wall of the oblong support receiving portion **152La** of the force applying member **152L**, to support the force applying member **152L** so as to be rotatable and movable in the oblong direction. Here, the swing center of the force applying member **152L** assembled to the non-driving side bearing **127** is the force applying member swing axis HC. As shown in FIG. **16**, the movement of the force applying member **152L** assembled to the non-driving side bearing **127** in the direction of the swing axis HE is restricted by the second retaining portion **127f** coming into contact with the separation holding member **151L**.

FIG. **31** is a view of the process cartridge **100** after being assembled with the separation holding member **151L** as viewed in the developing unit swing axis H direction. It is a view taken along a line CS with a part of the non-driving side cartridge cover member **117** omitted so that the fitting portion between the oblong support receiving portion **151La** of the force applying member **152L** and the cylindrical portion **127e** of the non-driving side bearing **127** can be seen. Here, the separation contact mechanism **150L** is provided with a tension spring **153** for urging the separation holding member **151L** to rotate in the direction of arrow B1 about the separation holding member swing axis H and for urging the force applying member **152L** in the direction of arrow B3. The arrow B3 direction is a direction substantially parallel to the longitudinal direction LH2 (see FIG. **29**) of the oblong support receiving portion **152La** of the force applying member **152L**. The tension spring **153** is assembled between the spring hooked portion **151Lg** provided on the separation holding member **151L** and the spring hooked portion **152Ls** provided on the force applying member **152L**. The tension spring **153** applies a force to the spring hooked portion **151Lg** of the separation holding member **151L** in the direction of arrow F2 in FIG. **31** to apply an urging force for rotating the separation holding member in the direction of arrow B1. Further, the tension spring **153** applies a force to the spring hooked portion **152Ls** of the force applying member **152L** in the direction of the arrow F1 to apply an urging force for moving the force applying member **152L** in the direction of the arrow B3.

The line connecting the spring hooked portion **151Lg** of the separation holding member **151L** and the spring hooked portion **152Ls** of the force holding member **152L** is GS. The line connecting the spring hooked portion **152Ls** of the force applying member **152L** and the force applying member swing axis HE is HS. A angle $\theta 3$ formed by the line GS and the line HE is selected to satisfy the following inequity (3) with the counterclockwise direction being positive about the spring hooked portion **152Ls** of the force applying member

33

152L. By this, the force applying member 152L is urged to rotate in the BA direction in the drawing about the force applying member swing axis HE.

$$0^\circ \leq \theta_3 \leq 90^\circ \quad (3)$$

In this embodiment, the mounting positions of the separation holding member 151L and the force applying member 152L are as follows. As shown in FIG. 29, in the direction of the swing axis K, the separation holding member 151L and the force applying member 152L are disposed on the side (longitudinal outside) where the non-driving side cartridge cover member 117 of the non-driving side bearing 127 is placed. However, the positions to be arranged are not limited to the examples, and they may be provided on the development frame 125 side (inside in the longitudinal direction) of the non-driving side bearing 127, and the separation holding member 151L and the force applying member 152L may be provided with the non-driving side bearing 127 interposed therebetween. Further, the arrangement order of the separation holding member 151L and the force applying member 152L may be interchanged.

The non-driving side bearing 127 is fixed to the development frame 125 to form the developing unit 109. As shown in FIG. 16, in the fixing method in this embodiment, a fixing screw 145 and an adhesive (not shown), but the fixing method is not limited to this example, and welding such as welding by heating or pouring and hardening of resin can be employed.

Part (a) of FIG. 32 and part (b) of FIG. 32 are sectional views in which a portion of the non-driving side cartridge cover member 117, the tension spring 153, and the separation holding member 151L is partially omitted by the partial sectional line CS. For the sake of explanation, in part (a) of FIG. 32 and part (b) of FIG. 32 the parts around the force applying member swing axis HE and the separation holding portion 151L of the force applying member 152L shown in FIG. 31 is enlarged.

In the force applying member 152L, the first restriction surface 152Lv of the force applying member 152L comes into contact with the second support portion 127e of the non-driving side bearing 127 by the urging force of the tension spring 153 in the arrow F1 direction. Further, as shown in part (b) of FIG. 32, the first pressing surface 152Lq of the force applying member 152L contacts the first pressed surface 127h of the non-driving side bearing 127 to be positioned in place. This position is referred to as an accommodation position (reference position) of the force applying member 152L. Further, the separation holding member 151L is rotated in the direction of the arrow B1 about the swing axis H of the separation holding member by the urging force of the tension spring 153 in the arrow F2 direction, and the contact surface 151Lp of the separation holding member 151L is brought into contact with the second pressing surface 152Lr of the force applying member 152L, by which it is positioned in place.

This position is referred to as a separation holding position (restricted position) of the separation holding member 151L. When the force applying member 152L moves to the projecting position which will be described hereinafter, the second pressed surface 151Le of the separation holding member 151L contacts the second pressing surface 152Lr of the force applying member 152L to be positioned at the separation holding position.

Further, FIG. 33 is an illustration in which the periphery of the separation holding portion 151L in FIG. 31 is enlarged for the sake of illustration, and the tension spring 153 is omitted. Here, the consideration will be made as to the case

34

where the process cartridge 100 including the separation contact mechanism 150L is dropped in the direction of arrow JA in FIG. 33 when the process cartridge 100 is transported. At this time, the separation holding member 151L receives a force of rotating in the direction of arrow B2 due to its own weight around the separation holding swing axis H. When the separation holding member 151L starts to rotate in the arrow B2 direction, for the above reason, the rotation prevention surface 151Ln of the separation holding member 151L comes into contact with the locking surface 152Lu of the force applying member 152L, and the separation holding member 151L receives the force in the direction F4 of suppressing the rotation in the arrow B2 direction. By this, it is possible to prevent the separation holding member 151L from rotating in the direction of the arrow B2 during transportation, and it is possible to prevent impairment of the state of separation between the photosensitive drum 104 and the developing unit 109.

In this embodiment, the tension spring 153 is mentioned as an urging means for urging the separation holding member 151L to the separation holding position and the force applying member 152L to the accommodation position, but the urging means is limited to this example. For example, a torsion coil spring, a leaf spring, or the like may be used as an urging means to urge the force applying member 152L to the accommodation position and to urge the separation holding member 151L to the separation holding position. Further, the material of the urging means may be metal, a mold, or the like, which has elasticity and can urge the separation holding member 151L and the force applying member 152L.

As described above, the developing unit 109 provided with the separation contact mechanism 150L is integrally coupled with the drum holding unit 108 by the non-driving side cartridge cover member 117 as described above (state in FIG. 30). As shown in FIG. 16, the non-driving side cartridge cover 117 of this embodiment has a contact surface 117c. The contact surface 117c is a surface parallel to the swing axis K. Further, as shown in FIGS. 16 and 30 when the non-driving side cartridge cover member 117 is assembled to the developing unit 109 and the drum holding unit 108, the contact surface 117c faces the separation holding surface 151Lc of the separation holding member 151L placed at a separation holding position.

Here, the process cartridge 100 includes a development pressure spring 134 as an urging member for bringing the developing roller 106 into contact with the photosensitive drum 104. The development pressure spring 134 is assembled between the spring hooked portion 117e of the non-driving side cartridge cover member 117 and the spring hooked portion 127k of the non-driving side bearing 127. The urging force of the development pressure spring 134 causes the separation holding surface 151Lc of the separation holding member 151L and the contact surface 117c of the non-driving side cartridge cover member 117 to contact each other. Then, when the contact surface 117cc and the separation holding surface 151Lc contact each other, the attitude of the developing unit 109 is positioned so that the developing roller 106 of the developing unit 109 and the photosensitive drum 104 are spaced by a gap P1. The state in which the developing roller 106 is spaced from the photosensitive drum 104 by the gap P1 by the separation holding member 151L is referred to as a separation position (retracted position) of the developing unit 109 (see part (a) of FIG. 35).

Here, referring to FIG. 35, the separated state and the contact state of the process cartridge 100 will be described

35

in detail. FIG. 35 is a side view of the process cartridge 100 as viewed from the non-driving side with the process cartridge 100 mounted inside the image forming apparatus main assembly 170. Part (a) of FIG. 35 shows a state in which the developing unit is separated from the photosensitive drum 104. Part (b) of FIG. 35 shows a state in which the developing unit 109 is in contact with the photosensitive drum 104.

First, in a state in which the separation holding member 151L is placed at the separation holding position and the developing unit 109 is placed at the separation position, the pushed portion 152Le of the force applying member 152L is pushed in the direction of arrow ZA. By this, the projecting portion 152Lh of the force applying member 152L projects from the process cartridge 100 (state of part (a) of FIG. 34). This position is referred to as a projecting position of the force applying member 152L. The second pressed surface 151Le of the separation holding member 151L is in contact with the second pressing surface 152Lr of the force applying member 152L by the tension spring 153 as described above. Therefore, when the second force receiving portion 152Ln is pressed in the direction of the arrow W42, the force applying member 152L rotates in the direction of the arrow BD about the force applying member swing axis HE to rotate the separation holding member 151L in the direction of the arrow B5. When the separation holding member 151L rotates in the direction of arrow B5, the separation holding surface 151Lc separates from the contact surface 117c, and the developing unit 109 becomes capable of rotating from the separation position in the direction of arrow V2 about the swing axis K.

That is, the developing unit 109 rotates in the V2 direction from the separated position, and the developing roller 106 of the developing unit 109 comes into contact with the photosensitive drum 104. Here, the position of the developing unit 109 in which the developing roller 106 and the photosensitive drum 104 contact each other is referred to as a contact position (development position) (state of part (b) of FIG. 34). The position where the separation holding surface 151Lc of the separation holding member 151L is separated from the contact surface 117c is referred to as a separation permission position (permission position). When the developing unit 109 is placed at the contact position, by the second restriction surface 151Lk of the separation holding member 151L contacting the second restriction surface 117d of the driving side cartridge cover 116, the separation holding member 151L is maintained at the separation permission position.

Further, the non-driving side bearing 127 of this embodiment has a first pressed surface 127h which is a surface perpendicular to the swing axis K. Since the non-driving side bearing is fixed to the developing unit 109, the developing unit 109 presses the first force receiving portion 152Lk of the force applying member 152L in the direction of the arrow 41 while the developing unit 109 is in the contact position. Then, by the first pressing surface 152Lq coming into contact with the first pressed surface 127h, the developing unit is rotated about the swing axis K in the direction of arrow V1 and moves to a separated position (state shown in part (a) of FIG. 34). Here, when the developing unit 109 moves from the contact position to the separated position, the direction in which the first pressed surface 127h moves is indicated by an arrow W41 in part (a) of FIG. 34 and part (b) of FIG. 34. Further, the direction opposite to the arrow W41 is indicated by the arrow W42, and the directions of the arrow W41 and the arrow W42 are substantially horizontal directions (X1, X2 directions). The second force receiving surface 152Lp of the force applying

36

member 152L assembled to the developing unit 109 as described above is placed on the upstream side of the first pressed surface 127h of the non-driving side bearing 127 in the direction of the arrow W41. In addition, the first pressed surface 127h and the second force receiving surface 151Le of the separation holding member 151L are arranged at positions where at least parts of them overlap in the W1 and W2 directions.

The operation of the separation contact mechanism 150L in the image forming apparatus main assembly 170 will be described below.

[Mounting of Process Cartridge to the Image Forming Apparatus Main Assembly]

Next, referring to FIGS. 35 and 36, the engagement between the separation contact mechanism 150R of the process cartridge 100 and the development separation control unit of the image forming apparatus main assembly 170 at the time when the process cartridge 100 is mounted on the image forming apparatus main assembly 170 will be described. For the sake of illustration, these Figures are sectional views in which a portion of the development cover member 128 and a portion of the non-driving side cartridge cover member 117 are partially omitted by the partial sectional line CS, respectively. FIG. 35 is a view as seen from the driving side of the process cartridge 100 when the process cartridge is mounted on the cartridge tray 171 (not shown) of the image forming apparatus M and the cartridge tray 171 is inserted into the first mounting position. In this Figure, the parts are omitted except for the process cartridge 100, the cartridge pressing unit 121, and the separation control member 196L.

As described above, the image forming apparatus main assembly 170 of this embodiment has separation control members 196L corresponding to respective process cartridges 100 as described above. The separation control member 196L is disposed on the lower surface side of the image forming apparatus main assembly 170 with respect to the separation holding member 151L when the process cartridge 100 is placed at the first inner position and the second inner position. The separation control member 196L has a first force applying surface 196La and a second force applying surface 196Lb which project toward the process cartridge and face each other across the space 196Rd. The first force applying surface 196Ra and the second force applying surface 196Rb are connected with each other by a connecting portion 196Rc on the lower surface side of the image forming apparatus main assembly 170. In addition, the separation control member 196R is supported by the control sheet metal 197 rotatably about rotation center 196Re as the center. The separating member 196R is normally urged in the E direction by the urging spring. In addition, the control sheet metal 197 is structured to be movable in the W41 and W42 directions by a control mechanism (not shown), so that the separation control member 196R is structured to be movable in the W41 and W42 directions.

As described above, in interrelation with the transition of the front door 11 of the image forming apparatus main assembly 170 from the open state to the closed state, the cartridge pressing unit 121 lowers in the direction of arrow ZA, and the first force applying portion 121a is brought into contact with the pressed surface 152Lf of the pressed surface 152L. Thereafter, when the cartridge pressing unit 121 is lowered to a predetermined position which is the second mounting position, the part 152Lh of the force applying member 152L moves to a projecting position where the

37

process cartridge 100 projects downward in the Z2 direction (state in FIG. 36). When this operation is completed, as shown in

FIG. 36, a gap T4 is formed between the first force applying surface 196La of the separation control member 196L and the first force receiving surface 152Lp of the force applying member 152L, and a gap T3 is formed between the second force receiving surface 152Lp and the second force applying surface 196Lb. Then, it is placed at the second mounting position where the separation control member 196L does not act on the force applying member 152L. This position of the separation control member 196L is referred to as a home position. At this time, the first force receiving surface 152Lp of the force applying member 152L and the first force applying surface 196La of the separation control member 196L are arranged so as to partially overlap in the W1 and W2 directions. Similarly, the second force receiving surface 152Lp of the force applying member 152L and the second force applying surface 196Lb of the separation control member 196L are arranged so as to partially overlap in the W1 and W2 directions.

[Contacting Operation of Developing Unit]

Next, referring to FIGS. 36 to 38, the operation of contacting the photosensitive drum 104 and the developing roller with each other by the separation contact mechanism 150L will be described in detail. For the sake of illustration, a part of the development cover member 128, a part of the non-driving side cartridge cover member 117, and a part of the non-driving side bearing 127 are partially omitted in the partial sectional line CS, respectively. It is a sectional view.

As described above, the development input coupling 32 receives a driving force from the image forming apparatus main assembly 170 in the direction of arrow V2 in FIG. 24, so that the developing roller 106 rotates. That is, the developing unit 109 including the developing input coupling 32 receives the torque in the arrow V2 direction about the swing axis K from the image forming apparatus main assembly 170. Further, the developing unit 109 also receives an urging force in the arrow V2 direction due to the urging force of the development pressure spring 134 described above.

As shown in FIG. 36, when the developing unit 109 is in the separated position and the separation holding member 151L is in the separated holding position, the developing unit receives this torque and the urging force by the development pressure spring 134. Even in this case, the separation holding surface 151Lc of the separation holding member 151L contacts the contact surface 117c of the non-driving side cartridge cover member 117, and the attitude of the developing unit 109 is maintained at the separation position (state of FIG. 36).

The separation control member 196L of this embodiment is structured to be movable from the home position in the direction of arrow W41 in FIG. 36. When the separation control member 196L moves in the W41 direction, the second force applying surface 196Lb of the separation control member 196L and the second force receiving surface 152Lp of the force applying member 152L are brought into contact with each other, and the force applying member 152L is rotated in the BD direction about the force applying member swing axis HD. Further, with the rotation of the force applying member 152L, the separation holding member 151L is rotated in the B5 direction, while the second pressing surface 152Lr of the force applying member 152L is in contact with the second pressed surface 151Le of the separation holding member 151L. Then, the separation holding member 151L is rotated by the force applying member 152L to the separation permission position where

38

the separation holding surface 151Lc and the contact surface 117c are separated from each other. Here, the position of the separation control member 196L for moving the separation holding member 151L to the separation permission position shown in FIG. 37 is referred to as a first position.

In this manner, the separation control member 196L moves the separation holding member 151L to the separation permission position. Then, the developing unit 109 rotates in the V2 direction by the torque received from the image forming apparatus main assembly 170 and the urging force of the development pressure spring 134, and moves to the contact position where the developing roller 106 and the photosensitive drum 104 are in contact with each other (state shown in FIG. 37). At this time, the separation holding member 151L urged in the direction of arrow B4 by the tension spring 153 is maintained at the separation permission position by the second regulated surface 151Lk contacting the second restriction surface 117d of the non-driving side cartridge cover member 117. Thereafter, the separation control member 196L moves in the direction of W42 and returns to the home position. At this time, the force applying member 152L is rotated in the BC direction by the tension spring 153, and the state changed toward the state in which the first pressing surface 152Lq of the force applying member 152L and the first pressed surface 127h of the non-driving side bearing 127 are in contact with each other (state shown in FIG. 38).

By this, the above-mentioned gaps T3 and T4 are formed again, and the separation control member 196L is placed at a position where the force applying member 152L does not act. The transition from the state of FIG. 37 to the state of FIG. 38 is performed without a delay. The position of the separation control member 196L in FIG. 38 is the same as that in FIG. 36.

As described above, with the structure of this embodiment, by moving the separation control member 196L from the home position to the first position, the force applying member 152L is rotated to move the separation holding member 151L from the separation holding position to the separation permission position. By this, the developing unit 109 can be moved from the separated position to the contacting position where the developing roller 9 and the photosensitive drum 104 are in contact with each other.

[Separating Operation of Developing Unit]

Next, the operation of moving the developing unit 109 from the contact position to the separation position will be described in detail referring to FIGS. 38 and 39. Note that FIG. 39 is a cross-section in which a portion of the development cover member 128, a portion of the non-driving side cartridge cover member 117, and a portion of the non-driving side bearing are partially omitted by the partial cross-section line CS, respectively.

The separation control member 196L in this embodiment is structured to be movable from the home position in the direction of arrow W42 in FIG. 38. When the separation control member 196L moves in the W42 direction, the first force applying surface 196Lb and the first force receiving surface 152Lm of the force applying member 152L come into contact with each other, and the force applying member 152L is rotated in the arrow BC centering about the force applying member swing axis HD. Since the first pressing surface 152Lq of the force applying member 152L is in contact with the first pressed surface 127h of the non-driving side bearing 127, the developing unit 109 is rotated from the contact position in the direction of arrow V1 about the swing axis K (state in FIG. 39). Here, the pressed surface 152Lf of the force applying member 152L has an arc shape, and the

center of the arc is placed so as to be aligned with the swing axis K. By this, when the developing unit 109 moves from the contact position to the separated position, the force received, from the cartridge pressing unit 121, by the pressed surface 152Lf of the force applying member 152L 5 faces the swing axis K direction. Therefore, the developing unit 109 can be operated so as not to hinder the rotation in the arrow V1 direction. In the separation holding member 151L, the second regulated surface 151Lk of the separation holding member 151L and the second restriction surface 117d of the non-driving side cartridge cover member 117 are separated, and the separation holding member 151L is rotated in the arrow B4 direction by the urging force of the tension spring 153. By this, the separation holding member 151L rotates until the second pressed surface 151Le comes 10 into contact with the second pressing surface 152LR of the force applying member 152L, and by the contact with the second pressing surface 152LR, the position shifts to the separation holding position. When the developing unit is moved from the contact position to the separation position by the separation control member 196L and the separation holding member 151L is placed at the separation holding position, A gap T5 is formed between the separation holding surface 151Lc and the contact surface 117c as shown in FIG. 39. Here, the position where the developing unit 109 is rotated from the contact position toward the separation position and the separation holding member 151 can be moved to the separation holding position is referred to as a second position of the separation control member 196L.

Thereafter, the separation control member 196L moves in the direction of the arrow W41 and returns from the second position to the home position. Then, while the separation holding member 151L is maintained at the separation holding position, the developing unit is rotated in the arrow V2 direction by the torque received from the image forming apparatus main assembly 170 and the urging force of the development pressure spring 134, and the separation holding surface 151Lc and the contact surface 117c are brought into contact with each other. That is, the developing unit 109 is in a state where the separation position is maintained by the separation holding member 151L, and the developing roller 106 and the photosensitive drum 104 are in a state where they are separated by a gap P1 (states in FIG. 36 and part (a) of FIG. 34. By this, the above-mentioned gaps T3 and T4 are formed again, and the separation control member 196L is placed at a position where the force applying member 152L does not act (state in FIG. 36). The transition from the state of FIG. 39 to the state of FIG. 36 is executed without a delay.

As described above, in the structure of this embodiment, by the movement of the separation control member 196L from the home position to the second position, the separation holding member 151L is moved from the separation permission position to the separation holding position. And, by the returning of the separation control member 196L from the second position to the home position, the developing unit 109 becomes in the state of maintaining the separation position by the separation holding member 151L.

So far, the operation of the separation mechanism placed on the driving side of the process cartridge 100 and the operation of the separation mechanism placed on the non-driving side have been described separately, but in this embodiment, they operate in interrelation with each other. That is, when the developing unit 109 is positioned at the separation position by the separation holding member R, the developing unit 109 is positioned at the separation position by the separation holding member L at substantially the same time, and the same applies to the contact position.

Specifically, the movements of the separation control member 121R and the separation control member 121L described in FIGS. 23 to 27 and 35 to 39 are integrally carried out by a connecting mechanism (not shown). By this, the timing at which the separation holding member 151R provided on the driving side is placed at the separation holding position, and the timing at which the separation holding member 151L provided on the non-driving side is placed at the separation holding position are substantially the same, and the timing at which the separation holding member 151R is placed at the separation permission position, and the timing at which the separation holding member 151L is placed at the separation permission position are substantially the same. These timings may be different between the driving side and the non-driving side, but in order to shorten the time from the start of the print job by the user until the printed matter is discharged It is desirable that at least the timings of positioning at least the separation permission positions are the same. In this embodiment, the separation holding member swing axes H of the separation holding member 151R and the separation holding member 151L are common, but it is sufficient that the timings of the separation holding member 151L and the separation holding member 151L are substantially the same as described above, and therefore the above-described example is not restrictive. Similarly, the force applying member swinging axis HC of the force applying member 152R and the force applying member swinging axis HE of the force applying member 152L are axes that do not match, but it will suffice if the timings of being placed at the separation permission positions are substantially the same as described above, and therefore, the above-described example is not restrictive.

As described above, the driving side and the non-driving side are provided with the same separation contact mechanisms, respectively, and they operate substantially at the same time. By this, even when the process cartridge 100 is twisted or deformed in the longitudinal direction, the amount of separation between the photosensitive drum 104 and the developing roller 9 can be controlled at the respective end portions in the longitudinal direction. Therefore, it is possible to suppress variations in the amount of separation in the longitudinal direction.

Further, according to this embodiment, by moving the separation control member 196R (L) between the home position, the first position, and the second position in one direction (arrows W41 and W42 directions), it is possible to control the contact state and the separation state between the developing roller 106 and the photosensitive member. Therefore, it is possible that the developing roller 106 is brought into contact with the photosensitive drum 104 only when the image is formed, and the developing roller 4 is maintained in a state of being separated from the photosensitive drum 104 when the image is not formed. Therefore, even if the image formation is not carried out for a long term, the developing roller 106 and the photosensitive drum 104 are not deformed, and a stable image can be formed.

Further, according to this embodiment, the force applying member 152R (L) acting on the separation holding member 151R (L) to rotate and move can be positioned at the accommodation position by the urging force of the tension spring 153 or the like. Therefore, it does not project out of the outermost shape of the process cartridge 100, when the process cartridge 100 is outside the image forming apparatus main assembly 170, and the process cartridge 100 per se can be downsized.

41

Similarly, the force applying member 152R (L) can be positioned at the accommodation position by the urging force of the tension spring 153 or the like. Therefore, when the process cartridge 100 is to be mounted to the image forming apparatus main assembly 170, the mounting of the process cartridge 100 can be completed by moving only in one direction. For this reason, it is not necessary to move the process cartridge 100 (tray 171) in the vertical direction. Accordingly, the image forming apparatus main assembly 170 does not require an additional space, and the main assembly can be downsized.

Further, according to this embodiment, when the separation control member 196R (L) is placed at the home position, the separation control member 196R (L) is not loaded from the process cartridge 100. Therefore, the rigidity required for the mechanism for operating the separation control member 196R (L) and the separation control member 196R (L) can be reduced, and the size can be reduced. Further, since the load on the sliding portion of the mechanism for operating the separation control member 196R (L) is also reduced, wear of the sliding portion and production of abnormal noise can be suppressed.

Further, according to this embodiment, the developing unit 109 can maintain the separated position only by the separation holding member 151R (L) included in the process cartridge 100. Therefore, the component tolerance can be eased and the spacing amount can be minimized by reducing the number of parts resulting in variations in the spacing amount between the developing roller 106 and the photosensitive drum 104. Since the amount of spacing can be reduced, when the process cartridge 100 is arranged in the image forming apparatus main assembly 170, the area occupied by the developing unit 109 when the developing unit 109 moves to the contact position and to the separated position can be made smaller, so that the image forming apparatus can be downsized. In addition, the space for the developer accommodating portion 29 of the developing unit 109 which moves to the contact position and to the separation position can be increased, and therefore, the downsized and large-capacity process cartridge 100 can be placed in the image forming apparatus main assembly 170.

Further, according to this embodiment, the force applying member 152R (L) can also be positioned at the accommodation position when the process cartridge 100 is mounted, and the developing unit 109 can maintain the separation position only by the separation holding member 151R (L) of the process cartridge 100. Therefore, when the process cartridge 100 is mounted to the image forming apparatus main assembly 170, the process cartridge 100 can be mounted by moving only in one direction. For this reason, it is not necessary to move the process cartridge 100 (tray 171) in the vertical direction. Accordingly, the image forming apparatus main assembly 170 does not require a space, and the main assembly can be downsized. Further, since the separation amount can be reduced, when the process cartridge 100 is placed in the image forming apparatus main assembly 170, the area occupied by the developing unit 109 when the developing unit 109 moves to the contact position and to the separation position can be made small, and therefore, the image forming apparatus can be downsized. In addition, since the space for the developer accommodating portion 29 of the developing unit 109 which moves to the contact position and to the separation position can be increased, the downsized and large-capacity process cartridge 100 can be placed in the image forming apparatus main assembly 170.

42

[Details of Arrangement of Separation Contact Mechanism]

Subsequently referring to FIGS. 40 and 41, the arrangement of the separation contact mechanisms R and L in this embodiment will be described in detail.

FIG. 40 is an enlarged view of the periphery of the separation holding member 151R as the process cartridge 100 is viewed from the driving side along the swing axis K (photosensitive drum axis direction) of the developing unit 109. In addition, for the sake of illustration, it is a sectional view in which a portion of the development cover member and a portion of the driving side cartridge cover member 116 are partially omitted by the partial sectional line CS. FIG. 41 is an enlarged view of the periphery of the separation holding member 151R as the process cartridge 100 is viewed from the non-driving side along the swing axis K of the developing unit 109 (along the axis in the photosensitive drum axis direction). In addition, for the sake of illustration, it is a sectional view in which a portion of the development cover member 128 and a portion of the driving side cartridge cover member 116 are partially omitted by the partial sectional line CS. Regarding the arrangement of the separation holding member and the force applying member described below, there is no distinction between the driving side and the non-driving side except for the part which will be described in detail hereinafter, and they are common, and therefore, the description will be made only for the driving side, the same applies to the non-driving side.

As shown in FIG. 40, the rotation center of the photosensitive drum 104 is a point M1, the rotation center of the developing roller 106 is a point M2, and the line passing through the points M1 and M2 is a line N. In addition, the contact region between the separation holding surface 151Rc of the separation holding member 151R and the contact surface 116c of the driving side cartridge cover member 116 is M3, and the contact region between the second pressed surface 151Re of the separation holding member 151R and the second pressing surface 152Rr of the second force applying member 152R is M4. Further, the distance between the swing axis K and the point M2 of the developing unit 109 is a distance e1, the distance between the swing axis K and the region M3 is e2, and the distance between the swing axis K and the point M4 is e3.

In the structure of this embodiment, the following positional relationship is a relationship when the developing unit 109 is in the separated position and the force applying member 152R (L) is in the projecting position. As viewed along the axial direction of the swing axis K shown in FIG. 40 (the axial direction of the photosensitive drum), at least a part of the contact region M3 between the separation holding member 151R and the driving side cartridge cover member is placed on a side opposite from the side in which the development coupling 32 center (swing axis K) exists, with respect to the line N passing through the center of the photosensitive drum 104 and the center of the developing roller. That is, the separation holding surface 151Rc of the separation holding member 151R is arranged such that the distance e2 is longer than the distance e1.

By arranging the separation holding member 151R and the separation holding surface 151Rc in this manner, it is possible to suppress variations in the attitude of the spaced position of the developing unit 109 when the positions of the separation holding surface 151Rc vary due to component tolerances and the like. That is, the influence of the variation of the separation holding surface 151Rc on the separation amount (gap) P1 (see part (a) of FIG. 42) between the developing roller 106 and the photosensitive drum 104 can be minimized, and the developing roller 106 can be accu-

rately spaced from the photosensitive member **104**. Further, it is not necessary to provide an additional space for permitting retraction when the developing unit **109** is separated, which leads to the downsizing of the image forming apparatus main assembly **170**.

Further, the first force receiving portion **152Rk** (Lk) and the second force receiving portion **152Rn** (Ln), which are the force receiving portions of the force applying member **152R** (L), are placed on a side opposite from the rotation centers of the development coupling **32** with respect to the extension line of the line N.

As described above, the force receiving portions **152Rk** (Lk) and **152Rn** (Ln) are provided at the end portions in the longitudinal direction. Further, as shown in FIG. **15** (FIG. **16**), a cylindrical portion **128b** (**127a**), which is a support portion of the developing unit **109**, is provided at the end portion in the longitudinal direction. Therefore, by disposing the force receiving portions **152Rk** (Lk) and **152Rn** (Ln) at positions opposite from the cylindrical portion **128b** (**127a**) (that is, the swing axis K) of the developing unit **109** with respect to the line N the functional elements can be arranged efficiently. That is, it leads to downsizing of the process cartridge **100** and the image forming apparatus M.

In addition, the force receiving portions **152Rk** and **152Rn** are placed at the longitudinal driving side end portions. Further, as shown in FIG. **15**, a development drive input gear **132** that receives a drive from the image forming apparatus main assembly **170** and drives the developing roller **106** is provided at the end portion on the driving side in the longitudinal direction. As shown in FIG. **40**, the force applying members **152Rk** and **152Rn** are placed on the side opposite from the rotation center K of the development drive input gear **132** (development coupling portion **132a**) shown by the broken lines with respect to the extension line of the line N. With this arrangement, the functional elements can be efficiently arranged. That is, it leads to downsizing of the process cartridge **100** and the image forming apparatus M.

Further, the contact portion between the separation holding member **151R** and the force applying member **152R** is arranged such that the distance e3 is longer than the distance e1. By this, the separation holding member **151R** and the driving side cartridge cover member **116** can be brought into contact with each other with a lighter force. That is, the developing roller **106** and the photosensitive drum **104** can be stably separated from each other.

[Detailed Description of Drive Transmission Mechanism for Photosensitive Drum]

A structure for transmitting a driving force from the image forming apparatus main assembly to the drum unit **103** of the cartridge **100** (see part (a) of FIG. **1** to drive (rotate) the drum unit will be described.

The drum unit **103** shown in FIGS. **1**, **13** and **55** to **58** is a unit including a photosensitive drum, a drum coupling (cartridge side coupling, coupling member) **143**, and a drum flange **142** (see FIG. **13**). The drum unit **103** is mountable to and dismountable from the image forming apparatus main assembly as a part of the cartridge **100**. By mounting the drum unit **103** to the main assembly of the apparatus, it can be connected with a drive transmission unit **203** (see FIGS. **43** and **44**, details will be described hereinafter) of the main assembly of the apparatus. The drum unit rotates in the direction of arrow A during image formation (see FIGS. **1**, **55** to **57**). In this embodiment, as the driving side of the drum unit **103** (the side where the drum coupling **143** is located) is viewed, that is, when the drum unit **103** is viewed along the arrow M1B direction, the rotational direction of the drum unit **103** corresponds to the clockwise direction (See FIG. **1**).

In other words, when the front surface of the drum coupling **143** is viewed, the rotational direction A of the drum coupling **143** corresponds to the clockwise direction.

The rotational direction A of the drum unit (drum coupling **143** and the photosensitive drum **104**) will be described below using the movement of the surface of the photosensitive drum **104** (see FIGS. **2** and **3**). In FIGS. **2** and **3**, unlike FIG. **1**, the cartridge is viewed from the non-driving side, and therefore, the rotational direction A of the drum unit **103** is counterclockwise.

As shown in FIG. **3**, the surface of the photosensitive drum **104** is charged inside the cartridge at a position near the charging roller **105** (around the position where it contacts the charging roller). Thereafter, the surface of the photosensitive drum **104** moves to a position where it receives the laser beam U, by which an electrostatic latent image is formed on the surface. Then, the surface of the photosensitive drum **104** moves to a position near the developing roller **106** (a position in contact with the developing roller in this embodiment), and a latent image formed on the surface of the photosensitive drum **104** developed into a toner image. After that, the surface of the photosensitive drum moves to a position exposed below the cartridge and outside the casing of the cartridge. Then, as shown in FIG. **2**, the surface of the photosensitive drum **104** exposed from the casing of the cartridge contacts the intermediary transfer belt **12a** provided in the image forming apparatus main assembly. By this, the toner image is transferred from the surface of the photosensitive drum **104** to the transfer belt **12a**. Thereafter, the surface of the photosensitive drum **104** returns, inside of the cartridge, to a position near the charging roller **105**.

In summary, when the photosensitive drum **104** rotates due to the driving force of the coupling **143**, a part of the surface of the photosensitive drum **104** moves from a position close to the charging roller **105** to a position close to the developing roller **106**. Thereafter, the part of the surface of the photosensitive drum **104** is exposed to the outside of the casing of the cartridge, and then returns to the inside of the casing of the cartridge and approaches the charging roller **105** again.

As described above, the cartridge **100** of this embodiment does not have a cleaning means for contacting the photosensitive drum **104** and removing the toner on the surface of the photosensitive drum **104** (see FIG. **3**). Therefore, the torque required to rotate the drum unit **103** (photosensitive drum **104**) inside the cartridge **100** is relatively small. In the case of such a structure, the drum unit **103** is easily affected by the surroundings when it is driven, and as a result, the drum unit **103** may be externally affected by the outside with the result of unstable rotation speed. For example, in this embodiment, the developing roller **106**, the charging roller **105**, and the transfer belt **12a** are in contact with the photosensitive drum **104**. If the magnitude of the frictional force generated between these means and the photosensitive drum **104** fluctuates, the speed of the drum unit **103** may fluctuate.

Therefore, in this embodiment, the structure is such that a torque a predetermined level or higher is required, when the drum drive coupling **180** of the drive transmission unit **203** (see FIG. **43**) provided in the main assembly of the apparatus rotates the drum unit (photosensitive drum **104**) of the cartridge. By this, the rotation of the drum unit **103** is relatively less influenced by the external factors, and its rotation speed is stable.

First, referring to part (a) of FIG. **1**, the drum coupling **143** of the process cartridge **100** will be described. Part (a) of FIG. **1** is a perspective view of the drum coupling.

The drum coupling **143** of this embodiment is manufactured by injection molding a polyacetal resin. As the material, a resin material such as a polycarbonate resin or polybutylene terephthalate resin, or a resin material provided by blending these with glass fiber, carbon fiber or the like may be used. Alternatively, a processing method such as die casting or cutting may be used with a metal material such as aluminum, iron, or stainless steel.

Next, referring to FIGS. **1**, **55** to **58**, the shape of the drum coupling **143** will be described.

In the following description of the drum coupling **143**, the direction (direction of arrow **M1A**) from the photosensitive drum **104** toward the drive transmission unit **230** (drum drive coupling **180**) along the axial direction is called outward (outward) in the axial direction. In addition, the direction opposite to the outward direction (the direction of the arrow **M1B**) is called inward direction in the axial direction.

In other words, in the drum coupling, the outward direction (**M1A** direction) in the axial direction is the direction from the non-driving side end portion **104b** of the photosensitive drum toward the driving side end portion **104a** (leftward in FIG. **80**). Alternatively, the outward direction (**M1A** direction) in the axial direction is the direction from the non-driving side cartridge cover **117** of the cartridge **100** toward the driving side cartridge cover **116** in FIG. **14**.

The inward direction in the axial direction (**M1B** direction) is the direction from the driving side end portion **104a** of the photosensitive drum **104** toward the non-driving side end portion **104b** (rightward in FIG. **80**). Alternatively, the inward direction (**M1B** direction) in the axial direction is the direction from the driving side cartridge cover **116** of the cartridge **100** toward the non-driving side cartridge cover **117** in Figure.

As shown in part (b) of FIG. **1**, the drum coupling **143** is mounted to one longitudinal end (driving side end) of the photosensitive drum **104**. As described above, the shaft portion **143j** shown in FIG. **1** is rotatably supported by the driving side cartridge cover member **116** (see FIG. **15**) which supports the photosensitive drum unit **103**. The drum unit **103** is structured to be rotatable in a predetermined rotational direction (direction of arrow **A**) during the image forming operation in which the latent image on the surface of the photosensitive drum is developed.

The drum coupling **143** receives a driving force for rotating the photosensitive drum **104** from the main assembly drive transmission unit **203** of the main assembly of the apparatus, and also receives a braking force for applying a load against the rotation of the photosensitive drum **104**, as well.

The drum coupling **143** is provided with a projections projecting outward in the axial direction from the surface of the end portion of the shaft portion **143j** (see FIGS. **1**, **52** to **57**). This projection has a driving force receiving portion **143b** as a first side surface (first side portion) for receiving the driving force from the driving transmission unit **203**. Further, the projection of the drum coupling **143** includes a braking force receiving portion **143c** as a second side surface (second side portion) for receiving the braking force from the drive transmission unit **203**.

The driving force receiving portion **143b** is a side surface (side portion) facing the upstream side in the rotational direction **A** of the drum unit. Further, the braking force receiving portion **143c** is a side surface (side portion) facing the downstream side in the rotational direction **A**.

In other words, one of the driving force receiving portion **143b** and the braking force receiving portion **143c** faces one

side in the circumferential direction of the drum unit, and the other faces the other side in the circumferential direction. That is, the driving force receiving portion **143b** and the braking force receiving portion **143c** are side surfaces (side portions) facing opposite to each other in the rotational direction and the circumferential direction.

Further, the projection of the drum coupling **143** has a helical slope (inclined portion, slope) **143d** as a top surface (upper surface, upper portion, upper portion). The slope (top surface) **143d** is a portion facing outward (arrow **MA1** direction) in the axial direction. That is, the slope **143d** is a portion facing toward the side opposite to the non-driving side end portion of the drum unit (that is, the end portion on the side where the drum flange **142** (FIG. **13**) is arranged). In other words, the helical slope (top surface) **143d** of the coupling **143** is a portion facing the side opposite to the side on which the photosensitive drum **104** exist.

The helical slope **143d** is inclined so as to be outward in the axial direction (arrow **MA1** direction) toward the upstream side in the rotational direction (upstream side in the arrow **A** direction). That is, the slope **143d** goes away from the non-driving side of the drum unit **103** as goes toward the upstream side in the rotational direction. In other words, the slope **143d** is inclined so as to go away from the photosensitive drum as goes toward the upstream side in the rotational direction.

In other words, the helical slope **143d** extends toward the non-driving end of the drum unit and the cartridge from upstream to downstream in the rotational direction. Namely, when the distance of the helical slope **143d** from the non-driving end of the cartridge is measured along the axial direction, the distance becomes shorter toward the downstream in the rotational direction.

The helical slope **143d** includes a downstream portion (downstream top surface, downstream inclined slope, downstream inclined portion, downstream guide) **143d1** sandwiched between the driving force receiving portion **143b** and the braking force receiving portion **143c** in the rotational direction of the drum unit. Further, the slope **143d** has an upstream portion (upstream side top surface, upstream side slope, upstream side inclined portion, upstream guide) **143d2**. The upstream portion **143d2** of the helical slope **143d** is provided upstream of the driving force receiving portion **143b** and the downstream portion **143d1** of the helical slope **143d** in the rotational direction (see FIGS. **55** to **58**).

Further, as the length of the slope **143d** is measured along the rotational direction of the drum unit, the length of the upstream side slope **143d2** is larger than the length of the downstream side slope **143d1**.

The upstream side portion (upstream side slope) **143d2** of the slope **143d** is provided inside (the side closer to the axis **L**) of the driving force receiving portion **143b** in the radial direction. That is, the upstream side portion (upstream side top surface, upstream side slope) **143d2** of the slope **143d** is provided closer to the axis **L** (part (a) of FIG. **1**) than the driving force receiving portion **143b**. The axis **L** (part (a) of FIG. **1**) is the axis (rotation axis) which is the center of rotation of the coupling **143** and the photosensitive drum **104**.

Further, the projection of the drum coupling **143** is provided with a circular hole portion **143a** as an opening for engaging with the positioning boss (positioning portion) **180i** of the drum drive coupling **180** and positioning each other's axes. The circular hole portion **143a** has a circular opening having a cross-section perpendicular to the axis **L** of the drum coupling **143**, and is extended along the axis **L**.

The projection of the drum coupling **143** includes a shaft portion **143p** (see FIG. **1**) formed along the axis L (see part (a) of FIG. **1**, and the circular hole portion **143a** is formed inside the shaft portion **143p**. The shaft portion **143p** is a portion for forming the circular hole portion **143a**.

The shaft portion **143p** and the circular hole portion **143a** are extended aligned with the axis L. By forming the circular hole portion **143a**, the space from the rotation axis L of the drum unit (see part (a) of FIG. **1** to the inner surface of the drum coupling **143** is an open space. The shaft portion **143p** has a diameter smaller than the shaft portion **143j** described above.

The drum coupling **143** described above has an axisymmetric shape (axisymmetric shape) with respect to the axis L (see part (a) of FIG. **1**. The driving force receiving portion **143b**, the braking force receiving portion **143c**, and the helical slope **143d** are arranged at two locations so as to be separated by 180° in the circumferential direction, respectively, thus providing a first coupling portion **143r** and a second coupling portion **143s** (see FIG. **58**).

Each coupling portion includes one driving force receiving portion **143b**, one braking force receiving portion **143c**, and one helical slope **143d**, and the first coupling portion **143r** and the second coupling portion **143s** are placed in position symmetrical with respect to the axis.

The driving force receiving portion **143b**, the braking force receiving portion **143c**, and the helical slope **143d** are arranged around the above-mentioned circular hole portion **143a** and the shaft portion **143p**. The driving force receiving portion **143b**, the braking force receiving portion **143c**, and the helical slope **143d** are located more remote than the circular hole portion **143a** and the shaft portion **143p** from the axis L of the drum unit.

Next, referring to FIGS. **43**, **44**, and **59**, the structure of the main assembly side drive transmission unit **203** provided on the main assembly side of the apparatus will be described. The drive transmission unit **203** is a unit for rotationally driving the drum coupling **143** by connecting (engaging) with the drum coupling **143**.

FIG. **43** is an exploded perspective view of the main assembly side drive transmission unit **203**. FIG. **59** is an enlarged perspective view of a portion shown in FIG. **43**. FIG. **44** is a sectional view of the main assembly side drive transmission unit **203**.

A drive gear **201** is rotatably supported by a support shaft **202** fixed to a frame (not shown) of the apparatus main assembly **170**, and a driving force is transmitted from a motor (not shown) to rotate the drive gear **201**. The drum drive coupling **180** includes a cylindrical portion **180c** and a flange portion **180a** provided at the end thereof, and the flange is fitted and supported by a fitting portion **201a** of the drive gear **201**. Further, the drum drive coupling **180** is provided with a rotation stop portion **180b** projecting from the flange portion **180a**, which receives a driving force when rotating in contact with the rotation stop portion **201b** of the drive gear **201**. The drive transmission unit **203** includes a plurality of components inside the cylindrical portion **180c** of the drum drive coupling **180**.

The parts arranged inside the cylindrical portion **180c** are as follows. There are a brake members **206** which is supported and stopped by the support shaft **202**, a brake transmission member **207** which is connected with the brake member **206** to transmit the braking force, and first and second braking engagement members **204** and **208** engaged with the braking force receiving surface **143c** of the drum coupling **143**, and, a brake engagement spring **211** and a drum drive coupling spring **210** which are arranged along

the axis **M1** and which generate an urging force in the direction of the axis **M1** (axis direction). The axis **M1** is a rotation axis of the main assembly side drive transmission unit **203**.

The shape of each of the parts arranged inside the main assembly drive transmission unit **203** will be described. The first braking engagement member **204** comprises a cylindrical portion **204d**, a flange portion **204a**, and a coupling engaging portion **204b** which projects like a claw and engages with the drum coupling **143**. A part of the cylindrical portion includes a rotation stop recess **204c** which engages with the rotation stop projection **208c** of the second braking engagement member **208**, which will be described hereinafter.

The second braking engagement member **208** includes a flange portion **208a**, a coupling engaging portion **208b** projecting in the form of a claw and engaging with the drum coupling **143**, and the rotation stop projection **208c** engaged with the rotation stop recess **204c** of the first braking engagement member **204**. Since the second braking engagement member **208** is stopped from rotating relative to the first braking engagement member **204**, the first and second braking engagement members **204** and **208** rotate integrally with each other. Further, the first and second braking engagement members **204** and **208** are connected so as to move integrally also in the axial direction.

Therefore, the first and second braking engagement members **204** and **208** may be collectively referred to simply as braking engagement members (**204**, **208**).

The first braking engagement member **204** is an outer braking engagement member disposed on the outer side in the radial direction, and the second braking engagement member **208** is an inner braking engagement member disposed on the inner side in the radial direction.

The brake transmission member **207** includes a flange portion **207a** and a shaft portion **207b**. The flange portion **207a** is provided with a projection **207e** which engages with the projection **204e** provided on the flange portion **204a** of the first braking engagement member **204**. The flange portion **207a** of the brake transmission member **207** is disposed between the flange portion **204a** of the first braking engagement member **204** and the flange portion **208a** of the second braking engagement member **208**, with a play (gap) **G** therebetween in the axial direction (FIG. **44**). In the axial direction **M1A**, when the brake transmission member **207** is in a position relative to the first brake engagement member **204** in which the projection **207e** of the brake transmission member **207** (see FIGS. **43** and **59**) is engaged with the projection **204e** of the first brake engagement member **204**, the first brake transmission member and the first and second braking engagement members **204** and **208** rotate integrally. On the other hand, when the brake transmission member **207** is in a position relative to the first braking engagement member **204** in the axial direction in which the projection **207e** does not engage with the projection **204e**, the brake transmission member **207** does not limit the rotation of the first and second engagement members **204**, **208**. That is, the first and second braking engagement members **204** and **208** are rotatable relative to the brake transmission member **207**. The shaft portion **207b** has a non-circular cross-section, and engages with the engagement hole **206c** of the brake member **206** which will be described hereinafter so that the brake transmission member **207** and the brake member **206** are integrally rotated.

The brake member **206** is divided into two portions, namely, a fixed side **206a** and a rotating side **206b**, but they are integrated in the axial direction by a retainer (not

shown). The fixed side **206a** is supported by the support shaft **202**, and the rotation about the shaft is also fixed. On the other hand, the rotating side **206b** can rotate around the support shaft **202**, but rotates while receiving a braking force (load) in the rotational direction from the fixed side **206a**. The method of producing the braking force can be appropriately selected from those using friction and viscosity.

The braking engagement members (**204**, **208**) are connected to the brake member **206** by way of the brake transmission member **207** as described above. Therefore, the rotational torque of the braking engagement members (**204**, **208**) increases due to the influence of the load (braking force) generated by the brake member **206**. The brake engagement spring **211** is a compression coil spring, and is provided so as to be sandwiched and compressed between the end surface **206d** of the brake member **206** and the flange portion **204a** of the first braking engagement member **204**. As a result, the spring **211** applies a repulsive force (urging force, elastic force) to each of the end surface **206d** of the brake member **206** and the flange portion **204a** of the first braking engagement member **204**.

The drum drive coupling spring **210** is a compression coil spring, and is provided so as to be sandwiched and compressed between the end surface **206d** of the brake member **206** and the flange portion **207a** of the brake transmission member **207**. As a result, the spring **210** applies a repulsive force (urging force, elastic force) to each of the end surface **206d** of the brake member **206** and the flange portion **207a** of the brake transmission member **207**.

The brake transmission member **207** directly receives the repulsive force of the drum drive coupling spring **210** while receiving the repulsive force of the brake engagement spring **211** by way of the flange portion **204a** of the first braking engagement member **204**. The projection **207f** at the end of the brake transmission member **207** in the axial direction M1A abuts against the contact surface **180f** of the drum drive coupling **180** (see FIG. 44).

By this, the drum drive coupling **180** also receives the force of the drum drive coupling spring **210** and the brake engagement spring **211** by way of the brake transmission member **207**. The drum drive coupling **180** tends to move due to the force of the springs **210** and **211**. Therefore, the movement of the drum drive coupling **180** in the arrow M1B direction is regulated (restricted) by the axial direction restricting portion **212** (see FIG. 44) so that the drum drive coupling **180** does not drop off the main assembly side drive transmission unit **203**. Specifically, when the drum drive coupling **180** moves to the arrow M1B by a certain distance, the flange portion **180a** (see FIG. 43) of the drum drive coupling **180** comes into contact with the restriction portion **212** (see FIG. 44). By this, the movement and drop-off of the drum drive coupling **180** can be suppressed.

When the drum drive coupling **180** receives a force in the arrow M1A direction from the outside in this state, the drum drive coupling **180** can move in the arrow M1A direction while compressing the springs **210** and **211**.

Further, when the braking engagement members (**204**, **208**) engage with the coupling **143**, the coupling engaging portions **204b**, **208b** may interfere with the coupling **143** (see FIG. 60, details will be described hereinafter). In such a case, the braking engagement members (**204**, **208**) can enter (retract) into the depth of the drive transmission unit **203** while compressing the springs **210** and **211** in the direction of the arrow M1A (see FIG. 61).

The braking engagement members (**204**, **208**) are disposed with a gap G from the brake transmission member **207** as described above (see FIG. 44). Within a range of the

width of the gap G, the braking engagement members (**204**, **208**) can move and retract in the M1A direction relative to the brake transmission member **207**. Similarly, the braking engagement members (**204**, **208**) can move in the direction of the arrow M1A within the range of the width of the gap G relative to the drum drive coupling **180**. When the braking engagement member (**204**, **208**) moves in the direction of the arrow M1A relative to the brake transmitting member **207** and the drum drive coupling **180**, the brake engagement spring **211** is compressed.

The brake transmitting member **207** is also moved in the direction of arrow M1A together with the braking engagement member (**204**, **208**), by the braking engagement member (**204**, **208**) contacting the brake transmitting member **207** which tends to move in the direction of the arrow M1A beyond the width of the gap G.

Together with the braking engagement members (**204**, **208**), the drum drive coupling **180** also moves in the direction of arrow M1A. As shown in FIG. 62, the drum drive coupling **180** and the first braking engagement member **204** are provided with a projecting engaging portion **180u** and an engaging portion **204u**, respectively. Therefore, when the braking engagement member **204** moves in the direction of the arrow M1A relative to the drum drive coupling **180** for a predetermined distance or more, the engaging portion **204u** pushes the engaging portion **180u** to retract the drive coupling **180** in the M1A direction. At this time, not only the spring **211** but also the spring **210** is compressed.

When the braking engagement member (**204**, **208**) moves in the direction of the arrow M1A relative to the brake transmission member **207**, the projection **207e** of the brake transmission member **207** and the projection **204e** of the first braking engagement member are disengaged. That is, the braking engagement members (**204**, **208**) are disconnected from the brake transmission member **207**, and the braking force is not transmitted from the brake transmission member **207**. The brake members (**204**, **208**) can rotate relative to the brake transmission member **207** without receiving the rotational load produced by the brake member **206**.

That is, by retracting the braking engagement members (**204**, **208**) in the direction of arrow M1A, the braking engagement members are movable from the position in which the brake member **206** receives the rotational load (braking force) during rotation to the position in which the rotational load is not received during rotation. The braking engagement members (**204**, **208**) are structured to reduce the own required torque by moving in the M1A direction relative to the brake transmission member **207** and to the drum drive coupling **180**.

FIG. 45 is a perspective view illustrating the positional relationship between the drum drive coupling **180** and the braking engagement members (**204**, **208**). Part (a) of FIG. 45 is a perspective view of only the drum drive coupling **180**, and part (b) of FIG. 45 shows a perspective view in which both the drum drive coupling **180** and the braking engagement member (**204**, **208**) are included. Parts (c) and (d) of FIG. 45 are illustrations in which the reinforcing cylindrical portion **180e** of the drum drive coupling **180** is not shown (invisible) for the sake of better illustration. The phases of the braking engagement members (**204**, **208**) differ between parts (c) and (d) of FIG. 45.

As shown in part (a) of FIG. 45, the drum drive coupling (driving force applying member) **180** includes a driving transmission surface **180d** provided at each of two positions which are away from each other by 180 degrees in the circumferential direction as a surface (driving force applying

51

portion) which engages with the coupling **143** to transmit the driving force. The drum drive coupling has an axisymmetric shape.

A through hole **180f** communicating in the direction of the axis **M1** is provided in a portion other than the drive transmission surface **180d**. Through the through hole **180f**, the coupling engaging portions **204b** and **208b** of the first braking engagement member **204** and the second braking engagement member **208** are exposed in the direction facing the coupling **143** (see FIG. 60).

Part (b) of FIG. 45 shows a state in which the coupling engaging portions **204b** and **208b** of the first braking engagement member **204** and the second braking engagement member **208** are exposed. The drum drive coupling **180** is provided with a reinforcing cylindrical portion **180e** in order to increase the rigidity of the drive transmission surface **180d**. Part (c) of FIG. 45 is an illustration in which the reinforcing cylindrical portion **180e** is not shown for the sake of better illustration. Part (c) of FIG. 45 shows a state in which the coupling engaging portions **204b** and **208b** and the drive transmission surface **180d** are in a close phase relationship in the rotational direction A. The size of the through hole **180f** is selected to be wider than the widths of the coupling engaging portions **204b** and **208b** in the circumferential direction. Therefore, the coupling engaging portions **204b** and **208b** can move within a predetermined range in the rotational direction in the drum drive coupling **180**.

Part (d) of FIG. 45 shows a state in which the coupling engaging portions **204b** and **208b** and the drive transmission surface **180d** are in a distant phase relationship in the rotational direction A.

Next, referring to FIGS. 1 and 43 to 51, a method of connecting the main assembly side drive transmission unit **203** of the drive transmission mechanism and the photosensitive member coupling **143** on the process cartridge **100** side will be described.

[Coupling Engagement Operation]

Next, the process of coupling between the main assembly side drum drive coupling **180** of the image forming apparatus main assembly **170** and the drum coupling **143** of the process cartridge **100** will be described.

FIG. 46 shows a sectional view of the image forming apparatus main assembly **170** around the main assembly side drum drive coupling **180**. Referring to FIG. 46, the outline of the movement of the drum drive coupling **180** on the main assembly side will be described.

When the user opens the front door **111** (FIG. 4) of the image forming apparatus main assembly to replace the process cartridge **100**, the drive transmission unit **203** is moved in the direction of the arrow **M1A** along the axis **M1** by a link mechanism (not shown) connected to the front door **111**. That is, the drive transmission unit **203** is in a state of being moved away from the process cartridge **100** and the drum coupling **143** (see FIG. 60).

When the user mounts the process cartridge **100** and closes the front door **111**, the action of the link described above disappears. Therefore, the drum drive coupling **180**, the brake engagement members **204**, **208**, and the brake transmission member **207** tends to move again in the direction of arrow **M1B** by the urging forces of the drum drive coupling spring and the brake engagement spring **211**. At this time, the drum coupling **143** of the process cartridge **100** stands by in the direction of the arrow **M1B** and interferes with the approaching drive transmission unit **203** (states

52

shown in FIGS. 61, 65, and 69). The drum coupling **143** and the drive transmission unit **203** are pressed against each other.

In these states, the drum coupling **143** and the drum drive coupling **180** of the drive transmission unit **203** are normally not engaged.

In order for the drum coupling **143** and the main assembly side drum drive coupling **180** to be in a normal engaged state, the drive transmission unit **203** is required to be further rotated from the above-mentioned pressing state. That is, it is necessary to advance the drive process of the drive transmission unit **203** until the drum drive coupling **180** on the main assembly side engages with the drum coupling **143**.

Further, the process until the engagement is completed may be carried out in different patterns, and therefore, the description will be made, dividing into a plurality of cases depending on the phase of the drum coupling **143** and the main assembly side drum drive coupling **180**.

Part (a) of FIG. 47 shows the drum coupling **143**, and part (b) of FIG. 47 shows the drive transmission unit, both as viewed in the axial direction. Referring to part (a) of FIG. 47, The shape of the coupling **143** will be further described. As for the profile of the coupling, the shape differs in the radial direction, depending on the functions to perform. The following structures are provided within the range of the radius indicated by **R1** in the Figure.

That is, the positioning hole (opening) **143a** which engages with the positioning boss (positioning portion) **180i** of the drive coupling **180**, a visor (visor portion) **143g** (see part (a) of FIG. 47 and FIG. 1) as a overhang portion for preventing the drive transmission unit **203** from entering in the axial direction and a part of the helical slope **143d** are provided. A part of the helical slope **143d** and a part of the braking force receiving surface **143c** are provided in the range between **R1** to **R2**. The braking force receiving surface **143c** is not visible in the line-of-sight direction of part (a) of FIG. 47 and is shown in FIG. 1. In the range between **R2** to **R3**, a part of the driving force receiving portion **143b**, a part of the helical slope **143d**, and a part of the braking force receiving surface **143c** are provided.

On the other hand, since the shape of the drive transmission unit **203** is also arranged in a shape including a different role in the radial direction, the same range as the coupling **143** is shown in part (b) of FIG. 47 using the same symbols **R1** to **R3**.

Within the range of the radius indicated by **R1** in part (b) of FIG. 47, the positioning boss **180i** that engages with the positioning hole **143a** of the drum coupling **143** and the second brake that comes into contact with the visor portion **143g** depending on the phase of the drum coupling **143**. An inward projection **208e**, which is a portion of the coupling engaging portion **208b** of the engaging member **208**, is arranged. Within the range indicated by **R1** to **R2**, the coupling engaging portion **208b** of the second braking engagement member **208** is arranged. The drive transmission surface **180d** and the first braking engagement member **204** are arranged within the range indicated by **R2** to **R3**.

FIG. 48 is a developed view of these portions developed around the rotation axis **M1**. FIG. 48 The process until the drum coupling **143** and the drive transmission unit **203** are engaged with each other will be described.

FIG. 48 shows the drive transmission unit **203** on the lower side and shows the process of approaching the drum coupling **143** while moving in the direction of the arrow **M1B** until the engagement is established. In this Figure, the structures provided within the radius **R1** shown in FIG. 47 are shown by broken lines, the structures provided within the

53

range between the radius R1 and the radius R2 are shown by solid lines, and further, the structures provided in the range between the radius R2 to radius R3 are shown by solid lines and hatching lines.

The drum coupling 143 includes two coupling portions 143s and 143r arranged 180° apart from each other, but only the coupling portion 143s will be described below for the sake of simplicity. The description of the coupling portion 143s also applies to the coupling portion 143r.

Part (a) of FIG. 48 shows a state in which the drive transmission surface 180d of the drive transmission unit 203 and the second braking engagement member 208 are in close to each other. As shown in part (a) of FIG. 48, the phases of the inclination start portion 143f of the drum coupling 143 and the inward projection 208e of the second braking engagement member 208 have the following relationship. That is, the inclination start portion 143f of the drum coupling 143 is on the upstream side of the projection 208e in the rotational direction (arrow A).

Part (b) of FIG. 48 shows a state in which the drive transmission unit 203 is further moved in the direction of arrow M1B from the position shown in part (a) of FIG. 48. The helical slope 143d is opposed to and is in contact with the inward projection 208e of the approaching first braking engagement member 204.

Part (c) of FIG. 48 shows a state in which the drive transmission unit 203 is further moved in the direction of the arrow M1B. The helical slope 143d stops the approaching second braking engagement member 208. By this, the movement of the second braking engagement member 208 in the M1B direction is suppressed. On the other hand, the portion excluding the second braking engagement member 208 (that is, the drum drive coupling 180 of the drive transmission unit 203, and so on) is moving in the direction of arrow M1B. In the drive transmission unit 203, the second braking engagement member 208 is in a state of being relatively pushed in the direction of the arrow M1A.

In this state reached, as described referring to FIG. 44, the second braking engagement member 208 can rotate without receiving a rotational load because of being disconnected from the brake member 206. At this time, the brake member 206 receives an elastic force F1 in the direction of the rotation axis M1 by the drum drive coupling spring 210 and the brake engagement spring 211 provided inside the drive transmission unit 203. The helical slope 143d moves the second braking engagement member 208, which becomes free of rotational load, in the direction of arrow C by the component force of the elastic force F1. That is, the second braking engagement member 208 moves to the downstream side in the rotational direction A along the helical slope 143d.

Part (d) of FIG. 48 shows a state immediately after the second braking engagement member 208 is moved to the downstream side in the rotational direction (direction of arrow A). The second braking engagement member 208 moves along the helical slope 143d of the drum coupling 143, and further moves in the M1B direction by the amount of the entire drive transmission unit 203 moving in the axial direction M1B, so that movement trace is as depicted by the arrow D. As a result, the second braking engagement member 208 moves away from the drive coupling 180 toward the downstream side in the rotational direction A to the position in which it is engageable with the braking force receiving portion 143c (second side surface, second side portion) of the drum coupling 143. That is, the helical slope 143d is a guide for guiding the braking engagement member toward the braking force receiving portion 143c. In this embodi-

54

ment, the helical slope (top surface) 143d, which is a guide, has a downstream portion 143d1 and an upstream portion 143d2. The downstream portion (downstream side slope, downstream side top surface, downstream side inclined portion) 143d1 is placed between the braking force receiving portion 143c and the driving force receiving portion 143b. The upstream side portion (upstream side slope, upstream side top surface, upstream side inclined portion) 143d2 is on the upstream side in the rotational direction (A direction) with respect to the driving force receiving portion 143b. Therefore, the second braking engagement member 208 can be smoothly guided from the upstream side portion 143d2 of the slope 143d to the braking force receiving portion 143c by way of the downstream side portion 143d1.

Part (e) of FIG. 48 shows a state in which the drum coupling 143 moves (rotates) in the direction of arrow A by the rotating drive transmission surface 180d, and as a result, the braking force receiving portion 143c contacts the second braking engagement member 208.

When the drive transmission unit 203 rotates in the direction of arrow A, the drive transmission surface 180d comes into contact with the drive force receiving portion 143b to transmit the drive force. The drive transmission surface 180d is a drive force applying portion which applies a drive force to the drum coupling 143.

The drum coupling 143 being rotated by receiving the driving force from the driving transmission surface 180d also receives the braking force by the braking force receiving portion 143c contacting (engaging) the second braking engagement member 208.

Parts (a) to (e) of FIG. 48 show only the second braking engagement member 208 out of the first and second braking engagement members 204 and 208 which are the braking engagement members. However, the first braking engagement member 204 (see FIG. 43) is connected to the second brake member 208 so as to move integrally with the second brake member 208. Therefore, in the process shown in part (a) of FIG. 48 to part (e) of FIG. 48, the first braking engagement member 204 also moves along the same line as the second brake member 208. In the state shown in part (e) of FIG. 48, the first braking engagement member 204 also engages with the braking force receiving portion 143c together with the second braking engagement member 208.

In part (a) to (e) of FIG. 48, only the engagement process of the braking engagement member (204, 208) and the drum drive coupling 180 with the coupling portion 143s are shown for simplicity of the description. Similarly to the coupling portion 143s, the coupling 143r also engages with the braking engagement member (204, 208) and the drum drive coupling 180. The engagement state of the braking engagement members (204, 208) and the drum drive coupling with respect to the coupling 143r is shown in part (a) of FIG. 76.

Here, in order to help the recognition of the process described so far, the description will be made again using the perspective views of FIGS. 60 to 64. In FIGS. 60 to 64, a part of the drum drive coupling 180 is not shown for better illustration, and the internal shapes are uncovered.

FIG. 60 is a perspective view illustrating the same state as in part (a) of FIG. 48 described above. That is, the inclination start portion 143f of the drum coupling 143 is on the upstream side of the projection 208e in the rotational direction (arrow A), and the drive transmission surface 180d of the drive transmission unit 203 and the second braking engagement member 208 are close to each other. FIG. 61 shows a state in which the drive transmission unit 203 has moved in the direction of arrow M1B from this state.

55

FIG. 61 shows a state corresponding to part (b) of FIG. 48, and the helical slope 143d is opposed to and is in contact with the inward projection 208e of the approaching second braking engagement member 208. The drive transmission unit 203 and the drum coupling 143 are relatively close to each other until they come into contact with each other, but the state inside the drive transmission unit 203 has not changed. FIG. 62 shows a state in which the drive transmission unit 203 is further moved in the direction of arrow M1B from this state.

FIG. 62 shows a state corresponding to part (c) of FIG. 48, in which the helical slope 143d stops the approaching second braking engagement member 208. By this, in the drive transmission unit 203, the second braking engagement member 208 is pushed in the direction of the arrow M1A

relative to the drum drive coupling 180. In this state, as described referring to FIG. 44, the second braking engagement member 208 can rotate without receiving a rotational load because of being disconnected from the brake member 206. At this time, the brake member 206 receives an elastic force F1 in the direction of the rotation axis M1 by the drum drive coupling spring 210 and the brake engagement spring 211 arranged inside the drive transmission unit 203. The helical slope 143d moves the second braking engagement member 208, which becomes free of rotational load, in the direction of arrow C by the component force of the elastic force F1. That is, the second braking engagement member 208 rotationally moves to the downstream side in the rotational direction A along the helical slope 143d.

FIG. 63 shows a state immediately after the second braking engagement member 208 moves to the downstream side in the rotational direction (direction of arrow A), and corresponds to part (c) of FIG. 48. The second braking engagement member 208 moves along the helical slope 143d of the drum coupling 143, and further moves in the M1B direction by the amount of movement of the entire drive transmission unit 203 in the axial direction M1B direction, the trace of the movement is as indicated by the arrow D. As a result, the braking engagement members (204, 208) move away from the drive coupling 180 toward the downstream side in the rotational direction A to the position in which they can engage with the second side surface (braking force receiving portion 143c) of the drum coupling 143. At this position reached, the braking engagement members (204, 208) return to a state where braking force can be produced.

FIG. 64 shows a state in which the drum coupling 143 is moved (rotated) in the direction of arrow A by the rotating drive transmission surface 180d, and as a result, the braking force receiving portion 143c contacts the second braking engagement member 208. FIG. 64 corresponds to part (d) of FIG. 48.

When the drum drive coupling 180 of the drive transmission unit 203 rotates in the direction of arrow A from the state of FIG. 64, the drive transmission surface 180d comes into contact with the drive force receiving portion 143b to transmit the drive force. The drum coupling 143 being rotated by receiving the driving force from the driving transmission surface 180d also receives the braking force by the braking force receiving portion 143c contacting (engaging with) the second braking engagement member 208 (see part (e) of FIG. 48).

In summary, through the processes shown in parts (a) to (e) of FIG. 48 and FIGS. 60 to 64, the braking engagement members (204, 208) are moved relative to the drum drive coupling 180 and the drum coupling 143 as follows.

56

The braking engagement member (204, 208) is moved from the position (part (a) of FIGS. 48 and 60 in which it is close to the drive transmission surface 180d to the position (part (d) of FIGS. 48 and 64) in which the drum coupling 143 is sandwiched between the drive transmission coupling 180d and the braking engagement member (204, 208).

When the drive transmission surface 180d rotates from the state shown in part (d) of FIG. 48 and FIG. 64, the drum coupling 143 also rotates together with the drive transmission surface 180d to reach the state shown in part (e) of FIG. 48. Then, the drum coupling 143 rotates in the direction of arrow A by the driving force received from the drum driving side coupling 180 while receiving an appropriate load (braking force) from the braking engagement member (204, 208). As a result, the torque required for the drum drive coupling 180 to rotate the drum unit is not too light and is appropriate, so that the rotational drive of the drum unit is stabilized.

Next, referring to part (a) to (e) of FIG. 49, another pattern of the engagement process of the drum drive coupling 180 and the braking engagement member (204, 208) with the drum coupling 143 will be described. The drum coupling 143 has two coupling portions 143s and 143r, but for the sake of simplicity, only the coupling portion 143s will be described.

As shown in part (a) of FIG. 49, a case where the phases of the inclination start portion 143f of the drum coupling 143 and the inward projection 208e of the second braking engagement member satisfy the following relationship will be described. That is, the case where the inclination start portion 143f of the drum coupling 143 is on the downstream side in the rotational direction (arrow A) with respect to the inward projection 208e.

Part (a) of FIG. 49 shows a state in which the drive transmission surface 180d of the drive transmission unit 203 and the second braking engagement member 208 are close to each other.

The visor portion 143g of the drum coupling 143 is in contact with the inward projection 208e of the second braking engagement member 208 approaching in the M1B direction.

Next, part (b) of FIG. 49 shows a state in which the visor portion 143g stops (blocks) the advancement of the approaching second braking engagement member 208. Here, the drum drive coupling 180, which is a component of the drive transmission unit 203, does not contact the visor portion 143g, and therefore, the advancement in the M1B direction cannot be stopped. That is, the visor portion 143g does not interfere with the shape of the drum drive coupling 180 because the position thereof is different in the radial direction. On the other hand, the second braking engagement member 208 has an inward projection 208e at the free end in the M1B direction. Since the inward projection 208e projects inward in the radial direction, it is in contact with the visor portion 143g of the drum coupling 143.

By the movement of only the drum drive coupling 180 in the M1B direction, the second braking engagement member 208 moves relative to the drum drive coupling 180 in the M1A direction. As described above, by this relative movement, the second braking engagement member 208 shifted to a state in which it can rotate without receiving a rotational load.

Then, part (c) of FIG. 49 shows a state in which the drive transmission unit 203 has started to rotate in the rotational direction A. First, when the drum drive coupling 180 starts rotating in the A direction, it is pushed by the drum drive coupling 180, and the second braking engagement member 208 also starts rotating in the A direction.

57

The helical slope **143d** of the drum coupling **143** moves the second braking engagement member in the direction of arrow C from the point where the inward projection **208e** of the second braking engagement member **208** passes the inclination start portion **143f**. That is, the second braking engagement member **208** moves toward downstream side in the rotational direction A and in the M1B direction.

Part (d) of FIG. 49 shows a state after the second braking engagement member **208** moves along the helical slope **143d** of the drum coupling **143** and passes the inclined surface **143d** as in part (d) of FIG. 48. At this time, the entire drive transmission unit **203** further moves in the axial direction M1B. As a result, the second braking engagement member also moves in the M1B direction. The first braking engagement member **204** moves along the line of arrow D.

Subsequent engagement operation is the same as in the description of part (d) of FIG. 48, and the subsequent engagement completion state is as shown in part (e) of FIG. 48. In this embodiment, visor portion **143g** is continuous with on the upstream side (upstream side slope, upstream side top surface) **143d2** of the helical slope **143d**. The inclination start portion **143f** is a boundary portion between the visor portion **143g** and the helical slope **143d**. Therefore, the second braking engagement member **208**, the movement of which has been blocked by the visor portion **143g**, can smoothly shift to a state of being in contact with the helical slope **143d**, as the drive transmission unit **203** rotates. However, the structure is not necessarily limited to this example structure, and a space may be provided between the visor portion **143g** and the slope **143d**.

Also in part (a) of FIG. 49 to part (d) of FIG. 49, only the second braking engagement member **208** of the braking engagement members (**204**, **208**) is shown. However, as described above, also in the process of part (a) of FIG. 49 to part (d) of FIG. 49, the first braking engagement member **204** (see FIG. 43) moves integrally with the second braking engagement member **208**.

Here, in order to help the recognition of the process described referring to part (a) of FIG. 49 to part (d) of FIG. 49, the description will be made again with reference to the perspective views of FIGS. 65 to 68. In FIGS. 65 to 68, a part of the drum drive coupling **180** is not shown for better illustration, and the internal shape is uncovered.

FIG. 65 shows a state in which the drive transmission surface **180d** of the drive transmission unit **203** and the second braking engagement member **208** are close to each other. At this time, the visor **143g** of the drum coupling **143** is in contact with the second braking engagement member **208** approaching in the M1B direction. FIG. 65 corresponds to part (a) of FIG. 49.

Next, FIG. 66 shows a state in which the drum drive coupling **180** has moved to the right side (M1B direction) along the axial direction relative to the second braking engagement member **208**. In FIG. 66, the visor portion **148g** is in a state of stopping (blocking) the advancement of the approaching second braking engagement member **208**.

FIG. 66 corresponds to part (b) of FIG. 49. The second braking engagement member **208** moves relative to the drum drive coupling **180** to the left side (M1A direction) in the axial direction. As described above, by this relative movement, the second braking engagement member **208** is shifted to a state in which it can rotate without receiving a rotational load.

Subsequently, FIG. 67 shows a state in which the drive transmission unit **203** has started to rotate in the rotational direction A. FIG. 67 corresponds to part (c) of FIG. 49. The helical slope **143d** of the drum coupling **143** moves the

58

second braking engagement member **208** in the direction of arrow C from the point where the second braking engagement member **208** passes the inclination start portion **143f**. FIG. 68 corresponds to part (d) of FIG. 49. In the state shown in FIG. 68, the first braking engagement member **204** moves along the helical slope **143d** of the drum coupling **143**, as in the state shown in part (d) of FIGS. 48 and 63. Further, the first braking engagement member **204** also moves in the M1B direction by the amount of the movement of the entire drive transmission unit **203** in the axial direction M1B direction. As a result, the first braking engagement member **204** moves along the trace of arrow D.

Then, as described above, the entire drive transmission unit **203** continues to rotate to complete the connection, resulting in the same state as in part (e) of FIG. 48.

Next, referring to part (a) of FIG. 50 to part (d) of FIG. 50, further pattern of the engagement process of the drum drive coupling **180** and the braking engagement member (**204**, **208**) with the drum coupling **143** will be described. The drum coupling **143** includes two coupling portions **143s** and **143r**, but for the sake of simplicity, only the coupling portion **143s** will be described.

As shown in part (a) of FIG. 50, a case where the phase of the inclination start portion **143f** of the drum coupling **143** and the inward projection **208e** of the second braking engagement member satisfy the following relationship will be described. That is, a case where the inclination start portion **143f** of the drum coupling **143** is on the downstream side in the rotational direction (arrow A) will be described.

Part (a) of FIG. 50 shows a state in which the drive transmission surface **180d** of the drive transmission unit **203** and the second braking engagement member **208** are separated from each other.

Next, part (b) of FIG. 50 shows a state in which the visor portion **143g** stops the advancement of the approaching second braking engagement member **208**. Here, the drum drive coupling **180**, which is a component of the drive transmission unit **203**, does not contact the visor portion **143g**, and therefore, the advancement cannot be stopped. By this, the second braking engagement member **208** moves relative to the drum drive coupling **180** in the M1A direction.

As described above, by this relative movement, the second braking engagement member **208** is shifted to a state in which it can rotate without receiving a rotational load. Here, the visor portion **143g** does not interfere with the shape of the drum drive coupling **180** because the position is different in the radial direction.

Then, part (c) of FIG. 50 shows a state in which the drive transmission unit **203** rotates in the rotational direction A and contacts the second braking engagement member. That is the state in which the second braking engagement member **208** does not start rotating by itself, so that it stops at that position, and the drum drive coupling **180** rotates and comes into contact with the second braking engagement member **208**. Thereafter, by further rotation, the second braking engagement member **208** and the drum drive coupling **180** rotate integrally.

Part (d) of FIG. 50 shows a state in which the second braking engagement member **208** is further rotated and has passed the inclination start portion **143f** of the drum coupling **143**. In this state reached, the second braking engagement member **208** moves in the direction of arrow C as described referring to part (c) of FIG. 48. The operation after this is the same as described above, and therefore, the description is omitted.

59

Also in part (a) of FIG. 50 to part (d) of FIG. 50, only the second braking engagement member 208 of the braking engagement members (204, 208) is shown. However, as described above, also in the process of part (a) of FIG. 50 to part (d) of FIG. 50, the first braking engagement member 204 (see FIG. 43) moves integrally with the second braking engagement member 208.

Here, in order to help the recognition of the process described referring to part (a) of FIG. 50 to part (d) of FIG. 50, the description will be made again with reference to the perspective views of FIGS. 69 to 72. In FIGS. 69 to 72, a part of the drum drive coupling 180 is not shown for better illustration, and the internal shape is uncovered.

FIG. 69 corresponds to part (a) of FIG. 50, and shows a state in which the drive transmission surface 180d of the drive transmission unit 203 and the second braking engagement member 208 are separated by a gap G1.

Next, FIG. 70 corresponds to part (b) of FIG. 50 and shows a state in which the entire drive transmission unit 203 has moved in the M1B direction. That is the state in which the visor portion 143g stops the advancement of the approaching second braking engagement member 208, and the drum drive coupling 180 has moved to the right side (M1B direction) in the axial direction beyond the second braking engagement member 208. At this time, the second braking engagement member 208 moves to the left side (M1A direction) relative to the drum drive coupling 180. As described above, by this relative movement, the second braking engagement member 208 is shifted to a state in which it can rotate without receiving a rotational load.

Then, FIG. 71 corresponds to part (c) of FIG. 50, and shows a state in which the drum drive coupling 180 of the drive transmission unit 203 is in contact with the second braking engagement member 208 by rotating in the rotational direction A.

Since the second braking engagement member 208 cannot rotate without receiving the rotational force from the drum drive coupling 180, the second braking engagement member 208 does not rotate immediately after the start of driving of the drive transmission unit 203 and remains at the initial position. That is, only the drum drive coupling 180 starts rotating in the A direction in advance. As a result, a state shown in FIG. 71 is reached in which the drum drive coupling 180 is in contact with the second braking engagement member 208.

FIG. 72 corresponds to part (d) of FIG. 50, and shows a state in which by the engagement between the drum drive coupling 180 and the second braking engagement member 208, not only the drum drive coupling 180 but also the second braking engagement member 208 start to rotate in the direction A. More specifically, that is the state in which by the second braking engagement member 208 being pushed by the drum drive coupling 180 to rotate in the A direction, the second braking engagement member 208 passes the inclination start portion 143f of the drum coupling 143. In this state reached, the second braking engagement member 208 is guided by the slope 143d and moves in the direction along the slope 143d (direction of arrow C), as described in part (c) of FIG. 48 and FIG. 62.

Subsequent operations are the same as those described above referring to part (c) of FIG. 48 to part (e) of FIG. 48 and FIGS. 62 to 64, and therefore, the description thereof are omitted here.

As described above, when the cartridge 100 is mounted on the image forming apparatus main assembly, the phase (arrangement) of the drive transmission unit 203 with respect to the drum coupling 143 is not predetermined (part

60

(a) of FIG. 48, FIG. 49 (a), part (a) of FIG. 50, FIG. 60, FIG. 65, FIG. 69). However, in any case, the drum coupling 143 can be connected to the drive transmission unit 203. The drive transmission unit 203 includes not only the drum drive coupling 180 but also the braking engagement members (204, 208), both of which the drum coupling 143 can be engaged with.

Next, referring to FIG. 51, the description will be made as to the structures for aligning the axes of the drive transmission unit 203 and the drum coupling 143, in the process of connecting them. FIG. 51 is a sectional view of the drive transmission unit 203 and the drum coupling 143, and part (a) of FIG. 51 shows the shapes in the connected state in this embodiment. The circular hole portion 143a of the drum coupling engages with the positioning boss 180i of the drum drive coupling 180 to align the axes with each other. Further, a conical guide surface 143h is provided at one end of the circular hole portion 143a. That is, the guide surface 143h has a conical shape as a part of the inner surface of the coupling 143. The guide surface 143h is provided so that when the drive transmission unit 203 is still separated in the axial direction M1B direction, the deviations from each other are eliminated upon starting engagement to align the axes with each other.

In addition to this embodiment, the circular hole portion 143a of the drum coupling 143 may be engaged with the positioning boss 180i without providing a guide surface, as shown in part (b) of FIG. 51. Further, as shown in part (c) of FIG. 6, the guide surface 143h can be enlarged to reduce the fitting between the circular hole portion 143a and the positioning boss 180i. Further, as shown in part (d) of FIG. 51, the diameter of the circular hole portion 143a can be increased. These arrangements can be selected depending on how to determine the relative position between the drive transmission unit 203 and the process cartridge 100 and the accuracy.

It is desirable that the circular hole portion 143a has a sufficient length to accommodate the positioning boss 180i. That is, as shown in FIG. 95, the positioning boss 180i enters at least the range of the region Pb on the axis L of the drum unit. The circular hole portion 143a is formed so as to include the entire region Pb. That is, the periphery of the axis L is open in the region Pb.

In FIG. 95, in this embodiment, on the axis L, the range occupied by the braking force receiving portion 143c, the helical slope (top surface) 143d, the visor portion 143g, and the driving force receiving portion 143b (not shown) is Pa which is included inside the region Pb.

The structure is such that projection area Pa when the braking force receiving portion 143c, the slope 143d, the visor portion 143g, and the driving force receiving portion 143b are projected onto the axis L at least partially overlap the projection region Pb of the circular hole portion 143a.

As described above, according to this embodiment, the coupling 143 of the cartridge receives the driving force from the drive transmission unit 203 of the image forming apparatus main assembly. Further, the coupling 143 operates the brake mechanism (brake member 206) inside the drive transmission unit 203 in accordance with receiving the driving force from the drive transmission unit 203. The drum coupling 143 can receive the braking force by way of the braking engagement member (204, 208).

With this brake mechanism, the load required to drive the cartridge can be set in an appropriate range. As a result, the cartridge 100 can be driven stably.

It is also possible to use the drum coupling 104 and the drive transmission unit 203 of this embodiment to rotate

61

members other than the photosensitive drum **104**, such as a developing roller and a toner feeding roller. However, the drum coupling **104** and the drive transmission unit **203** of this embodiment are particularly suitable for rotation of the photosensitive drum **104**, for the following reasons.

While the cartridge **100** of this embodiment includes the photosensitive drum **104**, it is not provided with a cleaning means contacting the photosensitive drum **104**. Therefore, the torque of the photosensitive drum **104** is relatively small, and the speed of the photosensitive drum **104** tends to fluctuate when it is affected by the surroundings during rotational driving thereof. For this reason, the drive transmission unit **203** rotates the photosensitive drum **104** with a constant load applied to the drum **104**. That is, the coupling **143** not only receives the driving force for rotating the photosensitive drum, but also receives the braking force for suppressing the rotation of the photosensitive drum from the drive transmission unit **203**. By simultaneously receiving two forces acting on the coupling in different rotational directions, the speed fluctuation of the photosensitive drum (drum unit **103**) is suppressed, and the rotation is stabilized.

The driving force can be inputted from the drive transmission unit **203** of this embodiment to the cartridge provided with the cleaning means by way of the coupling **143**. When the cartridge **100** is provided with a cleaning means, (for example, a cleaning blade) which contacts the surface of the photosensitive drum to remove toner from the photosensitive drum, a frictional force is produced between the photosensitive drum and the cleaning means. This frictional force increases the torque required to rotate the photosensitive drum **104**. However, even so, the torque required to rotate the photosensitive drum **104** may not be sufficiently large. At this time, as in this embodiment, if the coupling **143** can receive the driving force and the braking force from the drive transmission unit **203** at the same time, the torque required to rotate the photosensitive drum **104** increases, and therefore, the rotation of the photosensitive drum is stabilized. A cartridge provided with a cleaning means will be described in Embodiment 2 described hereinafter.

In this embodiment, the brake mechanism for applying an appropriate rotational load to the photosensitive drum is arranged not on the cartridge side but on the main assembly side of the image forming apparatus, more particularly, in the drive transmission unit **203**. Therefore, it is not necessary to provide the brake mechanism on the process cartridge which is the object (dismountably mountable unit) to be replaced after use. It can contribute to the downsizing and cost reduction of the process cartridge.

Further, the coupling **143** has such a shape that it can smoothly engage with both the driving force applying member (drum drive coupling **180**) and the braking force applying member (braking engagement member (**204**, **208**)) provided in the drive transmission unit **203**. For example, the coupling **143** is provided with a helical slope **143d** (inclined portion, guide, upper surface, upper portion) and a visor portion **143f**, so that it can be easily connected to the drive transmission unit **203** smoothly.

Hereinafter, the shape of the coupling **143** of this embodiment will be described in detail again referring to FIG. **79**.

The coupling **143** includes two coupling portions **143s** and **143r**, and each coupling portion includes an engaging portion **143i** and a guide forming portion **143j**. The engaging portion **143i** is a shaped portion for engaging with the driving force applying member (drum drive coupling **180**) or the braking force applying member (braking engagement member (**204**, **208**)). The engaging portion **143i** forms a

62

driving force receiving portion **143b**, a braking force receiving portion **143c**, and a downstream slope **143d1**.

The driving force receiving portion **143b** and the braking force receiving portion **143c** engage with the drum drive coupling **180** and the brake members (**204**, **208**), respectively. The driving force receiving portion (first side surface, first side portion) **143b** and the braking force receiving portion (second side surface, second side portion) **143c** are formed in a planar shape, but they are not limited to such a structure. They may be a curved surface-shaped portion or a portion having a small area, as long as they can receive a driving force and a braking force, respectively. For example, the edge (ridge line) formed by the engaging portion **143i** may form the driving force receiving portion (first side surface, first side portion) **143b** or the braking force receiving portion (second side surface, second side portion) **143c**.

Alternatively, the driving force receiving portion **143b** and the braking force receiving portion **143c** may be a portion formed by a plurality of separate regions. That is, the engaging portion **143i** may be a set of a plurality of shaped portions.

The driving force receiving portion **143b** and the braking force receiving portion **143c** are an upstream side portion and a downstream side portion of the engaging portion **143i**, respectively. That is, the driving force receiving portion **143b** is a side portion directed upstream in the rotational direction, and the braking force receiving portion **143c** is a side portion directed downstream in the rotational direction.

Further, the guide forming portion **143n** is a projection (extending portion) extending in the rotational direction toward the engaging portion **143i**. The top surface (upper part) of the guide forming portion **143n** is an upstream side slope (upstream side top surface, upstream side inclined portion) **143d2**. The upstream slope **143d2** is a guide (upstream guide, upstream guide) and an inclined portion for guiding the braking force applying member (braking engagement member (**204**, **208**)) toward the engaging portion **143i**.

That is, the guide forming portion **143n** is a projection for forming the upstream side slope **143d2** which is a guide (upstream side guide).

The guide forming portion **143n** is adjacent to the engaging portion **143i** and extends from the upstream to the downstream in the rotational direction toward the engaging portion **143i**. Further, the upstream slope **143d2** of the guide forming portion **143n** is inclined so as to approach the non-driving end of the photosensitive drum from the upstream to the downstream in the rotational direction (see FIG. **80**).

In FIG. **80**, the drum coupling **143** is placed in the neighborhood of the first end portion (driving side end portion) **104a** of the photosensitive drum **104**. That is, the first end portion **104a** of the photosensitive drum **104** is the end portion on the side for receiving the driving force from the drum coupling **143**.

The end on the opposite side of the photosensitive drum **104** with respect to the first end portion **104a** is the non-driving side end (second end) **104b**. The distances from the non-driving side end portion **104b** to the upstream side slope **143d2** are indicated by D1 and D2. The distance D1 is a distance measured from the non-driving side end portion **104b** of the photosensitive drum to the downstream end of the slope **143d2** along the axial direction parallel to the axis L. The distance D2 is a distance measured along the axial direction from the non-driving side end portion **104b** of the photosensitive drum to the upstream side end portion of the upstream side slope **143d2**.

Here, the distance D1 is shorter than the distance D2. That is, when the distance from the non-driving end portion 104b of the photosensitive drum to the upstream slope 143d2 is measured along the axial direction, the distance becomes shorter toward the downstream in the rotational direction.

That is, the upstream side slope 143d2 is inclined so as to approach the non-driving side end portion 104b of the photosensitive drum toward the downstream side in the rotational direction A. Not only the upstream slope 143d2 but also the downstream slope 143d1 is inclined in the same direction.

The distances D1 and D2 can also be regarded as the distances measured along the axial direction from the non-driving side end of the cartridge casing (that is, the non-driving side cartridge cover 117: see FIG. 14) to the upstream slope 143d2.

One of the guide forming portion 143n and the engaging portion 143i may be referred to as a first shape portion, and the other may be referred to as a second shape portion or the like.

In this embodiment, the first shape portion and the second shape portion (that is, the guide forming portion 143n and the engaging portion 143i) are adjacent to each other and are connected to each other. More specifically, the downstream side of the guide forming portion 143n in the rotational direction is connected to the engaging portion 143i. However, although the engaging portion 143i and the guide forming portion 143n are adjacent to each other, they may not be connected with a gap provided therebetween.

Further, in this embodiment, the top surface (downstream side slope) 143d1 of the engaging portion 143i is smoothly connected to the top surface (upstream side slope) 143d2 of the guide forming portion 143n to provide a one slope (top surface) 143d.

That is, the top surface (downstream side slope) 143d2 of the engaging portion 143i is a part of the guides having a function of guiding the braking engagement member (204, 208) to a position where it can engage with the braking force receiving portion 143c, similarly to the upstream side slope 143d1.

The downstream slope (downstream top surface) 143d2 does not necessarily have to be continuous with the upstream slope (upstream top surface) 143d1. Examples of the non-continuous form of the upstream slope 143d2 and the downstream slope 143d1 are as shown in part (a) of FIG. 81 and part (b) of FIG. 81. In part (a) of FIG. 81 and part (b) of FIG. 81, a modified example is shown in which the upstream slope 143d2 and the downstream slope 143d1 are provided with a step, and are separated in the axial direction, and the downstream slope 143d1 is changed to a flat surface. As described above, a part of the helical slope 143d which is a guide may be flat or may have a step.

As shown in part (c) of FIG. 48, part (c) of FIG. 49, part (d) of FIG. 50, FIG. 62, FIG. 67, and FIG. 72, the braking engagement members (204, 208) are brought into contact with the slope 143d to be guided in the direction of arrow C along the inclination direction of the slope 143. That is, the braking engagement member (204, 208) moves in the direction downstream in the rotational direction toward the non-driving side of the photosensitive drum (M1B direction).

After being guided by the slope 143d, the braking engagement member (204, 208) is further advanced in the axial direction (M1B) toward the space placed downstream of the braking force receiving portion (second side surface) 143c of the drum coupling 143 (See part (d) of FIG. 48, part (d) of FIG. 49, FIG. 63, FIG. 68). As a result, the braking

engagement members (204, 208) are enabled to engage with the braking force receiving portion 143c.

The braking engagement member (204, 208) being guided by the slope 143d, the braking engagement member (204, 208) moves to the downstream side in the rotational direction A so as to be away from the drum drive coupling 180. As a result, the gap is produced between the drum drive coupling 180 and the braking engagement members (204, 208). The engaging portion 143i of the drum coupling 143 enters the gap, so that the driving force receiving portion (side surface) 143b is enabled to engage with the drum drive coupling 180 (see part (d) of FIG. 48, part (e) of FIG. 48, part (d) of FIG. 49, FIG. 63, FIG. 64, FIG. 68).

The helical slope 143d also has a function of keeping the braking engagement members (204, 208) away from the drum drive coupling 180 so that the drum drive coupling 180 and the drive force receiving portion 143b can engage with each other.

The helical slope (top surface) 143d has not only the portion (downstream side guide, downstream guide, downstream side top surface, downstream side inclined portion) 143d1 arranged between the braking force receiving portion 143c and the driving force receiving portion 143b but also has the portion (upstream guide, upstream top surface, upstream inclined portion) 143d2 on the upstream side of the driving force receiving portion 143b (see part (a) of FIG. 48, FIG. 47, FIG. 56, and so on). By enlarging the area where the slope 143d is provided, the top surface 143d can reliably guide the braking engagement members (204, 208).

That is, even when the braking engagement member (204, 208) is placed on the upstream side of the driving force receiving portion 143b (see part (a) of FIG. 49) the braking engagement members (204, 208) can be moved to the space on the downstream side of the braking force receiving portion 143c (see part (c) of FIGS. 49 and 49 (d)), by passing the upstream slope 143d2.

In this embodiment, the entire slope 143d is the inclined portion. The downstream top surface 143d1 and the upstream side top surface 143d2 are both descending slopes which descend toward the downstream in the rotational direction.

However, it is also possible to incline only a part of the slope 143d which is the top surface. For example A structure is also conceivable (see part (a) of FIG. 81 and part (b) of FIG. 81) in which, the upstream side of the top surface is inclined as the upstream side slope 143d2, as described above, whereas the downstream side of the top surface (downstream side top surface 143d2) is not inclined and is a surface perpendicular to the axis of the drum unit. In the modified example of the drum coupling shown in part (a) of FIG. 81 and part (b) of FIG. 81, the braking engagement member (204, 208) is vigorously moved by the inclination of the upstream slope (upstream top surface) 143d2, and by utilizing the inertia (momentum) of the movement, it passes the flat downstream top surface 143d1.

Further, as a guide for guiding the braking engagement members (204, 208), it is conceivable that only the upstream side top surface (upstream side slope 143d2) is used and the downstream side top surface (downstream side slope 143d1) is not used. That is, it is conceivable that there is almost no portion corresponding to the downstream top surface, or that the portion is very short as compared with the upstream top surface. Such a structure will be described hereinafter referring to FIG. 74.

It is also conceivable that there is provided a partial ascending portion in the downhill helical slope 143d. Even in such a case, if the braking engagement member (204, 208)

can be sufficiently guided downstream in the rotational direction by the slope **143d**, the slope **143d** can be deemed as a downhill slope. That is, even if the slope is partially ascending, the helical slope **143d** can be regarded as a descending slope as a whole. In other words, the distance from the non-driving end of the cartridge to the helical slope **143d** can be considered as decreasing as the helical slope **143d** moves downstream in the rotational direction.

As an example of such, a structure is conceivable in which the ascending portion partially provided in the helical slope **143d** is sufficiently shorter than the other descending portions, or the ascending slope is less steep, and therefore, the ascending portion has a small influence on the descending portion.

Further, there is a case in which the helical slope **143d** has a curved surface shape or is divided into a plurality of sections. Furthermore, there is a case in which the width of at least a part of the slope **143d** is so small that the helical slope **143d** may be regarded as a ridge line (edge) rather than a surface. The helical slope **143d** has had a sector shape (helical shape) as the drum coupling **143** is viewed from the front side. However, the shape of the guide (top surface, inclined portion) to be provided on the drum coupling **143** is not limited to such a shape. For example, instead of using a sector-shaped (helical) slope **143d**, a linearly extending rectangular slope may be used. That is, as the inclined portion (guide, top surface) corresponding to the helical slope **143d**, it is possible to use a structure having a changed shape, size, extending direction, and the like. Some of such examples will be described hereinafter referring to FIG. **54** and so on.

The upstream slope (upstream top surface) **143d2** is structured to have a region narrower than the downstream slope (downstream top surface) **143d1** (see FIGS. **47** and **56**). Conversely, the downstream slope **143d1** has a region wider than the upstream slope **143d2**.

Here, the width of each slope is a length measured along the radial direction. Further, as shown in FIG. **79**, at least a part of the engaging portion **143i** is placed more remote than the guide forming portion **143n** with respect to the axis L of the drum unit in the radial direction of the drum unit. In other words, at least a part of the engaging portion **143i** is placed radially outside the guide forming portion **143n**.

The reason for such a dimensional relationship and such an arrangement relationship is that the driving force receiving portion **143b** of the engaging portion **143i** is disposed near the boundary between the guide forming portion **143n** and the engaging portion **143i**. That is, a part of the engaging portion **143i** overhangs outward in the radial direction from the guide forming portion **143n** so that the driving force receiving portion **143b** is formed. By this, the width of the downstream portion **143d1** of the slope (top surface) **143d** is larger than that of the upstream portion **143d2**.

The driving force receiving portion **143b** has a region placed radially outside (a position far from the axis L) with respect to the upstream slope **143d2**. Further, in the axial direction of the drum unit, the driving force receiving portion **143b** is disposed closer to the non-driving side end portion of the photosensitive drum than the upstream side slope **143d2**. In FIG. **80**, a state is shown in which the distance D3 measured along the axial direction from the non-driving side end portion **104b** of the photosensitive drum to the driving force receiving portion **143b** is shorter than the distance D1 measured along the same direction to the upstream top surface **143d2** from the non-driving side end portion **104b** of the photosensitive drum.

Conversely, at least a part of the upstream slope **143d2** is placed at a distance from the driving force receiving portion **143b** than the non-driving side end portion **104b** of the photosensitive drum in the axial direction. The upstream slope **143d2** is a free end portion placed closer to the free end of the drum coupling **143** than the driving force receiving portion **143b**.

The distances D1 and D3 can be regarded as being the distances measured from the non-driving side end of the cartridge (that is, the non-driving side cartridge cover **117**: see FIG. **14**) to the upstream slope **143d2** and the driving force receiving portion **143b**, in the axial direction.

The visor portion **143d** is a block portion (stopper) which suppresses (blocks) the movement of the braking engagement member (**204**, **208**) in the axial direction. That is, the visor portion **143d** blocks the braking engagement member (**204**, **208**) from approaching the drum coupling **143** and entering the region where it cannot engage with the braking force receiving portion **143c**. FIG. **66**, part (b) of FIG. **49**, FIG. **69**, part (a) of FIG. **50** show the blocked state.

In this embodiment, the visor portion (block portion) **143d** is further upstream in the rotational direction than the upstream slope **143d2**, and the visor portion **143d** is continuous with the top surface (upstream slope **143d2**) of the guide forming portion **143n** (See part (d) of FIG. **56**).

When the braking engagement member (**204**, **208**) enters the space upstream of the driving force receiving portion **143b** or the space downstream of the braking force receiving portion **143c** together with the drum drive coupling **180**, the braking engagement member (**204**, **208**) cannot engage with the braking force receiving portion **143c**. The visor portion **143g** blocks the movement of the braking engagement members (**204**, **208**) so as to prevent the occurrence of such a state.

In this embodiment, as the drum unit is viewed from the driving side along the axial direction (see part (a) of FIG. **47**), the visor portion **143g** of the first coupling portion **143s** is disposed such that it covers the space upstream of the drive force receiving portion **143b**. Further, the visor portion **143g** is provided so as to cover the space downstream of the braking force receiving portion **143c**.

Further, the visor portion **143d** has a width sufficient to cover at least a part of the downstream side portion (downstream side slope **143d1**) of the helical slope (top surface) **143d**. By this, the visor portion **143d** constrains the braking engagement member (**204**, **208**) from non-preferably entering the space on the upstream side of the driving force receiving portion **143b** and the space downstream of the braking force receiving portion **143c** together with the drum drive coupling **180**.

On the other hand, the visor portion **143g** is disposed so as to permit the braking engagement member (**204**, **208**) to enter the space on the downstream side of the braking force receiving portion independently of the drum drive coupling **180** (See part (d) of FIG. **50**, part (c) of FIG. **49**, part (c) of FIG. **48**).

That is, the braking engagement member (**204**, **208**) contacts the upstream slope **143d2** after passing the visor portion **143g**, and is guided along the slope **143d** toward the space on the downstream side of the braking force receiving portion **143c** (See part (c) of FIG. **49** and part (d) of FIG. **50**).

That is, when the braking engagement member (**204**, **208**) is enabled to contact a portion (upstream side top surface) **143d2** of the slope (top surface) **143d**, the visor portion **143g** releases the braking engagement member (**204**, **208**) from the blocked state.

67

The visor portion **143g** is adjacent to the upstream slope **143d2** and is upstream of the upstream slope **143d2**. In this embodiment, the top surface of the visor portion **143g** and the upstream slope **143d2** are continuous, but there may be a case in which the visor portion **143g** and the upstream slope **143d2** are adjacent to each other and a gap is formed between them.

Further, the top surface of the visor portion **143g** has a plane perpendicular to the axis L of the drum unit, but the shape is not limited to this example. For example, it is conceivable that the top surface of the visor portion **143g** is inclined in the same direction as with the upstream slope **143d2**. In such case, it can be considered that the visor portion **143g** forms a part of the upstream slope **143d2**. Alternatively, it can be considered that a part of the guide forming portion **143n** forms the visor portion **143g**.

Further, in this embodiment, the coupling **143** comprises two of the helical slopes **143d**, two of the visor portions **143g**, two of the driving force receiving portions **143b**, and two of the braking force receiving portions **143c**. That is, the coupling **143** has a shape symmetrical with respect to its axis, and comprises two coupling portions **143s** and **143r** (see FIG. 58). The coupling portion **143s** and the coupling portion **143r** each have the helical slope (inclined portion) **143d** or the like as the top surfaces. Then, the braking engagement member (**204**, **208**) and the drum driving member **180** engage with the coupling portion **143s** and the coupling portion **143r** as shown in part (a) of FIG. 76.

An example (modified example) of another shape of the coupling **143** will be described hereinafter.

The drive transmission unit **203** includes the first braking engagement member **204** and the second brake engagement member **208** as the braking force applying members (braking engagement members) which apply a braking force for imparting a load to the rotation of the photosensitive drum to the coupling **143**. There is a gap between the first braking engagement member and the second braking engagement member **208**, and the second braking engagement member provided radially inward is flexible slightly to move outward so as to approach to the first braking engagement member **204**. When the coupling and the drive transmission unit **203** are disengaged from each other, the second braking engagement member **208** can smoothly break the engagement with the coupling **143** by the flexing of the second braking engagement member **208**. For example, the second braking engagement member **208** can move over the visor portion **143g** by flexing and can be separated from the coupling **143**. [Various Modifications of Coupling and Cartridge Shown in Embodiment 1]

Modified examples (modified shape) in which the drum coupling **143** of the Embodiment 1 described above is partially modified will be described. Even when the above-described the visor portion **143g** is not provided on the drum coupling **143**, it can function properly, depending on the conditions.

FIG. 52 shows a perspective view of the drum coupling **143** in which the visor portion **143g** is not provided, and FIG. 53 shows a developed view illustrating the process of engagement.

The shape will be described referring to FIG. 52. FIG. 52 is a view illustrating one end of the drum unit, and shows a state in which the coupling member (drum coupling) **143** is mounted to the end portion of the photosensitive drum **104**. The drum coupling **143** includes the helical slope **143d** and a push-back surface **143k**, which will be described hereinafter, but does not have a visor shape.

68

Subsequently, the process of engaging with the drive transmission unit **203** will be described referring to FIG. 53.

The representation of the development view of FIG. 53 is the same as with the development view of FIG. 48. The drum coupling **143** comprises two coupling portions **143s** and **143r**, but only the coupling portion **143s** will be described for the sake of simplicity of explanation. The description of the coupling portion **143s** also applies to the coupling portion **143r**.

The case where the phases of the inclination start portion **143f** of the drum coupling **143** shown in part (a) of FIG. 53 and the inward projection **208e** of the second braking engagement member satisfy the following relationship will be described. That is, a case where the inclination start portion **146f** of the drum coupling **143** is on the downstream side in the rotational direction (arrow A) will be described.

Part (a) of FIG. 53 shows a state in which the drive transmission surface **180d** of the drive transmission unit **203** and the second braking engagement member **208** are close to each other.

Next, in part (b) of FIG. 53, since there is no such visor portion as described in embodiment 1, in the drum coupling **143**, the drum drive coupling and the second braking engagement member **208** advance into the space between the push-back surface **143k** and the helical slope **143d3**.

Part (c) of FIG. 53 shows a state in which the drive transmission unit **203** has started to rotate in the rotational direction A. When the drum drive coupling **180** and the second braking engagement member **208** rotate, the second braking engagement member **208** moves in the direction of arrow E along the slope by the function of the inclination $\theta 1$ of the push-back surface **143k** or the function of the inclination $\theta 2$ of the second braking engagement member **208**. As described referring to FIG. 48, the second braking engagement member **208** can rotate without receiving a rotational load.

As described above, when the braking engagement member (**204**, **208**) enters the region where it cannot engage with the braking force receiving portion, the push-back surface (push-back portion) **143k** applies a force to the second braking engagement member **208**. By this, the push-back surface **143k** pushes back the braking engagement members (**204**, **208**) toward the inside of the drive transmission unit **203** and moves it in the direction of arrow E.

However, the second braking engagement member **208** is urged by the spring **211** shown in FIG. 43 in the M1B direction in the Figure, and if the component force of the inclination $\theta 2$ of the second braking engagement member **208** is smaller than the spring force F1, the second braking engagement member **208** cannot be moved in the direction of arrow E. The component force changes depending on the load torque of the drum holding unit **108** and the angle of each slope ($\theta 1$ or $\theta 2$). It is preferable to set the magnitude relation of the force within the range in which the above function is performed in consideration of the component force and the frictional force.

Part (d) of FIG. 53 shows the movement of the second braking engagement member **208** which is no longer subjected to the rotational load. The drive transmission unit **203** has further rotated, and the second braking engagement member **208** is in a state of passing the inclination start portion **146f** of the drum coupling **146**. In this state reached, the second braking engagement member **208** moves in the direction of arrow C as described referring to part (c) of FIG. 48. The operation after this is the same as described above, and therefore, the description thereof will be omitted.

Although not shown in part (a) of FIG. 50 to part (d) of FIG. 50, the first braking engagement member 204 also moves together with the second braking engagement member 208 in these processes.

In the drum coupling 143 shown in the Embodiment 1 (see part (a) of FIG. 1, the braking engagement member (204, 208) is blocked by the visor portion 143g from entering the region in which it cannot engage with the braking force receiving portion. On the other hand, in the drum coupling 143 of this modified example, when the braking engagement member (204, 208) enters the region where the braking force receiving portion 143c cannot be engaged with the drum drive coupling 180, the braking engagement member (204, 208) is pushed back by the push-back surface (push-back) 143k. The push-back surface 143k is an inclined portion inclined in a direction different from that of the helical slope 143. More particularly, the helical slope 143 is a portion which inclines toward the non-driving side of the drum unit as it goes downstream in the rotational direction, whereas the push-back surface 143k is a portion of the drum unit which inclines toward the outside, that is, away from the non-driving side end portion 104b (see FIG. 80) of the photosensitive drum, as it goes downstream in the rotational direction A. If the helical slope 143 is regarded as a descending slope, the push-back surface 143k is an ascending slope. The push-back surface 143k is placed on the upstream side in the rotational direction with respect to the helical slope 143d, and is adjacent to the helical slope 43k.

The push-back surface 143k is also a guide (second guide) for guiding the braking engagement member (204, 208) toward the helical slope 143d. Further, the push-back surface 134k is a helical slope (second helical slope, second inclined portion) having a direction of inclination opposite to that of the helical slope 143d.

Further, another modified shape of the drum coupling 143 will be described. The inclined portion and the top surface (helical slope 143d) as the guide described in the Embodiment 1 are formed as smooth slopes, and guide the braking engagement members (204, 208) along such slope surfaces (See FIG. 56 and the like). However, the drum coupling 143 can also function even if the inclined portion has other shapes. An example thereof is shown in FIG. 54 in a perspective view.

First, the shape shown in part (a) of FIG. 54 is a reproduction of the shape described in the Embodiment 1. A gentle helical slope 143d is formed from the inclined starting portion 143f toward the braking force receiving portion 143c.

On the other hand, the shapes of part (b) of FIG. 54 and part (a) of FIG. 73 show modified examples. The height changes stepwise between the inclination start portion 147f and the braking force receiving portion 147c. That is, the top surface (inclined portion) has a stepped portion 147d, and the inclined portion is formed by the plurality of steps. Thus, the inclined portion (top surface) may not be a helical slope but may be a helical step shape providing an inclination which lowers in the direction of advancement of the second braking engagement member 208.

The stepped step portion 147d moves the second braking engagement member 208 by moving the stepped step portion 147d in the direction of the arrow C in part (a) of FIG. 73, whereby the same function as that of the helical slope 143d in part (a) of FIG. 54 is performed. While the inclined surface 143d is an inclined portion comprising continuously

inclined surfaces, the stepped portion 147d can be regarded as an inclined portion provided by stepwise structure of a plurality of surfaces.

If it is difficult to form a helical slope 143d on the coupling 143 due to restrictions on the structure of the mold for manufacturing the coupling 143, a stepped portion 147d may be used instead of the inclined surface 143d.

At this time, it is preferable that when the stepped portion 147d, which is the top surface, and the second braking engagement member 208 come into contact with each other, the second braking engagement member 208 is structured to be smoothly guided without being caught by the stepped portion 147d. For example, it is conceivable to sufficiently narrow the width of each surface of the stepped portion 147d. Further, in part (a) of FIG. 73, the top surface (inclined portion, guide) is formed in a stepped shape by combining a plurality of surfaces, but the top surface (inclined portion, guide) may be formed by combining a plurality of curved surfaces, and a similar function can be performed with such a structure. Similarly to the inclined surface 143d, the stepped portion 147d is a guide (inclined portion) for guiding the braking engagement member (204, 208) toward the braking force receiving portion by its own inclination.

Further, as shown in part (c) of FIG. 54 and part (b) of FIG. 73, the top surface is divided into an inclined surface (upstream side top surface, downstream side top surface) 148d1 and an inclined surface (downstream side top surface, downstream side guide, downstream side) 148d2 with a gap 148g therebetween. Also in this case, if the second braking engagement member 208 has such a shape that does not cause catching when it comes into contact with the top surface (148d1, 148d2), the top surface (148d1, 148d2) can function as a guide. Such a coupling can be used when there is a restriction in the structure of the mold for molding the coupling.

Further, part (d) of FIG. 54 and part (c) of FIG. 73 show a modified example in which the shape of each portion of the coupling 143 is formed by ribs. The top surface (inclined surface 149d) comprises the surfaces of a plurality of ribs 149p, and the top surface is divided into a plurality of ribs, and in such a case, the same function can be provided as well. That is, as shown in part (c) of FIG. 73, the guide forming portion 149n forming the upstream side top surface (upstream side guide, upstream side inclined portion) 149d2 is a projection (rib) projecting in the radial direction. Depending on the characteristics of the material used, it can be used when it is necessary to produce ribs without producing thick portions.

That is, with each structure of part (a) of FIG. 54 to part (d) of FIG. 54, each top surface (143d, 147f, 148d1, 148d2, 149d) guides the braking force of the braking engagement member (204, 208) toward the braking force receiving portion 143c regardless of its shape. In other words, each top surface is a guide (inclined portion) for guiding the braking engagement member (204, 208) toward the braking force receiving portion 143c regardless of its shape. At least a part of such a top surface (guide) is formed by the guide forming portion 143n.

Similar to the top surface, the push-back surface (push-back portion) 143k shown in FIG. 52 may have various shapes. For example, the push-back portion (push-back surface) 143k of this modification is a smoothly continuous helical slope, but the push-back portion may be inclined by a plurality of surfaces or steps. For example, the push-back portion 143k may be two surfaces including different inclinations, as in the push-back portion 143k of the Embodiment 1 shown in part (b) of FIG. 48 and part (d) of FIG. 56.

Further, although the push-back surface **143k** is ascending, a descending portion may be locally provided.

The drum coupling **143** may have either the visor portion **143g** or the push-back surface (push-back portion) **143k**, or may have both of them. As described above, the drum coupling **143** of the Embodiment 1 shown in part (b) of FIG. **48**, part (b) of FIG. **55** and part (d) of FIG. **56** has a structure in which not only the visor portion **143g** but also the push-back surface **143k** is provided. Normally, the drum coupling **143** can block improper entry and access of the braking engagement member (**204, 208**) by the visor portion **143g**, but in the unlikely event that it cannot be blocked, the push-back surface **143k** can function to push back the braking engagement members (**204, 208**) away from the coupling **143**.

The drum coupling **143** has a projection shape (push-back portion forming portion, second guide forming portion) **143m** that constitutes the push-back surface **143k** (see part (b) of FIG. **79** and part (c) of FIG. **79**).

The engaging portion **143i**, the guide forming portion **143n**, the projection shape **143m**, and the visor portion **143g** (see FIG. **79**) may be referred to as the first, second, third, and fourth shape portions in no particular order correspondence.

Referring to part (e) of FIG. **54** and part (d) of FIG. **73**, a modified example of the braking force receiving portion (second side surface) will be shown.

The braking force receiving portion **143c** described in Embodiment 1 shown in part (a) of FIG. **54** and part (a) of FIG. **1** and FIGS. **55** to **57**, and the other modified examples shown in FIG. **52** and part (b) of FIG. **54** to part (d) of FIG. **54** has a shape overhanging downstream in the rotational direction. This is because by the braking force receiving portion **143c** having a shape overhanging toward the downstream side in the rotational direction, the stability of engagement is increased when it is engaged with the braking engagement members (**204, 208**).

That is, because of this shape, when the braking force receiving portion **143c** engages with the braking engagement member (**204, 208**), a force is generated so as to attract then toward each other. The braking force receiving portion **143c** overhangs toward the downstream side in the rotational direction. Therefore, when the braking force engaging member (**204, 208**) contacts the braking force receiving portion **143ca** force is produced so that the braking force engaging member (**204, 208**) is attracted inward in the axial direction toward the drum coupling **143** or the photosensitive drum **104**. By this, the engaging state between the braking force receiving portion **143c** and the braking force engaging member (**204, 208**) is stabilized, and the engagement is not easily broken.

As described above, the braking engagement member (**204, 208**) is structured to be movable in the axial direction relative to the drum drive coupling **180** (see FIGS. **67** and **68**). However, if the braking engagement member (**204, 208**) moves in the axial direction while the drive transmission unit **203** is driving the drum coupling **143** there is a possibility that the engaged state with the braking force receiving portion **143c** is broken or becomes unstable. Therefore, it is preferable that the braking force receiving portion **143c** has a shape for stabilizing the engagement state with the braking engagement member (**204, 208**) to suppress the movement of the braking engagement member (**204, 208**) in the axial direction when the drum coupling **143** is driven.

However, when the braking force required to be applied to the braking force receiving portion is small, or when the

friction coefficient of the braking force receiving portion is high, the engagement between the braking force receiving portion and the braking engagement member (**204, 208**) tends to be stable. Therefore, it is possible to eliminate the overhang portion of the braking force receiving portion. Such a braking force receiving portion **144t** is shown in part (e) of FIG. **54** and part (d) of FIG. **73**. In the modified drum coupling shown in part (e) of FIG. **54** Figure and **73(d)**, the braking force receiving portion **144c** does not overhang toward the downstream side in the rotational direction (arrow A).

On the other hand, it is also conceivable to devise a device for stabilizing the engagement state with the braking engagement member (**204, 208**) even for the braking force receiving portion **144c** including such a shape.

In order to stabilize the engagement between the braking force receiving portion **144c** and the braking engagement member, It is also conceivable that an elastic member (elastic portion) **144t**, for example such as rubber is attached to the braking force receiving portion **144c**, or the elastic portion is integrally molded with to the braking force receiving portion **144c**. By increasing the friction coefficient of the braking force receiving portion **144t** or causing the braking engagement member (**204, 208**) to bite into the elastic portion of the braking force receiving portion **144t**, the engagement with the braking engagement member (**204, 208**) is less likely to break so that the engagement can be stabilized.

As a method of increasing the frictional force of the braking force receiving portion **144c**, it is conceivable to use an adhesive member (adhesive member) instead of using the elastic member **144t**. For example, if a double-sided tape (adhesive member) is attached to the surface of the braking force receiving portion **144c**, the frictional force between the braking force receiving portion **144c** and the braking engagement member (**204, 208**) increases due to the viscosity of the double-sided tape (adhesive member). In addition, it is conceivable to increase the friction coefficient of the braking force receiving portion **144c** by surface-treatment of braking force receiving portion **144c** without using the elastic member **144t**.

It is desirable that the helical slope **143d** (see FIG. **67**) for guiding the braking engagement member (**204, 208**) has a small friction coefficient in order to achieve smooth guiding. Therefore, even when a material having a high coefficient of friction is selected or surface treatment is applied to the braking force receiving portion **144c**, it is desirable that such a means is not used for the entire coupling, but the use of such material or such surface treatment is not applied to the helical slope **143d**. That is, it is desirable that the friction coefficient of the braking force receiving portion **144c** is higher than the friction coefficient of the helical slope **143d**.

The elastic portion **144t** may be provided on the braking force receiving portion **143c** of the drum coupling **143** as shown in part (a) of FIG. **54** to part (d) of FIG. **54**.

Next, referring to FIG. **101**, a preferable arrangement relationship and dimensional relationship of the drum coupling **143** will be described. FIG. **101** is a front view of the drum coupling **143** of the Embodiment 1, in which θ (theta) **11** is a value indicating the dimension of the engaging portion **143i** from the driving force receiving portion **143b** to the braking force receiving portion **143c** by an angle from the axis of the drum coupling. In other words, it is the angle of the region of the downstream inclined portion **143d1**.

Regarding the upper limit of θ **11**, it is desirable that θ **11** is 90° or less, more preferably 80° or less. The angle θ **11** corresponds to the gap created between the drum drive

73

coupling **180** and the braking engagement members (**204**, **208**) when the drum coupling engages the drive transmission unit **203** (see FIG. **64**). In order to securely sandwich the driving force receiving portion **143b** and the braking force receiving portion **143c** between the braking engagement members (**204**, **208**) and the drum drive coupling **180** of the apparatus main assembly, It is desirable that $\theta 11$ is 90° or less, more preferably 80° or less.

On the other hand, regarding the lower limit of $\theta 11$, if the strength of the engaging portion **143i** is increased by using metal as for the material of the engaging portion **143i** constituting the driving force receiving portion **143b** and the braking force receiving portion **143c**, the $\theta 11$ can be reduced. Although the details will be described hereinafter, in the modified example of the drum coupling shown in FIG. **74**, the thickness of the engaging portion **145i** corresponding to the engaging portion **143i** is made smaller than that in this embodiment, by forming the drum coupling **143** with metal. Considering such a structure, the preferable condition for the lower limit of $\theta 11$ (FIG. **101**) is that $\theta 11$ is 1° , more preferably 2° or still more preferably 8° or more. In this embodiment, $\theta 11$ is set to 30° or more, and $\theta 11$ is set to about 35° .

In order to increase the strength of the driving force receiving portion **143b** and the braking force receiving portion **143c** so that the force can be stably received, the angle $\theta 11$ corresponding to the thickness of the engaging portion **143i** is desirably in a certain range.

When $\theta 11$ is converted into a length, it becomes the thickness of the engaging portion **143i**, that is, the distance measured from the driving force receiving portion **143b** to the braking force receiving portion **143c** along the rotational direction. The desired range of this distance is 0.3 mm or more, more preferably 1 mm or more.

Further, in FIG. **101**, $\theta 12$ indicates a region occupied by the upstream slope (upstream guide, upstream slope) **143d2** by an angle. Regarding the lower limit of $\theta 12$, it is desirable that the value of $\theta 12$ is at least half the value of $\theta 11$, and more preferably the value of $\theta 12$ is not less than the value of $\theta 11$. This is because the upstream slope **143d2** needs to have a length in the rotational direction to the extent necessary for guiding the braking engagement member (**204**, **208**) to the braking force receiving portion **143c** by the upstream slope **143d2**.

As $\theta 11$ is smaller and the inclination angle of the upstream slope **143d2** is larger, the lower limit of $\theta 12$ can be made smaller.

As described above, the lower limit of $\theta 12$ depends on the value of $\theta 11$ and the angle of the upstream slope **143d2**, but when expressed numerically, $\theta 12$ is $^\circ$ or more, more preferably 2° or still more preferably 8° or more, even more preferably 30° or more. In this embodiment, $\theta 12$ is set to be 60° or more.

The upper limit of $\theta 12$ can be relatively large and can exceed 360° . However, preferably, $\theta 12$ is 360° or less, more preferably 270° or less, and it is 180° or less in this example. Specifically, $\theta 12$ is set to be approximately 67° .

A structure in which $\theta 12$ is larger than that of this embodiment will be described hereinafter referring to FIGS. **102** and **103**.

Angle $\theta 13$ is the sum of $\theta 11$ and $\theta 12$, and corresponds to the angle occupied by the entire helical slope **143d**. When $\theta 13$ is expressed numerically, it is desirable that $\theta 13$ is 2° or more, and more preferably 8° or more. Further, $\theta 13$ is preferably 360° or less, and more preferably 270° or less. In this embodiment, $\theta 13$ is set to 180° or less. Specifically, $\theta 13$ is set to be approximately 102° .

74

Referring to FIG. **74**, the shape of another modification of the coupling **143** will be described.

FIG. **74** is a perspective view and a front view as seen in two line-of-sight directions of the coupling in the modified example.

The coupling **143** of this modification includes an engaging portion **145i** including a driving force receiving portion **143b** and a braking force receiving portion **145b**, and a guide forming portion **145n** having a helical slope **145d**. The engaging portion **145i** and the guide forming portion **145n** correspond to the engaging portion **143i** and the guide forming portion **143n** of the coupling **143** shown in the Embodiment 1 (see FIG. **79**), but their shapes are partially different.

The coupling **143** of this modification includes the visor portion **143g** contacting the second braking engagement member **208** (not shown), and the helical slope **145d** is formed by a curved surface. This curved surface has a substantially arc shape, and is shaped so as to connect the braking force receiving portion **145c** from the inclination start point **143f**. In this modified example, since the braking force receiving portion **145c** does not have a shape overhanging to the downstream side in the rotational direction, the elastic member (elastic portion) **145i** may be attached to the braking force receiving portion **145c** as in the case of part (e) of FIG. **54**.

The helical slope **145d** in this modification (FIG. **74**) is a top surface corresponding to the upstream slope **143d2** of Embodiment 1 (FIG. **57**).

On the other hand, in this modification (FIG. **74**), the top surface (upper part) **145e** (part (b) of FIG. **74**) of the engaging portion **145i** corresponds to the downstream slope **143d1** of the Embodiment 1 (FIG. **57**), but it is not inclined unlike the downstream side slope **143d1**.

That is, the top surface **145e** provided downstream is connected to the top surface (helical slope **145d**) provided upstream, but the inclination angles of the surfaces thereof are different at the boundary. The top surface **145e** and the helical slope **145d** are not smoothly connected.

Further, since the distance between the driving force receiving portion **143b** and the braking force receiving portion **145c** is short, the length of the top surface **145e** measured along the rotational direction is smaller (shorter) than the length of the downstream slope **143d1** in FIG. **57**. Further, as described above, the top surface **145e** is not inclined. In this modification, it can be considered that the top surface **145e** is not used as a guide.

However, even with such a structure, the helical slope **145d**, which is a guide (inclined portion), can guide the braking engagement member (**204**, **208**) toward the braking force receiving portion **145c**.

A plane **145h** is adjacent to the upstream of the helical slope **145d**, and the helical slope **145d** and the plane **145h** are connected to each other. The plane **145h** can be inclined in the same direction as the helical slope **145d** to form a part of the helical slope **145d**. Further, the drum coupling of this modification may have the visor portion **143g** of the push-back surface **143k** described in embodiment 1 or another modification of the Embodiment 1 (see FIGS. **1**, **52**, and so on).

Further, regarding the shape of the drum coupling, the shape of the shaft portion **143j** shown in FIG. **1** can also be selected in view of design reasons. For example, FIG. **75** shows a shape of a modified example of the drum coupling. In the example of FIG. **75**, the diameter of the shaft portion **146j** is the same as the diameter of the photosensitive drum **104**. The shaft portion **146j** is rotatably supported by a

75

driving side cartridge cover member **116** (see FIG. **15**). The position restriction in the direction of the arrow MB1 can be performed using the shaft end surface **146s**, for example. In this manner, the shape of the shaft portion **146j** can be appropriately selected depending on the relationship with the peripheral portions and the manufacturing method.

Another modification of the drum coupling **143** is shown in part (b) of FIG. **76**, part (c) of FIG. **76**, part (a) of FIG. **78**, part (b) of FIG. **78**, part (c) of FIG. **78**, and part (d) of FIG. **78**. These Figures show drum couplings in which two coupling portions **143s** and **143r** have different shapes. Part (b) and (c) FIG. **76** are development views of the coupling **143**, and in part (c) of FIG. **76**, the drum drive coupling **180** and the braking engagement member **208** provided in the device main assembly side are also shown in the development view. Part (a) of FIG. **78** and part (b) of FIG. **78** are perspective views of the drum coupling **143**. Further, part (c) of FIG. **78** and part (d) of FIG. **78** show the engagement state of the braking engagement member (**204**, **208**) and the drum drive coupling with respect to the drum coupling **143**.

In the coupling **143** shown in these Figures, the engaging portion **143i** of one coupling portion **143s** is not provided with the braking force receiving portion **143c**, but includes only the driving force receiving portion **143b**. That is, the side surface **143y** provided on the engaging portion **143i** of the coupling portion **143s** does not engage with the braking engagement member (**204**, **208**). On the other hand, the engaging portion **143i** of the other coupling portion **143r** is provided only the braking force receiving portion **143c** and is not provided with the driving force receiving portion **143b**. The side surface **143x** of the engaging portion **143i** of the coupling portion **143r** does not engage with the drum drive coupling **180**.

An example of another asymmetrical coupling **143** is shown in part (d) of FIG. **76**. This coupling portion **143s** is an example in which the coupling portion **143s** does not have any side surface corresponding to the driving force receiving portion **143c**.

The modified example of the coupling **143** shown in part (b) of FIG. **76**, part (c) of FIG. **76**, part (a) of FIG. **78**, part (b) of FIG. **78**, part (c) of FIG. **78**, and FIG. **7** is a(d) receives a driving force at only one place and receives the braking force at only one place. Therefore, in order for the drum coupling to stably receive the driving force and the braking force, it is preferable to improve the fitting accuracy between the circular hole portion **143a** and the positioning boss **180i** of the drum drive coupling **180** (see FIG. **51**). That is, it is preferable to reduce the gap produced between them, thus improving, the positional accuracy of the drum coupling **143** relative to the drive transmission unit **203**, to stably and surely engage the drive transmission unit **203** and the drum coupling **143**.

Further, FIG. **77** shows another modification of the drum coupling including one driving force receiving portion and one braking force receiving portion. The drum coupling **143** shown in FIG. **77** has only one upstream side slope **143d2**, only one downstream side slope **143d1**, only one visor portion **143g**, only one driving force receiving portion **143b**, only one braking force receiving portion **143c**, and only one extrusion surface **143k**. Part (a) of FIG. **77** is a perspective view of the drum coupling, and part (b) of FIG. **77** is a front view thereof.

In the modified example of the drum coupling **143** as shown in FIG. **77**, arbitrary portions of the slope **143d**, the visor portion **143g**, the driving force receiving portion **143b**,

76

the braking force receiving portion **143c**, and the extrusion surface **143k** may be placed at a 180° position or positions (axisymmetrical).

For example, as shown in FIG. **96**, the drum coupling **143** visor portion **143g** shown in FIG. **77** may be moved to the 180° symmetric region **S143g**, or the extrusion surface **143k** may be moved to the symmetric region **S143k**.

This is because the drum drive coupling **180** and the braking engagement members (**204**, **208**) both have 180° symmetrical shape.

Therefore, regardless of which one of the two 180° symmetrical places is the place where one helical slope **143d** is disposed, the slope **143d** can act on the entire braking engagement member (**204**, **208**). Similarly, the extrusion surface **143k** may be placed at either of the two places which are ° symmetrical with respect to each other. The same applies not only to the visor portion **143g** and the extrusion surface **143k**, but also to the braking force receiving portion **143c**.

Further, the drum drive coupling **180** can engage with the drive force receiving portion **143b** regardless of whether the drive force receiving portion **143b** is placed at either of two 180° symmetrical positions.

The drum drive coupling **180** has two drive transmission surfaces **180d**, but the two drive transmission surfaces **180d** move integrally (part (a) of FIG. **45**). Further, the braking engagement members (**204**, **208**) have two coupling engaging portions **204b** and two each, and all of these coupling engaging portions move integrally (see part (b) of FIG. **45**).

As another modification in which the shape of the drum coupling **143** is made asymmetrical as described above, there is also a follow structure. That is, one coupling portion **143s** has an engaging portion **143i** but does not have a guide forming portion **143n**, and the other coupling portion **143r** has a guide forming portion **143n** but does not have an engaging portion **143i**. Such a structure is conceivable. Examples of such a structure are shown in parts (a) and (b) of FIG. **97**. Part (a) of FIG. **97** is a perspective view of a modified example of the drum coupling, and part (b) of FIG. **97** is a front view thereof.

In the modified example of the drum coupling shown in these Figures, the guide forming portion **343n** and the engaging portion **343i** have one. The guide forming portion **343n** forms a helical slope (guide, top surface, inclined portion) **343d2**. The engaging portion **343i** forms a driving force receiving portion **343b** and a helical slope (guide, top surface, inclined portion) **343d1**. The guide forming portion **343n** and the engaging portion **343i** are located on opposite sides of the axis L. Further, in this modification, the braking force receiving portion **343b** is not arranged at the engaging portion **343i**, but is arranged at the end portion downstream of the guide forming portion **343n** in the rotational direction. That is, the engaging portion **343i** engages with the driving force applying member (drum drive coupling) **180**, but does not engage with the braking force applying member (braking engagement members **204**, **208**).

Part (a) of FIG. **99**, (b), and (c) show the engagement process of the drum coupling and the braking engagement member (**204**, **208**) of this modified example in this order. For the sake of explanation, the drum drive coupling **180** of the drive transmission unit **203** is not shown.

As shown in part (a) of FIG. **99**, when the second braking engagement member **208** comes into contact with the slope **343d2** of the guide forming portion **343n**, the second braking engagement member **208** is on the downstream side in the rotational direction and in the axial direction. The movement is started so as to approach the photosensitive drum **104**.

As shown in part (b) of FIG. 99, when the second braking engagement member 208 reaches the neighborhood of the end of the upstream slope 343d2, the first braking engagement member 204 is brought into contact with the slope 343d1 which is the top surface of the engaging portion 343i. Thereafter, the braking engagement members (204, 208) continue to rotate, and, the free end of the first braking engagement member 204 enters the space downstream of the engaging portion 343i, as shown in part (c) of FIG. 99. The first braking engagement member 204 reaches a position where it can engage with the braking force receiving portion 343c (see part (b) of FIG. 97).

As described above, also in the drum coupling of the present modification shown in FIGS. 97 and 99, any portion thereof can be shifted to a 180° symmetrical position. For example, as shown in part (a) of FIG. 98, the engaging portion 343i and the driving force receiving portion 343b can be shifted to the positions S343i and S343b which are 180° symmetrical positions, respectively. The coupling in which the engaging portion 343i is shifted to S343i, is similar to the modified example of the drum coupling shown in FIG. 77. Conversely, when a portion of the drum coupling portion shown in FIG. 77 is shifted to a position symmetrical by 180°, the shape is similar to that of the drum coupling of this modification shown in FIG. 97.

As shown in part (a) of FIG. 98, in this modification, when the engaging portion 343i is imaginarily placed at the 180° symmetrical position S343i, the slope 343d2 is adjacent to the imaginarily arranged engaging portion S343i. The upstream side portion 343d2a of the slope 343d2 extends from the upstream to the downstream in the rotational direction toward the imaginarily arranged engaging portion S343i and the imaginarily arranged driving force receiving portion S343b.

Part (b) of FIG. 98 shows the angles $\theta 41$, $\theta 42$, $\theta 51$, and $\theta 52$ regarding the dimensions of each portion in this modification.

Angle $\theta 41$ is the angle of the region where the engaging portion 343i is arranged. $\theta 42$ is the angle of the region occupied by the helical slope 343d2 of the guide forming portion 343n. $\theta 51$ is an angle indicating a region from S343b in which the driving force receiving portion 343b is imaginarily arranged at 180° symmetrical positions to the braking force receiving portion 343c. $\theta 52$ is the angle of the region occupied by the portion 343d2a located on the helical slope 343d2 on the upstream side in the rotational direction from the position S343b of the imaginarily arranged driving force receiving portion.

Angle $\theta 41$ is preferably not less than 1°, further preferably not less than 2°, and even further preferably not less than 8°, from the stand point of assuring the strength of the driving force receiving portion 343b.

Angle $\theta 51$ corresponds to the angle of the gap between the braking engagement member (204, 208) and the drum drive coupling 180. Therefore, it is desirably not more than 80° as described above.

Further, since $\theta 51$ is larger than $\theta 41$, $\theta 51$ is preferably 1° or more, further preferably 2° or more, and even further preferably 8° or more. Furthermore, it is desirable that $\theta 41$ is 80° or less.

Angle $\theta 52$ is an angle corresponding to $\theta 12$ in FIG. 101, and the preferred range of $\theta 52$ is the same as that of $\theta 12$. Further, since $\theta 42$ is an angle corresponding to $\theta 13$ in FIG. 101, the preferable range of $\theta 42$ is the same as that of $\theta 13$.

Further, another modification of the asymmetrically shaped drum coupling is shown in part (a) of FIG. 100 and part (b) of FIG. 100. The structure is such that the upstream

slope 143d2 of the Embodiment 1 (see FIG. 58 and the like) is divided and arranged at two places. That is, the upstream slope 143d2 is divided into an upstream portion 143d2a and a downstream portion 143d2b. The engaging portion 143i is adjacent to the downstream portion 143d2b of the upstream side slope 143d2.

The dimensional relationship in this modified example is shown in part (b) of FIG. 100. The angle $\theta 21$ is the angle of the engaging portion 143i and corresponds to the angle $\theta 11$ in FIG. 101. The preferred angle of $\theta 21$ is the same as the angle $\theta 11$. $\theta 22b$ is an angle of the range occupied by the downstream portion 143d2b of the upstream side slope 143d2, and $\theta 22a$ is an angle occupied by the upstream portion 143d2a of the upstream side slope 143d2.

The region in which the downstream portion 143d2b of the upstream slope 143d2 is imaginarily moved to a position 180° symmetrical is the region S143d2b. At this time, the angle of the region occupied by the virtual region S143d2b and the upstream portion 143d2a is $\theta 32$. Since $\theta 32$ corresponds to the angle $\theta 12$ in FIG. 101, the preferred angle range of $\theta 32$ is equivalent to the preferred angle range of $\theta 12$.

The range of suitable angles of $\theta 22a$ and $\theta 22b$ is also based on $\theta 12$.

Further, a further modification of the drum coupling will be described. The helical slope 143d and the upstream slope 143d2 as the guide and the upstream guide can be changed to be longer than those the drum coupling of the Embodiment 1 (FIG. 1 and so on). Such an example is shown in FIGS. 102 and 103. In the drum couplings shown in these Figures, the helical slope 443d2 corresponding to the upstream slope 143d2 is extended to exceed 360°. That is, the helical slope 443d2 is extended more than one full circumference.

The engaging portion 443i corresponding to the engaging portion 143i of the Embodiment 1 is provided separately from the slope 443d2. The engaging portion 443i includes a braking force receiving portion 443c1 and a driving force receiving portion 443b. The braking force receiving portion 443c2 is also provided in the neighborhood of the end of the helical slope 443d2. The braking force receiving portion 443c1 and the braking force receiving portion 443c2 are arranged at positions 180° symmetrical.

In part (a) of FIG. 103, part (b) of FIG. 103, and part (c) of FIG. 103, the engagement process of the drum coupling and the braking engagement member in this modified example are shown in chronological order. The drum drive coupling 180 is not shown for the sake of illustration.

As illustrated in FIG. 103, the braking engagement members (204, 208) rotate one or more turns by being guided by the helical slope 443d2. In this manner, it is possible to increase the length of the helical slope 443d2, which is the guide and the inclined portion, beyond 360°. However, if the helical slope 443d2 is long, the time required for the braking engagement member (204, 208) to pass through the helical slope 443d2 is long, or the speed of the braking engagement member (204, 208) on the helical slope 443d2 is slow, as the case may be. In order to deal with this, when the drive transmission unit 203 and the coupling 143 are engaged with each other it may be necessary to take measures to secure sufficient time for the braking engagement member (204, 208) to pass the helical slope 443d2, by decreasing the rotation speed of the drive transmission 203, for example.

In order to smoothly engage the drive transmission unit 203 and the drum coupling 143 with each other while rotating the drive transmission unit 203 at high speed It is desirable to shorten the time required for the braking

engagement members (204, 208) to pass in the helical slope 443d2. From that standpoint, it is further preferable that the length of the helical slope (inclined portion, guide) 443d2 is 360° or less, and it is further preferable that the length is 270° or less.

As described above, it is also possible to use a modified example in which the drum coupling of the Embodiment 1 is changed to an asymmetrical shape.

However, as in the drum coupling 143 of the Embodiment 1 shown in FIGS. 1 and 58, It is further preferable that the coupling 143 includes the driving force receiving portion 143b and the braking force receiving portion 183c at 180° apart two positions, because then the engagement state of the drive transmission unit 203 with the coupling 143 and the transmission state of the drive force are stabilized. The coupling 143 receives the driving force at two symmetrically arranged points, and the braking force is also received at two symmetrically arranged points. Therefore, it becomes easy to maintain the balance of the force applied to the coupling 143.

Further, in the drum coupling 143 (see FIG. 1) of the Embodiment 1 described above, each shaped portion (engagement portion, guide forming portion, visor portion, and so on) of the coupling has a specific arrangement relationship. However, it is also conceivable to change these arrangement relationships by making any portion of the coupling 143 movable.

As an example of such a structure, FIGS. 104 to 106 show a structure in which the engaging portion 243i is movable relative to other portions of the drum coupling 143. And specifically, a structure in which the engaging portion 243i can advance and retract in the radial direction. As shown in FIG. 105, the drum coupling 143 is provided with two openings 243p, and the engaging portion 243i is partially exposed from the inside of the drum coupling through these openings 243p.

As shown in part (a) of FIG. 105, the two engaging portions 243i are supported by a guide 199a of a support member 199 provided inside the drum coupling. Further, in addition, the engaging portion 243i is structured to be movable in the radial direction along the guide 199a, but is urged inward in the radial direction by the tension spring 200.

Therefore, when the cartridge is not used, the two engaging portions 243i are retracted inside the drum coupling as shown in part (a) of FIG. 104 and part (c) of FIG. 104. On the other hand, when the cartridge is to be mounted to the image forming apparatus main assembly, the positioning boss 180i enters the inside of the drum coupling and comes into contact with the engaging portion 243i as shown in part (a) of FIG. 106. Further, when the positioning boss 180i enters the inside of the drum coupling 143, the engaging portion 243i is pushed outward in the radial direction by the positioning boss 180i. By this, as shown in part (b) of FIG. 104 and part (d) of FIG. 104, a part of the engaging portion 243i advances toward the outside of the drum coupling 143.

In this state, both side portions of the engaging portion 243i, that is, the driving force receiving portion 243b and the braking force receiving portion 243c are exposed, and the driving force and the braking force can be received from the image forming apparatus main assembly, respectively.

As described above, the arrangement relationship and shape of the coupling 143 are not constant and may vary or change. For example, it is conceivable the when the cartridge is not in use, the drum coupling portion which is vulnerable to external impact is retracted to be protected.

When a portion of the coupling 143 is movable, the state in which the coupling is actually used, that is, The state of the coupling 143 when the cartridge and the drum unit are mounted to the image forming apparatus main assembly and the coupling 143 engages with the drive transmission unit 203 may be regarded as a reference state, the shape of the coupling 143 and the arrangement relationship of each portion may be structured to satisfy the desired conditions as described above, in such a reference state.

Further, FIGS. 107 and 108 show another modified example of the drum coupling 143 structured so that a part of the drum coupling 143 is deformed and moved. In the above described modified example (see FIG. 105), the engaging portion 243i is structured to move in the radial direction, but in this modified example, the engaging portion 643i is structured to move in the axial direction. Part (a) of FIG. 107 shows a state in which the engaging portion 643i is retracted inside the drum coupling, and part (b) of FIG. 107 shows the engaging portion 643i moving toward the outside of the drum coupling and away from the photosensitive drum. Part (c) of FIG. 107 is an exploded perspective view of the drum unit in this modified example.

Part (a) of FIGS. 108 and 108 (b) show sectional views of the drum unit. Part (a) of FIG. 108 shows a state before the drum unit is mounted to the apparatus main assembly, and part (b) of FIG. 108 shows a state after the drum unit is mounted thereto.

When the drum unit is mounted to the main assembly of the apparatus, the positioning boss 180i provided on the drive transmission unit comes into contact with the working member of the drum coupling. Then, as shown in part (b) of FIG. 108, the operating member 698 moves inward in the axial direction (on the right side in the drawing). As the operating member 698 moves, the interlocking member 698 is pushed outward in the radial direction inside the drum coupling. As the interlocking member 698 moves outward in the radial direction, the engaging portion 643i is pressed outward in the radial direction by the interlocking member 698. As a result, the state is changed to the engaging portion 643i being partly exposed to the outside (part (b) of FIGS. 107 and 108 (b)) from the state of being retracted inside the drum unit (part (a) of FIG. 107 and part (a) of FIG. 108).

When a part of the drum coupling is movably provided in this manner, the moving direction may be the radial direction or the axial direction. A part of the drum coupling may move in both the radial direction and the axial direction, or may move in the rotational direction.

Next, referring to Figures and 110 another modification of the drum coupling will be described. Similarly to the above two modifications, the drum coupling 1043 of this modification is also structured so that a part thereof is deformed and moved.

Part (a) of FIG. 109 is an exploded perspective view of the drum unit of this modified example. Part (b) of FIG. 109 shows a state in which the engaging portion 1043i of the drum coupling has advanced toward the outside of the drum unit, and part (c) shows a state in which the engaging portion 1043i is partially retracted toward the inside.

In this modification, the engaging portion 1043i is in a projected (advanced) state as shown in part (b) of FIG. 109 before the drum unit is mounted on the apparatus main assembly. On the other hand, after the drum unit is mounted to the main assembly of the apparatus, the engaging portion 1043i changes to the retracted state as shown in part (c) of FIG. 109.

Part (a) of FIG. 110 and part (b) of FIG. 110 show sectional views of the drum unit. FIG. 110 (A) shows the

81

state before the drum unit is completely mounted on the apparatus main assembly, and part (b) shows the state after the mounting is completed.

As shown in part (a) of FIG. 109, the engaging member 1043 is provided inside the drum coupling so as to be movable in the axial direction. The engaging member 1043 is urged (pressed) to the outside in the axial direction by the pressing coil spring 1020 provided inside the drum coupling 143, and the engaging portion 1043i, which is a part of the engaging member 1043, is exposed to the outside of the drum coupling 143.

Then, the engaging member 1043 has an acting portion 1043p on its rotation axis. When the drum unit is mounted to the main assembly of the apparatus as shown in part (b) of FIG. 110, the engaging member 1043 and the engaging portion 1043i are retracted inward in the axial direction by the acting portion 1043p being pushed by the positioning boss 180i.

In the above three modified examples, an acting portion capable of receiving an action from the outside of the cartridge is provided inside the coupling 143, and this acting portion is operated by the positioning boss 180i to change the shape of the coupling 143. However, it is also conceivable to dispose an acting portion for changing the shape of the coupling 143 at a place other than the inside of the coupling 143.

As described above, the shape and pattern of the coupling can be selected depending on the design reason for arrangement, the manufacturing reason considering the mold for coupling production, and the purpose of protecting the coupling.

Further, in each of the three modified examples of the drum coupling described above, the engaging portion provided with the driving force receiving portion and the braking force receiving portion move relative to other portions. However, a portion such as a helical slope or a visor portion may be movable relative to the other portions.

Further, the cartridge 100 described above includes a photosensitive drum and a developing roller, but the structure of the cartridge 100 is not limited to such a structure. For example, the cartridge 100 may include a photosensitive drum but no developing roller. As an example of such a structure, a structure in which the cartridge 100 includes only the drum holding unit 108 (see FIG. 19) can be considered.

Further, in the Embodiment 1 and various modified examples thereof, the drum coupling 143 is placed in the neighborhood of one end (the end on the driving side) of the photosensitive drum 104, and it is press-fitted into the photosensitive drum 104. As a result, the driving force can be transmitted from the drum coupling 143 to the end of the photosensitive drum 104. However, the method of connecting the drum coupling 143 and the photosensitive drum 104 is not limited to press-fitting. Further, in the above described example, the drum coupling 143 and the photosensitive drum 104 are integrated to form the drum unit 103, but the drum coupling 143 and the photosensitive drum 104 may be separated from each other without constituting a drum unit.

That is, if the drum coupling 143 is operatively connected to the photosensitive drum 104, that is, if it is connected in a drive-transmittable manner, another connection method can be employed, and the coupling 143 and the photosensitive drum 104 may not constitute the same unit.

For example, one or more relay members may be interposed between the coupling 143 and the photosensitive drum 104. In such a case, it can be deemed that the drum coupling is indirectly connected to the driving side end of

82

the photosensitive drum 104 by way of the relay member. The drum coupling 143 operates the photosensitive drum 104 by way of the relay member by rotating itself.

For example, it is conceivable to mount a gear to the end of the photosensitive drum 104 and to form a gear portion on the outer peripheral surface of the drum coupling 143 as well. In this manner, the gear of the coupling 143 and the gear of the photosensitive drum 104 can be directly meshed with each other, or another idler gear can be interposed between the two gears to transmit the driving force to the photosensitive drum 104 from the drum coupling 143.

In addition to using the gear as a relay member, a method of connecting a drive transmission belt to the drum coupling 143 and the photosensitive drum 104 to use it as the relay member is also conceivable.

It is also conceivable to connect the end of the photosensitive drum 104 on the driving side and the drum coupling 143 by using an old dam coupling as a relay member. In this case, the drum unit 103 can be regarded as a unit including the photosensitive drum 104, the Oldham coupling (relay member), and the drum coupling 143.

As described above, the connection method between the photosensitive drum 104 and the drum coupling 143 may be a direct connection or an indirect connection. Further, the photosensitive drum 104 and the drum coupling 143 may be unitized to form the drum unit 103, or the photosensitive drum 104 and the drum coupling 143 may be provided apart from each other in the cartridge and may not constitute a unit.

However, if the coupling 143 and the photosensitive drum 104 form a drum unit 103 that can rotate integrally, or if the coupling 143 is directly connected to the end of the photosensitive drum 104, The driving (rotating) of the coupling 143 can be more accurately transmitted to the photosensitive drum 104. And therefore, doing so is further preferable.

In this embodiment, the axes of the drum coupling 143 and the photosensitive drum 104 are aligned. That is, the drum coupling 143 and the photosensitive drum 104 are aligned along the same rotation axis L (see FIG. 1). However, when the drum coupling 143 and the photosensitive drum 104 are indirectly connected, the positions of the axes may be different from each other.

In any case, the cartridge can be stably driven by engaging the coupling 143 with the drive transmission unit 203 provided in the main assembly of the apparatus.

An example in which the structure of the cartridge or the like is changed will be further described with reference to the Embodiment 2 in the following.

Embodiment 2

<Overall Structure of Image Forming Apparatus 800>

Referring to FIG. 82, the overall structure of the electrophotographic image forming apparatus 800 (hereinafter, image forming apparatus 800) according to this embodiment will be described. FIG. 82 is a schematic view of the image forming apparatus 800 according to this embodiment. In this embodiment, the process cartridge 701 and the toner cartridge 713 are mountable to and dismountable from the main assembly of the image forming apparatus 800.

In this embodiment, the structures and operations of the first to fourth image forming portions are substantially the same except that the colors of the formed images are different. Therefore, in the following, if no particular distinction is required, the subscripts Y to K will be omitted for general explanation.

The first to fourth process cartridges **701** are arranged side by side in the horizontal direction. Each process cartridge **701** includes a cleaning unit **704** and a developing unit **706**. The cleaning unit **704** includes a photosensitive drum **707** as an image bearing member, a charging roller **708** as a charging means for uniformly charging the surface of the photosensitive drum **707**, and a cleaning blade **710** as a cleaning means. The developing unit **706** includes a developing roller **711** and accommodates a developer T (hereinafter, toner), and includes a developing means for developing an electrostatic latent image on the photosensitive drum **707**. The cleaning unit **704** and the developing unit **706** are supported so as to be swingable relative to each other. The first process cartridge **701Y** contains yellow (Y) toner in the developing unit **706**. Similarly, the second process cartridge **701M** contains magenta (M) toner, the third process cartridge **701C** contains cyan (C) toner, and the fourth process cartridge **701K** contains black (K) toner.

The process cartridge **701** can be mounted to and dismounted from the image forming apparatus **800** by way of mounting means such as a mounting guide and a positioning member provided on the image forming apparatus **800**. Further, a scanner unit **712** for forming an electrostatic latent image is provided below the process cartridge **701**. Further, in the image forming apparatus **800**, the waste toner feeding unit **723** is provided behind the process cartridge **701** (downstream in the mounting/dismounting direction of the process cartridge **701**).

The first to fourth toner cartridges **713** are arranged horizontally below the process cartridge **701** in an order corresponding to the color of the toner contained in the respective process cartridges **701**. That is, the first toner cartridge **713Y** contains the yellow (Y) toner, similarly, the second toner cartridge **713M** contains the magenta (M) toner, the third toner cartridge **713C** contains the cyan (C) toner, and the fourth Toner cartridge **713K** contains the black (K) toner. Each toner cartridge **713** replenishes the process cartridge **701** containing the toner of the same color.

The replenishment operation of the toner cartridge **713** is carried out when a remaining amount detecting portion provided in the main assembly of the image forming apparatus **800** detects insufficient remaining amount of toner in the process cartridge **701**. The toner cartridge **713** can be mounted to and dismounted from the image forming apparatus **800** by way of mounting means such as a mounting guide and a positioning member provided in the image forming apparatus **800**. A detailed description of the process cartridge **701** and the toner cartridge **713** will be described hereinafter.

Below the toner cartridge **713**, first to fourth toner feeding devices **714** are arranged corresponding to each toner cartridge **713**. Each toner feeding device **714** transports the toner received from each toner cartridge **713** upward, and supplies the toner to each developing unit **706**.

An intermediary transfer unit **719** as an intermediary transfer member is provided above the process cartridge **701**. The intermediary transfer unit **719** is arranged substantially horizontally with the primary transfer unit (S1) side facing down. The intermediary transfer belt **718** facing each photosensitive drum **707** is a rotatable endless belt, which is stretched on a plurality of tension rollers. On the inner surface of the intermediary transfer belt **718**, a primary transfer roller **720** is provided as a primary transfer member at a position where the corresponding photosensitive drum **707** and primary transfer portion S1 are provided by way of the intermediary transfer belt **718**. Further, the secondary transfer roller **721**, which is a secondary transfer member,

contacts with the intermediary transfer belt **718**, and forms a secondary transfer portion S2 in cooperation with a roller on the opposite side by way of the intermediary transfer belt **718**. Further, in the left-right direction (the direction in which the secondary transfer portion S2 and the intermediary transfer belt are extended), the intermediary transfer belt cleaning unit **722** is provided on the side opposite to the secondary transfer portion S2.

A fixing unit **725** is provided above the intermediary transfer unit **719**. The fixing unit comprises a heating unit **726** and a pressure roller **727** which is press-contacted with the heating unit **726**. A discharge tray **732** is provided on the upper surface of the main assembly of the apparatus, and a waste toner collection container **724** is provided between the discharge tray **732** and the intermediary transfer unit **719**. Further, a sheet feed tray **702** for accommodating the recording material **703** is provided at the lowermost portion of the main assembly of the apparatus.

The recording material **703** is for receiving and being subjected to a toner image fixing operation on the surface thereof by the apparatus main assembly, and an example of the recording material **703** is paper.

<Image Forming Process>

Next, referring to FIGS. **82** and **83**, the image forming operation in the image forming apparatus **800** will be described.

During the image forming operation, the photosensitive drum **707** is rotationally driven at a predetermined speed in the direction of arrow A in FIG. **83**. The intermediary transfer belt **718** is rotationally driven in the direction of arrow B in FIG. **82** (forward with respect to the direction of rotation of the photosensitive drum **707**).

First, the surface of the photosensitive drum **707** is uniformly charged by the charging roller **708**. Then, the surface of the photosensitive drum **707** is scanned while being exposed to the laser beam emitted from the scanner unit **712**, so that an electrostatic latent image based on the image information is formed on the photosensitive drum **707**. The electrostatic latent image formed on the photosensitive drum **707** is developed into a toner image by the developing unit **706**. At this time, the developing unit **706** is pressed by a development pressure unit (not shown) provided in the main assembly of the image forming apparatus **800**. Then, the toner image formed on the photosensitive drum **707** is primarily transferred onto the intermediary transfer belt **718** by the primary transfer roller **720**.

For example, when forming a full-color image, the above-mentioned processes are sequentially performed in the image forming portions S701Y to S701K, which are the primary transfer units **1** to **4**, so that the toner images of respective colors are sequentially superimposed on the intermediary transfer belt **718**.

On the other hand, the recording material **703** stored in the sheet feed tray **702** is fed at a predetermined control timing, and is fed to the secondary transfer unit S702 in synchronization with the movement of the intermediary transfer belt **718**. Then, the four color toner images on the intermediary transfer belt **718** are collectively secondarily transferred onto the recording material **703** by the secondary transfer roller **721** which is in contact with the intermediary transfer belt **718** by way of the recording material **703**.

Thereafter, the recording material **703** now carrying the transferred toner image is fed to the fixing unit **725**. The toner image is fixed on the recording material **703** by heating and pressing the recording material **703** in the fixing unit **725**. After that, the recording material **703** is fed to the discharge tray **732** to complete the image forming operation.

Further, the primary untransferred residual toner (waste toner) remaining on the photosensitive drum **707** after the primary transfer step is removed by the cleaning blade **710**. The secondary untransferred residual toner (waste toner) remaining on the intermediary transfer belt after the secondary transfer step is removed by the intermediary transfer belt cleaning unit **722**. The waste toner removed by the cleaning blade **710** and the intermediary transfer belt cleaning unit **722** is fed by the waste toner feeding unit **723** provided in the main assembly of the apparatus and accumulated in the waste toner collection container **724**. The image forming apparatus **800** can also form a monochromatic or multicolored image by using only a desired single or several image forming portions.

<Process Cartridge>

Next, referring to FIGS. **83**, **84** and **85**, the overall structure of the process cartridge **701** mounted to the image forming apparatus **800** according to this embodiment will be described. FIG. **83** is a schematic sectional view of the process cartridge mounted on the image forming apparatus **800** and in a state (attitude) in which the photosensitive drum **707** and the developing roller **711** are in contact with each other, as viewed in the Z direction. FIG. **84** is a perspective view of the process cartridge **701** as viewed from the front (upstream side in the process cartridge mounting/dismounting direction). FIG. **85** is a perspective view of the process cartridge **701** as viewed from the rear (downstream side in the process cartridge mounting/dismounting direction).

The process cartridge **701** comprises the cleaning unit **704** and the developing unit **706**. The cleaning unit **704** and the developing unit **706** are swingably coupled around the rotation support pin **730**.

The cleaning unit **704** includes a cleaning frame **705** which supports various members in the cleaning unit **704**. Further, in the cleaning unit **704**, in addition to the photosensitive drum **707**, the charging roller **708**, and the cleaning blade **710**, a waste toner screw **715** extending in a direction parallel to the rotation axis direction of the photosensitive drum are provided. The cleaning frame **705** includes a cleaning bearing unit **733** which rotatably supports the photosensitive drum **707** and which includes a cleaning gear train **731** for transmitting driving force from the photosensitive drum **707** to the waste toner screw **715**, at both ends of the length.

The charging roller **708** provided in the cleaning unit **704** is urged toward the photosensitive drum **707** by a charging roller pressing springs **736** provided at both ends in the direction of arrow C. The charging roller **708** is provided so as to be driven by the photosensitive drum **707**, and when the photosensitive drum **707** is rotationally driven in the direction of arrow A during image formation, the charging roller **708** is rotated in the direction of arrow D (forward with respect to the rotation of the photosensitive drum **707**).

The cleaning blade **710** provided in the cleaning unit **704** comprises an elastic member **710a** for removing untransferred residual toner (waste toner) remaining on the surface of the photosensitive drum **707** after the primary transfer, and a support member **710b** for supporting the elastic member **710a**. The waste toner removed from the surface of the photosensitive drum **707** by the cleaning blade **710** is stored in the waste toner storage chamber **709** formed by the cleaning blade **710** and the cleaning frame **705**. The waste toner stored in the waste toner storage chamber **709** is fed toward the rear of the image forming apparatus **800** (downstream in the mounting/dismounting direction of the process cartridge **701**) by a waste toner feeding screw **715** provided in the waste toner storage chamber **709**. The fed waste toner

is discharged through a waste toner discharge portion **735** and is delivered to the waste toner feeding unit **723** of the image forming apparatus **800**.

The developing unit **706** includes a development frame **716** which supports various members in the developing unit **706**. The development frame **716** is divided into a developing chamber **716a** in which a developing roller **711** and a supply roller **717** are provided therein, and a toner storage chamber **716b** in which a toner is accommodated and in which a stirring member is provided.

In the developing chamber **716a**, the developing roller **711**, the supply roller **717**, and a developing blade **728** are provided. The developing roller **711** carries the toner, rotates in the direction of arrow E during image formation, and supplies the toner to the photosensitive drum **707** by contacting the photosensitive drum **707**. Further, the developing roller **711** is rotatably supported by the development frame **716** by way of the development bearing unit **734** at both ends in the longitudinal direction (rotational axis direction). The supply roller **717** is rotatably supported by the development frame **716** by way of the development bearing unit **734** while being in contact with the developing roller **711**, and rotates in the direction of arrow F during image forming operation. Further, a developing blade as a layer thickness regulating member which regulates the thickness of the toner layer formed on the developing roller **711** is provided so as to contact the surface of the developing roller **711**.

The toner storage chamber **716b** is provided therein with the stirring member **729** for stirring the accommodated toner T and for transporting the toner to the supply roller **717** through the developing chamber communication opening **716c**. The stirring member **729** is provided with a rotating shaft **729a** extending parallel to the rotation axis direction of the developing roller **711**, and a stirring sheet **729b** as a feeding member which is a flexible sheet. One end of the stirring sheet **729b** is mounted to the rotating shaft **729a**, and the other end of the stirring sheet **729b** is a free end, and The rotating shaft **729a** rotates and therefore the stirring sheet **729b** rotates in the direction of arrow G. By which the stirring sheet **729b** stirs the toner.

The developing unit **706** includes a developing chamber communication opening **716c** which communicates the developing chamber **716a** and the toner storage chamber **716b** with each other. In this embodiment, the developing chamber **716a** is placed above the toner storage chamber **716b** in the attitude in which the developing unit **706** is normally used (the attitude at the time of use). The toner in the toner storage chamber **716b** thrown up by the stirring member **729** is supplied to the developing chamber **716a** through the developing chamber communication opening **716c**.

Further, the developing unit **706** is provided with a toner receiving opening **740** at one end on the downstream side in the mounting/dismounting direction. Above the toner inlet **740**, an inlet seal member **745** and a toner inlet shutter **741** which can move in the front-rear direction are provided. The toner inlet **740** is closed by the inlet shutter **741** when the process cartridge **701** is not mounted to the image forming apparatus **800**. The reception shutter **741** is structured to be urged and opened by the image forming apparatus **800** in interrelation with the mounting/dismounting operation of the process cartridge **701**.

A receiving and feeding path **742** is provided so as to communicate with the toner receiving opening **740**, and a receiving and feeding screw **743** is provided therein. Further, a storage chamber communication opening **744** for supplying toner to the toner storage chamber **716b** is provided in

87

the neighborhood of the center of the length of the developing unit **706**, and communicates the receiving and feeding path **742** and the toner storage chamber **716b** with each other. The receiving and feeding screw extends in a direction parallel to the rotation axis directions of the developing roller and the supply roller **717**, and feeds the toner received from the toner receiving opening **740** to the toner storage chamber **716b** by way of the storage chamber communication opening **744**.

<Cleaning Unit>

Here, referring to FIG. **86**, the cleaning unit **704** will be described in detail.

As shown in FIG. **84**, the rotation axis direction of the photosensitive drum **707** is the Z direction (arrow **Z1**, arrow **Z2**), the horizontal direction in FIG. **82** is the X direction (arrow **X1**, arrow **X2**), and the vertical direction is the Y direction (arrow **Y1**, arrow **Y2**).

The side (**Z1** direction) on which the drum coupling (coupling member) **770** receives the driving force from the image forming apparatus main assembly is referred to as the driving side (back side), and the opposite side (**Z2** direction) is called the non-driving side (front side). At the end opposite to the drum coupling **770**, there is provided an electrode (electrode portion) which contacts the inner surface of the photosensitive drum **707**, to function as a ground by contacting the image forming apparatus main assembly.

A drum coupling **770** is mounted to one end of the photosensitive drum **707**, and a non-driving side flange member **769** is mounted to the other end to form the photosensitive drum unit **768**. The photosensitive drum unit **768** receives the driving force from a drive transmission unit **811** provided in the image forming apparatus main assembly **800** by way of the drum coupling **770**.

In the drum coupling **770**, the outer peripheral surface **771a** of the cylindrical portion **771** projecting from the photosensitive drum **707** as a supported portion is rotatably supported by the drum unit bearing member **733R**. Similarly, the non-driving side flange member **769** is rotatably supported by the drum unit bearing member **733L** at the outer peripheral surface **769a** of the cylindrical portion projecting from the photosensitive drum **707**. That is, the photosensitive drum **707** is rotatably supported by the casing of the cartridge (bearing members **733R**, **733L**) by way of the coupling **770** and the flange member **769**.

As shown in FIG. **86**, the drum unit bearing member **733R** abuts on the rear cartridge positioning portion **808** provided in the image forming apparatus main assembly **800**. Further, the drum unit bearing member **733L** abuts on the front cartridge positioning portion **810** of the image forming apparatus main assembly **800**. By this, the process cartridge **701** is positioned in the image forming apparatus **800**.

In the Z direction of this embodiment, the position where the drum unit bearing member **733R** supports the photosensitive drum unit **768** is close to the position where the drum unit bearing member **733R** is positioned by the back side cartridge positioning portion **808**. Therefore, in this embodiment, the free end side (**Z1** direction side) of the outer peripheral surface **771a** of the cylindrical portion **771** of the drum coupling is rotatably supported by the drum unit bearing member **733R**.

Similarly, in the Z direction, the position where the drum unit bearing member **733L** rotatably supports the non-driving side flange member **769** is close to the position where the drum unit bearing member **733L** is positioned by the front side cartridge positioning portion **810**.

88

By mounting the drum unit bearing members **733R** and **733L** to the respective sides of the cleaning frame **705**, the photosensitive drum unit **768** is rotatably supported by the cleaning frame **705**.

<Structure of Drive Transmission Unit>

Referring to FIGS. **87** and **88**, the structure of the drive transmission unit **811** provided in the image forming apparatus side will be described. FIG. **87** is an exploded perspective view of the drive transmission unit **811**. FIG. **88** is a sectional view of the drive transmission unit **811**.

A drum drive coupling gear **813** is rotatably supported by a supporting shaft **812** fixed to the frame of the image forming apparatus **800**, and the driving force is transmitted from the motor to rotate the drum drive coupling gear **813**. As is difference from the structure of the Embodiment 1, the drum drive coupling and the drive gear are integrated with each other in this embodiment. By integrating, the misalignment between the driving shaft axis on the main assembly side and the photosensitive drum shaft axis on the cartridge side is suppressed.

The drive transmission unit **811** includes a plurality of components inside a cylindrical portion of the drum drive coupling gear **813**. They are a brake member **816** which is supported and stopped in the rotation by a supporting shaft **812**, a brake transmission member **817** which is connected with the brake member **816** to transmit the braking force, a first and second braking engagement members **814**, **818** which engage with the braking force receiving surface of the drum coupling **770**, a brake engagement spring **821** and a drum drive coupling spring **820** which are extended along a axis **M1** and generate an urging force in the direction of the axis **M1**. The axis **M1** is the rotation axis of the drive transmission unit **811**.

The drum drive coupling spring **820** is provided so as to be sandwiched between the end surface of the brake member **816** and the brake transmission member **817**, and imparts a repulsive force to them. The brake transmission member **817** receives the repulsive force of the drum drive coupling spring **820** while receiving the repulsive force of the brake engagement spring **821** by way of the first braking engagement member **814**. As is different from the structure of the Embodiment 1, the stopper **815** is provided in this embodiment. The stopper **815** is assembled to the drum drive coupling gear **813**, and is fixed so as to move integrally with the drum drive coupling gear **813** in the axial direction. This prevents the drum coupling **770** from colliding with the first braking engagement member **814** and prevents the first braking engagement member **814** from disengaging out of the drum drive coupling gear **813** when the user mounts the cartridge with a strong force.

The other structures and functions are the same as those of the main assembly side drive transmission unit **203** shown in the Embodiment 1. And therefore the description thereof is omitted in this embodiment.

<Structure of Coupling Member>

The description will be made as to a structure for transmitting a driving force from the image forming apparatus main assembly to the drum unit **768** of the cartridge **701** to drive (rotate) the drum unit **768**.

The drum unit **768** shown in part (a) of FIG. **89** to part (c) of FIG. **89** is a unit including a photosensitive drum **707**, a drum coupling **770**, and a non-driving side flange member **769**. The drum unit **768** is structured to be connected to the drive transmission unit **811** provided in the main assembly by being mounted to the main assembly of the image forming apparatus.

During image formation, the drum unit **768** rotates in the direction of arrow A. In this embodiment, as the drum unit **768** is viewed from the driving side (the side where the drum coupling **770** is located), the rotational direction corresponds to the counterclockwise direction. That is, the rotational directions of the drum units of this embodiment and the Embodiment 1 are opposite to each other.

Therefore, the shape of the drum coupling **770** which engages with the drive transmission unit is a shape inverted (mirror shape) in the left-right with respect to the drum coupling **143** shown in the Embodiment 1. Similarly, the shape of the drive transmission unit **811** is also a left-right inverted shape of the drive transmission unit **203** in the Embodiment 1.

Referring to FIG. **83**, the rotational direction of the drum unit **768** of this embodiment will be described. FIG. **83** corresponds to a view of the drum unit as seen from the non-driving side. And therefore, the rotational direction A corresponds to the clockwise direction. When the drum unit is rotated in the A direction by the driving force received by the coupling member, the surface of the photosensitive drum **707** is structured to move as follows. The surface of the photosensitive drum **707** approaches to and contacts with the cleaning blade **710** inside the casing of the cartridge. Thereafter, the surface of the photosensitive drum **707** approaches to and contacts with the charging roller **708**. After that, the surface of the photosensitive drum **707** approaches to and contacts with the developing roller **711**. The surface of the photosensitive drum **707** is then exposed out of the casing of the cartridge above the cartridge. The surface of the exposed photosensitive drum **707** comes into contact with the intermediary transfer belt **718** of the main assembly of the apparatus (see FIG. **82**). Thereafter, the surface of the photosensitive drum **707** returns to the inside of the casing of the cartridge again and approaches to and contacts with the cleaning blade **710**.

Next, the drum coupling **770** will be described in detail. part (a) of FIG. **89** to part (c) of FIG. **89** are illustrations for explaining the detailed shape of the drum coupling **770**. Part (a) of FIG. **89** is a perspective view of the drum unit **768**, part (b) of FIG. **89** is a perspective view of another phase of part (a) of FIG. **89**, and part (c) of FIG. **89** is a front view of the drum unit **768** as viewed from the Z1 direction. The drum coupling **770** includes a positioning hole **770a**, a driving force receiving portion **770b**, a braking force receiving surface **770c**, a helical slope **770d**, and a visor portion **770g**.

The positioning holes **770a**, The driving force receiving portion **770b**, The braking force receiving surface **770c**, The helical slope **770d**, and the visor portion **770g** of this embodiment corresponding to the circular hole portion **143a**, the driving force receiving portion **143b**, the braking force receiving surface **143c**, the helical slope **143d**, and the visor portion **143g**, of the coupling member **143** of the Embodiment 1 shown in FIG. **1** and so on, respectively. The corresponding portions of the coupling members of this embodiment perform the same functions as in Embodiment 1.

As described above, the drum coupling **770** and the drum coupling **143** of the Embodiment 1 (see FIG. **1**) have a left-right symmetry (mirror symmetry) with each other except that the dimensions are partially different. Therefore, the shapes of the respective portions **770a**, **770b**, **770c**, **770d**, and **770g** of the drum coupling **770** are the same as those provided by substantially reversing the shapes of the respective portions **143a**, **143b**, **143c**, **143d**, and **143g** of the coupling member **143** (mirror image shapes). In this

embodiment, the drum coupling **770** rotates in the direction of arrow A shown in FIGS. **83** and **89** (a) to **89** (c) as described above. The rotational direction (arrow A direction) of the drum coupling **770** in this embodiment is a counterclockwise direction when the drum coupling **770** is viewed from the front (see part (c) of FIG. **89**).

The shape of the drum coupling **770** is not limited to this example. For example, the shape of the drum coupling **770** may have a left-right inverted shape (that is, a mirrored shape) of those of the modified example of the drum coupling **143** shown in FIG. **52**, part (b) of FIG. **54** through part (e) of FIG. **54**, FIGS. **74**, **75**, **77**, **78**, **81**, **97**, **100**, **102** to **110**, and so on.

<Mounting of Cartridge on Image Forming Apparatus Main Assembly>

Referring to FIGS. **90** and **91**, The mounting/dismounting of the process cartridge **701** relative to the image forming apparatus main assembly **800** will be described.

FIG. **90** is a perspective view illustrating mounting of the cartridge to the main assembly of the image forming apparatus. Further, FIG. **91** is a sectional view illustrating the operation of mounting the cartridge to the main assembly of the apparatus.

The image forming apparatus main assembly **800** of this embodiment employs a structure in which a cartridge can be mounted in a substantially horizontal direction. Specifically, the image forming apparatus main assembly **800** includes a space in which a cartridge can be mounted. A cartridge door **804** (front door) for inserting the cartridge into the above-mentioned space is provided on the front side (direction in which the user stands during use) of the image forming apparatus main assembly **800**.

As shown in FIG. **90**, the cartridge door **804** of the image forming apparatus main assembly **800** is provided so as to be openable and closable. When the cartridge door **804** is opened, the cartridge lower guide rail **805** which guides the cartridge **701** is provided on the bottom surface of the space, and the cartridge upper guide rail **806** is provided on the upper surface. The cartridge **701** is guided to the mounting position by the upper and lower guide rails (**805**, **806**) provided above and below the space.

Referring to Figure, The operation of mounting and dismounting the cartridge to and from the image forming apparatus main assembly **800** will be described below.

As shown in part (a) of FIG. **91**, the cleaning bearing unit **733R** and the photosensitive drum **707** in the cartridge **701** do not come into contact with the intermediary transfer belt **718** at the start of insertion. In other words, The dimensions are selected such that the photosensitive drum **707** and the intermediary transfer belt **718** do not come into contact with each other in the state that the end of the cartridge on the back side in the inserting direction is supported by the guide rail **805** under the cartridge.

Next, as shown in part (b) of FIG. **91**, the image forming apparatus main assembly **800** includes a rear side cartridge lower guide **807** projecting upward in the gravity direction from the cartridge lower guide rail **805** on the rear side in the inserting direction of the cartridge lower guide rail **805**. The rear side cartridge lower guide **807** is provided with a tapered surface **807a** on the front side in the inserting direction of the cartridge **701**. Upon insertion, the cartridge **701** rides on the tapered surface **807a** and is guided to the mounting position.

The position and shape of the back side cartridge lower guide **807** may be provided so that a portion of the cartridge does not rub against the image forming region **718A** of the intermediary transfer belt **718** when the cartridge is inserted

91

into the apparatus main assembly 800. Here, the image forming region 718A refers to a region on which the toner image transferred onto the recording material 703 of the intermediary transfer belt 718 is carried. Further, in this embodiment, among the cartridges which maintain the mounting attitude, the unit bearing member 733R provided on the back side in the inserting direction of the cartridge projects most upward in the gravity direction. Therefore, the arrangement and shape of each element may be appropriately selected such that the locus drawn by the innermost end of the drum unit bearing member 733R in the inserting direction at the time of insertion (hereinafter referred to as the insertion locus) and the image forming region 718A do not interfere with each other.

Thereafter, as shown in part (c) of FIG. 91, the cartridge 701 is further inserted into the back side of the image forming apparatus main assembly 800 from the state the cartridge 701 rides on the back side cartridge lower guide 807. Then, the drum unit bearing member 733R abuts on the rear side cartridge positioning portion 808 provided in the image forming apparatus main assembly 800. At this time, the cartridge 701 is tilted by about 0.5° to 2° with respect to the state in which the cartridge 701 is completely mounted to the image forming apparatus main assembly 800 (part (d) of FIG. 91).

Part (d) of FIG. 91 is an illustration of a state of the apparatus main assembly and the cartridge when the cartridge door 804 is closed. The image forming apparatus 800 includes a front side cartridge lower guide 809 on the front side of the cartridge lower guide rail 805 in the inserting direction. The front side cartridge lower guide 809 is structured to move up and down in interrelation with the opening and closing of the cartridge door (front door) 804.

When the cartridge door 804 is closed by the user, the front side cartridge lower guide 809 is raised. Then, the drum unit bearing member 733L and the front side cartridge positioning portion 810 of the image forming apparatus main assembly 800 come into contact with each other, and the cartridge 701 is positioned with respect to the image forming apparatus main assembly 800.

By the above-described operation, the cartridge 701 is completely mounted to the image forming apparatus main assembly 800.

Further, the removal operation of the cartridge 701 from the image forming apparatus main assembly 800 is in the reverse order in the above-mentioned insertion operation.

Since the oblique mounting structure is employed as described above, it is possible to suppress rubbing between the photosensitive drum 707 and the intermediary transfer belt when the cartridge 701 is mounted to the apparatus main assembly 800. Therefore, it is possible to suppress the occurrence of minute scratches (scratches) on the surface of the photosensitive drum 707 or on the surface of the intermediary transfer belt 718.

Further, with the structure disclosed in this embodiment, the structure of the image forming apparatus main assembly 800 can be simplified as compared with the structure in which the cartridge is horizontally moved and mounted on the apparatus main assembly and then the entire cartridge is lifted up.

<Process of Engaging Coupling Member with Main Assembly Driving Shaft>

Subsequently, referring to FIGS. 92 and 93, the engagement process between the drum coupling 770 and the drive transmission unit 811 will be described in detail. FIGS. 92 and 93 are sectional views illustrating the mounting operation of the drum coupling to the drive transmission unit 811.

92

Part (a) of FIG. 92 is a illustration of a state in which the drum coupling 770 has started engaging with the drive transmission unit 811, part (a) of FIG. 92 is a illustration of a state in which the process cartridge 701 is abutted to the back of the main assembly, and part (b) of FIG. 93 is a illustration of a state in which the front door of the main assembly is closed and the cartridge is lifted up. Part (a) of FIG. 93 is an illustration of a state in the middle of mounting/dismounting between part (b) of FIG. 93 and part (b) of FIG. 92. That is, the process cartridge 701 is mounted through the steps in the order of part (a) of FIG. 92, part (b) of FIG. 92, part (a) of FIG. 93, and part (b) of FIG. 93.

As shown in part (a) of FIG. 92, when the process cartridge is mounted to the inner side of the main assembly, the positioning hole 770a of the drum coupling 770 and the positioning boss 813i of the drum drive coupling gear 813 start to contact each other. As described referring to FIG. 91, when the drum coupling 770 starts engaging with the drive transmission unit 811, the process cartridge 701 is inserted in the state (part (b) of FIG. 91 to (c)) that it is tilted by about 0.5° to 2° by riding on the back side cartridge lower guide 807.

Therefore, the drum drive coupling gear 813 is guided by the positioning boss 813i moving along the positioning hole 770a of the drum coupling 770, and the drum drive coupling gear 813 is also tilted (see part (b) of FIG. 92). The chain lines in FIGS. 92 and 93 depict the horizontal direction by H, the rotation axis direction of the drum drive coupling gear 813 by A1, and the rotation axis direction of the drum coupling 770 by C1.

When the process cartridge is further inserted toward the back side of the main assembly from part (b) of FIG. 92, the side surface of the drum coupling 770 comes into contact with the drum drive coupling gear 813. When the cartridge is pushed further from the contact state, the drum drive coupling gear 813, the first braking engagement member 814, the second braking engagement member 818, the stopper 815 and the brake transmission member 817 are pushed toward the back side of the main assembly, until the process cartridge moves to the position where it abuts to the rear side plate of the main assembly. As a result, the process cartridge, the drum drive coupling gear 813, the first braking engagement member 814, the second braking engagement member 818, the stopper 815, and the brake transmission member 817 move to the positions shown in part (a) of FIG. 93. That is, the position of the gear end of the drum drive coupling gear 813 moves from U2 to U1.

Thereafter, when the front door of the main assembly is closed, the lower rail in the main assembly is lifted up and the inclination of the process cartridge is eliminated. That is, as shown in part (b) of FIG. 93, the inclinations of both the drum drive coupling gear 813 and the drum coupling 770 is eliminated, the axes thereof are aligned by the cooperation of the positioning boss 813i and the positioning hole 770a, and the mounting of the process cartridge 701 is completed.

After the axes of the drum drive coupling gear 813 and the drum coupling 770 are determined in the manner described above, the drive transmission unit 811 rotates so that the drum coupling 770 are brought into engagement with the drive transmission member, and the brake engaging member inside the drive transmission unit 811. The engagement operation is the same as the operation shown in the Embodiment 1 except that the rotational directions of the drive transmission unit 811 and the drum coupling 770 are reversed. Therefore, the description thereof is omitted in this embodiment.

93

In this embodiment and the above-mentioned Embodiment 1, the process cartridge includes a cleaning unit and a developing unit. That is, the process cartridge includes a photosensitive drum and a developing roller. However, the structure of the cartridge mounted to and dismounted from the image forming apparatus is not limited to such an example.

For example, as a modified example of this embodiment, a structure in which the cleaning unit **704** and the developing unit **706** are separately made into cartridges can be considered (see part (a) of FIGS. **94** and **94 (b)**).

The structure in which the cleaning unit **704** is in the form of a cartridge may be particularly referred to as a drum cartridge **704A**, and the structure in which the developing unit **706** is in the form of a cartridge may be particularly referred to as a developing cartridge **706A**.

In the case of such a modification, the drum cartridge **704A** has a photosensitive drum **707** and a drum coupling **770**. The drum cartridge **704A** can be regarded as a process cartridge including no developing unit **706**.

As described above, according to this embodiment, the drum coupling **770** of the process cartridge **701** receives the driving force from the drive transmission unit **811** of the image forming apparatus main assembly. Further, the drum coupling **770** receives a driving force from the drive transmission unit **811**, and at the same time operates the brake mechanism inside the drive transmission unit **811**. With this brake mechanism, the load required to drive the cartridge can be set in an appropriate range. By this, the process cartridge can be driven stably.

INDUSTRIAL APPLICABILITY

According to the present invention, there is provided an image forming apparatus and a cartridge and a drum unit capable of transmitting a driving force to a rotatable member of the cartridge and the drum unit.

The present invention is not limited to the above embodiments, and various modifications and modifications can be made without departing from the spirit and scope of the present invention. Therefore, the following claims are attached to make the scope of the present invention public.

This application claims priority based on Japanese Patent Application No. 2019-050355 filed on Mar. 18, 2019, and all the contents thereof are incorporated herein by reference.

The invention claimed is:

1. A cartridge comprising:

a casing having a first end portion and a second end portion opposite to the first end portion;

a photosensitive drum rotatably supported by the first end portion and the second end portion of the casing; and

a coupling operatively connected to the photosensitive drum so as to be capable of transmitting a driving force to the photosensitive drum, the coupling being positioned at the first end portion of the casing and being rotatable about an axis of the coupling,

wherein the coupling includes a guiding portion and an engaging portion,

wherein the guiding portion includes a remote portion at a position that is more remote from the second end portion of the casing than the engaging portion is from the second end portion of the casing,

wherein a distance measured from the second end portion of the casing to the remote portion of the guiding portion along a direction of the axis of the coupling decreases toward downstream in a rotational direction of the coupling,

94

wherein the engaging portion includes a first side surface at a position upstream in the rotational direction of the coupling and a second side surface at a position downstream in the rotational direction of the coupling,

wherein at least a part of the engaging portion is more remote from the axis of the coupling along a line that is perpendicular to the axis of the coupling than the remote portion of the guiding portion is from the axis of the coupling along a line that is perpendicular to the axis of the coupling, and

wherein at least a part of the second side surface of the engaging portion (i) overhangs an open space such that an undercut area is formed in the engaging portion or (ii) has a coefficient of friction that is greater than a coefficient of friction of at least a part of the guiding portion.

2. A cartridge according to claim 1, wherein at least a part of the first side surface of the engaging portion is more remote from the axis of the coupling along a line that is perpendicular to the axis of the coupling than the remote portion of the guiding portion is from the axis of the coupling along a line that is perpendicular to the axis of the coupling.

3. A cartridge according to claim 1, wherein the coupling is configured to transmit the driving force from the first side surface of the engaging portion to the photosensitive drum.

4. A cartridge according to claim 1, wherein the coupling is provided with an opening that is coaxial with the axis of the coupling.

5. A cartridge according to claim 4, wherein the opening of the coupling and the engaging portion are positioned such that when they are projected on the axis of the coupling, an area of the opening of the coupling and an area of the engaging portion at least partly overlap with each other.

6. A cartridge according to claim 4, wherein the opening of the coupling and the guiding portion are positioned such that when they are projected on the axis of the coupling, an area of the opening of the coupling and an area of the guiding portion at least partly overlap with each other.

7. A cartridge according to claim 4, wherein the guiding portion extends in the rotational direction of the coupling around the opening.

8. A cartridge according to claim 1, wherein the guiding portion is positioned upstream of the engaging portion in the rotational direction of the coupling and adjacent to the engaging portion.

9. A cartridge according to claim 1, wherein the remote portion of the guiding portion extends from upstream to downstream toward the engaging portion in the rotational direction of the coupling.

10. A cartridge according to claim 1, wherein at least a part of the second side surface of the engaging portion overhangs open space.

11. A cartridge according to claim 1, wherein the second side surface of the engaging portion includes an elastic portion.

12. A cartridge according to claim 1, wherein the remote portion of the guiding portion has an upper portion on a side opposite from the second end portion of the casing in the direction of the axis of the coupling, and

wherein a distance measured from the second end portion of the casing to the upper portion of the guiding portion along the direction of the axis of the coupling decreases toward downstream in the rotational direction of the coupling.

95

13. A cartridge according to claim 12, wherein the upper portion of the guiding portion is connected to an upper portion of the engaging portion.

14. A cartridge according to claim 1, wherein the coupling includes a visor portion projecting outwardly in a direction perpendicular to the axis of the coupling so as to cover a space downstream of the engaging portion in the rotational direction of the coupling.

15. A cartridge according to claim 1, wherein the coupling includes a visor portion projecting outwardly in a direction perpendicular to the axis of the coupling, the visor portion being upstream of the guiding portion in the rotational direction of the coupling and adjacent to the guiding portion.

16. A cartridge according to claim 1, wherein the coupling includes a first coupling portion and a second coupling portion,

wherein the first coupling portion includes the guiding portion and the engaging portion,

wherein the second coupling portion includes a guiding portion and an engaging portion,

wherein the guiding portion of the second coupling portion includes a remote portion at a position that is more remote from the second end portion of the casing than the engaging portion of the second coupling portion is from the second end portion of the casing,

wherein a distance measured from the second end portion of the casing to the remote portion of the guiding portion of the second coupling portion along the direction of the axis of the coupling decreases toward downstream in the rotational direction of the coupling,

wherein the engaging portion of the second coupling portion includes a first side surface at a position upstream in the rotational direction of the coupling and a second side surface at a position downstream in the rotational direction of the coupling, and

wherein at least a part of the engaging portion of the second coupling portion is more remote from the axis of the coupling along a line that is perpendicular to the axis of the coupling than the remote portion of the guiding portion of the second coupling portion is from the axis of the coupling along a line that is perpendicular to the axis of the coupling, and

wherein at least a part of the second side surface of the second coupling portion (i) overhangs an open space such that an undercut area is formed in the engaging portion of the second coupling portion or (ii) has a coefficient of friction that is greater than a coefficient of friction of at least a part of the guiding portion of the second coupling portion.

17. A cartridge according to claim 16, wherein the coupling further comprises a visor portion that includes a part that covers a space between the first coupling portion and the second coupling portion in the rotational direction of the coupling, with the visor portion projecting outwardly in a direction perpendicular to the axis of the coupling.

18. A cartridge according to claim 1, wherein the remote portion of the guiding portion includes an inclined portion, and

wherein a distance measured from the second end portion of the casing to the inclined portion of the guiding portion along the direction of the axis of the coupling decreases toward downstream in the rotational direction of the coupling.

19. A cartridge according to claim 18, wherein the inclined portion of the guiding portion includes a helical inclined surface.

96

20. A cartridge according to claim 18, wherein the inclined portion of the guiding portion includes a plurality of surfaces.

21. A cartridge according to claim 18, wherein the inclined portion of the guiding portion includes a step between a first surface of the guiding portion and a second surface of the guiding portion.

22. A cartridge according to claim 1, wherein, as viewed along the axis of the coupling from a position where the coupling is in front of the photosensitive drum, the rotational direction of the coupling is clockwise.

23. A cartridge according to claim 1, wherein the cartridge further includes:

toner accommodated in the casing,

a charging roller configured to charge the photosensitive drum, and

a developing roller configured to develop a latent image formed on a surface of the photosensitive drum with the toner, and

wherein, when the coupling rotates in the rotational direction, a point on the surface of the photosensitive drum moves inside of the casing from a position adjacent to the charging roller to a position adjacent to the developing roller, and then the point on the surface of the photosensitive drum moves to outside of the casing, and thereafter the point on the surface of the photosensitive drum returns into the casing to the position adjacent to the charging roller.

24. A cartridge according to claim 1, wherein the coupling is directly connected to the photosensitive drum.

25. A cartridge according to claim 1, wherein the coupling includes an inclination portion, and

wherein a distance measured from the second end portion of the casing to the inclination portion along the direction of the axis of the coupling increases toward downstream in the rotational direction of the coupling.

26. A cartridge according to claim 25, wherein the inclination portion is positioned upstream of the guiding portion in the rotational direction of the coupling and the inclination portion is positioned adjacent to the guiding portion.

27. A cartridge comprising:

a casing;

a photosensitive drum supported by the casing, the photosensitive drum being rotatable in a rotational direction about a rotational axis; and

a coupling operatively connected to the photosensitive drum so as to be capable of transmitting a driving force to the photosensitive drum, the coupling being rotatable in the rotational direction about the rotational axis, the coupling including:

a first projecting surface that extends away from the photosensitive drum in a direction of the rotational axis,

a second projecting surface that is downstream of the first projecting surface in the rotational direction, the second projecting surface extending in a direction that is (i) away from the photosensitive drum and (ii) inclined relative to the direction that the first projecting surface extends away from the photosensitive drum, and

a sloped surface extending about the rotational axis between a position upstream of the first projecting surface in the rotational direction to a position adjacent to the second projecting surface,

wherein a distance in the direction of the rotational axis from an end of the photosensitive drum to the sloped surface decreases along the sloped surface in the rotational direction, and

97

wherein a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to at least a part of the sloped surface that is upstream of the first projecting surface is less than a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to a part of the second projecting surface that is farthest from the rotational axis.

28. A cartridge according to claim 27, wherein a first end of the second projecting surface is positioned downstream in the rotational direction from a second end of the second projecting surface.

29. A cartridge according to claim 27, wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, a part of the sloped surface covers the second projecting surface.

30. A cartridge according to claim 27, further comprising an opening formed in the coupling about the rotational axis, wherein a distance along a line that is perpendicular to the rotational axis from the rotational axis to a surface forming the opening is less than a distance along a line that is perpendicular to the rotational axis from the rotational axis to the first projecting surface.

31. A cartridge according to claim 27, wherein an undercut area is formed in the coupling adjacent to the second projecting surface.

32. A cartridge according to claim 27, wherein the coupling includes a surface projecting outwardly in a direction perpendicular to the rotational axis so as to cover a space downstream of the second projecting surface in the rotational direction.

33. A cartridge according to claim 27, wherein the coupling includes a surface projecting outwardly in a direction perpendicular to the rotational axis so as to cover a space upstream of the sloped surface in the rotational direction.

34. A cartridge according to claim 27, wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, the rotational direction is clockwise.

35. A cartridge according to claim 27, wherein the cartridge further comprises:

- toner accommodated in the casing,
- a charging roller configured to charge the photosensitive drum, and
- a developing roller configured to develop a latent image formed on a surface of the photosensitive drum with the toner, and

wherein, when the coupling rotates in the rotational direction, a point on the surface of the photosensitive drum moves inside of the casing from a position adjacent to the charging roller to a position adjacent to the developing roller, and then the point on the surface of the photosensitive drum moves to outside of the casing, and thereafter the point on the surface of the photosensitive drum returns into the casing to the position adjacent to the charging roller.

36. A cartridge according to claim 27, wherein the sloped surface is a first sloped surface, and

wherein the coupling further comprises:

- a third projecting surface that extends away from the photosensitive drum in a direction of the rotational axis,
- a fourth projecting surface that is downstream of the third projecting surface in the rotational direction, the fourth projecting surface extending in a direction (i) away from the photosensitive drum and (ii) inclined relative

98

to the direction that the third projecting surface extends away from the photosensitive drum, and

- a second sloped surface extending about the rotational axis between a position upstream of the third projecting surface in the rotational direction to a position adjacent to the fourth projecting surface,

wherein a distance from an end of the photosensitive drum to the second sloped surface in the direction of the rotational axis decreases along the second sloped surface in the rotational direction,

wherein a distance, measured along a line that is perpendicular to the rotational axis from the rotational axis to at least a part of the second sloped surface that is upstream of the third projecting surface is less than a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to a part of the fourth projecting surface that is farthest from the rotational axis, and

wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, the first projecting surface, the second projecting surface, and the first sloped surface are on an opposite side of the coupling from the third projecting surface, the fourth projecting surface, and the second sloped surface.

37. A cartridge according to claim 27, wherein at least a part of the second projecting surface has a greater coefficient of friction than a coefficient of friction of another part of the coupling.

38. A cartridge according to claim 27, wherein at least a part of the second projecting surface has a greater coefficient of friction than a coefficient of friction of at least a part of the sloped surface.

39. A cartridge comprising:

- a casing;
- a photosensitive drum supported by the casing, the photosensitive drum being rotatable in a rotational direction about a rotational axis; and

- a coupling operatively connected to the photosensitive drum so as to be capable of transmitting a driving force to the photosensitive drum, the coupling being rotatable in the rotational direction about the rotational axis, the coupling including a planar surface extending in a direction perpendicular to a direction of the rotational axis and a projection extending from the planar surface, the projection including (i) a first surface extending in a direction perpendicular to the planar surface, (ii) a second surface that extends in a direction that is inclined relative to the planar surface, and (iii) a third surface extending in the rotational direction from a position upstream of the first surface in the rotational direction to a position adjacent to the second surface,

wherein a distance in the direction of the rotational axis from an end of the photosensitive drum to the third surface decreases along the third surface in the rotational direction, and

wherein a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to at least a part of the third surface that is upstream of the first surface is less than a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to a part of the second surface that is farthest from the rotational axis.

40. A cartridge according to claim 39, wherein a tip of the first surface is a farther distance from the planar surface in

the direction of the rotational axis than a tip of the second surface is from the planar surface in the direction of the rotational axis.

41. A cartridge according to claim 40, wherein the tip of the second surface is positioned downstream in the rotational direction from an end of the second surface that is opposite to the tip.

42. A cartridge according to claim 39, wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, a part of the third surface covers the second surface.

43. A cartridge according to claim 39, further comprising an opening formed in the coupling about the rotational axis, wherein a distance along a line that is perpendicular to the rotational axis from the rotational axis to a surface forming the opening is less than a distance along a line that is perpendicular to the rotational axis from the rotational axis to the first surface.

44. A cartridge according to claim 39, wherein an undercut area is formed in the coupling adjacent to the second surface.

45. A cartridge according to claim 39, wherein the coupling includes a surface projecting outwardly in a direction perpendicular to the rotational axis so as to cover a space downstream of the second surface in the rotational direction.

46. A cartridge according to claim 39, wherein the coupling includes a surface projecting outwardly in a direction perpendicular to the rotational axis so as to cover a space upstream of the third surface in the rotational direction.

47. A cartridge according to claim 39, wherein, as viewed along the rotational direction from a position where the coupling is in front of the photosensitive drum, the rotational direction is clockwise.

48. A cartridge according to claim 39, wherein the cartridge further comprises:

- toner accommodated in the casing,
- a charging roller configured to charge the photosensitive drum, and
- a developing roller configured to develop a latent image formed on a surface of the photosensitive drum with the toner, and

wherein, when the coupling rotates in the rotational direction, a point on the surface of the photosensitive drum moves inside of the casing from a position adjacent to the charging roller to a position adjacent to the developing roller, and then the point on the surface of the photosensitive drum moves to outside of the casing, and thereafter the point on the surface of the photosensitive drum returns into the casing to the position adjacent to the charging roller.

49. A cartridge according to claim 39, wherein projection further includes (i) a fourth surface extending in a direction perpendicular to the planar surface, (ii) a fifth surface that extends in a direction that is inclined relative to the planar surface, and (iii) a sixth surface extending in the rotational direction from a position upstream of the fourth surface in the rotational direction to a position adjacent to the sixth surface,

wherein a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to at least a part of the sixth surface that is upstream of the fourth surface is less than a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to a part of the fifth surface that is farthest from the rotational axis, and

wherein, as viewed along the rotational axis from a position where the coupling is in front of the photo-

sensitive drum, the first, second, and third surfaces are on an opposite side of the coupling from the fourth, fifth, and sixth surfaces.

50. A cartridge according to claim 39, wherein at least a part of the second surface has a greater coefficient of friction than a coefficient of friction of another part of the coupling.

51. A cartridge according to claim 39, wherein at least a part of the second surface has a greater coefficient of friction than a coefficient of friction of at least part of the third surface.

52. A cartridge comprising:

- a casing;
- a photosensitive drum supported by the casing, the photosensitive drum being rotatable in a rotational direction about a rotational axis; and
- a coupling operatively connected to the photosensitive drum so as to be capable of transmitting a driving force to the photosensitive drum, the coupling being rotatable in the rotational direction about the rotational axis, the coupling including:
 - a first surface configured to receive the driving force, the first surface extending away from the photosensitive drum in a direction of the rotational axis,
 - a second surface configured to receive a braking force that opposes the driving force, the second surface being positioned downstream of the first surface in the rotational direction, the second surface extending in a direction (i) away from the photosensitive drum and (ii) inclined relative to the direction that the first surface extends away from the photosensitive drum, and
 - a third surface extending about the rotational axis between a position upstream of the first surface in the rotational direction to a position adjacent to the second surface, wherein a distance from an end of the photosensitive drum to the third surface in the direction of the rotational axis decreases along the third surface in the rotational direction, and
- wherein a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to at least a part of the third surface that is upstream of the first surface is less than a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to a part of the second surface that is farthest from the rotational axis.

53. A cartridge according to claim 52, wherein a first end of the second surface is positioned downstream in the rotational direction from a second end of the second surface.

54. A cartridge according to claim 52, wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, a part of the third surface covers the second surface.

55. A cartridge according to claim 52, further comprising an opening formed in the coupling about the rotational axis, wherein a distance along a line that is perpendicular to the rotational axis from the rotational axis to a surface forming the opening is less than a distance along a line that is perpendicular to the rotational axis from the rotational axis to the first surface.

56. A cartridge according to claim 52, wherein an undercut area is formed in the coupling adjacent to the second surface.

57. A cartridge according to claim 52, wherein the coupling includes a visor surface projecting outwardly a direction perpendicular to the rotational axis so as to cover a space downstream of the second surface in the rotational direction.

101

58. A cartridge according to claim 52, wherein the coupling includes a visor surface projecting outwardly a direction perpendicular to the rotational axis so as to cover a space upstream of the first surface in the rotational direction.

59. A cartridge according to claim 52, wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, the rotational direction is clockwise.

60. A cartridge according to claim 52, wherein the cartridge further comprises:

toner accommodated in the casing,
a charging roller configured to charge the photosensitive drum, and

a developing roller configured to develop a latent image formed on a surface of the photosensitive drum with the toner, and

wherein, when the coupling rotates in the rotational direction, a point on the surface of the photosensitive drum moves inside of the casing from a position adjacent to the charging roller to a position adjacent to the developing roller, and then the point on the surface of the photosensitive drum moves to outside of the casing, and thereafter the point on the surface of the photosensitive drum returns into the casing to the position adjacent to the charging roller.

61. A cartridge according to claim 52, wherein the coupling further includes:

a fourth surface configured to receive a driving force, the fourth surface extending away from the photosensitive drum in the direction of the rotational axis,

a fifth surface configured to receive a braking force that opposes the driving force, the fifth surface being positioned downstream of the fourth surface in the rotational direction, the fifth surface extending in a direction (i) away from the photosensitive drum and (ii) inclined relative to the direction that the fourth surface extends away from the photosensitive drum, and

a sixth surface extending about the rotational axis between a position upstream of the fourth surface in the rotational direction to a position adjacent to the fifth surface,

wherein a distance from the end of the photosensitive drum to the sixth surface in the direction of the rotational axis decreases along the sixth surface in the rotational direction,

wherein a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to at least a part of the sixth surface that is upstream of the fourth surface is less than a distance, measured along a line that is perpendicular to the rotational axis, from the rotational axis to a part of the fifth surface that is farthest from the rotational axis, and

wherein, as viewed along the rotational axis from a position where the coupling is in front of the photosensitive drum, the first, second, and third surfaces are on an opposite side of the coupling from the fourth, fifth, and sixth surfaces.

62. A cartridge according to claim 52, wherein at least a part of the second surface has a greater coefficient of friction than a coefficient of friction of another part of the coupling.

63. A cartridge according to claim 52, wherein at least a part of the second surface has a greater coefficient of friction than a coefficient of friction of at least a part of the third surface.

64. A cartridge comprising:

a casing having a first end portion and a second end portion opposite to the first end portion;

102

a photosensitive drum rotatably supported by the first end portion and the second end portion of the casing; and a coupling operatively connected to the photosensitive drum so as to be capable of transmitting a driving force to the photosensitive drum, the coupling being positioned at the first end portion of the casing and being rotatable about an axis of the coupling,

wherein the coupling includes a guiding portion and an engaging portion,

wherein the guiding portion includes a remote portion at a position that is more remote from the second end portion of the casing than the engaging portion is from the second end portion of the casing,

wherein a distance measured from the second end portion of the casing to the remote portion of the guiding portion along a direction of the axis of the coupling decreases toward downstream in a rotational direction of the coupling,

wherein the engaging portion includes a first side surface at a position upstream in the rotational direction of the coupling and a second side surface at a position downstream in the rotational direction of the coupling,

wherein at least a part of the engaging portion is more remote from the axis of the coupling along a line that is perpendicular to the axis of the coupling than the remote portion of the guiding portion is from the axis of the coupling along a line that is perpendicular to the axis of the coupling, and

wherein at least a part of the second side surface of the engaging portion (i) overhangs an open space such that an undercut area is formed in the engaging portion or (ii) has a coefficient of friction that is greater than a coefficient of friction of another part of the coupling.

65. A cartridge according to claim 64, wherein at least a part of the first side surface of the engaging portion is more remote from the axis of the coupling along a line that is perpendicular to the axis of the coupling than the remote portion of the guiding portion is from the axis of the coupling along a line that is perpendicular to the axis of the coupling.

66. A cartridge according to claim 64, wherein the coupling is configured to transmit the driving force from the first side surface of the engaging portion to the photosensitive drum.

67. A cartridge according to claim 64, wherein the coupling is provided with an opening that is coaxial with the axis of the coupling.

68. A cartridge according to claim 67, wherein the opening of the coupling and the engaging portion are positioned such that when they are projected on the axis of the coupling, an area of the opening of the coupling and an area of the engaging portion at least partly overlap with each other.

69. A cartridge according to claim 67, wherein the opening of the coupling and the guiding portion are positioned such that when they are projected on the axis of the coupling, an area of the opening of the coupling and an area of the guiding portion at least partly overlap with each other.

70. A cartridge according to claim 67, wherein the guiding portion extends in the rotational direction of the coupling around the opening.

71. A cartridge according to claim 64, wherein the guiding portion is positioned upstream of the engaging portion in the rotational direction of the coupling and adjacent to the engaging portion.

103

72. A cartridge according to claim 64, wherein the remote portion of the guiding portion extends from upstream to downstream toward the engaging portion in the rotational direction of the coupling.

73. A cartridge according to claim 64, wherein at least a part of the second side surface of the engaging portion overhangs an open space.

74. A cartridge according to claim 64, wherein the second side surface of the engaging portion includes an elastic portion.

75. A cartridge according to claim 64, wherein the remote portion of the guiding portion has an upper portion on a side opposite from the second end portion of the casing in the direction of the axis of the coupling, and

wherein a distance measured from the second end portion of the casing to the upper portion of the guiding portion along the direction of the axis of the coupling decreases toward downstream in the rotational direction of the coupling.

76. A cartridge according to claim 75, wherein the upper portion of the guiding portion is connected to an upper portion of the engaging portion.

77. A cartridge according to claim 64, wherein the coupling includes a visor portion projecting outwardly in a direction perpendicular to the axis of the coupling so as to cover a space downstream of the engaging portion in the rotational direction of the coupling.

78. A cartridge according to claim 64, wherein the coupling includes a visor portion projecting outwardly in a direction perpendicular to the axis of the coupling, the visor portion being upstream of the guiding portion in the rotational direction of the coupling and adjacent to the guiding portion.

79. A cartridge according to claim 64, wherein the coupling includes a first coupling portion and a second coupling portion,

wherein the first coupling portion includes the guiding portion and the engaging portion,

wherein the second coupling portion includes a guiding portion and an engaging portion,

wherein the guiding portion of the second coupling portion includes a remote portion at a position that is more remote from the second end portion of the casing than the engaging portion of the second coupling portion is from the second end portion of the casing,

wherein a distance measured from the second end portion of the casing to the remote portion of the guiding portion of the second coupling portion along the direction of the axis of the coupling decreases toward downstream in the rotational direction of the coupling,

wherein the engaging portion of the second coupling portion includes a first side surface at a position upstream in the rotational direction of the coupling and a second side surface at a position downstream in the rotational direction of the coupling,

wherein at least a part of the engaging portion of the second coupling portion is more remote from the axis of the coupling along a line that is perpendicular to the axis of the coupling than the remote portion of the guiding portion of the second coupling portion is from the axis of the coupling along a line that is perpendicular to the axis of the coupling, and

wherein at least a part of the second side surface of the engaging portion of the second coupling portion (i) overhangs an open space such that an undercut area is

104

formed in the engaging portion of the second coupling portion or (ii) has a coefficient of friction that is greater than a coefficient of friction of the another part of the coupling.

80. A cartridge according to claim 79, wherein the coupling further comprises a visor portion that includes a part that covers a space between the first coupling portion and the second coupling portion in the rotational direction of the coupling, with the visor portion projecting outwardly in a direction perpendicular to the axis of the coupling.

81. A cartridge according to claim 64, wherein the remote portion of the guiding portion includes an inclined portion, and

wherein a distance measured from the second end portion of the casing to the inclined portion of the guiding portion along the direction of the axis of the coupling decreases toward downstream in the rotational direction of the coupling.

82. A cartridge according to claim 81, wherein the inclined portion of the guiding portion includes a helical inclined surface.

83. A cartridge according to claim 82, wherein the inclined portion of the guiding portion includes a plurality of surfaces.

84. A cartridge according to claim 82, wherein the inclined portion of the guiding portion includes a step between a first surface of the guiding portion and a second surface of the guiding portion.

85. A cartridge according to claim 64, wherein, as viewed along the axis of the coupling from a position where the coupling is in front of the photosensitive drum, the rotational direction of the coupling is clockwise.

86. A cartridge according to claim 64, wherein the cartridge further includes:

toner accommodated in the casing,

a charging roller configured to charge the photosensitive drum, and

a developing roller configured to develop a latent image formed on a surface of the photosensitive drum with the toner, and

wherein when the coupling rotates in the rotational direction, a point on the surface of the photosensitive drum moves inside of the casing from a position adjacent to the charging roller to a position adjacent to the developing roller, and then the point on the surface of the photosensitive drum moves to outside of the casing, and thereafter the point on the surface of the photosensitive drum returns into the casing to the position adjacent to the charging roller.

87. A cartridge according to claim 64, wherein the coupling is directly connected to the photosensitive drum.

88. A cartridge according to claim 64, wherein the coupling includes an inclination portion, and

wherein a distance measured from the second end portion of the casing to the inclination portion along the direction of the axis of the coupling increases toward downstream in the rotational direction of the coupling.

89. A cartridge according to claim 88, wherein the inclination portion is positioned upstream of the guiding portion in the rotational direction of the coupling and the inclination portion is positioned adjacent to the guiding portion.

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